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Preface

Education is perhaps the essential aspect of human development as it provides an avenue to help people to grow economically and broaden their understanding of cultural and social practices in the community. Most educated persons within a society that does not value the importance of education, often leave. This leads to massive brain drain, that if left unchecked could adversely affect the social and economic viability of the entire community.

Education gives the ability to think with reason, pursue dreams and aspirations in life and live a respectable life in the society. Education gives us a definite path to follow, to lead our lives by principles and gives us the freedom of expression. It frees our minds from the prejudices and motivates it to think with logic and reason. It is essential for the overall development of the human mind and brain. The literacy rate of a country determines its prosperity and economic health.

Developmental psychology is concerned not only with describing the characteristics of psychological change over time, but also seeks to explain the principles and internal workings underlying these changes. Understanding these factors is aided by the use of models. Developmental models are often computational, but they do not necessarily need to be. A model must simply account for the means by which a process takes place. This is sometimes done in reference to changes in the brain that may correspond to changes in behaviour over the course of the development. Computational accounts of development often use either symbolic, connectionist (neural network), or dynamical systems models to explain the mechanisms of development.

Knowledge of global dynamics refers to an understanding of the world as an interconnected system of complex traits and mechanisms and unanticipated consequences. A high level of sophistication on the part of the student is required because understanding these processes is difficult to achieve due to the unanticipated effects on the human condition.

The term 'human development' may be defined as an expansion of human capabilities, a widening of choices, 'an enhancement of freedom, and a fulfilment of human rights. At the beginning, the notion of human development incorporates the need for income expansion. However, income growth should consider expansion of human capabilities. Hence development cannot be equated solely to income expansion. Income is not the sum-total of human life. As income growth is essential, so are health, education, physical environment, and freedom. Human development should embrace human rights, socio-eco-politico freedoms. Based on the notion of human development. Human Development Index (HDI) is constructed. It serves as a more humane measure of development than a strictly .income-based benchmark of per capita GNP.

Human development is a lifelong process beginning before birth and extending to death. At each moment in life, every human being is in a state of personal evolution. Physical changes largely drive the process, as our cognitive abilities advance and decline in response to the brain's growth in childhood and reduced function in old age. Psychosocial development is also significantly influenced by physical growth, as our changing body and brain, together with our environment, shape our identity and our relationships with other people.

This book should be most useful to advanced students as a basic reference for courses in Education, Educational Psychology, and Human Development.

– Author

Chapter One

Education Development

Successful population policy is directly linked to successful education policy. Success in raising literacy rates and school enrolment rates while reducing dropout rates, especially for women, are closely correlated with the delayed onset of marriage and child birth, improved mortality for both mothers and children, and reduction in family size.

In fact, a successful education policy forms the bedrock of all fields of national development—political, economic, technical, scientific, social, and environmental. Education is the foundation for a vibrant democracy in which informed citizens exercise their franchise to support the internal growth of the nation and its constructive role in the world community. It is the foundation for growth in productivity, incomes and employment opportunities and for the development, application and adaptation of science and technology to enhance the quality of life.

Education is the foundation for access to the benefits of the information revolution that is opening up vistas on the whole world. Education is also the foundation for improved health care and nutrition. Literacy, the basis of all education, is as essential to survival and development in modern society as food is to survival and development of the human body. Literacy rates in India have arisen dramatically from 18 per cent in 1951 to 65 per cent in 2001, but these rates are still far from the UMI reference level of 95 per cent.

Literacy must be considered the minimum right and requirement of every Indian citizen. Vast differences also remain among different parts of the population. Literacy among males is nearly 50 per cent higher than females, and it is about 50 per cent higher in urban areas as compared to the rural areas. Literacy rates range from as high as 96 per cent in some districts of Kerala to below 30 per cent in some parts of Madhya Pradesh. Rates are also significantly lower among scheduled castes and tribes than among other communities. These differential rates of progress leave approximately 300 million illiterate adults in the country, the largest number of illiterates in the world.

For these people, the traditional avenues of knowledge dissemination through education and printed information are ruled out. These are the people who are most vulnerable to the challenges of development, because they are least equipped to rapidly expand their knowledge base. Since many of them are relatively young adults who will still be active in 2020, the country cannot afford to ignore them or leave them behind. The Government has already set a goal to achieve 75 per cent literacy by the end of the Tenth Five Year Plan.

A 100 per cent literate India is of paramount importance for realising the vision for the country in 2020 as presented in this document. Even when this goal is achieved, innovative approaches will be needed to increase knowledge dissemination through TV and other means, to literate adults with little or no formal education.

Literacy is an indispensable minimum condition for development, but it is not sufficient. In this increasingly complex and technologically sophisticated world, ten years of school education must also be considered as an essential prerequisite for citizens to adapt and succeed economically, avail of the social opportunities and develop their individual potentials. Education is the primary and most effective means so far evolved for transmitting practically useful knowledge from one generation to another. India's education system has expanded exponentially over the past five decades, but its current achievements are grossly inadequate for the nation to realise its potential greatness.

The net enrolment rate in primary schools is around 77 per cent and in secondary schools it is around 60 per cent. These compare with the 99.9 per cent primary and 69 per cent secondary enrolment for the UMI reference level. The drop out rate was 40 per cent at the primary level and 55 per cent at the upper primary level in 1999-2000. These high drop out rates from both primary and secondary school, combined with low enrolment rates at the higher levels deprive tens of millions of children of their full rights as citizens. Out of approximately 200 million children in the age group 6-14 years, only 120 million are in schools and net attendance in the primary level is only 66 per cent of the enrolment. Further, less than 7 per cent of the children ever pass the 10th standard public examination.

Apart from addressing the needs of a large illiterate population, India's knowledge strategy must also develop innovative approaches to enhance knowledge acquisition among the large community of school dropouts. Unless something is done to drastically reduce dropout rates, by the year 2016 there will be approximately 500 million people in the country with less than five years of schooling, and another 300 million that will not have completed high school. In other words, about two-thirds of the population will lack the minimum level of education needed to keep pace with and take advantage of the social changes occurring within the country and worldwide.

Extending the primary school system to over 500,000 villages in India has brought education to the masses. Unfortunately, this huge quantitative expansion has been accompanied by a tremendous dilution in the quality of schooling. High drop out rates in rural areas is one result of single room schools, with few teaching aids and inadequate instruction both in terms of quantity and quality. Qualitative improvement in the system can be accomplished by promoting centralised schools serving clusters of 10 or more villages, wherever distances and transportation links make that feasible.

This will permit greater investment in educational infrastructure, including introduction of computers. Achieving 100 per cent enrolment of all children in the 6 to 14 year age group is an ambitious but achievable goal for 2020. This must be coupled with efforts to increase the quality and relevance of school curriculum to equip students not only with academic knowledge but also with the values and practical knowledge needed for success in life. A business-as usual scenario for primary and secondary education in 2020, based on recent trends as well as an alternative scenario designed to radically enhance the quantity and quality of school education in the country. A tremendous expansion of schools and classrooms will be required to support a quantitative and qualitative improvement in the country’s school system. In order to achieve the best-case scenario total school enrolment would have to increase by 75 million or 44 per cent.

That will require a proportionate expansion in the number of classrooms. In addition, efforts to improve the quality of education by reducing the class size would require a further 20 per cent increase in the number of classrooms. Together, this will necessitate increasing the total number of classrooms by 65 per cent within 20 years.

Table. Education Scenarios in 2020

	1980 Actual	2000 Estimated as-usual	2020 Business-Scenario	2020 Best-case
Primary enrolment (1-5)	80%	89%	100%	100%
Elementary enrolment (1-8)	77%	79%	85%	100%
Secondary enrolment (9-12)	30%	58%	75%	100%
Dropout rate (1-5)	54%	40%	20%	0%
Dropout rate (1-8)	73%	54%	35%	0%

An enormous increase in the number of teachers will also be required to achieve the alternative scenario, *i.e.,* eliminating primary school dropouts and reducing the teacher-pupil ratio from the present high level of 1:42 down to around 1:20, which is the UMI reference level. Together, this will require an additional three million primary school teachers, more than twice the number currently employed. Similar increases will be required at middle and secondary school levels. The training of such large number of teachers will require the establishment of additional teacher training colleges and much larger budget allocations for teachers’ salaries. Qualitative improvements in education should reflect a change in pedagogical methods and lay emphasis on several dimensions, including:

- A shift from methods that emphasize passive learning to those that foster the active interest and ability of children to learn on their own.
- A shift from rote memorisation to development of children's capacity for critical thinking.
- A shift from traditional academic to practically relevant curriculum.
- A shift from imparting information to imparting life values such as independent thinking, self-reliance and individual initiative that are essential for success in any field of endeavour.

An important role of education is to foster in each child the attributes and values of a responsible, capable, active and healthy member of the family and society. The rigidity of curriculum, testing and teaching methods need to be relaxed so that innovative methods and new models of education can be evolved, tested and perfected. Vocational streams have to be developed and expanded to equip larger numbers of high school students with occupation-related knowledge and skills. Experimentation is needed with new methods for knowledge delivery. Television can be a very effective means for educating both school going and non-school going children and adults. It can deliver teaching materials in a more dynamic, entertaining, and interesting manner, utilising the nation's best teachers and multimedia teaching materials on each subject. A TV based curriculum can help slow learners to supplement classroom teaching, fast learners to learn at much faster rates than the rest of the class, dropouts to acquire knowledge they missed out in school, and adults to expand their level of education without returning to school.

New methods of delivery will be particularly necessary to augment access and improve delivery at higher levels of education. Although India's college and university network has expanded dramatically, it is able to accommodate only a tiny fraction of the college-age population. In addition, the quality of facilities, teaching and course materials leaves much to be desired. Recruitment and promotion are highly politicised; seniority rule precludes merit advancement; and the cost of delivery is beyond the means of the vast majority of young Indians.

India in 2020 must be a nation in which all those who aspire for higher education have access to college and university level courses.

Table. Growth of Higher Educational Institutions

Years	Colleges for General Education	Colleges for Professional Education	Universities
1951	370	208	27
1998	7199	2075	229

A national network of community colleges, similar to the highly successful American system, is needed to provide knowledge and job-oriented skills to millions of young people who lack interest in or capacity for more stringent academic studies. The advent of computer and the Internet-based educational methods offer an exciting new learning medium that can literally transform our concept of school and classroom from physical into virtual realities.

Studies in the USA project a radical reshaping of higher education over the next two decades as a result of the digital revolution. Many traditional colleges will close as more course works are delivered at a distance through alternative channels. The traditional boundaries between education and other sectors will fade, as publishers, for-profit and non-profit organisations, offer accredited, multimedia-enhanced courses directly to students, by-passing the university. The traditional classroom type of education, which is most useful for students that require personal attention and assistance and for subjects that involve hands-on experimentation, will no longer be the predominant model of education.

For all other purposes, it is very costly and not very efficient in the way it uses the time of both teachers and students. Experience shows that computer-based educational methods can lead to much faster rates and higher quality of learning, which is more interactive and motivating for students at all levels of education from pre-school to post-graduation. It is extremely effective for enhancing reading and language skills and general knowledge among the very young and even for some sophisticated professional courses such as medicine and engineering. Given the huge number of young students that will quest for all levels of higher education in the coming decades and the severe shortage of qualified instructors, and in the light of India's outstanding expertise in the IT industry, the country needs to embark on a massive programme to convert the entire higher educational curriculum into a multi-media, web-based format and to establish accreditation standards for recognition of the distance education so imparted. Our vision of India in 2020 is predicated on the belief that human resources are the most important determinants of overall development. As India's IT revolution has been fuelled by the availability of a very large reservoir of well-trained engineers, its future development in many different spheres will depend on commensurate development of sufficient and surplus capabilities.

Full development of India's enormous human potential will require a shift in national priorities, to commit a greater portion of the country's financial resources to the education sector. India currently invests 3.2-4.4 per cent of GNP on education. This compares unfavourably with the UMI reference level of 4.9 per cent, especially with countries such as South Africa, which invests 7.9 per cent of GNP on education. A near doubling of investments in education is the soundest policy for quadrupling the country's GDP per capita.

S&T CAPABILITIES

Literacy and general education form the base of the knowledge pyramid that is essential for rapid and sustained development in the 21st century. The continuous advancement of science and application of improved technology form the middle rung; and social ideals and spiritual values form the apex. Technical education, both vocational and professional, constitutes the foundation for development of science and technology. India is rightly proud of the international standing of its IITs, but a handful of world class technical institutions is not sufficient. A large number of the country's approximately 500 engineering colleges need to be upgraded to quality standards nearer to those of the IITs, and given similar autonomy. Private sector initiative and investment, whether from Indian corporates or NRIs or reputed foreign universities, need to be fully encouraged. Close links need to be fostered between technical institutions and industry. India's enormous manpower base of scientists and engineers is often coveted.

President Kalam observed that India's human resource base is one of its greatest core competencies. This is true in absolute terms, but as a percentage of the total population, we are at 1/100th of the US levels and 1/50th of the Korean level. Even China's manpower base of scientists and engineers as a percentage of its population exceeds ours by more than three times. Absolute numbers do count, but manpower alone is not sufficient to excel in the development and application of science and technology. In terms of total investment in R&D, India's expenditure is 1/60th of that of Korea, 1/250th of that of the USA, and 1/340th of that of Japan.

More significantly, atomic energy, space and defence research account for 71 per cent of all central spending on science and technology, which means that relatively little is left for investment in agriculture, energy, telecommunications and other crucial sectors within the sphere of science and technology. R&D expenditure even in India's fast-growing IT sector has been averaging around 3 per cent of sales turnover (STO), which is much lower as compared to the 14-19 per cent expended by internationally reputed software firms. These low figures reflect on our R&D performance.

India's share of global scientific output in 1998 was only 1.58 per cent of the world's total. Out of 500,000 new patent applications filed globally each year, China accounts for 96,000 and Korea accounts for 72,000, while India accounts for only 8,000. Of greater concern has been the country's inability to capitalise on our huge pool of manpower and extensive network of scientific research organisations for transferring proven technologies from the lab to the land and to the factory. Despite possessing the world's largest cadre of agricultural scientists, we have not been able to extend the momentum of Green Revolution to other regions and crops and to update the scientific practices of our farmers to levels comparable with most other nations. Crop productivity remains far below and production costs far above world averages. A similar gap exists in the application of science and technology for food processing and many other industries. High technology exports account for only 6 per cent of total manufacturing exports for India, compared to over 20 per cent for the UMI reference level.

India has a number of premier universities for scientific and technical education which produce world class scientists and engineers, but the national research system rarely produces world class results. One reason is the missing link between scientists in universities and research institutes and practitioners in agriculture and industry. Another is that non-teaching research institutes lack the continuous inflow of young talented researchers required to challenge assumptions and infiltrate new ideas and approaches.

Bureaucracy, tenure and lack of accountability minimise the pressure for practical application of scientific knowledge. Other countries have done a much better job of providing optimum incentives to their scientific community and creating close linkages between science and industry. The management of technology and the administration of government require different sets of skills and a different culture. Management of technology demands innovation, global competitiveness, and wide latitude for individual freedom, characteristics that are difficult to foster under an administrative regime. The objectives of many institutes and the mechanisms for achieving them need a radical reorientation to bring them in tune with the changing environment brought about by liberalisation of imports, technology tie-ups with foreign agencies and foreign direct investment in key sectors.

If the national R&D bodies do not reorganise themselves to face globalisation and global competition, nor take into consideration the nature of the factor endowment of the country along with many existing indigenous technologies so far neglected, then they risk becoming irrelevant to the mainstream activities of industrial development. The nature of service offered by these institutions has to undergo a sea change. Grass-root consultancy and turnkey project consultancy, including project implementation and monitoring are increasingly in demand. Another essential step is to improve the linkages between technology development and technology application by fostering close ties between basic research and business. In the USA, a large proportion of scientific research at the university level is financed by business, and most leading universities operate technology incubators designed to promote commercialisation of new products and processes.

India's recent experiment with the Science and Technology Entrepreneur Parks (STEPs) has been successful in promoting application of technologies, new business start-ups, and employment. The STEPs in combination with technology incubators need to be rapidly multiplied around the country. Effective R&D management backed by suitable incentives for commercialisation are essential for bridging the gap between the lab and the land or factory. R&D is not only essential for propelling growth in industrial fields. It can also become a growth industry on its own. Much of the work done by Indian software companies for overseas clients and parent companies rightly fall under the category of new product development. Vast potential exists for complementing India's strengths in IT with corresponding strengthening of fields such as biotechnology, pharmaceuticals, designer-made materials, and others. India's investment in the biotechnology sector is expected to increase five-fold by the end of the current decade largely as a result of growing collaboration between multinational corporations and indigenous research efforts. Huge opportunities also exist for R&D in the field of genomics, bioinformatics, DNA technologies, clinical studies and genetically modified crops. Rationalisation of procedure, backed by effective bio-safety regulations are needed to govern testing and approval of biotech products. India has the potential of converting its strong R&D infrastructure into a global research, design and development platform.

VOCATIONAL TRAINING

The knowledge and skill of our workforce will be a major determinant of India's future rate of economic growth as well as the type and number of jobs we create. The greater that knowledge and skill, the higher will be the productivity, the better the quality, and the lower the cost of the products and services we generate. Similarly, the better the quality and lower the cost, greater will be the comparative advantage and market potential. Currently only 5 per cent of the country's labour force in the 20-24 age category have undergone formal vocational training, compared with 28 per cent in Mexico, 60 to 80 per cent in most industrialised nations, and as much as 96 per cent in Korea.

A strategy to achieve full employment must include as an important component, a strategy to ensure that all new entrants to the workforce are equipped with the knowledge and skill needed for high productivity and high quality. India has over 4,200 industrial training institutes (ITI) imparting education and training in 43 engineering and 24 non-engineering trades. Of these, 1,654 are government run ITIs, while 2,620 are private. The total seating capacity in these ITIs is 6.28 lakh. Most of this training is conducted in classroom style in the form of one to two year diploma courses. In addition, about 1.65 lakh persons undergo apprenticeship vocational training every year in state-run enterprises. If a wider definition of applied courses is taken that includes agriculture, engineering and other professional subjects, the total number receiving job related training is about 17 lakh per annum, which still represents only 14 per cent of new entrants to the workforce. The nature of vocational skills makes it impossible for vocational schools to fully address the nation's needs.

The variety of skills needed by the workforce is far too great. The changes in technology and work processes are too rapid for training courses and their instructors to stay up-todate. The cost of training is also relatively high as it often demands full time enrolment for a prolonged period. Some vocational fields do not lend themselves to classroom or laboratory study at all. A comprehensive strategy is thus needed to enhance the nation's employable skills. It must begin by preparing a catalogue of the entire range of vocational skills needed to support the development of the country. The network of vocational training institutes and the range of vocational skills taught needs to be expanded substantially to impart those skills for which institutional training is most suitable. The private sector, which promoted the rapid proliferation of computer training institutes throughout the country, should be encouraged to recognise the commercial potential of vocational training in many other fields. In addition, it is essential to fashion more effective and efficient mechanisms for disseminating useful knowledge and skills, especially through the TV media and through computerised vocational training. The importance of computer has been widely recognised as a means to improve efficiency in business, government and formal education, but its application in vocational training is not fully appreciated.

Use of Computers in Vocational Training:

- Rates of learning on computer for both academic as well as vocational or skill-based subjects are four to ten times faster than they are in a classroom setting, and learning retention is likely to be much higher.
- Computers can provide multimedia, interactive, customised and individually paced learning with instantaneous feedback and testing.
- It eliminates the need for producing and deputing a large number of highly skilled instructors. Course contents can be rapidly modified to reflect changing needs.
- For many types of vocational skills, computerised training also offers specific advantages over the live delivery of skills in a classroom.
- Training can be delivered wherever computers are available. A nationwide network of 50,000 computerised vocational centres, run as private self-employed businesses similar to the STD booths and Internet cafes, can deliver low-cost, high-quality training to 10 million workers every year—more than five times the total number covered by existing programmes.

Although 58 per cent of Indians are engaged in agriculture, vocational training for farmers is one of the weakest links in the Indian educational system. Agricultural universities cater to the nation's need for agricultural scientists and extension officers. An extensive network of more than 300 Krishi Vignan Kendras (KVK) offers short and medium term courses for farmers on specialised subjects. The KVK network provides training to several hundred thousand farmers each year, but it can cater to the needs of only a miniscule percentage of all farmers. Agricultural education can be moved from the campus into the village by establishing a national network of farm schools, offering practical demonstration and training on lands leased from farmers in the local community. The prime objective of the schools would be to impart knowledge and skills designed to double yields on important commercial crops. Educated farmers can be trained as self-employed instructors to operate the farm schools as private enterprises on a commercial basis.

HEALTH FOR ALL

The health of a nation is difficult to define in terms of a single set of measures. At best, we can assess the health of the population by taking into account indicators like infant mortality and maternal mortality rates, life expectancy and nutrition, along with the incidence of communicable and non-communicable diseases. The health of the Indian population has improved dramatically over the past fifty years. Life expectancy has risen from 33 years to 64 years. The infant mortality rate (IMR) has fallen from 148 to 71 per 1000. The crude birth rate (CBR) has declined from 41 to 25 and the crude death rate (CDR) has fallen from 25 to under 9. The couple protection rate (CPR) and total fertility rate (TFR) have also improved substantially. Despite these achievements, wide disparities persist between different income groups, between rural and urban communities, and between different states and even districts within states. The infant mortality rate among the poorest quintet of the population is 2.5 times higher than that among the richest. Maternal mortality remains very high. More than one lakh women die each year due to pregnancy-related complications.

Like population growth and economic growth, the health of a nation is a product of many factors and forces that combine and interact with each other. Economic growth, per capita income, employment, levels of literacy and education—especially among females—age of marriage, birth rates, availability of information regarding health care and nutrition, access to safe drinking water, public and private health care infrastructure, access to preventive health care and medical care, health insurance, public hygiene, road safety, and environmental pollution are among the factors that contribute directly to the health of the nation. Communicable diseases such as malaria, kalaazar, tuberculosis and HIV infection remain the major causes of illness in India.

During the next five to ten years, existing programmes are likely to eliminate polio and leprosy and substantially reduce the prevalence of kalaazar and filariasis. However, TB, malaria and AIDS will continue to remain major public health problems.

India has about 1.5 million identified cases of TB that are responsible for more than 3,00,000 deaths annually. Improved diagnostic services and treatment can reduce the prevalence and incidence of TB by 2020. About 2 million cases of malaria are reported in India each year. Restructuring the "malaria workforce" and strengthening health infrastructure can reduce the incidence of this disease by up to 50 per cent within a decade. Assessing the impact of HIV epidemic is more difficult; there are about 4 million persons infected with HIV. The National Health Policy aims at achieving a plateau in the prevalence of HIV infection by 2007.

Childhood diarrhoea, another major cause of illness, is largely preventable through simple community action and public education, and deaths due to diarrhoea can be eliminated by 2010. Childhood under-nutrition, the other major area of concern, can be addressed by targeting children of low birth weights and employing low-cost screening procedures to detect under-nutrition at an early age. Given the projected improvement in living standards, food security, educational levels and access to health care among all levels of the general population, substantial progress can be made in reducing the prevalence of severe under-nutrition in children by 2020. China's remarkable success in combating disease over the past two decades is proof that a determined commitment to improving public health can dramatically reduce the incidence of infectious diseases within one or two decades. With the demographic and epidemiological transition taking place in the coming years, non-communicable diseases are also likely to emerge as major public health problems.

Modernisation of life styles will further aggravate health problems. The rapid proliferation of two and four wheel motor vehicles, increasing congestion on city roads and intercity highways have all contributed to an increasing number of deaths and serious injuries from traffic related accidents. Greater emphasis on education and enforcement of road safety rules by both drivers and pedestrians is an urgent need of the hour. As already noted, there will be a massive increase in population in the 15-64 age group. Reproductive and Child Health care programmes must meet the needs of this rapidly growing clientele. The population in this age group will be more literate and have greater access to information. They will have greater awareness and expectations about access to quality services for maternal and child health, contraceptive care, management of gynaecological problems, etc. A major focus of vision 2020 must be on improving access to health services to meet the health care needs of women and children. India's significant achievements in the field of health have been made possible by the establishment of a huge rural health infrastructure, along with the formation of a massive health care manpower consisting of over five lakh trained doctors working under plural systems of medicine, and a vast frontline of over seven lakh nurses and other health care workers; 25,000 primary and community health care centres; and 1.6 lakh sub-centres, complemented by 22,000 dispensaries and 2,800 hospitals practicing Indian systems of medicine and homeopathy. This infrastructure remains under-equipped, under-manned and under-financed to cope with the challenge of eradicating major threats to human life. The inadequacy of the current health care system is starkly showed by the fact that only 35 per cent of the population have access to essential drugs, while the UMI reference level is above 82 per cent.

Infant immunisation against measles and DPT for children under 12 years is only 60 per cent and 78 per cent compared to the UMI level of over 90 per cent for both diseases. As a larger proportion of the population reaches middle class standards of living, an increasing number of people will turn to private health care providers. This development is welcome, because it will permit the public health care system to concentrate more resources on meeting the needs of the poorer parts. But at the same time, the level of public expenditure on health care needs to rise about four-fold from the current level of 0.8 per cent of GDP to reach the UMI reference level of 3.4 per cent.

Rapid growth of the private health care system, however, requires the formulation of competence and quality standards to check and balance the increasing emphasis on health care as a business.

Criteria for a More Equitable and Effective Health Care System:

- Universal access and access to an adequate level of health care without financial burden.
- Fair distribution of financial costs for access and fair distribution of burden in rational care and capacity.
- Ensuring that providers have the competence, empathy and accountability for delivering quality care and for effective use of relevant research.
- Special attention to vulnerable groups such as women, children, the disabled and the aged.

Development plans for India's health care systems need to place greater emphasis on public health education and prevention. The wide dissemination of health and nutrition related information through traditional channels should be supplemented by an ambitious and persistent programme of public health education through the print, television, radio and electronic media. Health insurance can play an invaluable role in improving the overall health care system. The insurable population in India has been assessed at 250 million and this number will increase rapidly in the coming two decades. This should be supplemented by innovative insurance products and programmes by panchayats with reinsurance backup by companies and government to extend coverage to much larger parts of the population.

The life expectancy of the Indian population is expected to reach above 65 years in 2020, which compares favourably with the UMI reference level of 69 years. Mortality rates for infants is expected to decline to about 20 per 1000 in 2020, which compares favourably with the UMI reference level of 22.5.

VULNERABILITY

Economic growth, rising levels of education among the young, expansion of employment opportunities for the working age population, slower population growth, and declining infant mortality, however, will not eliminate and may even aggravate, inequalities between different age groups, the sexes, income groups, communities and regions, unless specific corrective steps are taken for levelling the different degrees of capacity and opportunity.

The Vision 2020 must have special focus on bridging the existing gaps in the various levels of development and endeavour its best to fulfil the Constitutional commitment of raising the status of the vulnerable groups vis-a-vis the rest of the society. During the next 20 years, the aged population in India will nearly double, placing much greater demand on the infrastructure of hospitals and nursing homes, while at the same time shifting the profile of health disorders from problems of the young to those of the aged.

The population above 65 years of age will increase from 45 to 76 million persons by 2020. Reference has already been made to the very high incidence of malnutrition among this group and the growing incidence of diseases associated with aging, such as cancer and cardiovascular diseases.

A disproportionate number of the aged population are illiterate and living below the poverty line. While majority of these people live in rural areas where the extended family system remains prevalent, increasing urbanisation and mobility will destabilise their situation in future. The problem of coping with a larger aged population will be partially solved by a big surge in the size of the working age labour force and a reduction in the dependency ratio, meaning that there will be a larger proportion of workers to support the aged.

Specific strategies will be needed to provide targeted assistance for the most vulnerable members of this group, including research on the disabilities of the aged, greater development of geriatric medicine, training of minders for the aged, and establishment of innovative support systems such as the highly successful Japanese model. Like the aged, the disabled persons also form another subgroup that is particularly vulnerable and unlikely to benefit directly from the general advance of society, unless specific provisions are made to address their particular needs, including a special system of schools, clinics and homebased learning programmes, combined with therapeutic and institutional care.

Increasing gender equity is an important challenge of the coming years. Literacy rates for females are 40 per cent lower than for males. Females represent only 43 per cent of all students at primary level, 37 per cent at higher secondary level, and 35 per cent of those in higher education. Drop out rates are also higher for girls. The UNDP's gender development index ranks India 108th among 174 countries in terms of gender equity. It is no coincidence therefore that countries ranking highest on this index are among the most prosperous in the world.

Gender equity and social development are inseparably interlinked. Reducing the disparity in nutritional, literacy and educational levels between the sexes is essential for realisation of the country's full potentials. Children being the supreme assets of the country, the 'Rights' based approach will continue to play an important role in ensuring their 'survival', 'protection' and 'development', with special attention to the girl child. Our vision for 2020 in this regard is to see a nation free from all forms of child labour. However, complete elimination of child labour through legal means is not only difficult to implement but may also prove to be counter productive in protecting the interest of these very children.

While legal enforcement can contain the demand for child labour, it may leave the poor households supplying child labour more vulnerable rather than addressing the underlying socio-economic compulsions generating the supply of such labour. Indeed, poverty eradication combined with educational reforms to provide free (or affordable) access to quality education with an interesting, innovative and job-oriented curriculum for all can effectively eliminate child labour once and for all. The other categories of the vulnerable groups include the scheduled castes (SC), scheduled tribes (ST), other backward classes (OBC) and minorities, constituting nearly three-fourths of the country's total population.

They require special attention in order to narrow down the disparities between them and the more privileged part of the population. Literacy rates for SCs and STs are 25 per cent lower than for other communities. Education enrolment rates among this group, especially for females, lag significantly behind that of other communities. Low levels of education lead to lower paying employment opportunities and lower incomes. Thus, it becomes exceedingly difficult for these communities to come out of the vicious circle. Large number of people in other communities also suffer from similar disparities. Regional disparities in rates of development are of similar concern. Regardless of the community or the region, the overall progress of the nation will depend to a large extent on its ability to provide increasing opportunities for the disadvantaged to take initiative for their self development. Otherwise, these disadvantaged will exact an increasing toll on the stability, well-being and development of the country as a whole. It is notable that rising levels of violence and crime in society are not so much associated with overall low levels of development, as they are with wide disparities between levels of development, which create frustration and resentment. The progress of the whole will ultimately depend on the progress of its weakest links. India's vision of 2020 must be one in which all levels and parts of the population and all parts of the country march forward together into a more secure and prosperous future.

EDUCATION IN INDIA

Education in India has a history stretching back to the ancient urban centres of learning at Taxila and Nalanda. Western education became ingrained into Indian society with the establishment of the British Raj. Education in India falls under the control of both the Union Government and the states, with some responsibilities lying with the Union and the states having autonomy for others. The various articles of the Indian constitution provide for education as a fundamental right. Most universities in India are Union or State Government controlled.

India has made a huge progress in terms of increasing primary education attendance rate and expanding literacy to approximately two thirds of the population. India's improved education system is often cited as one of the main contributors to the economic rise of India. Much of the progress in education has been credited to various private institutions. The private education market in India is estimated to be worth $40 billion in 2008 and will increase to $68 billion by 2012. However, India continues to face challenges. Despite growing investment in education, 40% of the population is illiterate and only 15% of the students reach high school. As of 2008, India's post-secondary high schools offer only enough seats for 7% of India's college-age population, 25% of teaching positions nationwide are vacant, and 57% of college professors lack either a master's or PhD degree. As of 2007, there are 1522 degree-granting engineering colleges in India with an annual student intake of 582,000, plus 1,244 polytechnics with an annual intake of 265,000. However, these institutions face shortage of faculty and concerns have been raised over the quality of education.

Three Indian universities were listed in the Times Higher Education list of the world's top 200 universities — Indian Institutes of Technology, Indian Institutes of Management, and Jawaharlal Nehru University in 2005 and 2006. Six Indian Institutes of Technology and the Birla Institute of Technology and Science-Pilani were listed among the top 20 science and technology schools in Asia by *Asiaweek*. The Indian School of Business situated in Hyderabad was ranked number 15 in global MBA rankings by the *Financial Times* of London in 2009 while the All India Institute of Medical Sciences has been recognized as a global leader in medical research and treatment.

HISTORY

Monastic orders of education under the supervision of a *guru* was a favoured form of education for the nobility in ancient India. The knowledge in these orders was often related to the tasks a section of the society had to perform. The priest class, the *Brahmins*, were imparted knowledge of religion, philosophy, and other ancillary branches while the warrior class, the *Kshatriya*, were trained in the various aspects of warfare. The business class, the *Vaishya*, were taught their trade and the lowest class of the *Shudras* was generally deprived of educational advantages. The book of laws, the *Manusmriti*, and the treatise on statecraft the *Arthashastra* were among the influential works of this era which reflect the outlook and understanding of the world at the time.

Apart from the monastic orders, institutions of higher learning and universities flourished in India well before the common era, and continued to deliver education into the common era. Secular Buddhist institutions cropped up along with monasteries. These institutions imparted practical education, *e.g.*, medicine. A number of urban learning centres became increasingly visible from the period between 200 BCE to 400 CE. The important urban centres of learning were Taxila and Nalanda, among others. These institutions systematically imparted knowledge and attracted a number of foreign students to study topics such as logic, grammar, medicine, metaphysics, arts and crafts.

By the time of the visit of the Islamic scholar Alberuni (973-1048 CE), India already had a sophisticated system of mathematics and science in place, and had made a number of inventions and discoveries.

With the arrival of the British Raj in India a class of Westernized elite was versed in the Western system of education which the British had introduced. This system soon became solidified in India as a number of primary, secondary, and tertiary centres for education cropped up during the colonial era. Between 1867 and 1941 the British increased the percentage of the population in Primary and Secondary Education from around 0.6% of the population in 1867 to over 3.5% of the population in 1941. However this was much lower than the equivalent figures for Europe where in 1911 between 8 and 18% of the population were in Primary and Secondary education. Additionally literacy was also improved. In 1901 the literacy rate in India was only about 5% though by Independence it was nearly 20%.

Following independence in 1947, Maulana Azad, India's first education minister envisaged strong central government control over education throughout the country, with a uniform educational system. However, given the cultural and linguistic diversity of India, it was only the higher education dealing with science and technology that came under the jurisdiction of the central government. The government also held powers to make national policies for educational development and could regulate selected aspects of education throughout India.

The central government of India formulated the National Policy on Education (NPE) in 1986 and also re-enforced the Programme of Action (POA) in 1986. The government initiated several measures the launching of DPEP (District Primary Education

Programme) and SSA (Sarva Shiksha Abhiyan, India's initiative for Education for All) and setting up of *Navodaya Vidyalaya* and other selective schools in every district, advances in female education, inter-disciplinary research and establishment of open universities. India's NPE also contains the National System of Education, which ensures some uniformity while taking into account regional education needs. The NPE also stresses on higher spending on education, envisaging a budget of more than 6% of the Gross Domestic Product. While the need for wider reform in the primary and secondary sectors is recognized as an issue, the emphasis is also on the development of science and technology education infrastructure.

OVERVIEW

The National Council of Educational Research and Training (NCERT) is the apex body for curriculum related matters for school education in India. The NCERT provides support and technical assistance to a number of schools in India and oversees many aspects of enforcement of education policies. In India, the various curriculum bodies governing school education system are:

- The state government boards, in which the majority of Indian children are enrolled.
- The Central Board of Secondary Education (CBSE) board.
- The Council for the Indian School Certificate Examinations (CISCE) board.
- The National Institute of Open Schooling (NIOS) board.
- International schools affiliated to the International Baccalaureate Programme and/or the Cambridge International Examinations.
- Islamic Madrasah schools, whose boards are controlled by local state governments, or autonomous, or affiliated with Darul Uloom Deoband.
- Autonomous schools like Woodstock School, Auroville, Patha Bhavan and Ananda Marga Gurukula.

In addition, NUEPA (National University of Educational Planning and Administration) and NCTE (National Council for Teacher Education) are responsible for the management of the education system and teacher accreditation.

PRIMARY EDUCATION

The Indian government lays emphasis to primary education up to the age of fourteen years (referred to as Elementary Education in India.) The Indian government has also banned child labour in order to ensure that the children do not enter unsafe working conditions. However, both free education and the ban on child labour are difficult to enforce due to economic disparity and social conditions. 80% of all recognized schools at the Elementary Stage are government run or supported, making it the largest provider of education in the Country. However, due to shortage of resources and lack of political will, this system suffers from massive gaps including high pupil teacher ratios, shortage of infrastructure and poor level of teacher training. Education has also been made free for children up to the age of 14 or class VIII under the Children's Right to Free and Compulsory Education Act 2009.

There have been several efforts to enhance quality made by the government. The District Primary Education Programme (DPEP) was launched in 1994 with an aim to universalize primary education in India by reforming and vitalizing the existing primary education system. 85% of the DPEP was funded by the central government and the remaining 15 percent was funded by the states. The DPEP, which had opened 160000 new schools including 84000 alternative education schools delivering alternative education to approximately 3.5 million children, was also supported by UNICEF and other international programmes. This primary education scheme has also shown a high Gross Enrolment Ratio of 93–95% for the last three years in some states. Significant improvement in staffing and enrollment of girls has also been made as a part of this scheme. The current scheme for universalization of Education for All is the Sarva Shiksha Abhiyan which is one of the largest education initiatives in the world. Enrollment has been enhanced, but the levels of quality remain low.

SECONDARY EDUCATION

The National Policy on Education (NPE), 1986, has provided for environment awareness, science and technology education, and introduction of traditional elements such as Yoga into the Indian secondary school system. Secondary education covers children 14-18 which covers 88.5 million children according to the Census, 2001. However, enrolment figures show that only 31 million of these children were attending schools in 2001-02, which means that two-third of the population remained out of school. A significant feature of India's secondary school system is the emphasis on inclusion of the disadvantaged sections of the society. Professionals from established institutes are often called to support in vocational training. Another feature of India's secondary school system is its emphasis on profession based vocational training to help students attain skills for finding a vocation of his/her choosing. A significant new feature has been the extension of SS to secondary education in the form of the Madhyamik Shiksha Abhiyan.

A special Integrated Education for Disabled Children (IEDC) programme was started in 1974 with a focus on primary education., but which was converted into Inclusive Education at Secondary Stage Another notable special programme, the *Kendriya Vidyalaya* project, was started for the employees of the central government of India, who are distributed throughout the country. The government started the *Kendriya Vidyalaya* project in 1965 to provide uniform education in institutions following the same syllabus at the same pace regardless of the location to which the employee's family has been transferred.

TERTIARY EDUCATION

Our university system is, in many parts, in a state of disrepair…In almost half the districts in the country, higher education enrolments are abysmally low, almost two-third of our universities and 90 per cent of our colleges are rated as below average on quality parameters… I am concerned that in many states university appointments, including that of vice-chancellors, have been politicised and have become subject to caste and communal considerations, there are complaints of favouritism and corruption. – *Prime Minister Manmohan Singh in 2007*

India's higher education system is the third largest in the world, after China and the United States. The main governing body at the tertiary level is the University Grants Commission (India), which enforces its standards, advises the government, and helps coordinate between the centre and the state. Accreditation for higher learning is overseen by 12 autonomous institutions established by the University Grants Commission.

As of 2009, India has 20 central universities, 215 state universities, 100 deemed universities, 5 institutions established and functioning under the State Act, and 13 institutes which are of national importance. Other institutions include 16000 colleges, including 1800 exclusive women's colleges, functioning under these universities and institutions. The emphasis in the tertiary level of education lies on science and technology. Indian educational institutions by 2004 consisted of a large number of technology institutes. Distance learning is also a feature of the Indian higher education system.

Some institutions of India, such as the Indian Institutes of Technology (IITs), have been globally acclaimed for their standard of education. The IITs enrol about 8000 students annually and the alumni have contributed to both the growth of the private sector and the public sectors of India.

Besides top rated universities which provide highly competitive world class education to their pupil, India is also home to many universities which have been founded with the sole objective of making easy money. Regulatory authorities like UGC and AICTE have been trying very hard to extirpate the menace of private universities which are running courses without any affiliation or recognition. Students from rural and semi urban background often fall prey to these institutes and colleges.

TECHNICAL EDUCATION

From the first Five Year Plan onwards India's emphasis was to develop a pool of scientifically inclined manpower. India's National Policy on Education (NPE) provisioned for an apex body for regulation and development of higher technical education, which came into being as the All India Council for Technical Education (AICTE) in 1987 through an act of the Indian parliament. At the level of the centre the Indian Institutes of Technology and the Indian Institutes of Information Technology are deemed of national importance. The Indian Institutes of Management are also among the nation's premier education facilities. Several Regional Engineering Colleges (REC) have been converted into National Institutes of Technology. The UGC has inter-university centres at a number of locations throughout India to promote common research, *e.g.*, the Nuclear Science Centre at the Jawaharlal Nehru University, New Delhi.

LITERACY

2001 government statistics hold the national literacy to be around 64.84%. Government statistics of 2001 also hold that the rate of increase of literacy is more in rural areas than in urban areas. Female literacy was at a national average of 53.63% whereas the male literacy was 75.26%. Within the Indian states, Kerala has shown the highest literacy rates of 90.02% whereas Bihar averaged lower than 50% literacy, the lowest in India. The 2001 statistics also indicated that the total number of 'absolute non literates' in the country was 304 million..

ATTAINMENT

World Bank statistics found that fewer than 40 percent of adolescents in India attend secondary schools. *The Economist* reports that half of 10-year-old rural children could not read at a basic level, over 60% were unable to do division, and half dropped out by the age 14.

Only one in ten young people have access to tertiary education. Out of those who receive higher education, *Mercer Consulting* estimates that only a quarter of graduates are "employable".

An optimistic estimate is that only one in five job-seekers in India has ever had any sort of vocational training.

PRIVATE EDUCATION

According to current estimates, 80% of all schools are government schools making the government the major provider of education. However, because of poor quality of public education, 27% of Indian children are privately educated.

According to some research, private schools often provide superior results at a fraction of the unit cost of government schools. However, others have suggested that private schools fail to provide education to the poorest families, a selective being only a fifth of the schools and have in the past ignored Court orders for their regulation In their favour, it has been pointed out that private schools cover the entire curriculum and offer extra-curricular activities such as science fairs, general knowledge, sports, music and drama. The pupil teacher ratios are much better in private schools (1:31 to 1:37 for government schools and more teachers in private schools are female. There is some disgreement over which system has better educated teachers. According to the latest DISE survey, the percentage of untrained teachers (paratechers) is 54.91% in private, compared to 44.88% in government schools and only 2.32% teachers in unaided schools receive inservice training compared to 43.44% for government schools. The competition in the school market is intense, yet most schools make profit. Even the poorest often go to private schools despite the fact that government schools are free. A study found that 65% of schoolchildren in Hyderabad's slums attend private schools.

Private schools are often operating illegally. A 2001 study found that it takes 14 different licenses from four different authorities to open a private school in New Delhi and could take years if done legally. However, operation of unrecognized schools has been made illegal under the Children's Right to Free and Compulsory Education Act

WOMEN'S EDUCATION

Women have much lower literacy rate. Compared to boys, far fewer girls are enrolled in the schools, and many of them drop out. According to a 1998 report by U.S. Department of Commerce, the chief barrier to female education in India are inadequate school facilities (such as sanitary facilities), shortage of female teachers and gender bias in curriculum (majority of the female characters being depicted as weak and helpless)

The number of literate women among the female population of India was between 2-6% from the British Raj onwards to the formation of the Republic of India in 1947. Concerted efforts led to improvement from 15.3% in 1961 to 28.5% in 1981. By 2001 literacy for women had exceeded 50% of the overall female population, though these statistics were still very low compared to world standards and even male literacy within India. Recently the Indian government has launched Saakshar Bharat Mission for Female Literacy. This mission aims to bring down female illiteracy by half of its present level.

Sita Anantha Raman outlines the progress of women's education in India:

Sita Anantha Raman also maintains that while the educated Indian women workforce maintains professionalism, the men outnumber them in most fields and, in some cases, receive higher income for the same positions.

RURAL EDUCATION

Following independence, India viewed education as an effective tool for bringing social change through community development. The administrative control was effectively initiated in the 1950s, when, in 1952, the government grouped villages under a Community Development Block—an authority under national programme which could control education in up to 100 villages. A Block Development Officer oversaw a geographical area of 150 square miles which could contain a population of as many as 70000 people.

Setty and Ross elaborate on the role of such programmes, themselves divided further into *individual-based*, *community based*, or the *Individual-cum-community-based*, in which microscopic levels of development are overseen at village level by an appointed worker:

The community development programmes comprise agriculture, animal husbandry, cooperation, rural industries, rural engineering (consisting of minor irrigation, roads, buildings), health and sanitation including family welfare, family planning, women welfare, child care and nutrition, education including adult education, social education and literacy, youth welfare and community organisation. In each of these areas of development there are several programmes, schemes and activities which are additive, expanding and tapering off covering the total community, some segments, or specific target populations such as small and marginal farmers, artisans, women and in general people below the poverty line.

Despite some setbacks the rural education programmes continued throughout the 1950s, with support from private institutions. A sizable network of rural education had been established by the time the *Gandhigram Rural Institute* was established and 5, 200 Community Development Blocks were established in India. Nursery schools, elementary schools, secondary school, and schools for adult education for women were set up. The government continued to view rural education as an agenda that could be relatively free from bureaucratic backlog and general stagnation. However, in some cases lack of financing balanced the gains made by rural education institutes of India. Some ideas failed to find acceptability among India's poor and investments made by the government sometimes yielded little results. Today, government rural schools remain poorly funded and understaffed. Several foundations, such as the Rural Development Foundation (Hyderabad), actively build high-quality rural schools, but the number of students served is small.

ISSUES

One study found out that 25% of public sector teachers and 40% of public sector medical workers were absent during the survey. Among teachers who were paid to teach, absence rates ranged from 15% in Maharashtra to 71% in Bihar. Only 1 in nearly 3000 public school head teachers had ever dismissed a teacher for repeated absence. A study on teachers by Kremer, etc. found that 'only about half were teaching, during unannounced visits to a nationally representative sample of government primary schools in India.'.

Modern education in India is often criticized for being based on rote learning rather than problem solving. *BusinessWeek* denigrates the Indian curriculum saying it revolves around rote learning and ExpressIndia suggests that students are focused on cramming.

A study of 188 government-run primary schools found that 59% of the schools had no drinking water and 89% had no toilets. 2003-04 data by National Institute of Educational Planning and Administration revealed that only 3.5% of primary schools in Bihar and Chhattisgarh had toilets for girls. In Madhya Pradesh, Maharashtra, Andhra Pradesh, Gujarat, Rajasthan and Himachal Pradesh, rates were 12-16%.

Fake degrees are a problem. One raid in Bihar found 0.1 million fake certificates. In February 2009, the University Grant Commission found 19 fake institutions operating in India.

Only 16% of manufacturers in India offer in-service training to their employees, compared with over 90% in China.

INITIATIVES

Following India's independence a number of rules were formulated for the backward Scheduled Castes and the Scheduled Tribes of India, and in 1960 a list identifying 405 Scheduled Castes and 225 Scheduled Tribes was published by the central government. An amendment was made to the list in 1975, which identified 841 Scheduled Castes and 510 Scheduled Tribes. The total percentage of Scheduled Castes and Scheduled Tribes combined was found to be 22.5 percent with the Scheduled Castes accounting for 17 percent and the Scheduled Tribes accounting for the remaining 7.5 percent. Following the report many Scheduled Castes and Scheduled Tribes increasingly referred to themselves as *Dalit*, a Marathi language terminology used by B. R. Ambedkar which literally means "oppressed".

The Scheduled Castes and Scheduled Tribes are provided for in many of India's educational programmes. Special reservations are also provided for the Scheduled Castes and Scheduled Tribes in India, *e.g.*, a reservation of 15% in *Kendriya Vidyalaya* for Scheduled Castes and another reservation of 7.5% in *Kendriya Vidyalaya* for Scheduled Tribes. Similar reservations are held by the Scheduled Castes and Scheduled Tribes in many schemes and educational facilities in India. The remote and far-flung regions of North East India are provided for under the Non Lapsable Central pool of Resources (NLCPR) since 1998-1999. The NLCPR aims to provide funds for infrastructure development in these remote areas.

The government objective for the *Sarva Shiksha Abhiyan* (SSA), started in 2001, is to provide education to children between 6–14 years by 2010. The programme focuses specially on girls and children with challenged social or financial backgrounds. The SSA also aims to provide practical infrastructure and relevant source material in form of free textbooks to children in remote areas. The SSA also aims at widening computer education in rural areas. SSA is currently working with Agastya International Foundation-an educational NGO-to augment its efforts in making science curriculum current and exciting. However, some objectives of the SSA, *e.g.*, enrollment of all children under the scheme in schools by 2005 remain unfulfilled. Education Guarantee Scheme and Alternative and Innovative Education are components of the SSA.

Women from remote, underdeveloped areas or from weaker social groups in Andra Pradesh, Assam, Bihar, Jharkhand, Karnataka, Kerala, Gujarat, Uttar Pradesh, and Uttarakhand, fall under the *Mahila Samakhya Scheme*, initiated in 1989. Apart from provisions for education this programme also aims to raise awareness by holding meetings and seminars at rural levels. The government allowed 340 million rupees during 2007–08 to carry out this scheme over 83 districts including more than 21, 000 villages.

Currently there are 68 *Bal Bhavans* and 10 *Bal Kendra* affiliated to the *National Bal Bhavan*. The scheme involves educational and social activities and recognising children with a marked talent for a particular educational stream. A number of programmes and activities are held under this scheme, which also involves cultural exchanges and participation in several international forums.

India's minorities, especially the ones considered 'educationally backward' by the government, are provided for in the 1992 amendment of the Indian National Policy on Education (NPE). The government initiated the Scheme of Area Intensive Programme for Educationally Backward Minorities and Scheme of Financial Assistance or Modernisation of Madarsa Education as part of its revised Programme of Action (1992). Both these schemes were started nationwide by 1994. In 2004 the Indian parliament allowed an act which enabled minority education establishments to seek university affiliations if they passed the required norms.

Chapter Two

Modern Systems of Education

But let us first realise two fundamental differences between Ancient and Modern Systems of Education in their relation to the State, one of them prevailing alike in India and in Britain, and the other peculiar to India. In the Ancient System of India, Education and Culture were self-controlled, and while the State, the organised Nation, profited by them and from them drew its dignity, its religion, its morality, its effectiveness, and its consequent efficiency, the Legislative and Executive Departments of its Government exercised over them no control, and did not interfere with their management. Kings built Universities and bestowed on them wealth, but claimed in them no authority. A Monarch might enter into the Convocation of a University, but no one rose to greet him and he took his seat like any other visitor; but on the entrance of its Head, the "Venerable of Venerables," all rose and turned their faces towards him and in silence awaited his words. The University was the Temple of Learning, and the learned were its only Hierophants. When Learning visited Royalty, when a Wise One entered a Court, even Shri Krishna descended from His throne and bowed at the feet of the Sage.

In the Modern System, Education is under the control of a Government Department, the Legislature makes laws for it, the Executive appoints its Directors, or the Ministers, who are really its masters, sends its Inspectors into its Schools and Colleges, and puts the Educators into a steel-frame, which it misnames efficiency, This is now alike in East and West. But in India, where Kings had been its nursing fathers and had poured out their treasures at its feet, the foreign Government ignored the Ancient System, and, as its Rule spread, Education and Culture died of starvation in the kingdoms which became provinces. The splendid incheritance from the Indian Past—Hindu, Buddhist and Muslim—disappeared, leaving only the Schools of Pandits, maintained by Indian Princes, or by the reverent charity of the Hindus, till but one University, that of Nadiya, survived; the Temple and Musjid schools remained for a while and the muffasal village school—that which the East India Company, on being compelled by the British Parliament to spend a lakh on Education, called in 1814, "this venerable and benevolent institution of the Hindus," after the testimony of Sir Thomas Munro in 1813, that there were "schools established in every village".

The E.I.Co. ascribed to these "the general intelligence of the natives as scribes and accountants". Dr. John Matthai, in his *Village Administration in British India,* says that "when the British took possession of the country," they found in most parts of the country (except western and central India) that "there existed a widespread system of national Education". Even in 1838, Adam's Reports show a similar state of things in Bengal. He reports the results of an enquiry, held in 1835-1838, made in typical districts of the Presidency, and found both Toles and Madrasahs (High Schools) and Pathashalas and Maktabs (schools attached to Temples and Musjids).

The colleges were found, he writes, in "all the large villages as in the towns. The age of the scholars was from about five or six to sixteen. The curriculum included reading, writing, the composition of letters, and elementary arithmetic and accounts, either commercial, or agricultural, or both". I may add that in the village schools "Elementary Arithmetic" included multiplication tables not of only 12 × 12, but up to 20 × 20. The schools however continued to diminish in number. The *Quinquennial Review* for 1907-1912, shows 2,051 Madrasahs in 1907 against 1,446 in 1912, and 10,504 Musjid Schools in 1907 against 8,288 in 1912.

Let me pause for a moment on the age of the scholars mentioned above. In the old days, the education of the child up to the age of seven seems to have been more in the home than in the school. From seven to sixteen, the boy was to be taught and trained in school, and then to pass on to the University. The stage of infancy ends at seven, and up to that age, the body should be the first care, and lessons should be in the form of play, and great freedom of choice should be given to the little ones. No care in later life can restore the stamina of the body ill nourished, or unwisely nourished during those first seven years of life.

With the joint family system there were children enough in the household, including those of the dependents, to make a society for the children, in which they learned unconsciously lessons of kindness, of courtesy, of gentle manners and refined speech, of little sacrifices born of love, of mutual helpfulness and mutual service. With the narrowing of the home circle, the playing school is in many ways better and the children are happier in the merry games and the gay company of their little comrades. But the school must be well chosen, the teachers tender and helpful, songs, stories and play that exercises and trains the senses, the hand and the eye, and teaches graceful harmonious movements, are enough.

From seven to fourteen are the years for training the memory and the emotions, for the stories of heroism and of virtue that inspire, drawn from the history of the Motherland, and great men and women; stories too of other countries; of all that can arouse enthusiasm and inspire to service. Thus will the children have their minds and emotions so trained as to fit them to cross in safety the perilous bridge between childhood and youth. From fourteen to twenty-one is the time for hard mental study. By sixteen, the special capacities will have shown themselves, and will mark out the best avocation for the future life, and specialised Education may safely begin. This is but the barest indication of the broad stages in the preparation for manhood and womanhood, the Ideal of the Student Order of the well regulated life. But the knell of popular education was struck in 1854, when Sir Charles Wood tried, and the Government supported, the singular experiment of teaching the people in a foreign tongue, with the result that after seventy years, 3.4 per cent of the people receive primary education. So we have three stages in Education in India in relation to the State: I. Lavish help from Rulers and complete liberty of Education, paid for by the wealthy and free to the poor, who, in exchange, served their teachers and performed household duties; II. Entire neglect for 97 years, with an interval of a lakh a year spent on it; III. The Government English-speaking Schools, and Colleges, and later Universities with, of recent years, partial and grudging introduction of the vernaculars.

How shall we apply the Indian Ideals to the salvation of Modern Education and Culture in India? That is the question which Indian Universities alone can solve, and before they can answer it, may, before they can even begin the task, the old relationship must be recreated between the State and the Universities. Learning must again be inspired with the Ancient Ideals, and these will be embodied in new forms. And in order that these new forms shall be expressions of India's life, and not strait jackets to confine her, the old freedom must be restored to Education and Culture.

Government should assign to educational and cultural institutions the material means for their support, gifts of land, grants of money for buildings, and for the necessary equipment, so that they may be able to give to the Nation the priceless assets of learned and skilled men and women of high character, to carry on the work in every department of national life. Money given to Education by the Nation is not a gift, but an investment. It returns high interests to the Nation as well as power and happiness to the individual. Learned men produce literature which raises the Nation in the eyes of the world and, far more important, spreads knowledge over the earth, literature which ennobles and inspires not only contemporaries, but generations yet unborn. Science makes discoveries which add to human knowledge, increase man's power over the forces on Nature, and—if it tread only righteous paths—will preserve, uplift any strengthen human life and human happiness. By Education and Culture of man's spiritual, intellectual, emotional and physical nature can he be lifted from the savage to the Sage and the Saint, can poverty be abolished, can society be made fraternal instead of barbarous, can crime, the fruit of ignorance, be gotten rid of, and international and social peace replace war and the strife of classes. Avidya is the mother of poverty, of sorrow, of misery. It is the darkness which the Sun of Vidya must chase away.

A generation of really educated people, with a proportion of the cultured, will change the face of India. Japan educated her people in forty years. As rapid as was the destruction may be the recovery, and each successive generation will show an improved result. Already Indian Ministers have made Primary Education free in seven Provinces and compulsory in three, compulsion to be introduced as rapidly as possible in the other four. When India gains her own political freedom, may she be wise enough to restore freedom to Education and Culture, and, once more, the highest honour to Learning.

After Freedom in the Educational and Cultural field is won, for it is not possible until this freedom is possessed, the very first thing must be the restoration of the Mother-tongues of India to their proper place in that field. Nothing so denationalises a people as the imposition upon them of a foreign tongue, dominating their life and thought. When Germany, Russia and Austria rent Poland into three fragments, each banned the Polish tongue in the schools and imposed its own.

Macaulay, with the most generous feeling and the most utter ignorance, urged the substitution of the English language, literature and civilisation for those which he regarded as heathen and superstitious. The Mother-tongues were despised, and a gulf was dug between the English-educated minority and the learned in the ancient Mother-language and the middle classes education in tongues derived from it. The free Universities will use the languages of the century throughout all schools and colleges, with English as a second language, and probably other tongues as well. So far, the Universities have given little culture; that has been gained by individuals for themselves. But free Universities will have curricula which shall give both Education and Culture. Students will, as of old, be surrounded with Beauty in the Schools, the Colleges, the Universities.

The second basic difference between the Ancient System and the Modern English one, as imposed on India, is the absence of religious and moral education. In Britain itself, the religion of the country and the morality based on it are taught in the schools as an integral part of education; lately, as Nonconformity and Free Thought spread, a conscience clause has been introduced exempting children, whose parents objected to the Anglican form of Christianity or to Christianity itself, from compulsory attendance at the religious services and lessons.

But when the rule of the East India Company spread, and English Education was introduced into India, the Government schools dropped religious and moral teaching, since, on the one hand, a Christian Government could not teach heathen religions, and, on the other, as there were several religions in India, the Government must treat them all equally, and therefore remain neutral in regard to them.

Thus Indians must pay the taxes which keep up Government and other schools, and must further send their children to these, or to Missionary schools where an alien religion is taught, or open their own schools and teach any religion they belong to, Government giving them grants-in-aid. Modern Education in India had practically confined itself to the training of the mental and intellectual nature, and has ignored the unfolding of the spiritual nature, the evoking and training of the emotional nature, and, until lately, the development and training of the physical body to a high state of efficiency.

The result has been, in the older generations, the overstrain of the nervous system, the enfeebling of the physical health, the shortening of the period of vigorous maturity, often a sudden breakdown, or, at best, the premature appearance of debility and old age. Further, the exclusive development of the intelligence and the neglect of the emotions has overstimulated the self-regarding instincts, and has largely destroyed the feeling of Social and National Dharma, of duty to Society and to the Nation; hence the decay of public spirit, of social service, of responsibility and of sacrifice for the common weal, which characterise the good *citizen* as distinguished from the good *man.*

These were prominent in the result of the Ancient System; as Shri Krishna said:

> Janaka and other indeed attained to perfection by action; having an eye to the welfare of the world, thou also shouldst perform action. Whatsoever a great man death, that other men also do; the standard he setteth up, by that the people go...As the ignorant act from attachment to action. O Bharata, so should the wise act without attachment, desiring the welfare of the world...He who on earth doth not follow the wheel thus revolving, sinful of life and rejoicing in the senses, he, O Partha, liveth in vain.

This brings us to a very serious question, which has to be decided before you can settle the grading of your Education and Culture: that which in the West is called "Vocational Education". This is founded on the realisation of the fact that in modern days Society is no longer a cosmos, but has fallen into chaos, into anarchy, and that this disorder must be remedied if modern civilisation is to survive.

As Society in the Ancient India Ideal was a community of national beings, not a fortuitous concourse of atoms, it was regarded as an organism, a body politic with definite organs, each discharging a definite function, for the benefit and health of the whole community. This system was called Caste, and it was necessarily built up by Cast Education. The qualities of each pupil point to his natural avocation in the Nation. The lad who loves the open air and the care of animals, should not be an accountant, or a Clerk in city office. Nor should the quiet youth who seeks study and loves figures be sent off to a farm or a market gardener's. This is recognised in the "learned professions": Law, Medicine, Engineering demand and have separate instruction. A sturdy athletic lad fond of games is not tied down to a stool in a Bank, but is made an Engineer, to plan out railways, or enters some other active occupation.

A budding philosopher must not be sent to a factory, nor a poet to a coal-mine. While a general level of Education and Culture should be reached, so that mingling of different types should be useful and agreeable, specialisation is necessary after this is attained. At Takshasila, it was not thought unreasonable that a poor student with an aptitude for some branch of learning, should meet the cost of his board and lodging by cutting firewood and helping in domestic affairs. In studying he was on equal terms with a student whose father paid one thousand pieces for his education. No student was allowed to have any money, and a King's son was as poor as the son of a Brahamana peasant. Outcastes, however, were not received, for two Chandalas, who disguised themselves as Brahamanas but betrayed themselves by coarse language and manners when one of them burned his mouth, were beaten and sent away.

Students there were taught according to their caste. The Brahamana followed Literature as a rule, while the Kshattriya learned less Literature, but became skilled in the use of arms. Medicine and Surgery and Anatomy were there for the future physician, Mathematics for the astronomer. The courses include so much that to follow them all the manifestly impossible.

As most progressive people, hypnotised by words, object to caste, because it has been abused, if you wish to avoid prejudice, you can drop the word and call it Vocation. But, as Shri Krishna pointed out:

> The four castes were emanated by Me, by the different distribution of qualities and action.

This is the essence of Cast: the utilisation of physical heredity to provide bodies suitable for the manifestation of the qualities was an advantage, but unessential, and could only be secured by the co-operation of Devas with men, the men following the Dharma laid down for each caste and thus preserving a sub-type of physical body, to which the Devas guided the appropriate egos, *i.e.*, the agos who had evolved the given "distribution of qualities". The group of qualities was that which fitted the ego to discharge one of the functions of one of the fundamental organs of the body politic: Education, spiritual, intellectual, moral, physical; Government; Organisation of Production and Distribution; Production. In each there are many sub-divisions, as Government would include Kings, Assemblies, Judges, Lawyers, Police, etc. These are the predominant and essential groupings of qualities, whether they are called Castes or Vocations.

In the Aryan Race, the four great groups were called Castes, and Casts was a scientific system of Social Service according to the inborn qualities of the individual, birth being a convenient, but not essential, concomitant. While it remained on these lines it was honoured. It became a matter of National and Social Privileges, and is now therefore resented and, in its present form, it is doomed to disappear. Sub-castes arose sometimes from guilds of artisans, like goldsmiths, who now form a fairly powerful sub-caste in Southern India. Families carrying on the same occupation tended to live together in a particular area in a village, and made a "cheri," of their own. Others arose on religious points, or different customs. But those connected with occupations were the most numerous. Under the Ancient System, youths were trained for their future functions, National and Social, and this is reappearing in the West, as Specialised and Vocational Training, no longer confined to the learned professions, such as Law and Medicine, but extending over all avocations, commercial, trading, industrial and manual, turning the unskilled into the skilled, and thus increasing the value of each to the Nation, each with his own vocation, necessary and honourable, because a function of the organised National life.

It is remarkable that Johan Ruskin, with his far- reaching vision as artist and poet, as well as Auguste Comet, with his encyclopedic knowledge and keen and lucid intelligence, both recognised the necessity of rescuing Europe from, its anarchic social condition, if it were to survive. John Ruskin, in his *Unto This Last,* says:

> *Five great intellectual professions, relating to daily necessities of life, have hitherto existed in every civilised Nation:*
> *The Soldier's profession is to defend it.*
> *The Pastor's to teach it.*
> *The Physician's to keep it in health.*
> *The Lawyer's to enforce justice in it.*
> *The Merchant's to provide for it.*
> *And the duty of all these men is, on due occasion, to die for it.*
> *"On due occasion," namely:*
> *The Soldier, rather than leave his post in battle.*
> *The Physician, rather than leave his post in plague.*
> *The Pastor, rather than teach Falsehood.*
> *The Lawyer, rather than countenance Injustice.*
> *The Merchant—what is his "due occasion" of death?*
> *It is the main question for the Merchant, as for all of us. For, truly, the man who does not know how to die, does not know how to live.*

Ruskin then proceeds to discuss the Ideal Merchant, and, doubtless quiet unconsciously, he describes the Ideal Vaishya. But I must not follow him further on this line, as it would lead me away from Education.

Auguste Comte's classification is not so good, as it is based on a separation of Capital and Labour, and on a rigid barrier of birth instead of on a distribution of qualities.

It is, however, worthy of note that two thinkers, one purely intellectual, the other artistic, should both revert to what is supposed to be an outworn superstition, and that the intuition of the artist has carried him to the truth of the existence of a law of Nature of essential importance of Society, the disregard of which is menacing civilisation. That law unites length of days and general prosperity with the assignment of human beings to the National function for which their qualities fit them. For the proper discharge of that function they must also be fitted by a suitable Education.

India must once more have an Ideal whereby to shape an Education suited to her needs, and to her coming lofty position among the Nations of the world. Can she find a loftier Ideal than that which was her Pole Star in the past, and which preserved her from an antiquity the history of which remains alone in the "Memory of Nature," in the archives of her Rishis, in her own literature, an antiquity which cannot be checked by what is called history, for so far none exists earlier than her own, and archeological researches extend it ever further and further back, and so far tend to confirm her claim to an immense antiquity. All we can say is that history as recognised in Europe, shews nothing contrary to it, and that Europe-recognised history has never known her save as learned, wealthy, prosperous, great in her commerce, her trade, her arts and her crafts, in the magnificence of her courts and the skill of her artificers and her agriculturists, her people brave and gentle, courteous and hospitable to strangers, until the interlude of which the charter signed by Elizabeth of English was the embryo, and which will close when she is again Mistress in her own household.

I have spoken of the honour paid to Learning in India; whether it was Ancient. Middle or Modern India, whether in the Hindu, Buddhist, or Muslim Period. Learning was sought for its own sake as the mark of the highest human development, that of Man, the Thinker, short only of the supreme achievement of the Paravidya, Self-Realisation. Even to that, Jnana was one of the paths, as we shall see tomorrow.

It is worthy of notice that, in India, Education spread downwards; it was not built up from below. Indian Civilisation was a product of the country not of the town, of the forest not of the city. Greek Civilisation evolved in her cities and reached its highest point in the City-Stage. But as Rabindranath Tagore has said:

> A most wonderful thing that we notice in India is that here the forest not the town is the fountainhead of all its civilisation.... It is the forest that has nurtured the two great Ancient Ages of India, the Vaidic and the Buddhistic. As did the Vaidic Rishis, Lord Buddha also showered His teaching in many woods of India. The royal palace had no room for Him, it is the forest that took Him into its lap. The current of civilisation that flowed from its forests inundated the whole of India.

Here is an Indian Ideal that it would be well to revive, for this planting of Universities in the midst of great cities is European, not Indian. Oxford and Cambridge alone in England have kept the tradition of their Aryan forefathers. The modern "Civic Universities," as they are called, are planted in the midst of the most tumultuous, hurrying, noisy cities in England.

Not from them will come sublime philosophies or artistic masterpieces; but they will doubtless produce men of inventive genius, miracles of machinery, new ways of annihilating space. But for a country in which a man is valued for what he is, not for what he has, in which a man's life consisteth not in the abundance of the things which he possesseth, the Indian Ideal is the more suitable.

The essence of that Ideal is not the forest as such, but the being in close touch with Nature; to let her harmonies permeate the consciousness, and her calm soothe the restlessness of the mind. Hence, it was the forest, which best suited the type and the object of the instruction in the days which evolved Rishis; instruction which aimed at profound rather than at swift and alert thought; which cared not for lucid exposition by the teacher, but presented to the pupil a kernel of truth in a hard shell, which he must crack unassisted with his own strong teeth if he would enjoy the kernel; if he could not break the shell, he could go without the fruit: instruction which thought less of an accumulation of facts poured out into the pupil's memory than of the drawing out in him the faculty which could discover a truth, hidden beneath a mass of irrelevancies; of such fruitful study the Hindu Ashram in the forest is the symbol.

It must have a few representatives, at least, in India, if she is to rise to her former level in supreme intellectual and spiritual achievement, some places in which the three Margas may be taught and Yoga may be practised, until the Yogi is fit, as of old, to go out into the world of human activity, as the wise man who lives that which the *Bhagavad-Gita* teaches. This was learnt by some of the adults in the Ashrama and the Vihara, where also under the then conditions the youth of the Nation could be trained in any of the Vijjas (branches of learning) and the Shilpas (Art and Crafts) without sharing in the studies of the elders and the ascetics, yet sharing in the atmosphere they created, which radiated from them. A few "forests" should exist in India, for those who seek the Paravidya, that She may again become the spiritual Teacher of the World.

The Buddhist Vihara obtained similar results by founding the University in a spot of natural beauty, and enclosing a huge space with a high wall, pierced as in Nalanda with but one gate, in Vikramasila by six, in all cases carefully guarded by a Dvara Pandita. Within were not only splendid buildings—"Towers, domes and pavilions stood amidst a paradise of trees, gardens and fountains." There were flower-strewn lakes and blossom-laden shrubs. Well was understood the influence of natural beauty. The sacred books of Hindus and Buddhists were studied; the curriculum included anatomy and medicine, and it will be remembered that Ashoka in the third century B.C., established hospitals both for men and animals, and Mr. Dutt speaks of these being "established all over the country". One list of the subjects studied gives the five Siddhantas, Logic, Grammar, Philosophy and Metaphysics, History, Arithmetic, Geometry, Astronomy, Sanskrit, Pali, Music and Tantric medicine. Dr. Macdonnell states that in Science, Phonetics, Grammar, Mathematics, Anatomy, Medicine and Law, the attainment of Indians was far in advance of what was achieved by the Greeks.

In the *Chhandogyopanishat* we read how Narada resorted to the Lord Sanat Kumara, and prayed to be instructed by Him, and He asked what he knew already. And Nara da gives a list which reminds one of the curricula of the Universities which we know, and which evidently existed in the Ancient Hindu Age. For Narada replied:

> O Lord, I have read the Rig Veda, the Yajur Veda, the Same Veda, fourth the Atharva Veda, fifth the Ithihasa and Purana, Grammar, Rituals, the Science of Numbers, Physics, Chronology, Logic, Polity, Technology, the Science cognate to the Vedas, the Science of Bhutas, Archery, Astronomy, the Science of Antidotes, and the fine arts. [Shankara annotaies the last as the Science of making essences, of dancing, singing, music, architecture, painting, etc. (Shilpa)]... Unto him said Sanat Kumara: *"All these that you have learned are merely nominal."*

And then He leads him on step by step.

Thus "did the Lord Sanat Kumara explain what is beyond darkness".

The lists given may avail to shew why men remained in the forest, or in a monastery which was also a University for youth, into quite late maturity.

During the whole course in school as in college, strict Brahmacharya was enjoined. Here, again, is an Ideal which must be restored. The rule of Manu for the student was strictly observed: *simple dress, plain food, hard bed, the vow of the Brahmachari.* There were no exceptions, prince, noble, commoner, all were treated alike. Not is Ancient, as in Modern, India were young princes allowed to live softly, luxuriously, and they lived to a healthy old age. Now, we have boys at school who are fathers, and the seeds are sown of premature old age.

Nor must we forget how the lack of Brahmacharya in the student reacts on the child wife. Happily now young men are demanding educated brides, and hence the period of Education is being prolonged. I am not going to argue as to the orthodox view of pre-puberty marriage: Pandits find texts for and against; but this I say: if you will look at the registered death rates at different ages, you will find that the curve of the death rate of married girls shoots up suddenly at the age of 15; silent but terrible witness to the superstition which cuts short the thread of girl-life, and sacrifices the fairest and sweetest women in the world on the altar of child-marriage.

I have not found in connection with the Buddhist Universities the same attention to physical exercises as one *reads in the Jatakas in relation to Takshasila. There student practised archery, the use of the sword and the javelin, and there were military, medical and* law schools. We read also that young nobles, trained in Arts and Crafts, used to visit on their travels, after leaving the University, artists and craftsmen, and see that a high level was maintained. Thus the University re-acted on the villages, and preserved the artistic capacities and traditions of the people.

In the Muslim Period, there was a remarkable development of Architecture, an art in which the Musalmans excelled, as Arabia, Spain and India testify. The courts of the Musalman Rulers were sanctuaries of learned men, of painters, poets and musicians. Their use of jewels in architecture was extraordinarily skilful, giving richness without being meretricious. As with the Hindus and their temples, schools were attached to the Musjids, giving primary education, while Madrasahs afforded the higher education, Whether in Hinduism, Buddhism or Islam we find a similar care for Vocational Education among the higher social classes, supplying the Nation with the professions necessary for the healthy functioning of the National Life, maintaining the high level of Literature and the Arts, as well as the training of the statesman, the Minister, the military and civil organisation and administration.

The manual labour classes were equally well provided for by general instruction in reading, writing, arithmetic, accountancy, and careful training in the simple and more artistic crafts, the first for home use, the second for sale to local and export merchants. The teaching of religion and morality was universal, and much was done for the adult culture of villagers by the wandering Sannyasis who travelled on foot from village to village, and in the evenings related stories from the sacred books and chanted stotras and legends.

Taking a bird's-eye view, we may perhaps say that the Ashramas were dominated by Philosophy and Metaphysics, while not neglecting the Sciences and the Arts; the Viharas, were dominated by Science, while again not neglecting Philosophy and Arts; the Madrasahs were dominated by Art, with a divided allegiance to Science. Such classifications are, however, somewhat arbitrary, and all poured rich knowledge into the national life. To the people all were closely related, for they spread that love and reverence for Learning which shed abroad, by the stimulating force of example the superiority of Learning to Wealth, the value of Voluntary Poverty and of Sacrifice consecrated to Social Service, a Social Order which conduced to mutual usefulness, and a Beauty which, as in Japan today, is said by Mr. E.B. Havell to be "not a luxury for the rich, but the basis of National Education". He goes on:

> Poetry has done as much for National Culture in Japan as it did formerly in Greece, and, until the nineteenth century, in India also. Poetical tournaments are still a favourite form of popular entertainment in Japan, and even among the poorest classes any occasion of domestic importance, either joyful or sad, is marked by poems composed by the people themselves. In the spring mornings in Japan the working classes, the poorest of the poor, and not only the well-to-do, will rise by hundreds to watch the opening of the lotus flowers; the flowering of the plum and cherry trees in the early summer are days of national rejoicing. India need not cease to take delight in Beauty, and to have faith in the inspiration of Nature which her ancient Rishis taught, because she has become poor. It is far worse to be poor in spirit than to be poor in worldly goods. Modern science and English education are not sufficient substitutes for Art. It will not profit India to gain the whole world and lose her won soul.

The disappearance of Indian Ideals was as sudden as it was disastrous; invasions and even the establishment of a foreign Empire and foreign kingdoms previous to the invasion and triumph of the East India Company in 1757, had not touched the Soul or the Spirit of India. She had been invaded but she had assimilated the invaders and had enriched her own Culture by theirs.

Portions of her land had been conquered and occupied, but she turned the conquerors into Indians. But the East India Company not only drained her of her accumulated wealth and reduced her to poverty, but despised her Learning and her Art, crushed her with ignorance, and filled the palaces of her Princes with Brummagen imitations and glass-legged sofas and chairs.

It destroyed her self-respect and jeered at her religion and her traditions. It consummated her degradation by imposing on her an Education in a foreign language, till her educated people talked it better than their Mother-tongue. Having destroyed the Schools which had given it clerks and accountants, *it wanted English-knowing men to* *fill the lower ranks of its administration, so introduced its new system.* It got them, but the corollaries thereof were unexpected and disconcerting. It taught them English history and they became interested in English struggles for Liberty. It gave them the masterpieces of English Literature, and they studied Milton's *Areopagitica,* and declaimed Shelley's *Masque of Anarchy.* They admired the Ideals held up, and desired to find Liberty among the "blessing of British Rule". They found it not, and thirty years after the introduction of Sir Charles Wood's educational measure, they met in Madras and decided to create an Indian National Congress.

Forty years later, having revived Indian religions and started Muslaman and Hindu colleges and schools, and having meanwhile studied Indian history and assimilated its lessons, we have resolved to revive the Ancient Ideals of Indian Education and Indian Culture, to teach our children *in their Mother-tongue, to make Indian Ideals the basis of Indian Civilisation, renouncing the hybrid and sterile ideals of anglicised-Indianism, and to adept them to a new form, instinct with the Ancient Life, and moulding it into a glorious new body for the Ancient Spirit.* India will then lead the world into a new era of Literature and Beauty, Brotherhood and Peace.

THE EDUCATION SYSTEM

Education in India is the joint responsibility of the central and state governments, and educational rights are provided for within the Constitution. Following the recommendations of the National Policy on Education 1968 and subsequently by NPE 1986, attempts are being made to adopt a common structure of schooling across the country. The general pattern adopted at the national level, commonly known as the 10+2+3 pattern, envisages a broad-based general education for all pupils during the first ten years of schooling.

Diversification of courses takes place only at the higher secondary level and is reliant on students successfully completing the secondary school examination at the end of grade 10. Successful completion of the public examination at the end of grade 12 qualifies the student for university entry. Of these twelve years of schooling, the first eight years are termed 'elementary education', and this should broadly correspond to the compulsory education period of 6-14 years of age.

At the operational level, elementary school is generally divided into two parts with five years of primary schooling followed by three years of upper primary or middle school. While the description gives the general picture found in national level, actual decisions regarding the organization and structure of school education are the prerogative of state governments.

Consequently, considerable variations are found in the organizational patterns of schooling across the different states of India. Several states follow patterns in which elementary schooling consists of seven years, divided into four years of primary followed by three years of upper primary.

Thus, even while grade 8 is part of the compulsory education age range, it is part of the secondary school cycle. Correspondingly, the length of secondary schooling also varies, 'while in 22 states/UTs, secondary stage consists of classes IX and X, it consists of classes VIII, IX and X in 13 states/UTs'. Variation is also found at the higher secondary level; in some states the higher secondary stage is part of collegiate education known as junior college.

PRIMARY EDUCATION SYSTEM IN INDIA

The Indian government lays emphasis on primary education up to the age of fourteen years, referred to as elementary education in India. The Indian government has also banned child labour in order to ensure that the children do not enter unsafe working conditions. However, both free education and the ban on child labour are difficult to enforce due to economic disparity and social conditions. 80 per cent of all recognized schools at the elementary stage are government run or supported, making it the largest provider of education in the country.

However, due to a shortage of resources and lack of political will, this system suffers from massive gaps including high pupil to teacher ratios, shortage of infrastructure and poor levels of teacher training. The Indian government in 2011 show that there were 5,816,673 elementary school teachers in India. As of March 2012 there were 2,127,000 secondary school teachers in India. Education has also been made free for children for 6 to 14 years of age or up to class VIII under the Right of Children to Free and Compulsory Education Act 2009.

There have been several efforts to enhance quality made by the government. The District Education Revitalization Programme (DERP) was launched in 1994 with an aim to universalize primary education in India by reforming and vitalizing the existing primary education system. 85 per cent of the DERP was funded by the central government and the remaining 15 per cent was funded by the states. The DERP, which had opened 160000 new schools including 84000 alternative education schools delivering alternative education to approximately 3.5 million children, was also supported by UNICEF and other international programmes. This primary education scheme has also shown a high Gross Enrolment Ratio of 93–95 per cent for the last three years in some states. Significant improvement in staffing and enrolment of girls has also been made as a part of this scheme. The current scheme for universalization of Education for All is the Sarva Shiksha Abhiyan which is one of the largest education initiatives in the world. Enrolment has been enhanced, but the levels of quality remain low.

Private Education

In India and Sri Lanka, due to the British influence, a *public school* implies a non-governmental, historically elite educational institution, often modelled on British public schools which are in certain cases governmental. The most well known public school in Sri Lanka is Royal College Colombo. Although it is a governmental school it has much autonomy. S. Thomas' College located in Mount Lavinia and its branches are located in Kollupitiya, Gurutalawa, Bandarawella and Trinity College, Kandy are the most prominent private schools in the island.

Apart from this Ladies' College, Colombo; Bishop's College, Colombo and Hillwood College, Kandy are the well known private school for ladies. There are privately owned and managed schools, many of whom have the appellation "Public" attached to them, *e.g.,* the Delhi Public Schools, National Public Schools or Frank Anthony Public Schools. Most middle-class families send their children to such schools, which might be in their own city or distant boarding school such as Rajkumar College, Rajkot, the oldest public school in India. The medium of education is English, but Hindi and/or the state's official language is also taught as a compulsory subject. Preschool education is mostly limited to organised neighbourhood nursery schools with some organised chains.

According to current estimates, 80 per cent of all schools are government schools making the government the major provider of education. However, because of poor quality of public education, 27 per cent of Indian children are privately educated. With more than 50 per cent children enrolling in private schools in urban areas, the balance has already tilted towards private schooling in cities; even in rural areas, nearly 20 per cent of the children in 2004-5 were enrolled in private schools. According to some research, private schools often provide superior results at a multiple of the unit cost of government schools. However, others have suggested that private schools fail to provide education to the poorest families, a selective being only a fifth of the schools and have in the past ignored Court orders for their regulation.

In their favour, it has been pointed out that private schools cover the entire curriculum and offer extra-curricular activities such as science fairs, general knowledge, sports, music and drama. The pupil teacher ratios are much better in private schools (1:31 to 1:37 for government schools and more teachers in private schools are female. There is some disgreement over which system has better educated teachers.

According to the latest DISE survey, the percentage of untrained teachers (paratechers) is 54.91 per cent in private, compared to 44.88 per cent in government schools and only

2.32 per cent teachers in unaided schools receive inservice training compared to 43.44 per cent for government schools. The competition in the school market is intense, yet most schools make profit.

However, the number of private schools in India is still low - the share of private institutions is 7 per cent (with upper primary being 21 per cent and secondary 32 per cent - *source : fortress team research*). Even the poorest often go to private schools despite the fact that government schools are free. A study found that 65 per cent of schoolchildren in Hyderabad's slums attend private schools.

Home Schooling

Homeschooling is legal in India, though it is the less explored option. The Indian Government's stance on the issue is that parents are free to teach their children at home, if they wish to and have the means. HRD Minister Kapil Sibal has stated that despite the RTE Act of 2009, if someone decides not to send his/her children to school, the government would not interfere.

Secondary Education

The National Policy on Education (NPE), 1986, has provided for environment awareness, science and technology education, and introduction of traditional elements such as Yoga into the Indian secondary school system. Secondary education covers children 14–18 which covers 88.5 million children according to the Census, 2001.

A significant feature of India's secondary school system is the emphasis on inclusion of the disadvantaged sections of the society. Professionals from established institutes are often called to support in vocational training. Another feature of India's secondary school system is its emphasis on profession based vocational training to help students attain skills for finding a vocation of his/her choosing.

A significant new feature has been the extension of SSA to secondary education in the form of the Madhyamik Shiksha Abhiyan A special Integrated Education for Disabled Children (IEDC) programme was started in 1974 with a focus on primary education, but which was converted into Inclusive Education at Secondary Stage Another notable special programme, the *Kendriya Vidyalaya* project, was started for the employees of the central government of India, who are distributed throughout the country. The government started the *Kendriya Vidyalaya* project in 1965 to provide uniform education in institutions following the same syllabus at the same pace regardless of the location to which the employee's family has been transferred.

Higher Education

After passing the Higher Secondary Examination (the grade 12 examination), students may enrol in general degree programmes such as bachelor's degree in arts, commerce or science, or professional degree programmes such as engineering, law or medicine.

India's higher education system is the third largest in the world, after China and the United States. The main governing body at the tertiary level is the University Grants Commission (India), which enforces its standards, advises the government, and helps coordinate between the centre and the state. Accreditation for higher learning is overseen by 12 autonomous institutions established by the University Grants Commission. In India, education system is reformed. In future, India will be one of the largest education hub. As of 2009, India has 20 central universities, 215 state universities, 100 deemed universities, 5 institutions established and functioning under the State Act, and 33 institutes which are of national importance. Other institutions include 16,000 colleges, including 1,800 exclusive women's colleges, functioning under these universities and institutions. The emphasis in the tertiary level of education lies on science and technology. Indian educational institutions by 2004 consisted of a large number of technology institutes. Distance learning is also a feature of the Indian higher education system.

Some institutions of India, such as the Indian Institutes of Technology (IITs), have been globally acclaimed for their standard of undergraduate education in engineering. The IITs enrol about 10,000 students annually and the alumni have contributed to both the growth of the private sector and the public sectors of India. However the IIT's have not had significant impact on fundamental scientific research and innovation. Several other institutes of fundamental research such as the Indian Association for the Cultivation of Science (IACS), Indian Institute of Science IISC), Tata Institute of Fundamental Research (TIFR), Harishchandra Research Institute (HRI), are acclaimed for their standard of research in basic sciences and mathematics.

However, India has failed to produce world class universities both in the private sector or the public sector. Besides top rated universities which provide highly competitive world class education to their pupils, India is also home to many universities which have been founded with the sole objective of making easy money. Regulatory authorities like UGC and AICTE have been trying very hard to extirpate the menace of private universities which are running courses without any affiliation or recognition. Indian Government has failed to check on these education shops, which are run by big businessmen & politicians. Many private colleges and universities do not fulfil the required criterion by the Government and central bodies (UGC, AICTE, MCI, BCI, etc.) and take students for a ride.

For example, many institutions in India continue to run unaccredited courses as there is no legislation strong enough to ensure legal action against them. Quality assurance mechanism has failed to stop misrepresentations and malpractices in higher education. At the same time regulatory bodies have been accused of corruption, specifically in the case of deemed-universities. In this context of lack of solid quality assurance mechanism, institutions need to step-up and set higher standards of self-regulation. The Government of India is aware of the plight of higher education sector and has been trying to bring reforms, however, 15 bills are still awaiting discussion and approval in the Parliament. One of the most talked about bill is Foreign Universities Bill, which is supposed to facilitate entry of foreign universities to establish campuses in India.

The bill is still under discussion and even if it gets passed, its feasibility and effectiveness is questionable as it misses the context, diversity and segment of international foreign institutions interested in India. One of the approaches to make internationalization of Indian higher education effective is to develop a coherent and comprehensive policy which aims at infusing excellence, bringing institutional diversity and aids in capacity building. Three Indian universities were listed in the Times Higher Education list of the world's top 200 universities — Indian Institutes of Technology, Indian Institutes of Management, and Jawaharlal Nehru University in 2005 and 2006. Six Indian Institutes of Technology and the Birla Institute of Technology and Science – Pilani were listed among the top 20 science and technology schools in Asia by *Asiaweek*. The Indian School of Business situated in Hyderabad was ranked number 12 in global MBA rankings by the *Financial Times* of London in 2010 while the All India Institute of Medical Sciences has been recognized as a global leader in medical research and treatment.

Technical Education

The number of graduates coming out of technical colleges increased to over 700,000 in 2011 from 550,000 in FY 2010. However, according to one study, 75 per cent of technical graduates and more than 85 per cent of general graduates are unemployable by India's most demanding and high-growth global industries, including information technology. Nevertheless, that still means that India offers the largest pool of technically skilled graduates in the world. From the first Five Year Plan onwards India's emphasis was to develop a pool of scientifically inclined manpower.

India's National Policy on Education (NPE) provisioned for an apex body for regulation and development of higher technical education, which came into being as the All India Council for Technical Education (AICTE) in 1987 through an act of the Indian parliament.

At the federal level, the Indian Institutes of Technology, the Indian Institute of Space Science and Technology, the National Institutes of Technology and the Indian Institutes of Information Technology, Rajiv Gandhi Institute of Petroleum Technology are deemed of national importance. The Indian Institutes of Technology are among the nation's premier education facilities.

Since 2002, Several Regional Engineering Colleges (RECs) have been converted into National Institutes of Technology giving them Institutes of National Importance status. The Rajiv Gandhi Institute of Petroleum Technology : The Ministry of Petroleum and Natural Gas (MOP&NG), Government of India set up the institute at Jais, Rae Bareli district, Uttar Pradesh through an Act of Parliament.

RGIPT has been accorded "Institute of National Importance" along the lines of the Indian Institute of Technology (IIT), Indian Institute of Management (IIM) and National

Institute Of Technology (NIT). With the status of a Deemed University, the institute awards degrees in its own right.

The UGC has inter-university centres at a number of locations throughout India to promote common research, *e.g.,* the Nuclear Science Centre at the Jawaharlal Nehru University, New Delhi.

Besides there are some British established colleges such as Harcourt Butler Technological Institute situated in Kanpur and King George Medical University situated in Lucknow which are important centre of higher education.

Central Universities such as Banaras Hindu University, Jamia Millia Islamia University, Delhi University, Mumbai University, University of Calcutta, etc. are too pioneers of technical education in the country. Efforts towards the enhancement of technical education are supplemented by a number of recognized Professional Engineering Societies such as:

- Institution of Mechanical Engineers (India)
- Institution of Engineers (India)
- Institution of Chemical Engineering (India)
- Institution of Electronics and Tele-Communication Engineers (India)
- Indian Institute of Metals
- Institution of Industrial Engineers (India)
- Institute of Town Planners (India)
- Indian Institute of Architects
- Birla Institute of Technology and Science, Pilani.

That conduct Engineering/Technical Examinations at different levels (Degree and diploma) for working professionals desirous of improving their technical qualifications.

Management and Administration of Primary Schools

The education sector has become a major focus of attention amongst development partners in the last two years, driven primarily by the desire to assist India in achieving the United Nations (UN) Millennium Development Goals (MDGs) for education. The Government aims to improve national competitiveness and reduce poverty through its Medium Term Strategic Plan. It is committed to the achievement of the MDGs and the Education for All (EFA) goals, and if current efforts are sustained, these key education goals are set to be reached by 2010.

Key challenges for the Government include widening access to education without compromising quality, improving quality in a newly decentralised environment, and ensuring that the newly increased funds for education are well spent. To address this, the Ministry of Education is working through its strategic plan for 2005-2009, which has three pillars to its strategy, including (1) Improving access, (2) Increasing quality, and (3) Improving the governance and management of education. Globally, the European Commission (EC), along with a number of development partners, seeks where possible to provide support to partner governments through SWAPs and budget support. This approach is based on two key principles: one, that programmes are led by partner governments, and two, that they have the common goal of improving the efficiency and effectiveness with which internal and external resources are used. This reflects a mutual concern to improve results of government and donor spending both by focusing resources on the priorities stated in national planning documents and by improving the quality of spending.

A number of development partners are currently supporting programmes in basic education, including: Australian Agency for International Development (Ausaid), United States Agency for International Development (USAID), Asian Development Bank (ADB), Japan International Cooperation Agency (JICA), UN Children's Fund (UNICEF) and UN Educational, Scientific and Cultural Organisation (UNESCO). Large amounts of development assistance funds are being made available to support education reform and development.

There is growing interest in moving towards a SWAP among this group, which formalised inter-donor coordination in 2005 with the establishment of the Education Sector Working Group (ESWG), currently co-chaired by the EC and the Netherlands for one year. This group is taking forward the Paris Declaration on Aid Effectiveness, and is seeking to work with the Government in support of achievement of MDGs. World Bank support to basic education is moving into a new phase towards teacher quality support and district level SWAPs, joining forces with the Netherlands and the EC, meanwhile both the ADB and Ausaid have developed major programmes aimed at working towards a SWAP in Indonesia.

For development assistance through a SWAP to become effective, the Government needs to take full ownership of the process, to take a lead in donor coordination and harmonisation, and establish a platform for policy dialogue at sector level which will allow development partners to monitor developments in education and to engage productively with the Government. Alongside this, a medium term expenditure framework will allow development partners' assistance to fill financing gaps, and will give the Government much clearer information than it has at present on how development assistance is being used and can be used in the future.

Professional Developments

Professional development refers to skills and knowledge attained for both personal development and career advancement. Professional development encompasses all types of facilitated learning opportunities, ranging from college degrees to formal coursework, conferences and informal learning opportunities situated in practice.

It has been described as intensive and collaborative, ideally incorporating an evaluative stage. There are a variety of approaches to professional development, including consultation, coaching, communities of practice, lesson study, monitoring, reflective supervision and technical assistance.

CLASSROOM MANAGEMENT

Classroom management is a term used by teachers to describe the process of ensuring that classroom lessons run smoothly despite disruptive behaviour by students. The term also implies the prevention of disruptive behaviour. It is possibly the most difficult aspect of teaching for many teachers and indeed experiencing problems in this area causes some to leave teaching altogether. In 1981 the US National Educational Association reported that 36% of teachers said they would probably not go into teaching if they had to decide again. A major reason was "negative student attitudes and discipline". (Wolfgang and Glickman)

According to Moskowitz and Hayman (1976), once a teacher loses control of their classroom, it becomes increasingly more difficult for them to regain that control (Moskowitz & Hayman, 1976). Also, research from Berliner (1988) and Brophy & Good (1986) shows that the time that teacher has to take to correct misbehavior caused by poor classroom management skills results in a lower rate of academic engagement in the classroom (Berliner, 1988; Brophy & Good, 1986).

Classroom management is closely linked to issues of motivation, discipline and respect. Methodologies remain a matter of passionate debate amongst teachers; approaches vary depending on the beliefs a teacher holds regarding educational psychology. A large part of traditional classroom management involves behaviour modification, although many teachers see using behavioural approaches alone as overly simplistic. Many teachers establish rules and procedures at the beginning of the school year.

They also try to be consistent in enforcing these rules and procedures. Many would also argue for positive consequences when rules are followed, and negative consequences when rules are broken.

There are newer perspectives on classroom management that attempt to be holistic. One example is affirmation teaching, which attempts to guide students towards success by helping them see how their effort pays off in the classroom. It relies upon creating an environment where students are successful *as a result of their own efforts*...

Management of School Building

During The Construction Period: Liaison with the local authority and the contractor:

1. Before construction activities commence on site, it is important for the school to be aware of the project communication arrangements between the local authority project manager and the contractor. Regular site meetings are likely to be held, and it may be appropriate for the school to be represented, or to be briefed on progress.
2. In many projects, day-to-day communication should be encouraged between the school co-ordinator and the contractor's site manager regarding 'housekeeping' matters which will affect the running of the school.

3. As well as having a clear understanding of the design and contract programme, the school should also be aware of what terms and conditions, if any, were included in the contract regarding the contractor's working restrictions, health and safety, security and so on. The school itself will have no authority to enforce the terms of the contract. It is therefore essential to establish a single point of contact with the local authority for all communication regarding the project.

Keeping Staff and Pupils Informed

4. The flow of information to and from school users is very important throughout the project. This is usually an element of the school co-ordinator's role.
5. Existing communication arrangements within the school such as regular departmental and whole staff meetings should be used wherever possible rather than setting up separate project meetings. However, staff should be able to raise individual issues directly with the school co-ordinator.
6. Pupils are usually kept informed about the project through school assemblies and feedback is often provided through the pupil council or through 'house structures' where these operate in secondary schools.
7. The project is also likely to feature heavily on the agenda of any School Board and PTA meetings. The HT and school co-ordinator, as well as the contractor, may well be expected to make regular progress reports at these meetings.
8. *One secondary school undergoing a major refurbishment set up a specific project users group which met monthly throughout the duration of the project. The group consisted of members representing staff, pupils, parents and others from the school community. Meetings were chaired by the HT and were attended by the local authority project manager and the contractor.*

Managing Hazards and Disruptions

9. The contractor will manage the health and safety and security aspects of the construction activities in any project. However, this in itself does not prevent the school from having to consider a number of health and safety matters or having to prepare at certain times for considerable disruption to the normal day-to-day running of the school.

Health and Safety/security

10. The school will need to monitor its health and safety management procedures throughout the project and carry out additional risk assessments where appropriate. In particular, any alterations to fire escape routes and gathering

points, access arrangements and site boundaries should be clearly identified. Arrangements should be clearly displayed in the school with signage amended as appropriate. Fire drills should be undertaken each time escape routes are changed.

11. Prior to construction commencing on site, all school users should be briefed on the health and safety arrangements for the project. Presentations to school assemblies by the contractor, often incorporating protective clothing and some basic statistics about construction site safety, are generally considered to have more impact than if these were delivered by the school management team.
12. As school holidays approach further advice should be issued to pupils regarding the hazards of building sites. Security monitoring will be provided by the contractor but in some instances it may also be appropriate to alert the local police who may patrol the site at high risk periods.
13. Where work needs to be carried out in occupied buildings, contractor's staff may require to gain access to parts of the construction site through operational parts of the school. These situations generally place greater responsibilities on the contractor's staff and school users. These responsibilities need to be understood by all.
14. It may be considered appropriate in these situations for the contractor's staff to wear agreed forms of identification and to be prepared to be challenged by staff on school premises. It may also be considered necessary to caution contractor's staff to avoid initiating contact with pupils. It has also been known for school pupils to abuse construction workers, so the conduct of school users must also be considered and managed.
15. Despite contractual agreements, breaches in health and safety procedures can still occur in these situations. For example contractor's staff may leave doors unlocked or materials and equipment unattended in circulation routes. School users should be particularly vigilant about possible hazzards and procedures should be in place to report such incidents.

Managing Disruptions

16. A certain degree of inconvenience is unavoidable during a major construction project. The level of disruption likely to be experienced by the school will depend upon a variety of factors such as the scale and type of building operations, their proximity to the school activities, the time at which the works are carried out, and the constraints of the existing buildings and site.
17. The following examples are the most common disruptions reported by schools during construction projects:
    — Dust and dirt : Apart from being particularly uncomfortable, airborne dust and dirt can give rise to medical complaints leading to staff and pupil absences. Locating classrooms which require natural ventilation away

from construction activities, and insisting on the constructing and maintaining of seals in affected areas can help to minimise the ingress of dust and dirt. In some cases, it may be appropriate to arrange additional cleaning for the school during the project.

— Noise : Health and Safety regulations ensure that noise levels will not be hazardous to health, but they may still be extremely distracting. Where construction works are in close proximity to the school, the contractor may be excluded from undertaking certain noisy activities during particular periods such as exams. It may also be possible for the school to request the contractor, on an informal basis, to reduce noise levels for short periods from time to time.

— Distractions from increased traffic, both vehicular and personnel : In order to avoid continuing distraction it may be possible to locate particularly sensitive school classes and activities away from the main site access and construction works. Ensure site activities are appropriately screened and avoid allowing contractor's staff access through pupil areas.

— Frequent changes to access points and circulation routes : the need to advise school users of continuing changes in access arrangements can be disruptive and resource intensive. This should be considered when agreeing the sequence of phasing of the construction works.

— Planned and unplanned interruptions to water, gas, power, ICT services; prior consultation and contingency planning with staff about the consequences of a particular service failure or disconnection will allow the school to better manage these situations when they occur.

— Reduced playground space, loss of playing fields and car parking : The loss of amenities during the construction process is often an unavoidable source of inconvenience to school users. However, early consultation with those affected, provides the opportunity to investigate and implement alternatives.

Need of Value Education in 21 century

Education is a methodical effort towards learning basic facts about humanity. And the core idea behind **value education** is to cultivate essential values in the students so that the civilization that teaches us to manage complexities can be sustained and further developed. It begins at home and it is continued in schools. Everyone accepts certain things in his/her life through various mediums like society or government.

Value education is important to help everyone in improving the value system that he/she holds and put them to use. Once, everyone has understood their values in life they can examine and control the various choices they make in their life. One has to frequently uphold the various types of values in his life such as cultural values, universal values, personal values and social values.

Thus, **value education** is always essential to shape one's life and to give him an opportunity of performing himself on the global stage. The need for **value education** among the parents, children, teachers, etc., is constantly increasing as we continue to witness increasing violent activities, behavioural disorder, lack of unity in the society, etc.

The family system in India has a long tradition of imparting value education. But with the progress of modernity and fast changing role of the parents it has not been very easy for the parents to impart relevant values in their wards. Therefore many institutes today conduct various value education programmes that are addressed to rising problems of the modern society. These programmes concentrate on the development of the children, young adults, etc. focusing on areas like happiness, humility, cooperation, honesty, simplicity, love, unity, peace, etc.

Values

Values are those things that really matter to each of us... the ideas and beliefs we hold as special. Caring for others, for example, is a value; so is the freedom to express our opinions.

Most of us learned our values-or morals, if you prefer-at home, at church or synagogue, at school. But, where are our children learning their values? Maybe from parents, teachers and religious leaders, but society has changed. Too often young people today are most influenced by what they see and hear on television or on the street.

For this reason, the Boy Scouts of America-the nation's largest youth development organization-introduced new tools to help young people-from Cub Scouts through Exploring-develop positive values while learning to make ethical decisions.

The Scout Oath and Law express a well-defined code of ethical and moral conduct. If you think about it, you'll see that these abstract ideas-trustworthy, loyal, helpful, friendly, courteous, kind, obedient, cheerful, thrifty, brave, clean, reverent-can become very concrete goals for young people.

No, these ideas aren't "new and improved." They've been around since the very beginning of Scouting. These new tools provide a means for teaching today's young people how to apply these abstract ideas in everyday situations. Scouting is designed to keep young people busy and involved in all sorts of projects. Scouts learn by doing, and learning values is as action-oriented as other Scouting programmes. And all are designed to help young people acquire the values that they will strive towards well beyond their Scouting years.

After years of talk about the "moral decay" in just about every area of American life, our society seems to be turning back to the traditional values that guided this nation to greatness. To pass these values on to our children through Scouting relies upon three components: caring adults, age-appropriate and purposeful activities, and meaningful roles in the community

Leadership training and the literature of Scouting have been revised to help caring adults become better, more effective Scout leaders... to recognize that young people develop physically, mentally, socially-and, yes, ethically-at different rates... to identify community service projects that drive home the message that young people, by interacting with people in their community, can have a positive impact on the world around them. Building upon Scouting's long history of youth development, makes this heritage even more relevant in meeting the needs of contemporary youth.

In ethics, **value** is a property of objects, including physical objects as well as abstract objects (*e.g.*, actions), representing their degree of importance.

Ethic value denotes something's degree of importance, with the aim of determining what action or life is best to do or live, or at least attempt to describe the value of different actions. It may be described as treating actions themselves as abstract objects, putting value to them. It deals with right conduct and good life, in the sense that a highly, or at least relatively highly, valuable action or may be regarded as ethic "good" (adjective sense), and an action of low, or at least relatively low, value may be regarded as "bad". What makes an action valuable may in turn depend on the ethic values of the objects it increases, decreases or alters.

An object with "ethic value" may be termed an "ethic or philosophic good" **personal and cultural value** is a relative ethic value, an assumption upon which implementation can be extrapolated. A *value system* is a set of consistent values and measures that are not true. A *principle value* is a foundation upon which other values and measures of integrity are based. Values are considered subjective, vary across people and cultures and are in many ways aligned with belief and belief systems. Types of values include ethical/moral values, doctrinal/ideological (religious, political) values, social values, and aesthetic values. It is debated whether some values are intrinsic.

Differing Global Perspectives

Perhaps the most important step in understanding and incorporating global education in classrooms and communities is to understand and relate to the themes of global awareness as presented by experts in the field. Hanvey (1976), one of the first scholarly experts to give a comprehensive definition of the concept "global awareness", proposes five dimensions that prepare students to achieve global awareness. These include perspective consciousness, state-of-the-planet awareness, cross-cultural awareness, knowledge of global dynamics, and awareness of human choices. Haavenson, Savukova, and Mason (1998/99) conducted their research on United States and Russian perspectives on teacher education reform and global education and found that these dimensions form the first level known as attitude formation upon which global education can be implemented. The second level is the development of cognition skills and the third level is an integrated view of the world. An explanation of Hanvey's five dimensions, paraphrased by Kirkwood (2001) and Haavenson *et al.* (1998/99) is provided below and an explanation of the other two levels will be identified.

Perspective Consciousness

Perspective consciousness refers to an awareness of and appreciation for other images of the world and that a person's worldview is neither universally shared, nor necessarily right, yet may be profoundly different. It is the realization that an individual's worldview is both a matter of conscious opinions and ideas and more importantly to subconscious evaluations, conceptions and unexamined assumptions.

Perspectives are shaped by ethnic, religious, differences in age, sex, and social status, among many other factors. These differences, as stated by Haavenson *et al.* (1998/99), "have been one of the main causes of conflict and confrontation in the history of mankind" (p.38).

The authors go on to say that, "It is important to teach students to look upon a certain phenomenon or event from different perspectives so as to encourage respect and appreciation for beliefs, customs, and values different from their own" (p. 38). It is not only about racial and cultural differences, instead, a pluralistic view needs to be taken when looking at global perspectives.

State-of-the-planet Awareness

State-of-the-planet awareness requires comprehension of prevailing world conditions, developments, trends, and problems that are confronting the world community. It includes an in-depth understanding of global issues such as population growth, migrations, economic disparities, depletion of resources, and international conflicts, that require global learners to be aware of the world around them. Children need to be made aware that what affects the world affects them as well. In elementary school, students can be taught to make decisions about ways to prevent disaster by studying the consequences of environmental illiteracy.

Cross-cultural Awareness

This dimension includes the diversity of ideas and practices in human societies and how these ideas and practices are found in human societies around the world, including concepts of how others might view one's own society as perceived from other vantage points. According to Hanvey (1976), this dimension is the most difficult to attain most likely because it refers to the highest level of global cognition. The misconception about cross-cultural awareness is that people consider it no more than a set of stereotypes that do more harm than good as superficial knowledge engenders prejudice.

An effective way to promote cross-cultural awareness, as explained by Haavenson *et al.* (1998/99), is by showing videos and then having discussions with students about these films to help them in separating stereotypical views from those that are more authentic.

KNOWLEDGE OF GLOBAL DYNAMICS

Knowledge of global dynamics refers to an understanding of the world as an interconnected system of complex traits and mechanisms and unanticipated consequences. A high level of sophistication on the part of the student is required because understanding these processes is difficult to achieve due to the unanticipated effects on the human condition. It includes a consciousness of global change and cannot be acquired through mass media. Haavenson *et al.* (1998/99) explain that "[s]tudents learn to identify subtle cause-effect relationships, anticipate side effects, model processes and make decisions about eliminating or altering undesirable consequences". Students may be asked to create webs of the factors influencing the issue, to suggest feasible solutions, and to foresee possible side effects of such actions.

Awareness of Human Choices

Hanvey (1976) challenges global thinkers to realize the problems of choice confronting individuals and nations as consciousness and knowledge of global systems expand. It is related to global dynamics in such a way that it focuses on making choices and develops a sense of responsibility for making decisions made which affect future generations. It also includes an awareness of the interconnectedness of individual, national, and international settings. It fosters a sense of responsible citizenship on the local and global levels. Students may be introduced to alternatives on thought and behaviour by looking at relationships and interactions between man and the world. Students are asked to account for their choices and are taught to be tolerant towards the view of others.

In their study with teachers, Haavenson *et al.* (1998/99) posits that the second level of global education implementation is cognition focused. This means that life demands both a thorough knowledge of a domain combined with a broad perspective of the world. This is similar to the 'interconnections' theme that Werner & Case (1997) identify and develop which explores both the international and inter-system linkages and conclude that we live in an interconnected world. Therefore students must be encouraged to see the different ways in which one situation is influenced by and influences others.

Further exploration in the topic is explained by Haavenson *et al.* (1998/99) that the brain often searches for common patterns and relationships and seeks to connect new knowledge with prior experiences that result in the fact that cognition operates in all concepts. The traditional approach of filling the minds with facts and information that students are simply asked to memorize and reproduce does nothing to promote global awareness and teachers must keep this in mind when working to plan curriculum. Instead, students need experience in critical thinking, in taking part in cross-cultural experiences, and to make decisions and substantiate them. In the study by Haavenson *et al.* (1998/99), students are taught to think for themselves and to be able to stand their ground. The authors advise that the atmosphere created by the teacher is very important.

The third level of global education implementation is an integrated view of the world as explained by Haavenson *et al.* (1998/99). They state that “the third level aims to create a specific picture of the world where geographical, physical and linguistic features all fit into a complex pattern”. This means that all discipline-focused world perspectives need to overlap due to the interdependence of facts, events, and phenomena. For instance, university interdisciplinary courses may be the most effective way to create a cross-disciplinary perspective.

Global Awareness Elements

Case (1993) identifies five key substantive elements that keep people informed of a range of global topics. The first element describes the universal values and cultural practices, and the second includes global interconnections, which refers to the study of the workings of the four major interactive global systems: economic, political, ecological, and technological.

The third presents worldwide concerns and conditions such as development and peace issues while the fourth forms the origins and past patterns of worldwide affairs such as global history and geography. The last presents alternative future directions in worldwide affairs. In addition to these substantive elements, he proposes perceptual elements that should be addressed which include open mindedness, resistance to stereotyping, anticipation of complexity, empathy, and nonchauvinism.

Kirkwood (2001) analyses Case’s elements and explains that the substantive elements listed above “includes the objects of global education that incorporate the contemporary events, conditions and locations in the world that Hanvey (1976) addresses within the state of the planet awareness dimension”.

The perceptual elements focus on the development of world mindedness and empathy and resistance to prejudicial thinking as well as stereotyping and cross-cultural knowledge. These elements are similar to the Hanvey dimensions of perspective consciousness and cross-cultural awareness.

Case (1993) and Hanvey (1976) provide similar definitions for global awareness even though the terminology they use is different. Merry M. Merryfield, one of the leading scholars in the field of global education, combines the definitions of other scholars and provides us with a current framework in this field today.

Kirkwood (2001) lists Merryfield’s eight elements which include: human beliefs and values global systems, global issues and problems, cross-cultural understanding, awareness of human choices, global history, acquisition of indigenous knowledge, and development of analytical, evaluative, and participatory skills. Kirkwood (2001) concludes that “Merryfield’s work contributes significantly in reducing, if not eliminating, the definitional ambiguities that still linger in the field”.

New Understandings of Global Awareness

Kirkwood (2001) presents another dimension to the definition of global education that Lamy (1987) identifies as the acquisition of knowledge transmitted by indigenous people. He concludes that a global education must include knowledge about the contributions of native people who are representing the views of their world. In his words, "The teaching of historical and contemporary events must be balanced by listening to indigenous voices".

To provide further elaboration in regards to listening to Indigenous voices, Battiste and Henderson (2000) think of globalization as a new threatening transformation that is emerging. In the introduction of their book, they state that, "Globalization with its cognitive and linguistic imperialism is the modern force that is taking our heritages, knowledge, and creativity".

For Indigenous people it is not just physical survival that concerns them; it is an issue of "maintaining Indigenous worldviews, languages, and environments". It is ironic that the world looks to Indigenous people for help in order to solve the world's crisis that its worldview has created. Battiste and Henderson (2000) state that "in view of the history of relations between the colonizers and the colonized, this is an extraordinarily bold request."

The work of David Selby and Graham Pike has brought new understandings in regards to ecological awareness or 'state of the planet awareness' as outlined by Hanvey. They have been influential in the global education field in the 1980's and are known nation wide. Influenced mainly by Richardson and Hanvey in the 1970's they have worked mainly with secondary schools. Hicks (2003) in his review of global education discusses the work of Selby & Pike.

He explains that in 1988 they further developed the conceptual map of the field and highlighted what they called 'the four dimensions of globality'. These dimensions make up the core elements of global education. The first is 'issues dimension', which embraces five major problem areas and solutions to them: inequality/equality, injustice/justice, conflict/peace; environmental damage/care; alienation/participation.

The second is 'spatial dimension' which emphasizes exploration of the local-global connections that exist in relation to these issues, including the nature of both interdependency and dependency. The third is 'temporal dimension' that emphasizes exploration of the interconnections that exist between past, present, and future in relation to such issues and in particular scenarios of preferred futures. The fourth is the 'process dimension' that emphasizes a participatory and experiential pedagogy which explores differing value perspectives and leads to politically aware local-global citizenship. Selby and Pike then relate this to both individual subjects in the curriculum and whole-school case studies.

Hicks (2003) further explains that each of these four elements needs to be present before one can claim to be involved in global education. Both Selby & Pike have written extensively on the importance of ecological thinking in global education and this is evident within the four-dimensional model that they propose for global education. It needs to be stressed that the environmental health of the world is just as important as taking care of all humanity and that the two must work together simultaneously. For example when explaining the 'spatial dimension' Selby (1998) writes that "this dimension also concerns the cycles and systems of nature and the relationships between human society and the environment".

Hicks (2003) explains that the 'temporal dimension' is a futures perspective that "looks at how global issues affect and are affected by interrelationships between past, present and future". He goes on to say that "this works to help young people think more critically and creatively about the future, especially in relation to creating more just and sustainable futures".

EDUCATIONAL SCENARIO IN INDIA TODAY

More and more parents, therefore, were turning towards private schools even in villages and wayside slums. The PROBE team discovered that private schools, even in remote villages, were very active in teaching and getting good results for their pupils. In the private schools the teachers did not have job security and consequently, their continuation depended on their performance, watched both by the owners who wanted a fair name for their schools and the parents, who could shop around for a school with a good name anyway. That motivates the teachers to work hard and put their heart and soul into teaching. Those village schools charged very reasonable sum of 35-50 rupees per month; in the city slums the going rate was 60-100 rupees per month.

These private schools were more popular because they all taught English, in addition. In the slums around Charminar in Hyderabad alone there were as many as 500 such schools belonging to a single Federation, serving predominantly the poorer sections of society like the Rickshaw pullers, daily wage earners, vegetable sellers, fisher women, and the like. They charged very modest fee of around a thousand rupees an year. None of them depended on government subsidy! These schools also had an altruistic motive in reserving a quarter of their seats for the poorest of the poor in the locality giving them away free. Similar experiments were going on in many other developing countries like Thailand, Columbia, Tanzania, and Chile. This method could be easily replicated elsewhere. In fact, Prof. Tooley recommends many of these methods to the inner city area schools that starve for want of funds and good teachers.

This self-help idea was the one that prompted the first ever private medical college in India in Manipal by a thinking man, Late Dr. T. M. A. Pai, who wanted that the motivated students should be provided with an opportunity to pursue their interest, assisted by their parents chipping in their lot to sustain the institutions. The Manipal Academy of Higher Education, a Deemed to be University, is the result of his initial effort. Independent surveys have given us very high rating not only in India but abroad. We have students from thirty English speaking countries. We have been assessed as one of top four colleges in medical education.

Institutions do not depend for their growth on either the government or other owners. Educational institutions get their name and fame because of the men and women who struggle to keep up the highest academic standards as also the ethical values. Our founder's motto was to nurture the best of teachers. Having been in these institutions for forty years I could vouch for that truth. Universities should be proud of their faculty and not their brick and mortar or equipment as much. Our powers-that-be do not seem to realize this naked truth.

It is only in our country that private effort in education is being looked down upon and treated badly. People outside governmental control have founded some of the great institutions all over. A banker, like Dr. TMA Pai, founded the University of Edinburgh, way back in the eighteenth century. That was the time of the Scottish enlightenment when Edinburgh was considered to be the Athens of the north. Except the University of Georgia Medical School, most of the American medical institutions of repute are in the private sector. Harvard, Yale, Stanford, Johns Hopkins, Mayo Clinic, Cleveland Clinic and most of the others belong to that class. The Guy brothers, business tycoons of those days, founded the famous Guy's Hospital Medical School in London. They are not untouchables in their countries, but are venerated very much as centres of excellence and repute.

This pioneering Manipal experiment, the first of its kind in India, unfortunately has been badly replicated by many others to make it into a big "business" in higher education. These kind of unscrupulous methods are being abetted and nurtured by politicians and their goons for their own benefit. That is no reason why the original idea of Dr. T.M.A. Pai should be found fault with by our thinkers. Many of the later institutions do not have even the bare minimum necessities, not to speak of the all-important faculty. It would be shocking to know that a sizable percentage of them thrive only on visiting faculty, who rarely visit. People in authority would overlook all these so long as their machinery is well oiled and greased! Thanks to the munificence of our greedy powers-that-be, such institutions thrive better in India. It is only the honest and the meritorious that suffer in this environment.

What is the need of the hour? Many of our journalists with the holier-than-though attitude towards private efforts at higher education and many of our armchair intellectuals who live in ivory towers, having no touch with reality and without any personal experience in the field, think that education, outside the government setting, is getting commercialized in India. They feel that this is the greatest sin and should be curbed at any cost. We would be happy to host them here to have first hand knowledge of the trials and tribulations of running excellent educational institutions.

Similar arguments were put forward for "socialism" of the Russian variety in the '50s, despite the fact the Mahatma Gandhi had strongly advocated the cottage industry and village development as the need of the hour. Now even the champions of the mega things in development are convinced that "small is beautiful". The country is suffering from the fall out of that sin at an enormous cost to the people. Governments got involved in every aspect of the common man's life, starting from transport, electricity, water supply, food distribution, industry, health care, hotels, and what have you. None of them function properly even after half a century! It has now dawned on the politicians that they should get out of these as soon as possible. Education is another field where the government has miserably failed. Sooner they realize this the better for our future generations.

We must nurture and develop private schools and even professional colleges that keep up excellent standards of education. There should be no compromise on standards. Let us learn the lessons of the PROBE study. Let private institutions depend on their excellence for their very existence! Allow them a level playground without throttling them. Let there be an independent accreditation body of independent people of integrity that periodically publicizes the standards for students and parents to know.

Let there be survival of the fittest without any outside agency bringing in unreasonable restrictions. Let the buyer, the well-informed student, take the pick. The bad institutions would die a natural death in the process. That is the exact way how American medical education was cleaned of the unscrupulous medical schools of which there were more than two hundred in the fifties. Flexner Committee, an independent body, rated all of them. Their findings were made public by the government repeatedly in national dailies over a period of a month or so. Only seventy-two colleges survived, as the rest died a natural death for want of student aspirants to fill their seats. No amount of regulations and rules would work that effectively, since the crooked owners could twist many of the latter.

This reminds me of a lecture given by Mrs. Margaret Thatcher, the then Prime Minister of Britain, in one of the conferences on "quality control". She got up to speak and started by narrating her own experience, as the grocer's daughter, at her father's shop, when she was young. "Whenever the vegetables we sold were good the customers came back; when the vegetables were bad the vegetables came back and the customers went away to other shops! That is quality control." How true! The institutions should survive on their merit alone and not on rules and controls in the open market. That would be healthy for higher educational institutions as well. The really bad institutions will die a natural death under those conditions, as no student would want to go there.

The private institutions should be allowed to grow without the usual constraints of the "licence-raj" let loose on them both by the politicians in the government and their henchmen. The above categories do not want to loosen their grip on the private institutions. The corrupt people's bread and butter are these institutions. That is precisely why they want to have their complicated regulations and rules to have an absolute control over private effort at education. Our biggest curse is this "license-raj" system. The root of all corruption starts there. The more rules that the government puts forward, the better for the greedy in power. Every single rule is an opportunity for corruption.

This country, or for that matter, even the developed countries, would not be able to subsidize primary education, leave alone higher education, in the new millennium. The recipient should pay for higher education. If he is poor or hails from an economically backward community, the government could help him raise a loan from the Educational Development Bank to be repaid only after he gets a job and the interest thereon could be waived or reduced depending on his economic status.

Most governments would not be able to service the interest on their foreign borrowings in the next decade even with all their revenue collection. Many of them would not have the money to pay their staff! In that situation education could only be in the private sector. Our netas should not be under the delusion, that by selling the idea to the gullible public that commercialization in education is bad for the country, they feel that they could continue to use this "milch cow" to fill their coffers for elections and also to hoard money for their progeny. They seem to forget that they can not take the hoarded money with them when they go to meet their maker.

Even the famous "Unnikrishnan" judgement of the Supreme Court subscribes to this view that private education is bad in principle. I am sure with the present scenario one could go back to the Supreme Court for a revision of that judgement in the background of the wisdom from the PROBE investigations. With more evidence accumulated against the view, our adversarial system of justice might agree with the new wisdom.

As of now the private institutions face great hostility from the government. They are viewed with suspicion as thieves. While there are thieves in every field of human endeavour, there are also excellent institutions run on ethical principles. Of course, there is no free lunch anywhere. The private institutions should be allowed to get back a reasonable return on their investments and also be able to break even to sustain themselves. A logical financial assessment of the needs vis-à-vis their expenditure could be made by an independent body whose members' credentials could be bared for public scrutiny before being appointed, but the committee should not have any kind of restrictions, either from the governments or the watchdog bodies. I am sure there are many in this country that are honest and ethical. In fact, they are in the majority, but they are a silent majority not noticed by our media and also the merit award giving bodies of the country! Let them be given at least this thankless job of overseeing private initiative in educating our masses in higher studies.

Many of our leaders in the government behave as if they belong to a higher race. I am reminded of what French President Giscard d'Estaing wrote in his memoirs in 1981. "In my country" he said, "there is the idea that those who govern belong to another race."

Earlier we change the set up and clean up the stables the better for the future generation. With more than five hundred million young men and women looking for higher education in the country in the next fifty years, we would create chaos in the near future unless we set our house in order. There is no desirable pattern of non-governmental educational efforts in higher education prescribed so far. Consequently, the unscrupulous would want to commercialize education. Unfortunately, they are the ones that get all the perks from the powers-that-be as they supply the needs of the latter. Those who want to be honest and authentic get all sorts of hurdles put on their way.

Every rule is made to be used to either make money for the rulers and their ilk or for having some control over the hapless organizations in the field of higher education. A survey of the present facilities for higher education in the existing set up would throw up worse things than what was brought out by PROBE team in primary education.

While the government set-ups have very little infrastructure, the private ones are made to achieve the impossible. I get a feeling that there must be a deep-rooted conspiracy to keep up the governmental hegemony in education for the politician and the bureaucrats to make money perpetually. This reminds me of the nice study published from the Cato Institue in Washington edited by Doug Bandow and Ian Vasquez entitled Perpetuating Poverty, wherein the authors have systematically shown how the World Bank and the International Monetary Fund have promoted poverty in the developing world for the ultimate good of the big powers, so that they could keep the former under their thumb perpetually. Writing a commentary on the book, Melvyn B. Krauss, professor of economics at the NewYork University, has this to say: "This book destroys the myth that the multilateral aid agencies are forces for good and shows they are instead a major fraud perpetuating poverty in the developing countries…………..The editors are to be congratulated for editing this precise, cold-eyed, and extremely important collection of essays that merits a wide audience."

Chapter Three

Social Development

Social development is a process which results in the transformation of social structures in a manner which improves the capacity of the society to fulfil its aspirations. Society develops by consciousness and social consciousness develops by organization.

The process that is subconscious in the society emerges as conscious knowledge in pioneering individuals. Development is a process, not a programme. Its power issues more from its subtle aspects than from material objects.

Not all social change constitutes development. It consists of four well-marked stages — survival, growth, development and evolution, each of which contains the other three within it. The quantitative expansion of existing activities generates growth or horizontal expansion.

Development implies a qualitative change in the way the society carries out its activities, such as through more progressive attitudes and behaviour by the population, the adoption of more effective social organizations or more advanced technology which may have been developed elsewhere. The term evolution refers to the original formulation and adoption of qualitative and structural advances in the form of new social attitudes, values, behaviours, or organizations.

While the term is usually applied to changes that are beneficial to society, it may result in negative side-effects or consequences that undermine or eliminate existing ways of life that are considered positive.

FACTORS OF SOCIAL DEVELOPMENTS

Discrimination: Sometimes there are social or cultural factors that hold back poor countries. Discrimination is one of these. If there are certain people groups that are discriminated against, the country's overall productivity can suffer. This may be a tribe, a caste, a racial category or minority language group. I have already mentioned Cameroon, which has both French speaking and English speaking regions. All the infrastructure happens in the French speaking part. French speakers in Canada complain of the opposite. Welsh speakers in Britain, or Catalans in Spain, have historically faced similar problems. Racial discrimination may be an issue, excluding certain groups from economic activity, either deliberately or not. Racial minorities regularly have poorer exam results and economic prospects than the majority. More serious forms of exclusion would be apartheid South Africa, or the Asian communities driven out of Uganda under Idi Amin, which was disastrous for Uganda's economy.

Another division may be the role of women. Jeffrey Sachs talks about this in The End of Poverty: 'Cultural or religious norms may block the role of women… leaving half the population without economic or political rights and without education, thereby undermining half of the population in its contribution to overall development.' If you don't believe that women should work, you have effectively halved the earning potential of your country.

Population: Closely linked to this is the population issue. If women see staying at home and bringing up children as their chief role, they will have more children than those who work. There is nothing wrong with having lots of children, as long as you can provide for them. Jeffrey Sachs again: 'With fewer children, a poor household can invest more in the health and education of each child, thereby equipping the next generation with the health, nutrition, and education that can lift living standards in future years.'

As Paul has talked about here before, world population has exploded. What is interesting is that the countries where this has happened are often those where women do not play a role in business or society. When women are educated and given a choice, some will stay at home and look after children, and others will pursue careers or start small businesses.

This is an important factor, as some countries have seen their population double or triple without their economies keeping pace. That leaves more mouths to feed, and just not enough to go around.

CULTURE

I've already mentioned the role of women, but culture can have hidden effects in business, trade and development. China may be a major power now, but it was the world's most developed country in the middle ages, and stagnated, or even went backwards, for centuries. Part of this was cultural, a pride and sense of self-sufficiency that led to a closing of China's borders. 'China seems to have long been stationary', Adam Smith wrote in 1776, in his Wealth of Nations. 'A country which neglects or despises foreign commerce… cannot transact the same quantity of business which it might do with different laws and institutions.' That's changed, but nationalism, suspicion, or radical philosophy still has some countries closed down to outside involvement – communism in North Korea, or extremist Islam in Taliban Afghanistan, locking countries out of development.

This the far end of the spectrum, but culture works in subtler ways too. Some cultures believe in a greater good, in unity, in the rule of law. They are optimistic, hopeful, ambitious and ready to pull together. Others can be paranoid, fragmented, uncertain of their place in the modern world, angry, resistant to change. Rich countries can be overconfident and brash. Poor countries can see themselves as victims and become despondent. In his The Wealth and Poverty of Nations, economic historian David Landes says 'If we learn anything from the history of economic development it is that culture makes all the difference.'

SOCIAL MATURITY: ERIKSON'S STAGES OF SOCIAL DEVELOPMENT

What we've just gone over is a sort of abstracted version of Kegan's social maturity theory without any real detail shown. Of course, it will help to have that detail along with some concrete examples of what he is talking about to make this all comprehensible, so that is what I will now try to supply. Kegan is suggesting that as babies grow into adults, they develop progressively more objective and accurate appreciations of the social world they inhabit. They do this by progressing through five or more states or periods of development which he labelled as follows:

- Incorporative
- Impulsive
- Imperial
- Interpersonal
- Institutional.

In their beginnings, babies are all subjective and have really no appreciation of anything objective at all, and therefore no real self-awareness. This is to say, at first, babies have little idea how to interpret anything, and the only perspective they have with which to interpret things is their own scarcely developed perspective. They can recognize parent's faces and the like, but this sort of recognition should not be confused with babies being able to appreciate that parents are separate creatures with their own needs. This key recognition doesn't occur for years.

Kegan describes this earliest period as Incorporative. The sense of self is not developed at this point in time. There is no self to speak of because there is no distinction occurring yet between self and other. To the baby, there is not any reason to ask the question, "who am I" because the baby's mind is nothing more and nothing less than the experience of its senses as it moves about. In an important sense, the baby is embedded in its sensory experience and has no other awareness.

Babies practice using their senses and reflexes a lot and thus develop mental representations of those reflexes. At some point it occurs to the baby that it *has* reflexes that it can use and senses that it can experience. Reflex and sensation are thus the first mental objects; the first things that are understood to be distinct components of the self. The sense of self emerges from the knowledge that there are things in the world that aren't self (like reflexes and senses); things that I am not. To quote Kegan,

> "Rather than literally being my reflexes, I now have them, and "I" am something other. "I" am that which coordinates or mediates the reflexes..."

Kegan correspondingly refers to this second period of social appreciation development as Impulsive, to suggest that the child is now embedded in impulses – which are those things that coordinate reflexes. The sense of self at this stage of life would be comfortable saying something like, "hungry", or "sleepy", being fully identified with these hungers. Though babies are now aware that they can take action to fulfil a need, they still are not clear that other people exist yet as independent creatures. From the perspective of the Impulsive mind, a parent is merely another reflex that can be brought to bear to satisfy impulses.

The objectification of what was previously subjective experience continues as development continues. Kegan's next developmental leap is known as the Imperial self. The child as "little dictator" is born. In the prior impulsive self, the self literally is nothing more and nothing less than a set of needs. *There isn't anyone "there" having those needs yet. The needs alone are all that exists.* As awareness continues to rise, the child now starts to become aware that "it" is the very thing that has the needs. Because the child is now aware that it *has* needs (rather than *is* needs), it also starts to become aware that it can consciously manipulate things to get its needs satisfied. The impulsive child was also manipulative, perhaps, but in a more unaware animal manner. The imperial child is not yet aware that other people have needs too. It only knows at this stage that it has needs, and it doesn't hesitate to express them.

The Interpersonal period that follows next starts with the first moment when the child comes to understand that there are actually other people out there in the world whose needs need to be taken into account along side their own. The appreciation of the otherness of other people comes about, as always by a process of expanding perspectives. The child's perspective in this case expands from its own only to later include both its own and those of other important people around it. It is the child's increasingly sophisticated understanding of the idea that people have needs itself which cause the leap to occur. To quote Kegan again,

> "I" no longer am my needs (no longer the imperial I); rather I have them. In having them I can now coordinate, or integrate, one need system with another,

> and in so doing, I bring into being that need-mediating reality which we refer to when we speak of mutuality."

In English then, the interpersonal child becomes aware that "not only do I have needs, *other people do too!*" This moment in time is where conscience is born and the potential for guilt and shame arises, as well as the potential for empathy. Prior to this moment, these important aspects of adult mental life don't exist except as potentials.

The interpersonal child is aware that other people have needs which it needs to be taken into account if it is to best satisfy its own needs. There is no guiding principle that helps the interpersonal child to determine which set of needs is most important – its own, or those of the other people. Some children will conclude that their own needs are most important to satisfy, while others will conclude that other's needs should be prioritized, and some children will move back and forth between the two positions like a crazy monkey.

As the child's sense of self continues to develop, at some point it becomes aware that a guiding principle can be established which helps determine which set of needs should take precedence under particular circumstances. This is the first moment that the child can be said to have values, or commitments to ideas and beliefs and principles which are larger and more permanent than its own passing whims and fears. Kegan refers to this new realization of and commitment to values as the Institutional period, noting that in this period, the child's idea of self becomes something which can be, for the first time, described in terms of institutionalized values, such as being honest. "I'm an honest person. I try to be fair. I strive to be brave." are the sorts of things an institutional mind might say. Values, such as the Golden Rule (*e.g.*, "Do unto others as you would have them do unto you"), start to guide the child's appreciation of how to be a member of the family and of society. The moral, ethical and legal foundations of society follow from this basic achievement of an Institutional self. Further, children (or adults) who achieve this level of social maturity understand the need for laws and for ethical codes that work to govern everyone's behaviour. Less socially mature individuals won't grasp why these things are important and cannot and should not simply be disregarded when they are inconvenient.

For many people, social maturity seems to stop here at the Institutional stage. Kegan himself writes that this stage is the stage of conventional adult maturity; one that many (but not all) adults reach, and beyond which most do not progress. However, the potential for continued development continues onwards and upwards. The next evolution of self understanding occurs when the child (by now probably an adult) starts to realize that there is more than one way of being "fair" or "honest" or "brave" in the world. Whereas before, in the interpersonal mindset, there is only one possible right way to interpret a social event (*e.g.*, in accordance with one's own value system), a newly developed Inter-industrial mindset starts to recognize a diversity of ways that someone might act and still be acting in accordance with a coherent value system (though not necessarily one's own value system).

For example, let's consider how someone with an Institutional mindset and someone with an InterIndividual mindset might judge someone who has become a "draft dodger" so as to avoid military duty. There are precisely two ways that an Institutionally minded person might look at such an action. If he or she is of the mainstream institutional mindset, draft dodging is a non-religious sort of heresy and a crime which should be punishable. If, on the other hand, he or she is of a counter-cultural institutional mindset, then judgements are reversed and draft dodging is seen as a brave action which demonstrates individual courage in the face of massive peer pressure to conform. An institutionally minded person can hold *one or the other* of these perspectives but not both, because he or she is literally embedded in one or the other of those perspectives and cannot appreciate the other except as something alien and evil.

A person who has achieved InterIndividual social maturity is able to hold both mainstream and counter-cultural value systems in mind at the same time, and to see the problem of draft dodging from both perspectives. This sort of dual-vision will appear to be the worst kind of wishy-washiness and flip-floppery to someone stuck in a conventional Institutional mindset and maturity level. However, if you are following the progression of social maturity states, and how one states' embedded subjective view becomes something which is seem objectively alongside other points of view as social maturity progresses, you will see that such dual-vision is indeed the logical next step; what a more socially mature sort of human being might look like.

Kegan thinks of the achievement of InterIndividual social maturity, what might be considered "post-maturity", as a dubious thing. In a wonderful interview published by "What is Enlightenment Magazine "and available online here, Kegan comments on the danger that this state poses:

> "... You have to think about what it means to actually be more complex than what your culture is currently demanding. You have to have a name for that, too. It's almost something beyond maturity, and it's usually a very risky state to be in. I mean, we loved Jesus, Socrates, and Gandhi—after we murdered them. While they were alive, they were a tremendous pain in the ass. Jesus, Abraham Lincoln, Martin Luther King, Jr.—these people died relatively young. You don't often live a long life being too far out ahead of your culture."

I'm not going to comment on whether or not Kegan's social maturity theory is accurate. Whether or not it is accurate, it is still a very useful and interesting way of thinking about how social maturity develops. If we can agree to accept this theory as basically correct, for a moment, a whole lot of mental problems and disorders that are otherwise difficult to talk about suddenly start to make some sense; start to "click into place". This is already a rather long essay and I don't want to belabour it, but I need to give at least one example, and to my mind there is no better example for my purpose than Narcissistic Personality Disorder.

Narcissists are (typically) arrogant, self-important and even grandiose people who consider themselves special and "above the law", lacking in empathy and compassion, willing to exploit innocents, and who are consumed by visions of dramatic personal success and power to which they quite passionately believe they are entitled for no apparent reason. Narcissists use other people if it suits their purpose to use them, and discard or attack them if they are in the way.

Everyone knows a few people who fit the narcissist mold to one degree or another. It is generally not at all clear what the heck is wrong with narcissists that causes them to act in this obnoxious fashion, so most people tend to think of them in more simple terms, as “jerks” or “a holes”.

Now, think for a moment what an adult might look like who never left the Imperial self stage of social maturity development. Keep in mind that there are many different types of maturity and that we are only suggesting an adult whose growth has been stunted in this particular social-emotional manner. Other aspects of maturity (*e.g.*, cognitive and intellectual maturity, knowledge, and of course age) are unaffected. That particular hypothetical adult is pretty much going to look like a narcissist, huh? A “little dictator” writ into adult form.

For example, the adult narcissist lacks empathy for the same reason that the normal imperial child lacks empathy; as an Imperially minded individual, he cannot conceive of any perspective that has any meaning other than his own. He or she is literally embedded in an inadequate and inaccurate representation of social reality; one in which only his own needs and impulses are important and no one else is important. The existence of other human beings with separate needs may be partially understood by such a fellow, but there is no recognition that those other people’s needs have equivalent weight and reality to the Imperial-minded narcissists’ own.

A similar argument can be made to explain how to account for sociopaths; antisocial personality disorders (who similarly lacks in empathy, guilt and remorse for criminal actions which harm other people), and also for the childish, immature and often reckless behaviour that is frequently displayed by otherwise normal people who have been abusing drugs and/or alcohol since they were small and have only recently become sober. Such people’s behaviour can be made more comprehensible if you think of them as developmentally delayed in this dimension of social-emotional maturity.

Just because we can use Kegan’s theory to explain why people act like jerks, doesn’t for a second excuse jerky behaviour. Or criminal behaviour, for that matter. Adult narcissists and antisocials may be akin to little children in terms of their social maturity development, but they are not typically retarded in other aspects of maturity. They often have the full compliment of adult intellectual capabilities and may even have very good social skills. They often know right from wrong in some abstract manner even if they can’t conceive it like a more socially mature person might. They are accountable for their actions even if they possess real handicaps that lead them to act in unacceptable ways.

Ask any therapist and he or she will tell you-it is quite difficult to do effective therapy with people who have social immaturity problems. If Kegan is right in his thinking, the reason for this would not be that these people are fundamentally resistant to the therapy process (which is how many therapists see the problem), but rather in large part because they cannot *comprehend* the therapy process, which is after all, very much a social process that requires a certain level of social maturity on the part of patients before they can benefit. It isn't enough to simply teach a set of skills to such people, because all such people will be capable of doing is aping those skills. They won't be able to fully appreciate the meaning of those skills and thus generalize from them to a more abstract (and mature) way of being with other people. Teaching social skills might actually work for some such people. Some people might actually learn the skills and that will be enough to trigger their growth. However, there ought to be a better, more direct way to make this sort of social maturity growth occur.

Actually, the question of how to help adult people become more socially mature when they aren't is a huge unanswered question that this theory leaves us with. Is there a way to help the Narcissists and the Antisocials and the Pedophiles out there? What about the rest of the people out there who are not quite at the level of social maturity that society demands of them? Kegan explores these questions in his follow-on book, In Over Our Heads" which is also very good and worth reading. If there is interest in this essay (we'll see), I'll find a way to go over his conclusions from that book in a future essay.

Hopefully, those of you who have stuck this out will have comprehended what I've been trying to say and it will have been a worthwhile expenditure of your time. If not, all I can say in my defence is that I tried my best (grin!). My aim is to educate, and this stuff is worth knowing about, even if it isn't something Oprah or Dr. Phil would do a show about. As per usual, I'm happy to answer questions and post non-trollish comments. Be well.

SOCIAL DEVELOPMENT THEORY

Importance of Theory: The formulation of valid theory possesses enormous power to elevate and accelerate the expansion and development of human capabilities in any field, leading to fresh discoveries, improvement of existing activities and capacity for greater results. Science is replete with examples of theoretical formulations that have led to important breakthroughs, such as the discoveries of Neptune and Pluto, electromagnetic waves, subatomic particles, and new elements on the periodic table. Today scientists are discovering new substances on computer by applying the laws of quantum mechanics to predict the properties of materials before they synthesize them. In fact, a broad range of technological achievements in this century has been made possible by the emergence of sound theoretical knowledge in fields such as physics, chemistry and biology.

As management expert Peter Drucker put it, "There is nothing more practical than a good theory." Valid theory can tell us not only what should be done, but also what can be done and the process by which it can be achieved. Social development can be summarily described as the process of organizing human energies and activities at higher levels to achieve greater results. Development increases the utilization of human potential.

In the absence of valid theory, social development remains largely a process of trial and error experimentation, with a high failure rate and very uneven progress. The dismal consequences of transition strategies in most Eastern Europe countries, the very halting progress of many African and Asian countries, the increasing income gap between the most and least developed societies, and the distressing linkage between rising incomes, environmental depletion, crime and violence reflect the fact that humanity is vigorously pursuing a process without the full knowledge needed to guide and govern it effectively.

Advances in development theory can enhance our social success rate by the same order of magnitude that advances in theoretical physics have multiplied technological achievements in this century. The emergence of a sound theoretical framework for social development would provide the knowledge needed to address these inadequacies. It would also eventually lead us to the most profound and practical discovery of all – the infinite creative potentials of the human being.

Hierarchy of Learning: Social development consists of two interrelated aspects – learning and application. Society discovers better ways to fulfil its aspirations and it develops organizational mechanisms to express that knowledge to achieve its social and economic goals. The process of discovery expands human consciousness. The process of application enhances social organization. Society develops in response to the contact and interaction between human beings and their material, social and intellectual environment. The incursion of external threats, the pressure of physical and social conditions, the mysteries of physical nature and complexities of human behaviour prompt humanity to experiment, create and innovate.

The experience resulting from these contacts leads to learning on three different levels of our existence. At the physical level, it enhances our control over material processes. At the social level, it enhances our capacity for effective interaction between people at greater and greater speeds and distances. At the mental level, it enhances our knowledge.

While the learning process takes place simultaneously on all these planes, there is a natural progression from physical experience to mental understanding. Historically, society has developed by a trial and error process of physical experimentation, not unlike the way children learn through a constant process of physical exploration, testing and even tasting. Physically, this process leads to the acquisition of new physical skills that enable individuals to utilize their energies more efficiently and effectively. Socially, it leads to the learning and mastery of organizational skills, vital attitudes, systems and institutions that enable people to manage their interactions with other people and other societies more effectively. Mentally, it leads to organization of facts as information and interpretation of information as thought.

The outcome of this learning process is the organization of physical skills, social systems, and information, which are then utilized to improve the efficiency and effectiveness of human activities. It is a cyclical process in which people are continuously learning from past experiences and then applying that learning in new activities. This learning process culminates in a higher level of mental effort to extract the essence and common principles or ideas from society's organized physical experiences, social interactions and accumulated information and to synthesize them as conceptual knowledge. This abstract conceptual knowledge has the greatest capacity for generalization and application in other fields, times and places. The conceptual mind is the highest, most conscious human faculty. Conceptual knowledge is the organization of ideas by the power of mind. That conceptual knowledge becomes most powerful when it is organized into a system. Theory is a systematic organization of knowledge.

A comprehensive theory of social development would provide a conceptual framework for discovering the underlying principles common to the development process in different fields of activity, countries and periods. It would also provide a framework for understanding the relationships between the accumulated knowledge generated by many different disciplines. If pursued to its logical conclusions, it would lead to not just a theory of social development, but a unifying theory of knowledge—which does not yet exist in any field of science or art.

Search for a Social Operating System: Rapid advancement in computer technology and application has primarily been the result of dramatic progress in two parallel but interrelated fields–development of the processing capacity of the silicon chip and development of more advanced operating systems that enable users to utilize the chip's greater computing power. Chip development increases the *potential power* of the computer. Development of more powerful, intuitive and easier to use operating systems increases the *practical power* of the technology.

As a parallel, advances in scientific and technical knowledge have vastly increased the potential productivity and developmental achievements of society. But full utilization of this potential requires the capacity to consciously direct and accelerate social development processes. The discovery of methods to genetically engineer improved varieties of food crops or to control population growth through improved medical devices would have little practical value unless we also possessed the know-how to promote dissemination and adoption of these advanced technologies.

Historically, advances in our understanding of material and biological process have far outstripped advances in our understanding of social processes. As a result, vast social potential has been created, but society has not yet acquired the capacity to fully utilize it for its own development. A theory of development should aim at a knowledge that will enable society more consciously and effectively to utilize its development potentials.

Why a Framework has not yet Emerged: A question naturally arises. If such a framework is possible, why with all the attention focused on development for so many decades has it not yet emerged?

Social development theory has been elusive for several reasons. First, because of the very practical importance of this issue, attention in this field has very largely focused on the material results of development and on those strategies that have proven most effective for achieving those results, rather than on abstract principles or theoretical concepts. Rapid economic progress in North America and Europe after the Second World War, which was followed by even more stunning achievements in Japan and other East Asian nations, imbued governments and the international community with the confidence that development was primarily a question of money, technology, industrialization and political will. Confident that the lessons of early achievers provided all the knowledge necessary for those that were to follow, there was an urge for concerted action and an expectation of results, rather than a quest for theoretical knowledge.

In most discussions, development was conceived in terms of a set of desirable results—higher incomes, longer life expectancy, lower infant mortality, more education. Recently emphasis has shifted from the results to the enabling conditions, strategies and public policies for achieving those results—peace, democracy, social freedoms, equal access, laws, institutions, markets, infrastructure, education and technology. But still little attention has been placed on the underlying social process of development that determines how society formulates, adopts, initiates, and organizes, and few attempts have been made to formulate such a framework.

Second, a very large number of factors and conditions influence the process. In addition to all the variables that influence material and biological processes, social processes involve the interaction of political, social, economic cultural, technological and environmental factors as well. Development theorists have not only to cope with atoms, molecules, material energy and various life forms. They must also cope with the near infinite variety and complexity of human beliefs, opinions, attitudes, values, behaviours, customs, prejudices, laws, social institutions, etc. Third, the timeframe for social development theory cannot be confined to the modern day or even the past few centuries. Human development has been occurring for millennia. The basic principles of development theory must be as applicable to the development of early tribal societies as they are to the emergence of the post-modern global village. Development theory must be a theory of how human society advances through space and time.

Looking beyond the Instruments: Fourth, the instruments of development—science and technology, capital and infrastructure, social policies and institutions—are so compellingly powerful in their action, that they are often mistaken for its cause and source. Most efforts to understand the development process have focused on the central importance of one or a few of these instruments—primarily on money, markets, the organization of production and technological innovation. Some efforts have also been made to describe what has been learned about the contribution of education, skills, laws, public policies, strategies, social systems and institutions. While it is evident that all of these instruments can and do play an important role in social development, it has not been adequately explained what determines the development of these instruments themselves or the extent to which they are utilized by society or the process by which they can be made to generate maximum results.

Obviously, the ultimate determinants of development cannot be the instruments themselves, for none of them exists independently from society. To understand the central principles of development, we must look beyond these instruments to the creator of the instruments. Human beings fashion technology, invent money, erect infrastructures, establish policies, build institutions and adopt values to serve their needs and aspirations. Although humanity exhibits a strong tendency to mistake these instruments for primary determinants rather than created products of its own initiative, the ultimate power of determination must lie with the human beings who create and use these instruments, rather than with the instruments themselves.

Money and technology do have useful power, including a power of organization and efficiency, a power to increase the velocity of production and transactions. But they do not possess an intrinsic living power for growth or development, a source of aspiration or energy that compels their own advancement. Moore's Law describing advances in the speed of microprocessors is not driven by material forces—the microprocessor does not increase its own speed—it is driven by humanity's quest for greater productive power. The surge in value of financial markets is not driven by impersonal physical or mathematical laws governing the growth of money, but by the quest of human beings for greater material prosperity. This self-existent power for growth is an endowment of human beings, living organisms compelled to develop by a pressure within themselves, which in turn gives life and energy to the growth of the instruments and systems they create.

What has been lacking is an organized theoretical framework that describes the role of each of these instruments as aspects of a greater whole and shows each in its proper relation to the others or the greater whole of which they are all parts. To arrive at such a framework, we have to shift our focus from the instruments of development to their creator; from the role of money and technology to the role of human beings that invent new forms of money and technology and harness them for productive purposes. The theory has to place human beings at the centre and view all other aspects of development from the perspective of and in relation to human motivation and action. This conceptual knowledge of the development process should enable every society to better utilize the available instruments better, in order more fully to tap its developmental potential.

Development as a Spherical Whole: A theory of social development should generate a framework around which all knowledge of the factors, instruments, conditions, agencies and processes of development can be integrated. Rather than singling out a specific set of determinants or giving primacy to a limited set of instruments, it would reveal the nature of the relationships and processes that govern the interaction of all these elements to generate developmental results. Rather than generate a linear formula or 'right' perspective, it would make it possible to view the whole field and phenomenon of development from multiple perspectives that are integrated and unified ways of knowing the whole, rather than divided and separate ways of viewing the parts.

The modern tendency to divide scientific enquiry into an increasing number of specialized fields of study has made the emergence of an integrated perspective very difficult. Philosopher Stephen Toulmin mourns the absence of broader conceptual thinking in physics over the past few centuries and argues the need for grand cosmological visions of the universe to unify and integrate the discoveries of many different disciplines. Comparatively, the need for synthesis is even greater for the study of human social development than for understanding the physical and chemical evolution of the universe. For in human development, we must not only grapple with four material dimensions in space and time that preoccupy the physicist and chemist, but also integrate the dimensions of life and mind—including physical, genetic and biological determinants; social behaviours, skills, attitudes, customs, traditions, systems, formal organizations, non-formal institutions, and cultural values; and linguistic determinants, data, facts, information, beliefs, opinions, systems of thought, ideas, theories, and spiritual values—all of which interact and influence each other to impact the course of human development.

The quest for theory in social development cannot lead to any linear or logarithmic equation that adequately explains and predicts human progress. The reality we seek to understand is not of that type. It is not linear or uni-dimensional or even a combination of several dimensions. It is a complex, many-dimensional whole that evolves in many interrelated directions simultaneously. The development of society is best represented to our minds as an expansion from a point to a sphere, rather than as movement along a single line or along multiple lines of progress. Social development is the gradual discovery and unfolding of the potential of a complex, integrated whole, a living organization, a living social organism.

From Unconscious Experience to Conscious Knowledge: Finally, social development theory remains elusive because the very nature of social learning is a subconscious seeking by the collective that leads ultimately to conscious knowledge. We experience first and understand later. Our mental comprehension perpetually lags behind physical experience and struggles to catch up with it.

Our view is that the very intensive, concentrated global experience of the past five decades provides fertile soil for the formulation of a more synthetic conceptual framework for social development. Such a framework can vastly accelerate the transfer and replication of developmental achievements around the world and make possible more conscious and rapid progress even for the most advanced societies in the world.

Basic Premises: These observations suggest a starting point for formulation of a comprehensive conceptual framework:

- Social development theory should focus on underlying processes rather than on surface activities and results, since development activities, policies, strategies, programmes and results will always be limited to a specific context and circumstance, whereas social development itself encompasses a potentially infinite field in space and time.

- The theory should recognize the inherent creativity of individuals and of societies by which they fashion instruments and direct their energies to achieve greater results. It should view development as a human creative process, rather than as the product of any combination of external factors or objective instruments that are created and utilized as the process unfolds and whose results are limited to the capacity of the instruments. Society will discover its own creative potentials only when it seeks to know the human being as the real source of those potentials.
- The implication of this view is that even though it may be influenced, aided or opposed by external factors, society develops by its own motive power and in pursuit of its own goals. No external force and agency can develop a society. (Paul Hoffman, the Administrator of the Marshall Plan for European Recovery who later became the first head of the United Nations Development Programme, said it succinctly: "Technical assistance cannot be exported. It can only be imported." The aspiration of the collective expressed through the initiative of pioneering individuals is the determinant and driving force for a society's own development.

DEVELOPMENT AS SELF-CONCEPTION

Material and biological sciences focus on the interaction of physical conditions, materials and forces to generate results. The tendency to view social development in the same way has led to a host of mathematical equations seeking to define and predict the consequences of combining different external variables in different proportions and under different conditions. The underlying assumption of this approach is that social development is determined by external conditions.

The hypothesis on which our attempt at theory is based is that social development is determined by human beings, not external conditions. External conditions certainly can and do influence the process. People may even act and react in predictable ways to a given set of external conditions. But the results of any development equation cannot be reliably predicted on the basis of external factors. Human development is determined by human responses based on choices made by people. To our knowledge, external forces alone have never unleashed a process of social development, but there are countless instances in which external agents have failed to do so.

Human development is a function of human awareness, aspirations, attitudes and values. Like all human creative processes, it is a process of self-conception. As the writer, artist, composer, political visionary and businessman conceive of unrealized possibilities and pour forth their creative energies to give expression to them, the social collective evolves a conception of what it wants to become and by expressing its creative energies through myriad forms of activity seeks to transform its conception into social reality. The only major difference is that while the individual sometimes (but not always) is conscious of the conception he or she is trying to express, the society is usually (not always) unconscious of the idea and the urge that move it to create something more out of its own latent potential. Society is a subconscious living organism which strives to survive, grow and develop. Individual members of society express conscious intention in their words and acts, but these are only surface expressions of deeper subconscious drives that move the society-at-large. The consciousness of a true collective organism is not merely the sum of its individual parts, but acquires its own identifiable character and personality. This is why the USA has been able to assimilate such large numbers of immigrants, yet retain its distinctive (but constantly changing) national character. Immigrants are moved by the values of the collective to share in the national aspiration for greater individual freedom, practical organization and material progress. In a similar vein, the feverish collective behaviour of the stock market, fashions and pop culture are subconscious social collectives that acquire their own distinct personalities.

Role of the Individual: Society has no direct means to give conscious expression to its subconscious collective aspirations and urges. That essential role is played by pioneering conscious individuals–visionary intellectuals, political leaders, entrepreneurs, artists and spiritual seekers who are inspired to express and achieve what the collective subconsciously aspires and is prepared for. Where the aspiration and action of the leader do not reflect the will of the collective, it is ignored or rejected. Where it gives expression to a deeply felt collective urge, it is endorsed, imitated, supported, and systematically propagated. This is most evident at times of war, social revolution or communal conflict. India's early freedom fighters consciously advocated the goal of freedom from British rule long before that goal had become a felt aspiration of the masses. The leaders spent decades urging a reluctant population to conceive of itself as a free nation and to aspire to achieve that dream. When finally the collective endorsed this conception, no foreign nation had the power to impose its will on the Indian people.

Process of Value Creation: During the World Academy of Art and Science's meeting on development theory in Washington DC in May 1999, there was a broad consensus of participants that the formation of values was a critical aspect of the development process. In this paper, we propose to re-examine the process of development as a process of value formation. If gross physical actions are the most visible and tangible form of human initiative, the creation of values is the most subtle and intangible. Yet human existence is powerfully determined by the nature of its values. Physical skills, vital attitudes, mental opinions and values represent a gradation of internal organizing principles that direct human energies and determine the course of individual and social development.

All human creative processes release and harness human energy and convert it into results. The process of skill formation involves acquiring mastery over our physical-nervous energies so that we can direct our physical movements in a precisely controlled manner. In the absence of skill, physical movements are clumsy, inefficient and unproductive, like the stumbling efforts of a child learning to walk. Human beings acquire social behaviours in a similar manner. Here, apart from the physical skills required for communication and interaction with other people, vital attitudes are centrally important. Each social behaviour expresses not just a movement, but an attitude and intention of the person. Acquiring social behaviours requires gaining control over our psychological energies and channelling them into acceptable forms of behaviour. Change the attitude and the behaviour changes. The developmental achievements of modern society are founded upon such intangible social attitudes as confidence in the government, trust in other people, tolerance and cooperation. Without such attitudes, our money would become valueless paper and our institutions would cease to function.

The same process takes place at the mental level. The mind's energy naturally flows as thought in many different directions without any structure to contain or organize it. The acquisition of knowledge involves construction of a mental structure of understanding that is analogous to the structure of skills and attitudes that govern expression of our physical and vital energy. It forms an organizational framework for learning and application of what is learned. Human values are formed by a similar process and act in a similar manner. Although the word is commonly used with reference to ethical and cultural principles, values are of many types. They may be physical (cleanliness, punctuality), organizational (communication, coordination), psychological (courage, generosity), mental (objectivity, sincerity), or spiritual (harmony, love, self-giving). Values are central organizing principles or ideas that govern and determine human behaviour.

Unlike the skill or attitude that may be specific to a particular physical activity or social context, values tend to be more universal in their application. They express in everything we do. Values can be described as the essence of the knowledge gained by humanity from past experiences distilled from its local circumstances and specific context to extract the fundamental wisdom of life derived from these experiences. Values give direction to our thought processes, sentiments, emotional energies, preferences and actions.

Centuries of experience have been distilled by society into essential principles. Values such as hard work, sense of responsibility, integrity in human relations, tolerance and respect for others are not just noble ideas or ideals. They are pragmatic principles for accomplishment which society has learned and transmitted to successive generations as a psychological foundation for its further advancement. The values of a society are a crucial aspect of its people's self-conception of what they want to become. Because values are intangible to our senses and their formation is the result of a very long process, we tend to overlook their central role in development. Social values constitute the cultural infrastructure on which all further social development is based. In this sense, values are the ultimate product of past development and the ultimate determinant of its future course.

Chapter Four

Moral Development

IMPACTING MORAL AND CHARACTER DEVELOPMENT

Campbell and Bond (1982) state there are four major questions to be addressed when focusing on character development:

1. What is good character;
2. What causes or prevents it;
3. How can it be measured so that efforts at improvement can have corrective feedback; and
4. How can it best be developed?

As previously discussed, good character is defined in terms of one's actions. Character development traditionally has focused on those traits or values appropriate for the industrial age such as obedience to authority, work ethic, working in group under supervision, etc. However, as discussed in the SCANS report (1991) and Huitt's (1997) critique, modern education must promote character based on values appropriate for the information age: truthfulness, honesty, integrity, individual responsibility, humility, wisdom, justice, steadfastness, dependability, etc.

In terms of what influences character development, Campbell and Bond (1982) propose the following as major factors in the moral development and behaviour of youth in contemporary America:

1. Heredity
2. Early childhood experience
3. Modelling by important adults and older youth
4. Peer influence
5. The general physical and social environment
6. The communications media
7. What is taught in the schools and other institutions
8. Specific situations and roles that elicit corresponding behaviour.

These sources of influence are listed in approximate order of least tractable to most tractable in order to suggest why we often seek solutions to social problems through schools. It is important to realize that while schools do and should play a role in the development of character, families, communities, and society in general also have an important influence (Huitt, 1999).

The measurement of character has proven difficult since character, by definition, involves behaviour, but character is often defined in terms of traits (*i.e.*, honesty, integrity, etc.). Some possible measures suggested by Campbell and Bond (1982) are:

1. Student discipline;
2. Student suicide rates;
3. Crimes: assault, burglary, homicides;
4. Pregnancy rates of teenage girls; and
5. Prosocial activities.

Bennett (1993) proposed a list of cultural indicators that he believes could be used as measures of the character of our society. In addition, he cited a number of social trends that he believes have impacted these indicators. The following table provides an overview of how these have changed from 1960 to 1990.

EARLIER THEORETICAL MODELS: PSYCHOANALYSIS AND BEHAVIOURISM

Theories have approached morality differently. Sigmund Freud (1856–1939) described *Oedipus complex* to explain the origins of moral conscience, called the *superego*. The Oedipus complex occurs when a child loves the opposite sex parent and, in order to avoid the anxiety and fear of punishment that this causes, the child identifies with the same sex parent. The child incorporates the same sex parent's prohibitions, starting with "Do not love (sexually) your parent."

For behaviourist theorists, such as Robert R. Sears, Robert Grinder, and Albert Bandura (1982), conscience or morality was considered analogous to the phenomenon of resistance to extinction. H. Hartshorne and M. A. May, at the end of the 1920s, were pioneers in this line of research. Later, Robert Sears, Eleanor Maccoby, and Harry Levin (1957) and other researchers studied the influence of maternal and paternal disciplines upon development of the conscience.

These studies found that warm and affectionate parents, who reason with their children rather than punish them physically, are more successful in having their children assimilate the moral values of the culture. Cognitive behaviourists have added other dimensions to this process, such as *expectancies* (what the child expects is going to happen), *incentive value* (how much the child wants something), *hypothesis testing* ("If I do this, then that will happen"), and *self-efficacy* (one's capacity and confidence on doing something) (Bandura 1977, 1978). In the psychoanalytic and behaviourist models, morality seems to be something that comes from outside, from society, which is internalized.

THEORIES OF MORAL DEVELOPMENT

Evaluating theories of moral development is difficult due to lack of a consistent definition. For some, moral development is seen principally as ability to reason about universal principles of justice and fairness (moral judgement). For others, it is a matter of ability to empathize with and act to alleviate others' suffering (compassion). Both reasoning and compassion are necessary in formulating moral actions; however, it is the relative importance of each that distinguishes different theories.

Two of the main modern theories of moral development, Kohlberg's (1984) and Gilligan's (1982), are based on long standing, underlying philosophical arguments about the basis of moral development. Each is a stage theory in which people develop from one stage to the next as they grow in ability to make complex judgements. Where the theories differ is on how people make judgements.

Kohlberg's (1984) theory focuses more on the use of reason to draw conclusions about what ought to be done to achieve justice and fairness in a particular situation. Altruism, compassion and empathy are less important than principles of justice, and are not the main part of the reasoned process of coming to moral decisions. The moral reasoner is one who knows that a moral decision is required, understands that principles need to be applied universally, thinks of the greatest good for the most people, and then makes a decision based on abstract principles of justice and fairness. Kohlberg's theory follows Piaget's and Inhelder's (1969) thinking about the stages of mental development. Because Kohlberg's (1984) theory is based on the ability to reason abstractly, young children are not seen as being able to reason about moral issues yet; they are pre-moral. Another major avenue of exploration of moral issues is based on altruism. Modern philosophers such as Blum (1987) and Matthews (1994) suggested that childhood responsiveness to others is a primary moral characteristic. Responsiveness requires both a cognitive and affective grasp of a situation. It does not require true empathy yet, nor does it require that children be aware of why they act as they do, only that the act has been done. What is required for an act to be moral is a recognition of the emotions of the other, and of the action needed to change the situation. What changes from childhood to adulthood is the amount and type of experience the person brings to the situation. For example, the adult might offer advice the young child might not yet know about. This model of morality has some similarity to the model of care developed by Gilligan (1982). In Gilligan's model it is the interrelationships among people that are important, and these are based on empathic responses between people. Responding to another's pain or difficulty is the basis of moral action. This requires an empathic attitude to others, a sensitivity to others' needs, and a wish to act with these needs in mind. This type of thinking reflects sensitivity towards others rather than a focus on reasoning about what is a principled or unprincipled act.

While Kohlberg's (1984) theory has been directly tested with gifted children, especially through use of the Defining Issues Test (Rest, 1979), there has been little direct application of Gilligan's (1982) theory to the gifted. Like many other researchers, Gilligan appears to use some data from gifted children, but does not distinguish between data obtained from gifted children and more average subjects. For example, an extensive study at the Emma Willard School illustrated the moral decisions made in the context of relationships with other girls (Gilligan, Lyons, & Hanmer, 1989). Many of the girls described appear to be quite gifted.

A third approach was suggested by Dabrowski whose work has been described and interpreted by Piechowski (1986; 1991). Dabrowski specifically studied gifted youth and adults and developed a theory of emotional development that was based on observations of his gifted subjects. Piechowski (1991), in interpreting the theory, noted that gifted youth, like gifted adults, feel a deep longing for ideals in life, such as justice, fairness, honesty, and responsibility. Gifted children also expect that adults ought to be able to do something to right the wrongs of the world and may be profoundly disappointed by their lack of doing so.

Dabrowski's theory (Piechowski, 1986) does not specifically discuss young gifted children. Dabrowski focused on adolescents and adults; thus, his theory required both life experience and ability to evaluate concepts in order for people to develop further in emotional and moral complexity.

Like the theories of Kohlberg (1984) and Gilligan (1982), Dabrowski's theory suggests stages of development (five), with growth towards an ideal of self-actualization in the final stage, realized by few.

For young gifted children, the applicability of Dabrowski's theory lies in his description of exceptional emotional sensitivity and intensity that can accompany giftedness. As the child grows into adolescence and identity is formed through evaluation of personal values, the person may develop through Dabrowski's stages.

Piechowski (1991) described a number of adolescents who appeared to show potential for the kind of inner growth in both emotional self-awareness and moral sensitivity described by Dabrowski's theory.

For these adolescents, the development of self is accompanied throughout life by growth in moral sensitivity, integration of universal concepts of justice and fairness and universal compassion.

KOHLBERG'S STAGES OF MORAL DEVELOPMENT

Lawrence Kohlberg's stages of moral development constitute an adaptation of a psychological theory originally conceived of by the Swiss psychologist Jean Piaget. Kohlberg, while a psychology postgraduate student at the University of Chicago expanded and developed throughout the course of his life. The theory holds that moral reasoning, the basis for ethical behaviour, has six identifiable developmental stages, each more adequate at responding to moral dilemmas than its predecessor. Kohlberg followed the development of moral judgement far beyond the ages studied earlier by Piaget, who also claimed that logic and morality develop through constructive stages.

Expanding on Piaget's work, Kohlberg determined that the process of moral development was principally concerned with justice, and that it continued throughout the individual's lifetime, a notion that spawned dialogue on the philosophical implications of such research. Kohlberg relied for his studies on stories such as the Heinz dilemma, and was interested in how individuals would justify their actions if placed in similar moral dilemmas. He then analyzed the form of moral reasoning displayed, rather than its conclusion, and classified it as belonging to one of six distinct stages.

There have been critiques of the theory from several perspectives. Some argue that it emphasizes justice to the exclusion of other moral values, such as caring; or that there is such an overlap between stages that they should more properly be regarded as separate domains; or that evaluations of the reasons for moral choices are mostly *post hoc* rationalizations (by both decision makers and psychologists studying them) of essentially intuitive decisions. Nevertheless, an entirely new field within psychology was created as a result of Kohlberg's theory, and according to Haggbloom *et al.*'s study of the most eminent psychologists of the 20th century, Kohlberg was the 16th most frequently cited psychologist in introductory psychology textbooks throughout the century, as well as the 30th most eminent overall.

STAGES

Kohlberg's six stages can be more generally grouped into three levels of two stages each: pre-conventional, conventional and post-conventional. Following Piaget's constructivist requirements for a stage model, as described in his theory of cognitive development, it is extremely rare to regress backward in stages—to lose the use of higher stage abilities.

Stages cannot be skipped; each provides a new and necessary perspective, more comprehensive and differentiated than its predecessors but integrated with them.

Level 1 (Pre-Conventional)

1. Obedience and punishment orientation

   (*How can I avoid punishment?*)

2. Self-interest orientation
   (*What's in it for me?*)
   Level 2 (Conventional)
3. Interpersonal accord and conformity
   (*Social norms*)
   (*The good boy/good girl attitude*)
4. Authority and social-order maintaining orientation
   (*Law and order morality*)
   Level 3 (Post-Conventional)
5. Social contract orientation
6. Universal ethical principles
   (*Principled conscience*).

PRE-CONVENTIONAL

The pre-conventional level of moral reasoning is especially common in children, although adults can also exhibit this level of reasoning. Reasoners at this level judge the morality of an action by its direct consequences. The pre-conventional level consists of the first and second stages of moral development, and is solely concerned with the self in an egocentric manner.

In Stage one (obedience and punishment driven), individuals focus on the direct consequences of their actions on themselves. For example, an action is perceived as morally wrong if the perpetrator is punished. "The last time I did that I got spanked so I will not do it again." The worse the punishment for the act is, the more "bad" the act is perceived to be. This can give rise to an inference that even innocent victims are guilty in proportion to their suffering. It is "egocentric", lacking recognition that others' points of view are different from one's own. There is "deference to superior power or prestige".

Stage two (self-interest driven) espouses the "what's in it for me" position, in which right behaviour is defined by whatever is in the individual's best interest.

Stage two reasoning shows a limited interest in the needs of others, but only to a point where it might further the individual's own interests. As a result, concern for others is not based on loyalty or intrinsic respect, but rather a "you scratch my back, and I'll scratch yours" mentality.

The lack of a societal perspective in the pre-conventional level is quite different from the social contract (stage five), as all actions have the purpose of serving the individual's own needs or interests. For the stage two theorist, the world's perspective is often seen as morally relative.

CONVENTIONAL

The conventional level of moral reasoning is typical of adolescents and adults. Those who reason in a conventional way judge the morality of actions by comparing them to society's views and expectations. The conventional level consists of the third and fourth stages of moral development. In Stage three (interpersonal accord and conformity driven), the self enters society by filling social roles. Individuals are receptive to approval or disapproval from others as it reflects society's accordance with the perceived role.

They try to be a "good boy" or "good girl" to live up to these expectations, having learned that there is inherent value in doing so. Stage three reasoning may judge the morality of an action by evaluating its consequences in terms of a person's relationships, which now begin to include things like respect, gratitude and the "golden rule".

"I want to be liked and thought well of; apparently, not being naughty makes people like me." Desire to maintain rules and authority exists only to further support these social roles. The intentions of actions play a more significant role in reasoning at this stage; "they mean well ...".

In Stage four (authority and social order obedience driven), it is important to obey laws, dictums and social conventions because of their importance in maintaining a functioning society. Moral reasoning in stage four is thus beyond the need for individual approval exhibited in stage three; society must learn to transcend individual needs.

A central ideal or ideals often prescribe what is right and wrong, such as in the case of fundamentalism. If one person violates a law, perhaps everyone would—thus there is an obligation and a duty to uphold laws and rules. When someone does violate a law, it is morally wrong; culpability is thus a significant factor in this stage as it separates the bad domains from the good ones. Most active members of society remain at stage four, where morality is still predominantly dictated by an outside force.

POST-CONVENTIONAL

The post-conventional level, also known as the principled level, consists of stages five and six of moral development. There is a growing realization that individuals are separate entities from society, and that the individual's own perspective may take precedence over society's view. Because of this level's "nature of self before others", the behaviour of post-conventional individuals, especially those at stage 6, can be confused with that of those at the pre-conventional level.

In Stage five (social contract driven), individuals are viewed as holding different opinions and values. Similarly, laws are regarded as social contracts rather than rigid dictums. Those which do not promote the general welfare should be changed when necessary to meet "the greatest good for the greatest number of people". This is achieved through majority decision, and inevitable compromise. Thus democratic government is ostensibly based on stage five reasoning.

In Stage six (universal ethical principles driven), moral reasoning is based on abstract reasoning using universal ethical principles. Laws are valid only insofar as they are grounded in justice, and a commitment to justice carries with it an obligation to disobey unjust laws. Rights are unnecessary, as social contracts are not essential for deontic moral action. Decisions are not reached hypothetically in a conditional way but rather categorically in an absolute way, as in the philosophy of Immanuel Kant. This involves an individual imagining what they would do in another's shoes, if they believed what that other person imagines to be true.

The resulting consensus is the action taken. In this way action is never a means but always an end in itself; the individual acts because it is right, and not because it is instrumental, expected, legal, or previously agreed upon. Although Kohlberg insisted that stage six exists, he found it difficult to identify individuals who consistently operated at that level.

FURTHER STAGES

In his empirical studies of individuals throughout their life Kohlberg observed that some had apparently undergone moral stage regression. Faced with the option of either conceding that moral regression could occur or revising his theory, Kohlberg chose the latter, postulating the existence of sub-stages in which the emerging stage has not yet been fully integrated into the personality. In particular Kohlberg noted a stage 4½ or 4+, a transition from stage four to stage five, that shared characteristics of both. In this stage the individual is disaffected with the arbitrary nature of law and order reasoning; culpability is frequently turned from being defined by society to viewing society itself as culpable. This stage is often mistaken for the moral relativism of stage two, as the individual views those interests of society which conflict with their own as being relatively and morally wrong. Kohlberg noted that this was often observed in students entering college.

Kohlberg suggested that there may a seventh stage—Transcendental Morality, or Morality of Cosmic Orientation—which linked religion with moral reasoning. Kohlberg's difficulties in obtaining empirical evidence for even a sixth stage, however, led him to emphasize the speculative nature of his seventh stage.

TYPES AND FACTORS FOR MORAL DEVELOPMENT

Moral Development

Definition

Moral development is the process through which children develop proper attitudes and behaviours towards other people in society, based on social and cultural norms, rules, and laws.

Description

Moral development is a concern for every parent. Teaching a child to distinguish right from wrong and to behave accordingly is a goal of parenting. Moral development is a complex issue that—since the beginning of human civilization—has been a topic of discussion among some of the world's most distinguished psychologists, theologians, and culture theorists. It was not studied scientifically until the late 1950s.

Piaget's Theory of Moral Reasoning

Jean Piaget, a Swiss psychologist, explored how children developed moral reasoning. He rejected the idea that children learn and internalize the rules and morals of society by being given the rules and forced to adhere to them. Through his research on how children formed their judgements about moral behaviour, he recognized that children learn morality best by having to deal with others in groups. He reasoned that there was a process by which children conform to society's norms of what is right and wrong, and that the process was active rather than passive.

Piaget found two main differences in how children thought about moral behaviour. Very young children's thinking is based on how actions affected them or what the results of an action were. For example, young children will say that when trying to reach a forbidden cookie jar, breaking 10 cups is worse than breaking one. They also recognize the sanctity of rules. For example, they understand that they cannot make up new rules to a game; they have to play by what the rule book says or what is commonly known to be the rules. Piaget called this "moral realism with objective responsibility." It explains why young children are concerned with outcomes rather than intentions.

Older children look at motives behind actions rather than consequences of actions. They are also able to examine rules, determining whether they are fair or not, and apply these rules and their modifications to situations requiring negotiation, assuring that everyone affected by the rules is treated fairly. Piaget felt that the best moral learning came from these cooperative decision-making and problem-solving events. He also believed that children developed moral reasoning quickly and at an early age.

Kohlberg's Theory of Moral Development

Lawrence Kohlberg, an American psychologist, extended Piaget's work in cognitive reasoning into adolescence and adulthood. He felt that moral development was a slow process and evolved over time. Still, his six stages of moral development, drafted in

1958, mirrors Piaget's early model. Kohlberg believed that individuals made progress by mastering each stage, one at a time. A person could not skip stages. He also felt that the only way to encourage growth through these stages was by discussion of moral dilemmas and by participation in consensus democracy within small groups. Consensus democracy was rule by agreement of the group, not majority rule. This would stimulate and broaden the thinking of children and adults, allowing them to progress from one stage to another.

Preconventional Level

The child at the first and most basic level, the preconventional level, is concerned with avoiding punishment and getting needs met. This level has two stages and applies to children up to 10 years of age.

Stage one is the Punishment-Obedience stage. Children obey rules because they are told to do so by an authority figure (parent or teacher), and they fear punishment if they do not follow rules. Children at this stage are not able to see someone else's side.

Stage two is the Individual, Instrumentation, and Exchange stage. Here, the behaviour is governed by moral reciprocity. The child will follow rules if there is a known benefit to him or her. Children at this stage also mete out justice in an eye-for-an-eye manner or according to Golden Rule logic. In other words, if one child hits another, the injured child will hit back. This is considered equitable justice. Children in this stage are very concerned with what is fair.

Children will also make deals with each other and even adults. They will agree to behave in a certain way for a payoff. "I'll do this, if you will do that." Sometimes, the payoff is in the knowledge that behaving correctly is in the child's own best interest. They receive approval from authority figures or admiration from peers, avoid blame, or behave in accordance with their concept of self. They are just beginning to understand that others have their own needs and drives.

Conventional Level

This level broadens the scope of human wants and needs. Children in this level are concerned about being accepted by others and living up to their expectations. This stage begins around age 10 but lasts well into adulthood, and is the stage most adults remain at throughout their lives.

Stage three, Interpersonal Conformity, is often called the "good boy/good girl" stage. Here, children do the right thing because it is good for the family, peer group, team, school, or church. They understand the concepts of trust, loyalty, and gratitude. They abide by the Golden Rule as it applies to people around them every day. Morality is acting in accordance to what the social group says is right and moral.

Stage four is the Law and Order, or Social System and Conscience stage. Children and adults at this stage abide by the rules of the society in which they live. These laws and rules become the backbone for all right and wrong actions. Children and adults feel compelled to do their duty and show respect for authority. This is still moral behaviour based on authority, but reflects a shift from the social group to society at large.

PARENTAL CONCERNS

When to Call the Doctor

Every child misbehaves and will sometimes act selfishly and hurtfully. It is when these acts increase, impulses cannot be controlled, or authority defiance becomes troublesome, that parents may need to seek professional help. Lack of impulse control and authority defiance can be symptoms of medical conditions and psychological disorders. Self-centred behaviour, coupled with lack of acceptance of wrongdoing that continues into older childhood and adolescence, may be a problem that requires family or individual counseling. Risky behaviours such as speeding, drinking, smoking, doing drugs, or engaging in sexual behaviour may be related to peer pressure and wanting to conform to the group or may be a way to defy authority. These behaviours, though deemed morally wrong by most societies, may also be symptoms of deeper psychological troubles.

Of extreme concern is the rare child who acts with no remorse, and appears to have to conscience. This is usually signaled by early violent outbursts, destructive behaviour, or by acts of cruelty to pets or other children. After each incident, the child has a flat affect (no emotion) or fails to admit that there was anything wrong with the his or her actions. These children need intervention immediately. Behaviours such as these may be indicators of sociopathic disorders.

Kohlberg's Stages of Moral Development

Kohlberg identified three levels of moral reasoning: pre-conventional, conventional, and post-conventional. Each level is associated with increasingly complex stages of moral development.

Level 1: Preconventional

Throughout the preconventional level, a child's sense of morality is externally controlled. Children accept and believe the rules of authority figures, such as parents and teachers. A child with pre-conventional morality has not yet adopted or internalized society's conventions regarding what is right or wrong, but instead focuses largely on external consequences that certain actions may bring.

Stage 1: Obedience-and-Punishment Orientation

Stage 1 focuses on the child's desire to obey rules and avoid being punished. For example, an action is perceived as morally wrong because the perpetrator is punished; the worse the punishment for the act is, the more "bad" the act is perceived to be.

Stage 2: Instrumental Orientation

Stage 2 expresses the "what's in it for me?" position, in which right behaviour is defined by whatever the individual believes to be in their best interest. Stage two reasoning shows a limited interest in the needs of others, only to the point where it might further the individual's own interests. As a result, concern for others is not based on loyalty or intrinsic respect, but rather a "you scratch my back, and I'll scratch yours" mentality. An example would be when a child is asked by his parents to do a chore. The child asks "what's in it for me?" and the parents offer the child an incentive by giving him an allowance.

Level 2: Conventional

Throughout the conventional level, a child's sense of morality is tied to personal and societal relationships. Children continue to accept the rules of authority figures, but this is now due to their belief that this is necessary to ensure positive relationships and societal order. Adherence to rules and conventions is somewhat rigid during these stages, and a rule's appropriateness or fairness is seldom questioned.

Stage 3: Good Boy, Nice Girl Orientation

In stage 3, children want the approval of others and act in ways to avoid disapproval. Emphasis is placed on good behaviour and people being "nice" to others.

Stage 4: Law-and-Order Orientation

In stage 4, the child blindly accepts rules and convention because of their importance in maintaining a functioning society. Rules are seen as being the same for everyone, and obeying rules by doing what one is "supposed" to do is seen as valuable and important. Moral reasoning in stage four is beyond the need for individual approval exhibited in stage three. If one person violates a law, perhaps everyone would—thus there is an obligation and a duty to uphold laws and rules. Most active members of society remain at stage four, where morality is still predominantly dictated by an outside force.

Level 3: Postconventional

Throughout the postconventional level, a person's sense of morality is defined in terms of more abstract principles and values. People now believe that some laws are unjust and should be changed or eliminated. This level is marked by a growing realization that individuals are separate entities from society and that individuals may disobey rules inconsistent with their own principles. Post-conventional moralists live by their own ethical principles—principles that typically include such basic human rights as life, liberty, and justice—and view rules as useful but changeable mechanisms, rather than absolute dictates that must be obeyed without question. Because post-conventional individuals elevate their own moral evaluation of a situation over social conventions, their behaviour, especially at stage six, can sometimes be confused with that of those at the pre-conventional level. Some theorists have speculated that many people may never reach this level of abstract moral reasoning.

Stage 5: Social-Contract Orientation

In stage 5, the world is viewed as holding different opinions, rights, and values. Such perspectives should be mutually respected as unique to each person or community. Laws are regarded as social contracts rather than rigid edicts. Those that do not promote the general welfare should be changed when necessary to meet the greatest good for the greatest number of people. This is achieved through majority decision and inevitable compromise. Democratic government is theoretically based on stage five reasoning.

Stage 6: Universal-Ethical-Principal Orientation

In stage 6, moral reasoning is based on abstract reasoning using universal ethical principles. Generally, the chosen principles are abstract rather than concrete and focus on ideas such as equality, dignity, or respect. Laws are valid only insofar as they are grounded in justice, and a commitment to justice carries with it an obligation to disobey unjust laws. People choose the ethical principles they want to follow, and if they violate those principles, they feel guilty. In this way, the individual acts because it is morally right to do so (and not because he or she wants to avoid punishment), it is in their best interest, it is expected, it is legal, or it is previously agreed upon. Although Kohlberg insisted that stage six exists, he found it difficult to identify individuals who consistently operated at that level.

Critiques of Kohlberg's Theory

Kohlberg has been criticized for his assertion that women seem to be deficient in their moral reasoning abilities when compared to men. Carol Gilligan (1982), a research assistant of Kohlberg, criticized her former mentor's theory because it was based so narrowly on research using white, upper-class men and boys. She argued that women are not deficient in their moral reasoning and instead proposed that males and females reason differently: girls and women focus more on staying connected and maintaining interpersonal relationships.

Kohlberg's theory has been criticized for emphasizing justice to the exclusion of other values, with the result that it may not adequately address the arguments of those who value other moral aspects of actions. Similarly, critics argue that Kohlberg's stages are culturally biased—that the highest stages in particular reflect a westernized ideal of justice based on individualistic thought. This is biased against those that live in non-Western societies that place less emphasis on individualism.

Another criticism of Kohlberg's theory is that people frequently demonstrate significant inconsistency in their moral judgements. This often occurs in moral dilemmas involving drinking and driving or business situations where participants have been shown to reason at a lower developmental stage, typically using more self-interest driven reasoning (*i.e.*, stage two) than authority and social order obedience driven reasoning (*i.e.*, stage four). Critics argue that Kohlberg's theory cannot account for such inconsistencies.

DISCIPLINE –TYPES AND MERITS AND DEMERITS

Child Discipline

Child discipline is the methods used to prevent future behavioural problems in children. The word discipline is defined as imparting knowledge and skill, in other words, to teach. In its most general sense, discipline refers to systematic instruction given to a disciple. *To discipline* means to instruct a person to follow a particular code of conduct.

Discipline is used by parents to teach their children about expectations, guidelines and principles. Children need to be given regular discipline to be taught right from wrong and to be maintained safe. Child discipline can involve rewards and punishments to teach self-control, increase desirable behaviours and decrease undesirable behaviours. While the purpose of child discipline is to develop and entrench desirable social habits in children, the ultimate goal is to foster sound judgement and morals so the child develops and maintains self-discipline throughout the rest of his/her life.

Because the values, beliefs, education, customs and cultures of people vary so widely, along with the age and temperament of the child, methods of child discipline vary widely. Child discipline is a topic that draws from a wide range of interested fields, such as parenting, the professional practice of behaviour analysis, developmental psychology, social work, and various religious perspectives. In recent years, advances in the understanding of attachment parenting have provided a new background of theoretical understanding and advanced clinical and practical understanding of the effectiveness and outcome of parenting methods.

In Western society, there has been debate in recent years over the use of corporal punishment for children in general, and increased attention to the concept of "positive parenting" where good behaviour is encouraged and rewarded. Consistency, firmness and respect are all important components of positive discipline. The goal of positive discipline is to teach, train and guide children so that they learn, practice self-control and develop the ability to manage their emotions, and make wise choices regarding their personal behaviour.

Cultural differences exist among many forms of child discipline. Shaming is a form of discipline and behaviour modification. Children raised in different cultures experience discipline and shame in various ways. This generally depends on whether the society values Individualism or Collectivism.

History

Historical research suggests that there has been a great deal of individual variation in methods of discipline over time.

Medieval Times

Nicholas Orme of the University of Exeter argues that children in medieval times were treated differently from adults in legal matters, and the authorities were as troubled about violence to children as they were to adults. In his article, "Childhood in Medieval England," he states, "Corporal punishment was in use throughout society and probably also in homes, although social commentators criticized parents for indulgence towards children rather than for harsh discipline." Salvation was the main goal of discipline, and parents were driven to ensure their children a place in heaven. In one incident in early 14th-century London, neighbours intervened when a cook and clerk were beating a boy carrying water. A scuffle ensued and the child's tormentors were subdued. The neighbours didn't even know the boy, but they firmly stood up for him even when they were physically attacked, and they stood by their actions when the cook and clerk later sued for damages.

Colonial Times

During colonial times in the United States, parents were able to provide enjoyments for their children in the form of toys, according to David Robinson, writer for the *Colonial Williamsburg Journal*. Robinson notes that even the Puritans permitted their young children to play freely. Older children were expected to swiftly adopt adult chores and accountabilities, to meet the strict necessities of daily life. Harsh punishments for minor infractions were common. Beatings and other forms of corporal punishment occurred regularly; one legislator even suggested capital punishment for children's misbehavior.

Corporal Punishment

In many cultures, parents have historically had the right to spank their children when appropriate. A 2006 retrospective study in New Zealand, showed that physical punishment of children remained quite common in the 1970s and 1980s, with 80% of the sample reporting some kind of corporal punishment from parents, at some time during childhood. Among this sample, 29% reported being hit with an empty hand. However 45% were hit with an object, and 6% were subjected to serious physical abuse. The study noted that abusive physical punishment tended to be given by fathers and often involved striking the child's head or torso instead of the buttocks or limbs.

Attitudes have changed in recent years, and legislation in some countries, particularly in continental Europe, reflect an increased skepticism towards corporal punishment. As of December 2017, domestic corporal punishment has been outlawed in 56 countries around the world, most of them in Europe and Latin America, beginning with Sweden in 1966. Official figures show that just 10 percent of Swedish children had been spanked or otherwise struck by their parents by 2010, compared to more than 90 percent in the 1960s. The Swedish law does not actually lay down any legal punishment for smacking but requires social workers to support families with problems.

Even as corporal punishment became increasingly controversial in North America, Britain, Australia and much of the rest of the English-speaking world, limited corporal punishment of children by their parents remained lawful in all 50 states of the United States. It was not until 2012 that Delaware became the first state to pass a statute defining "physical injury" to a child to include "any impairment of physical condition or pain."

Cultural Differences

A number of authors have emphasized the importance of cultural differences in assessing disciplinary methods. Baumrind argues that "The cultural context critically determines the meaning and therefore the consequences of physical discipline..." (Baumrind, 1996; italics in original). Polite (1996) emphasizes that the "debate over whether or not to use corporal punishment rages in many ethnic communities." Larzelare, Baumrind and Polite assert that "After ignoring decades of cultural differences in the effects of spanking, these 2 ARCHIVES [1997] studies and 2 other studies in the past year have each found significantly different effects for African Americans than for non-Hispanic European Americans. The effects of spanking in African American families are generally beneficial to children, unless it is used excessively, either in severity or in frequency." (Larzelere *et al.*, 1998; references to other articles omitted). Our results confirm the serious differences of opinion on discipline, even in a relatively homogenous ethnic community.

Child discipline is often affected by cultural differences. Many Eastern countries typically emphasize beliefs of collectivism in which social conformity and the interests of the group are valued above the individual. Families that promote collectivism will frequently employ tactics of shaming in the form of social comparisons and guilt induction in order to modify behaviour. A child may have their behaviour compared to that of a peer by an authority figure in order to guide their moral development and social awareness. Many Western countries place an emphasis on individualism. These societies often value independent growth and self esteem. Disciplining a child by contrasting them to better-behaved children is contrary to the individualistic societies value of nurturing children's self-esteem. These children of individualistic societies are more likely to feel a sense of guilt when shame is used as a form of behaviour correction. For the collectivist societies, shaming corresponds with the value of promoting self improvement without negatively affecting self esteem.

Parenting Styles

There are different parenting styles which parents use to discipline their children. Four types have been identified: authoritative parents, authoritarian parents, indulgent parents, and indifferent parents.

Authoritative parents are parents who use warmth, firm control, and rational, issue-oriented discipline, in which emphasis is placed on the development of self-direction. They place a high value on the development of autonomy and self-direction, but assume the ultimate responsibility for their child's behaviour. "You live under my roof, you follow my rules!" is a cliché, but one that parents may often find themselves speaking - and it probably most closely mimics the authoritative parenting style.

Authoritarian parents are parents who use punitive, absolute, and forceful discipline, and who place a premium on obedience and conformity. If parents exhibit good emotional understanding and control, children also learn to manage their own emotions and learn to understand others as well. These parents believe it is their responsibility to provide for their children and that their children have no right to tell the parent how best to do this. Adults are expected to know from experience what is really in the child's best interest and so adult views are allowed to take precedence over child desires. Children are perceived to know what they want but not necessarily what is best for them.

Indulgent parents are parents who are characterized by responsiveness but low demandingness, and who are mainly concerned with the child's happiness. They behave in an accepting, benign, and somewhat more passive way in matters of discipline.

Indifferent parents are parents who are characterized by low levels of both responsiveness and demandingness. They try to do whatever is necessary to minimize the time and energy they must devote to interacting with their child. In extreme cases, indifferent parents may be neglectful. They ask very little of their children. For instance, they rarely assign their children chores. They tend to be relatively uninvolved in their children's lives. It's not that they don't love their children. It's just that they believe their children should live their own lives, as free of parental control as possible.

A fifth type of parenting style is connectedness. Connected parents are parents who want to improve the way in which they connect with their children using an empathetic approach to challenging or even tumultuous relationships. Using the 'CALM' technique, by Jennifer Kolari, parents recognize the importance of empathy and aspire to build capacity in their children in hopes of them becoming confident and emotionally resilient. The CALM acronym stands for: Connect emotionally, match the Affect of the child, Listen to what your child is saying and Mirror their emotion back to show understanding.

Non-physical Discipline

Non-physical discipline consists of both punitive and non-punitive methods but does not include any forms of corporal punishment such as hitting or spanking. The regular use of any single form of discipline becomes less effective when used too often, a process psychologists call habituation. Thus, no single method is considered to be for exclusive use. Non-Physical discipline is used in the concerted cultivation style of parenting that comes from the middle and upper class, concerted cultivation is the method of parenting that includes heavy parental involvement, and use reasoning and bargaining as disciplinary methods.

Time-outs

A common method of child discipline is sending the child away from the family or group after misbehavior. Children may be told to stand in the corner ("corner time") or may be sent to their rooms for a period of time. A time-out involves isolating or separating a child for a few minutes and is intended to give an over-excited child time to calm down.

Alternatively, time-outs have been recommended as a time for parents to separate feelings of anger towards the child for their behaviour and to develop a plan for discipline.

When using time-outs as a discipline strategy, individuals must also take into consideration the temperaments of the child if one decides to use time-outs. If a child, for example, has a feisty temperament, or a temperament that expresses emotion in a highly intense way, then discipline strategies of using time-outs would be ineffective because of the clash of discipline strategy to the child's temperament trait.

If an individual decides to use the time-out with a child as a discipline strategy, the individual must be unemotional and consistent with the undesired behaviour. Along with taking into consideration the child's temperament, the length of the time-out needs to also depend on the age of the child. For example, the time-out should last one minute per year of the child's age, so if the child is five years old, the time-out should go no longer than five minutes.

Several experts do not recommend the use of time-out or any other form of punishment. These authors include Thomas Gordon, Alfie Kohn, and Aletha Solter.

Grounding

Grounding is a form of discipline, usually, for older children, preteens and teenagers, that restricts their movement outside of the home, such as visiting friends or using the car and they are not allowed to go anywhere but school and few required places. Sometimes it is combined with the withdrawal of privileges for computer, video games, telephone or television.

Scolding

Scolding involves reproving or criticizing a child's negative behaviour and/or actions.

Some research suggests that scolding is counter-productive because parental attention (including negative attention) tends to reinforce behaviour.

Non-punitive Discipline

While punishments may be of limited value in consistently influencing rule-related behaviour, non-punitive discipline techniques have been found to have greater impact on children who have begun to master their native language. Non-punitive discipline (also known as empathic discipline and positive discipline) is an approach to child-rearing that does not use any form of punishment. It is about loving guidance, and requires parents to have a strong relationship with their child so that the child responds to gentle guidance as opposed to threats and punishment. According to Dr. Laura Markham, the most effective discipline strategy is to make sure your child wants to please you.

Non-punitive discipline also excludes systems of "manipulative" rewards. Instead, a child's behaviour is shaped by "democratic interaction" and by deepening parent-child communication. The reasoning behind it is that while punitive measures may stop the problem behaviour in the short term, by themselves they do not provide a learning opportunity that allows children the autonomy to change their own behaviour. Punishments such as time-outs may be seen as banishment and humiliation. Consequences as a form of punishment are not recommended, but natural consequences are considered to be possibly worthwhile learning experiences provided there is no risk of lasting harm.

Positive discipline is both non-violent discipline and non-punitive discipline. Criticizing, discouraging, creating obstacles and barriers, blaming, shaming, using sarcastic or cruel humour, or using physical punishment are some negative disciplinary methods used with young children. Any parent may occasionally do any of these things, but doing them more than once in a while may lead to low self-esteem becoming a permanent part of the child's personality.

Authors in this field include Aletha Solter, Alfie Kohn, Pam Leo, Haim Ginott, Thomas Gordon, Lawrence J. Cohen, and John Gottman.

Essential Aspects

In the past, harsh discipline has been the norm for families in society. However, research by psychologists has brought about new forms of effective discipline. Positive discipline is based on minimizing the child's frustrations and misbehavior rather than giving punishments. The main focus in this method is the "Golden Rule", treat others the way you want to be treated. Parents follow this when disciplining their children because they believe that their point will reach the children more effectively rather than traditional discipline.

The foundation of this style of discipline is encouraging children to feel good about themselves and building the parent's relationship with the child so the child wants to please the parent. In traditional discipline, parents would instill fear in their child by using shame and humiliation to get their point across. However, studies show that this type of punishment ultimately causes the children to have more psychological problems in their adolescence and adulthood.

Physical and harsh punishment shows the child that violence and negative treatment is acceptable in some circumstances, where, positive discipline demonstrates the opposite. In positive discipline the parents avoid negative treatment and focus on the importance of communication and showing unconditional love. Feeling loved, important and well liked has positive and negative effects on how a child perceives themselves.

The child will feel important if the child feels well liked and loved by a person. Other important aspects are reasonable and age-appropriate expectations, feeding healthy foods and providing enough rest, giving clear instructions which may need to be repeated, looking for the causes of any misbehavior and making adjustments, and building routines. Children are helped by knowing what is happening in their lives. Having some predictability about their day without necessarily being regimental will help reduce frustration and misbehavior. Not only are the children taught to be open-minded, but the parents must demonstrate this as well.

METHODS

Praise and Rewards

Simply giving the child spontaneous expressions of appreciation or acknowledgement when they are not misbehaving will act as a reinforcer for good behaviour. Focusing on good behaviour versus bad behaviour will encourage appropriate behaviour in the given situation. According to B. F. Skinner, past behaviour that is reinforced with praise is likely to repeat in the same or similar situation.

In operant conditioning, schedules of reinforcement are an important component of the learning process. When and how often we reinforce a behaviour can have a dramatic impact on the strength and rate of the response. A schedule of reinforcement is basically a rule stating which instances of a behaviour will be reinforced. In some case, a behaviour might be reinforced every time it occurs. Sometimes, a behaviour might not be reinforced at all. Either positive reinforcement or negative reinforcement might be used, depending on the situation. In both cases, the goal of reinforcement is always to strengthen the behaviour and increase the likelihood that it will occur again in the future. In real-world settings, behaviours are probably not going to be reinforced each and every time they occur.

Example of Operant Conditioning

Positive Reinforcement: Whenever he is being good, cooperative, solves things non-aggressively, immediately reward those behaviours with praise, attention, goodies. Punishment: If acting aggressively, give immediate, undesired consequence (send to corner; say "NO!" and couple with response cost).

Response Cost: Most common would be "time-out". Removing sources of attention by placing in an environment without other people. Careful: This can become (aversive) punishment, depending on how done. To be response cost, it can only simply be taking away a desirable thing; not adding a negative one.

Negative Reinforcement: One example would be to couple negative reinforcement with response cost—after some period of time in which he has acted cooperatively or calmly while in the absence of others, can bring him back with others. Thus, taking away the isolation should reinforce the desired behaviour (being cooperative).

Extinction: Simply ignoring behaviours should lead to extinction. Note: that initially when ignored, can expect an initial increase in the behaviour—a very trying time in situations such as a child that is acting out.

It is common for children who are otherwise ignored by their parents to turn to misbehavior as a way of seeking attention. An example is a child screaming for attention. Parents often inadvertently reward the bad behaviour by immediately giving them the attention, thereby reinforcing it. On the other hand, parents may wait until the child calms down and speaks politely, then reward the more polite behaviour with the attention.

Natural Consequences

Natural consequences involve children learning from their own mistakes. In this method, the parent's job is to teach the child which behaviours are inappropriate. In order to do this, parents should allow the child to make a mistake and let them experience the natural results from their behaviour. For instance, if a child forgets to bring his lunch to school, he will find himself hungry later. Using natural consequences would be indicative of the theory of accomplishment of natural growth, which is the parenting style of the working class and poor.

The accomplishment of natural growth focuses on separation between children and family. Children are given directives and expected to carry them out without complaint or delay. Children are responsible for themselves during their free time, and the parent's main concern is caring for the children's physical needs. In order for this method to be effective the parents cannot shield their child from harm or from getting in trouble. They must allow for the mistake to occur in order for the child to learn the consequences. For example, a basic natural consequence is that if the child touches a hot pot he will get burned. The consequence is usually immediate, and the parent may have little control when protecting the child. However, the pain is the consequence of touching the pot which will teach the child to not do that again.

Obedience and Internal Discipline

Show respect is a part of the Elements of Discipline distinguished from obedience-based with a respectful intervention and effective discipline model, state "When setting a limit, caregivers should avoid commenting on a child's motives, intentions or overall patterns of behaviour.".

Accept feelings is stated as; "caregivers should provide symbolic [experiential] outlets for the child's expression of feelings, even when setting a limit on an overt behaviour stemming from that feeling.". A child's feelings is never a problem, even when action is. Assert needs can be stated as: "Caregivers have a right and a responsibility to set a limit whenever a child does something, dangerous, destructive or that violates the caregiver's standards of acceptability."

Providing respect to the childs needs and feelings/emotions, expecting likewise in return while being careful to prevent providing praise when the caregiver doesn't like the behaviour, and to refrain from punishing behaviour the caregiver does like. This umbrella of principles is called warmth, being warm is respectful, encouraging, giving room for experience, firm ground rules and praise. Warmth, tolerance and influence are the three forming this framework of child development and psychological practical application in a schooling context.

The Pros and Cons of Child Discipline

Parents discipline their children in an effort to teach them appropriate ways to behave. Their morals and values are also conveyed through appropriate discipline techniques. There are many different styles in which to parent; some parents seem to fit the mold of one particular style while others are a combination of two or more. Children thrive on the guidelines that parental discipline provides them; behavioural issues may increase if secure boundaries aren't in place, according to the article, "About Discipline- Helping Children Develop Self-Control," published by the Child Study Centre at the NYU Langone Medical Centre.

One Way, My Way

Strict or authoritarian parents teach their children through obedience and punishment. Children in these families aren't encouraged to discuss issues with their parents; their attempts may be thwarted. Negotiating and compromising is minimal or non-existent. The pros of a strict home environment include respect for their elders as well as children who excel in academics. The cons may be decreased creativity, an increased risk of low self-esteem and stress levels as well as inflexible thinking; this is due to excessive pressure from their parents, according to the article, "Effects of Parenting Styles on Children's Behaviour," Published on Education.com.

Attached at the Hip

Attachment parenting begins at birth by breastfeeding your baby, sharing your bed and wearing your baby in a sling or baby carrier. Through close and continuous contact between parent and baby, this type of parenting relies on watching your child for cues as to when he needs to eat, sleep or try something new, according to the article, "Your Parenting Style: Are You an Extreme Parent?" published on the Family Education website.

Attachment parenting advocates breastfeeding which is nutritionally and emotionally healthy and encourages parents to discipline in an age-appropriate manner. On the flip side, feeling the need to be with your baby all of the time can provoke guilt and anxiety when a mother has a need or desire to return to the workforce. The American Academy of Pediatrics warns against sharing a bed with your baby.

Running Free

Permissive parents have little control over their children. The children in these families generally play by their own rules and create their own schedules. Parental expectation of permissive parents aren't very high. Parents who support this parenting style may believe that they are giving their kids the opportunity to learn from their mistakes in the real world without fearing that something bad may happen to them, according to Family Education. The cons of permissive parenting is that children are not mini-adults. This type of parenting can put children in danger and it may border on neglect.

The Spanking Debate

The subject of discipline often arouses controversy regarding the spanking of children as a method of discipline. The manner in which you choose to discipline your child is a personal one. Every child is different and parenting involves finding what works for your child. That said, while some people advocate that spanking works, the American Academy of Pediatrics is opposed to the spanking of children, according to the HealthyChildren.org website. Effective alternatives to physical punishment exist, such as natural and logical consequences, losing privileges and time-out.

Chapter Five

Emotional Development

MEANING—POSITIVE AND NEGATIVE EMOTIONS

Research and theory on the role of emotion and regulation in morality have received considerable attention in the last decade. Much relevant work has concerned the role of moral emotions in moral behaviour. Research on differences between embarrassment, guilt, and shame and their relations to moral behaviour is reviewed, as is research on the association of these emotions with negative emotionality and regulation. Recent issues concerning the role of such empathy-related responses as sympathy and personal distress to prosocial and antisocial behaviour are discussed, as is the relation of empathy-related responding to situational and dispositional emotionality and regulation. The development and socialization of guilt, shame, and empathy also are discussed briefly. In addition, the role of non-moral emotions (*e.g.*, anger and sadness), including moods and dispositional differences in negative emotionality and its regulation, in morally relevant behaviour, is reviewed.

THEORIES OF EMOTION

Several perspectives help explain the role of emotion in development. Some theorists emphasize that emotions occur during events involving self and environment, but that events must be cognitively appraised before an emotion is experienced; this appraisal occurs with reference to one's goals.

The *social constructivist* approach (*e.g.*, Saarni 1999) also highlights appraisal, but focuses on emotions as social products based on cultural beliefs. In contrast, *Differential Emotions Theory* asserts that different emotions are already present at birth (Izard 1991). Keith Oatley and Jennifer Jenkins (1996) assimilate these divergent views, holding that emotions derive from a universal biological core, but also contain an appraisal/semantic component that is largely a product of social construction.

EMOTIONAL COMPETENCE

Both Susanne Denham (1998) and Carolyn Saarni (1990, 1999) have written about children's emotional competence; they agree that, although there are no overarching stages for emotional development, children become increasingly sophisticated in their expression and experience, understanding, and regulation of emotions. These early foundations of emotional competence contribute to mental health throughout the lifespan.

SOCIALIZATION OF EMOTIONAL COMPETENCE

Because emotions are inherently social, skills of emotional competence are vividly played out during interaction and within relationships with others. As noted by Joseph Campos and Karen Barrett (1984), emotions provide useful information for self and others. This entry focuses on the emotional transactions between parent and child and on parents' contributions to emotional competence.

Amy Halberstadt (1991) has highlighted three possible mechanisms of parents' socialization of emotional competence: Modelling, reactions to children's emotions, and teaching about emotions. The theories of psychologists like Sylvan Tomkins (1963, 1991), as well as empirical findings from late twentieth-century research, predict that parents' positive emotional expression and experience, accepting and helpful reactions to children's emotions, and emphasis on teaching about emotions in the family, contribute to young children's more sophisticated emotional competence.

Over and above these mechanisms for the socialization of emotion, cultural issues are paramount. Parents socialize their children based on specific cultural values and norms, but cross-cultural similarities and differences remain to be delineated. In both Japan and the United States, people often agree on the antecedents and evaluative components of emotional experience, and even on some primitive aspects of appraisal. Nevertheless, they differ markedly on some of the more advanced aspects of appraisal, including control of and responsibility for emotion. In the United States, people might state, "I have to show this emotion," or even "I am not responsible for this emotion," whereas the Japanese might say "I should not show this emotion" and "I am responsible for this emotion". Given these differences, the goals of emotion socialization surely differ across the two cultures.

Regarding Modelling, children observe parents' ever-present emotions, and incorporate this learning into their expressive behaviour. Parents' expressiveness also teaches children which emotions are acceptable in which contexts. Their emotional displays tell children about the emotional significance of differing events, behaviours that may accompany differing emotions, and others' likely reactions. A mostly positive emotional family climate makes learning about emotions accessible to children. Thus, parents' expressiveness is associated with children's understanding of emotions as well as their expressiveness.

However, several factors suggest possible negative contributions of parents' expressiveness to children's emotional competence. Parents' frequent and intense negative emotions may disturb children, making emotional learning more difficult. Further, parents whose expressiveness is generally limited impart little information about emotions to their children.

Parents may cultivate some emotional expressions, but not others. Western cultures urge children to separate self from others and express themselves, but many non-Western cultures view people as fundamentally connected, with the goal of socialization attunement or alignment of one's actions and reactions with that of others'. Thus, in Japan, the public display of emotions is mostly discouraged because it is seen as disruptive, leading us to expect Japanese parents to model mostly low intensity emotions (Ujiie 1997).

Moreover, there is a qualitative difference in the emotions modelled. Valued emotions accompanying interdependence—friendliness, affiliation, calmness, smoothness, and connectedness—would be most available for observation by Japanese children. In contrast, anger, regarded as extremely negative in Japan because it disturbs interdependence, would be modelled less (Ujiie 1997). Research on these culturally unique aspects of socialization of emotions, however, is still largely lacking.

Parents' contingent reactions to children's emotional displays are also linked to children's emotionally competent expression, experience, understanding, and regulation of emotions. Contingent reactions include Behavioural and emotional encouragement or discouragement of specific emotions. Parents who dismiss emotions may actively punish children for showing emotions, or they may want to be helpful, but ignore their child's emotions in an effort to "make it better." Children who experience negative reactions are distressed by their parents' reactions as well as the events that originally elicited emotion. Positive reactions, such as tolerance or comforting, convey a very different message—that emotions are manageable, even useful. Good emotion coaches, at least in the United States, accept children's experiences of emotion and their expression of emotions that do not harm others; they empathize with and validate emotions. In fact, emotional moments may be opportunities for parent-child intimacy.

Japanese parents' reactions to children's emotions differ from U.S. parents', although not at every age or in every situation. In general, U.S. parents *Children observe parents' emotions and incorporate this learning into their expressive behaviour. Parents' emotional displays tell children about the significance of events, behaviours that may accompany emotions, and others' likely reactions.*

OWEN FRANKEN/CORBIS see expression of emotions as legitimate and part of healthy self-assertion. Japanese mothers also respond positively to their infants and young children's emotions, but gradually emphasize, more than U.S. parents, parenting goals of *inhibitory self-regulation* and *acquisition of good manners.* Thus, for children older than about three years, Japanese mothers react most positively to children's suppression of emotion and demonstration of empathy. Compared to U.S. parents, they especially discourage negative emotional expression (Kojima 2000).

Socializers' tendencies to discuss emotions, if nested within a warm parent-child relationship, assist the child acquiring emotional competence (Kochanoff 2001). Parents directly teach their children about emotions, explaining its relation to an observed event or expression, directing attention to salient emotional cues, and helping children understand and manage their own responses.

Parents who are aware of emotions and talk about them in a differentiated manner (*e.g.*, clarifying and explaining, rather than "preaching") assist their children in experiencing and regulating their own emotions. Children of such parents gradually formulate a coherent body of knowledge about emotional expressions, situations, and causes.

Late twentieth-century research suggests that Japanese mothers also talk to their preschoolers about emotions. They use emotion language for similar reasons as U.S. mothers; what differs is the content of their conversations, which focus on aspects of emotion relevant for Japanese culture.

Thus, positive elements of emotion socialization seem clear. Moreover, there is some evidence that parents' support of one another also helps to ensure such positive elements (Denham and Kochanoff, in press). However, do beneficial aspects of parents' socialization of emotion differ across children's ages, or across parents? Although more research is needed in this area, it is predicted that these socializing techniques would occur across development and parents, albeit with different emphasized emotions, and different aspects yielding positive child outcomes. In part, however, these questions require an elucidation of children's changing skills of emotional competence.

EXPRESSION AND EXPERIENCE OF EMOTIONS

An important element of emotional competence is emotional expressiveness, the sending of affective messages. Emotions must be expressed in keeping with the child's goals, and in accordance with the social context; goals of self and others must be coordinated. Thus, emotional competence includes *expressing emotions* in a way that is advantageous to moment-to-moment interaction and relationships over time.

First, emotionally competent individuals are aware that an affective message needs to be sent in a given context. But what affective message should be sent, for interaction to proceed smoothly? Children slowly learn which expressions of emotion facilitate specific goals. Second, children also come to determine the appropriate affective message, one that works in the setting or with a specific playmate. Third, children must also learn how to send the affective message convincingly. Method, intensity, and timing of an affective message are crucial to its meaning, and eventual success or failure.

After preschool, children learn that their goals are not always met by freely showing their most intense feelings. For example, grade-schoolers regulate anger in anticipation of the negative consequences they expect in specific situations or from specific persons. Along with the *cool rule* mandating more muted emotions within most social settings, older children's emotional messages become more complex, with the use of more blended signals, and better differentiated expressions of social emotions.

These general tenets of competent experience and expression of emotion may be universal, but children from different cultures differ in the emotions they express. For example, Japanese preschoolers show less anger and distress in conflict situations than U.S. children, even though the two groups' prosocial and conflict *behaviours* do not differ. These differences fit the Japanese taboo on publicly displayed negative emotions.

Japanese toddlers and preschoolers' expressiveness is not always different from that of their U.S. peers'. For example, they express empathy in response to others' distress. Even this similarity, however, may arise from differing cultural imperatives. Japanese youngsters are encouraged to feel as one with their group, whereas Western children are encouraged to feel the state of another as part of their increasingly autonomous regulation of emotional states. The import of these subtle differences needs further exploration.

UNDERSTANDING EMOTIONS

Emotion knowledge predicts later social functioning, such as social acceptance by peers. By preschool, most children can infer basic emotions from expressions or situations, and understand their consequences. Preschoolers gradually come to differentiate among the negative emotions, and become increasingly capable of using emotional language. Furthermore, young children begin to identify other peoples' emotions, even when they may differ from their own.

Grade-schoolers become more aware of emotional experience, including multiple emotions, and realize that inner and outer emotional states may differ. By middle school, children comprehend the time course of emotions, display rules associated with emotional situations, and moral emotions. They now have an adult-like sense of how different events elicit different emotions in different people, and that enduring personality traits may impact individualized emotional reactions.

These general tenets of competent emotion knowledge seem similar for Japanese children. For example, even two-year-olds use some emotion language; by the end of preschool, their understanding of culturally appropriate emotion language is acute (Clancy 1999; Matsuo 1997). They begin to understand dissemblance of emotion (Sawada 1997). As in U.S. research, however, there is a relative dearth of research on older children.

EMOTION REGULATION

Emotion regulation is necessary when the presence or absence of emotional expression and experience interferes with a person's goals. Negative or positive emotions can need regulating, when they threaten to overwhelm or need to be amplified. Children learn to retain or enhance those emotions that are relevant and helpful, to attenuate those that are relevant but not helpful, and to dampen those that are irrelevant. These skills help them to experience a greater sense of well-being and maintain satisfying relationships with others (Thompson 1994).

Early in preschool, much of this emotion regulation is biobehavioural (*e.g.*, thumb sucking), and much is supported by adults. Important cognitive foundations of emotion regulation contribute to the developmental changes observed in emotional competence from preschool to adolescence. Preschoolers gradually begin to use independent coping strategies for emotion regulation, and grade-schoolers refine these strategies—problem solving, support-seeking, distancing, internalizing, externalizing, distraction, reframing/redefining, cognitive "blunting" (*i.e.*, convincing oneself that one's distress is minimal), and denial.

Older children are uniquely aware of the multiple strategies at their command, and know which are adaptive in specific situations. They also use more cognitive and problem-solving, and fewer support-seeking, strategies. Adolescents appraise the controllability of emotional experiences, shift thoughts intentionally, and reframe situations to reach new solutions (Saarni 1997).

Japanese children, as noted above, are initially very close to their mothers, who assist them in emotion regulation even more than Western mothers. Some researchers have noted, however, that once emotionally distressed, Japanese children find it harder to regain their equilibrium (Kojima 2000). It could be that extended maternal coregulation, coupled with stricter cultural display rules, make it more difficult for these children to self-regulate once distressed. More research is needed to follow up on these findings.

Place of Emotions in Life

An emotion is the meaning we give to our felt states of arousal. Psychologists consider emotions to be complex states involving diverse aspects. On the one hand an emotion is a physiological state of arousal; on the other, it also involves an object as having a certain significance or value to the individual. Emotions are dynamically fed by our drives and dispositions; they are also interlocked with other emotions, related to an individual's beliefs, a wide-ranging network of symbols and the "cultural ethos" of a society.

Emotions basically involve dispositions to act by way of approach or withdrawal. Let us take an example to illustrate this.

A man who walks a long distance across a forest track feels thirsty, he is attracted by the sight of water in a passing stream and he approaches; but there is a fierce animal close to the stream and he is impelled to withdraw or fight; if he withdraws he might then have a general feeling of anxiety, and if he gets back home safely he will be relieved.

Thus perception of objects and situations is followed by a kind of appraisal of them as attractive or harmful. These appraisals initiate tendencies to feel in a certain manner and an impulse to act in a desirable way. All states of appraisal do not initiate action; for instance, in joy we like a passive continuation of the existing state and in grief we generally give up hope.

Though there may be certain biologically built-in patterns of expressing emotions, learning plays a key role. Learning influences both the type and intensity of arousal as well as the control and expression of emotions.

The emotional development of people has been the subject of serious study. There are significant differences in the emotional development of people depending on the relevant cultural and social variables. In fact, certain societies are prone to give prominence to certain types of emotions (a dominant social ethos). There are also differences regarding the degree of expressiveness and control of emotions. The important point is that each of us develops a relatively consistent pattern of emotional development, coloured by the individual's style of life.

Emotions Organize and Motivate Action

Emotions prepare for and motivate action. There is an action urge connected to specific emotions that is hard-wired. "Hard-wired" means it is an automatic, built-in part of our behaviour. For example, if you see your two-year old child in the middle of the street and a car coming, you will feel an emotion, fear, and this emotion will prompt you to run to save your child. You don't stop to think about it.

You just do it. Your emotion has motivated your behaviour without you having to take the time to think. Emotions can also help us overcome obstacles in our environment. This anxiety, though it's uncomfortable, helps to motivate you to study so you will do well on the test. Anger may motivate and help people who are protesting injustices. The anger may override the fear they might feel in a demonstration or protest. Guilt may keep someone who is dieting stick to her diet.

Exercises

See if you can come up with a couple of examples where your emotion prompted you to take action before you thought about it. See if you can come up with a situation where an emotion helps you overcome an obstacle in your environment, where it makes it easier for you to get something done, for example. It may not be a pleasant emotion but it does help get the job done. During the week, notice when your emotions motivate your action, save you time, or help you get something done.

Emotions can be Self-Validating

Emotions can give us information about a situation or event. They can signal to us that something is going on. Sometimes signals about a situation will be picked up unconsciously, and then we may have an emotional reaction, but not be sure what set off the reaction. Feeling "something doesn't feel right about this" or "I had a feeling something was going to happen and it did" are some of the signals we might get. Think of some times when your feel for a situation turned out to be right. Is there some time when you felt anxiety or apprehension that turned out to be justified? Or that you had a good feeling about someone that turned out to be right? When dealing with our feelings this way is carried to extremes, though, we may think of the emotion as fact. "I love him, so he's a good person." If I feel stupid, I am stupid." While our emotions are always valid, it doesn't necessarily make them facts. This is difficult for people with Borderline Personality Disorder and others, because one of our biggest issues is that we have been in invalidating environments—so much that we don't trust our emotions.

If our emotions are minimized or invalidated, it's hard to get our needs taken seriously. So we may increase the intensity of our emotions in order to get our needs met. And then if we decrease the intensity of our emotions, we may find again that we are not taken seriously. Think of some times when emotions are self-validating. For example: I am going to a party, but I feel uneasy about it, as if something is going to happen. At the party, a friend and I have an argument and I leave. My feeling about something happening is right. I am at work, and there seems to be a lot of tension. I sense that something is up.

At lunch, my coworkers hold a surprise birthday party for me. Again my emotion is validated. I am home alone and feeling very lonely. I am getting more and more anxious and angry. I call friends and try to get someone to come and stay with me. No one will come. So this intense negative emotion also validates my feeling that I am lonely and no one cares. Think of some examples of your own. Remember that we are not evaluating or judging anyone's feelings or behaviour. We are just trying to look at how emotions function.

Exercises

Fill out an emotion diary for several days. For each day choose your strongest emotion, or the one that lasted the longest, or was the most difficult or painful. Describe the prompting event, the event that caused or triggered the emotion.

And describe the emotion's function:

- To communicate to others
- To motivate action
- To communicate to yourself

Production of Emotion

Emotion Production is a Serbian media company. Based in the Belgrade municipality of Stari Grad and registered as a limited liability company, Emotion produces television content. The company's ownership is divided between two parent entities: IMGS and Multikom Group.

IMGS is owned by Goran Stamenkoviæ while Multikom Group is 50% owned by influential Serbian businessman and politician Dragan Ðilas, currently the mayor of Belgrade. In its portfolio it has some of the highest rated reality programmes that air throughout the Balkans such as: Leteæi start, 48 sati svadba, Veliki brat, Sve za ljubav, Veliki brat VIP, Operacija trijumf, Uzmi ili ostavi, Menjam •enu, 1 protiv 100, Ruski rulet, etc.

Additionally, the company is responsible for putting various individual personalities on the map such as Milan Kaliniæ, Ana Mihajlovski, Maca Marinkoviæ, etc. who have become household names throughout the Balkan region as a result of their appearances in Emotion programmes.

Emotional Intelligence

Emotional intelligence is the ability, capacity, skill; or, in the case of the trait EI model, a self-perceived ability to identify, assess, and control the emotions of oneself, of others, and of groups. Different models have been proposed for the definition of EI and there is disagreement about how the term should be used. Despite these disagreements, which are often highly technical, the ability-EI and trait-EI models enjoy support in the literature and have successful applications in various domains.

History

The earliest roots of emotional intelligence can be traced to Darwin's work on the importance of emotional expression for survival and second adaptation. In the 1900s, even though traditional definitions of intelligence emphasized cognitive aspects such as memory and problem-solving, several influential researchers in the intelligence field of study had begun to recognize the importance of the non-cognitive aspects.

For instance, as early as 1920, E.L. Thorndike used the term social intelligence to describe the skill of understanding and managing other people. Similarly, in 1940 David Wechsler described the influence of non-intellective factors on intelligent behaviour, and further argued that our models of intelligence would not be complete until we can adequately describe these factors.

In 1983, Howard Gardner's *Frames of Mind: The Theory of Multiple Intelligences* introduced the idea of multiple intelligences which included both interpersonal intelligence and intrapersonal intelligence. In Gardner's view, traditional types of intelligence, such as IQ, fail to fully explain cognitive ability. Thus, even though the names given to the concept varied, there was a common belief that traditional definitions of intelligence are lacking in ability to fully explain performance outcomes.

The first use of the term "emotional intelligence" is usually attributed to Wayne Payne's doctoral thesis, *A Study of Emotion: Developing Emotional Intelligence from 1985*. However, prior to this, the term "emotional intelligence" had appeared in Leuner.

Greenspan also put forward an EI model, followed by Salovey and Mayer and Goleman. The distinction between trait emotional intelligence and ability emotional intelligence was introduced in 2000.

Definitions

Substantial disagreement exists regarding the definition of EI, with respect to both terminology and operationalizations. There has been much confusion about the exact meaning of this construct. The definitions are so varied, and the field is growing so rapidly, that researchers are constantly re-evaluating even their own definitions of the construct.

Currently, there are three main models of EI:

- Ability EI model
- Mixed models of EI
- Trait EI model

Different models of EI have led to the development of various instruments for the assessment of the construct. While some of these measures may overlap, most researchers agree that they tap different constructs.

Ability Model

Salovey and Mayer's conception of EI strives to define EI within the confines of the standard criteria for a new intelligence. Following their continuing research, their initial definition of EI was revised to "The ability to perceive emotion, integrate emotion to facilitate thought, understand emotions and to regulate emotions to promote personal growth." The ability-based model views emotions as useful sources of information that help one to make sense of and navigate the social environment. The model proposes that individuals vary in their ability to process information of an emotional nature and in their ability to relate emotional processing to a wider cognition. This ability is seen to manifest itself in certain adaptive behaviours.

The model claims that EI includes four types of abilities:

(1) *Perceiving emotions*: The ability to detect and decipher emotions in faces, pictures, voices, and cultural artifacts—including the ability to identify one's own emotions. Perceiving emotions represents a basic aspect of emotional intelligence, as it makes all other processing of emotional information possible.

(2) *Using emotions*: The ability to harness emotions to facilitate various cognitive activities, such as thinking and problem solving. The emotionally intelligent person can capitalize fully upon his or her changing moods in order to best fit the task at hand.

(3) *Understanding emotions*: The ability to comprehend emotion language and to appreciate complicated relationships among emotions. For example, understanding emotions encompasses the ability to be sensitive to slight variations between emotions, and the ability to recognize and describe how emotions evolve over time.

(4) *Managing emotions*: The ability to regulate emotions in both ourselves and in others. Therefore, the emotionally intelligent person can harness emotions, even negative ones, and manage them to achieve intended goals.

The ability EI model has been criticized in the research for lacking face and predictive validity in the workplace.

Measurement of the Ability Model

The current measure of Mayer and Salovey's model of EI, the Mayer-Salovey-Caruso Emotional Intelligence Test is based on a series of emotion-based problem-solving items. Consistent with the model's claim of EI as a type of intelligence, the test is modelled on ability-based IQ tests. By testing a person's abilities on each of the four branches of emotional intelligence, it generates scores for each of the branches as well as a total score. Central to the four-branch model is the idea that EI requires attunement to social norms.

Therefore, the MSCEIT is scored in a consensus fashion, with higher scores indicating higher overlap between an individual's answers and those provided by a worldwide sample of respondents. The MSCEIT can also be expert-scored, so that the amount of overlap is calculated between an individual's answers and those provided by a group of 21 emotion researchers. Although promoted as an ability test, the MSCEIT is most unlike standard IQ tests in that its items do not have objectively correct responses.

Among other problems, the consensus scoring criterion means that it is impossible to create items that only a minority of respondents can solve, because, by definition, responses are deemed emotionally "intelligent" only if the majority of the sample has endorsed them. This and other similar problems have led cognitive ability experts to question the definition of EI as a genuine intelligence.

In a study by Føllesdal, the MSCEIT test results of 111 business leaders were compared with how their employees described their leader.

It was found that there were no correlations between a leader's test results and how he or she was rated by the employees, with regard to empathy, ability to motivate, and leader effectiveness. Føllesdal also criticized the Canadian company Multi-Health Systems, which administers the MSCEIT test.

The test contains 141 questions but it was found after publishing the test that 19 of these did not give the expected answers. This has led Multi-Health Systems to remove answers to these 19 questions before scoring, but without stating this officially.

Mixed Models

The model introduced by Daniel Goleman focuses on EI as a wide array of competencies and skills that drive leadership performance.

Goleman's model outlines four main EI constructs:

(1) *Self-awareness*: The ability to read one's emotions and recognize their impact while using gut feelings to guide decisions.

(2) *Self-management*: Involves controlling one's emotions and impulses and adapting to changing circumstances.

(3) *Social awareness*: The ability to sense, understand, and react to others' emotions while comprehending social networks.

(4) *Relationship management*: The ability to inspire, influence, and develop others while managing conflict.

Goleman includes a set of emotional competencies within each construct of EI. Emotional competencies are not innate talents, but rather learned capabilities that must be worked on and can be developed to achieve outstanding performance.

Goleman posits that individuals are born with a general emotional intelligence that determines their potential for learning emotional competencies. Goleman's model of EI has been criticized in the research literature as mere "pop psychology".

Measurement of the Emotional Competencies (Goleman) Model

Two measurement tools are based on the Goleman model:

(1) The Emotional Competency Inventory which was created in 1999, and the Emotional and Social Competency Inventory which was created in 2007.

(2) The Emotional Intelligence Appraisal, which was created in 2001 and which can be taken as a self-report or 360-degree assessment.

Bar-On Model of Emotional-Social Intelligence

Bar-On defines emotional intelligence as being concerned with effectively understanding oneself and others, relating well to people, and adapting to and coping with the immediate surroundings to be more successful in dealing with environmental demands. Bar-On posits that EI develops over time and that it can be improved through training, programming, and therapy. Bar-On hypothesizes that those individuals with higher than average EQs are in general more successful in meeting environmental demands and pressures. He also notes that a deficiency in EI can mean a lack of success and the existence of emotional problems. Problems in coping with one's environment are thought, by Bar-On, to be especially common among those individuals lacking in the subscales of reality testing, problem solving, stress tolerance, and impulse control. In general, Bar-On considers emotional intelligence and cognitive intelligence to contribute equally to a person's general intelligence, which then offers an indication of one's potential to succeed in life.

Measurement of the ESI Model

The Bar-On Emotional Quotient Inventory is a self-report measure of EI developed as a measure of emotionally and socially competent behaviour that provides an estimate of one's emotional and social intelligence. The EQ-i is not meant to measure personality traits or cognitive capacity, but rather the mental ability to be successful in dealing with environmental demands and pressures. One hundred and thirty three items are used to obtain a Total EQ and to produce five composite scale scores, corresponding to the five main components of the Bar-On model. A limitation of this model is that it claims to measure some kind of ability through self-report items. The EQ-i has been found to be highly susceptible to faking.

Trait EI Model

Petrides and colleagues proposed a conceptual distinction between the ability based model and a trait based model of EI. Trait EI is "a constellation of emotional self-perceptions located at the lower levels of personality". In lay terms, trait EI refers to an individual's self-perceptions of their emotional abilities. This definition of EI encompasses behavioural dispositions and self perceived abilities and is measured by self report, as opposed to the ability based model which refers to actual abilities, which have proven highly resistant to scientific measurement. Trait EI should be investigated within a personality framework. An alternative label for the same construct is trait emotional self-efficacy.

The conceptualization of EI as a personality trait leads to a construct that lies outside the taxonomy of human cognitive ability. This is an important distinction in as much as it bears directly on the operationalization of the construct and the theories and hypotheses that are formulated about it.

Measurement of the Trait EI Model

There are many self-report measures of EI, including the EQ-i, the Swinburne University Emotional Intelligence Test and the Schutte EI model. None of these assess intelligence, abilities, or skills but rather, they are limited measures of trait emotional intelligence. One of the more comprehensive and widely researched measures of this construct is the Trait Emotional Intelligence Questionnaire which is an open-access measure that was specifically designed to measure the construct comprehensively and is currently available in many languages.

The TEIQue provides an operationalization for the model of Petrides and colleagues, that conceptualizes EI in terms of personality. The test encompasses 15 subscales organized under four factors: Well-Being, Self-Control, Emotionality, and Sociability. The psychometric properties of the TEIQue were investigated in a study on a French-speaking population, where it was reported that TEIQue scores were globally normally distributed and reliable.

The researchers also found TEIQue scores were unrelated to nonverbal reasoning which they interpreted as support for the personality trait view of EI.

As expected, TEIQue scores were positively related to some of the Big Five personality traits as well as inversely related to others. A number of quantitative genetic studies have been carried out within the trait EI model, which have revealed significant genetic effects and heritabilities for all trait EI scores.

Two recent studies involving direct comparisons of multiple EI tests yielded very favourable results for the TEIQue Alexithymia and EI Alexithymia from the Greek words lexis and thumos is a term coined by Peter Sifneos in 1973 to describe people who appeared to have deficiencies in understanding, processing, or describing their emotions.

Viewed as a spectrum between high and low EI, the alexithymia construct is strongly inversely related to EI, representing its lower range. The individual's level of alexithymia can be measured with self-scored questionnaires such as the Toronto Alexithymia Scale or the Bermond-Vorst Alexithymia Questionnaire or by observer rated measures such as the Observer Alexithymia Scale.

The Effects of Emotional Biases

Its effects can be similar to those of a cognitive bias, it can even be considered as a sub-category of such biases. The specificity is that the cause lies in one's desires or fears, which divert the attention of the person, more than in one's reasoning.

Neuroscience experiments have shown how emotions and cognition, which are present in different areas of the persons human brain, interfere between each other in the decision making process, resulting often on a primacy of emotions over reasoning.

This might explain some irrational and damaging reactions and moves that might take place when those emotions are biased.

Optimism

Optimism is "an inclination to put the most favourable construction upon actions and events or to anticipate the best possible outcome". It is the philosophical opposite of pessimism. Optimists generally believe that people and events are inherently good, so that most situations work out in the end for the best.

Alternatively, some optimists believe that regardless of the external world or situation, one should choose to feel good about it and make the most of it.

This kind of optimism doesn't say anything about the quality of the external world; it's an internal optimism about one's own feelings. A common conundrum illustrates optimism-versus-pessimism with the question, does one regard a given glass of water, filled to half its capacity, as *half full* or as *half empty*? Conventional wisdom expects optimists to reply, "Half full," and pessimists to respond, "Half empty" (assuming that "full" is considered good, and "empty", bad).

Psychology

Overoptimism, naive optimism or strong optimism, is the overarching mental state wherein people believe that things are more likely to go well for them than go badly. Compare this with the valence effect of prediction, a tendency for people to overestimate the likelihood of good things happening rather than bad things. Optimism bias is the demonstrated systematic tendency for people to be over-optimistic about the outcome of planned actions.

Personal optimism correlates strongly with self-esteem, with psychological well-being and with physical and mental health. Optimism has been shown to be correlated with better immune systems in healthy people who have been subjected to stress.

Martin Seligman, in researching this area, criticizes academics for focusing too much on causes for pessimism and not enough on optimism. He states that in the last three decades of the 20th century journals published 46,000 psychological papers on depression and only 400 on joy.

Popular culture has reflected the link between optimism and well-being with works like the fable "The Moth and the Star", and Barack Obama's speech and book, *The Audacity of Hope*. Ideologically convinced optimists may defend failures in their hoped-for outcomes by discussing "misplaced optimism" rather than abandoning optimism altogether.

A number of scholars have suggested that, although optimism and pessimism might seem like opposites, in psychological terms they do not function in this way. Having more of one does not mean you have less of the other. The factors that reduce one do not necessarily increase the other. On many occasions in life we need both in equal supply.

Antonio Gramsci famously called for "pessimism of the intellect, optimism of the will": the one the spur to action, the other the resilience to believe that such action will result in meaningful change even in the face of adversity. Hope can become a force for social change when it combines optimism and pessimism in healthy proportions. John Braithwaite, an academic at the Australian National University, suggests that in modern society we undervalue hope because we wrongly think of it as a choice between hopefulness and naivete as opposed to skepticism and realism.

Rationality

In philosophy, rationality and reason are the key methods used to analyse the data gathered through systematically gathered observations. In economics, sociology, and political science, a decision or situation is often called rational if it is in some sense optimal, and individuals or organizations are often called rational if they tend to act somehow optimally in pursuit of their goals. Thus one speaks, for example, of a rational allocation of resources, or of a rational corporate strategy. In this concept of "rationality", the individual's goals or motives are taken for granted and not made subject to criticism, ethical or otherwise. Thus, rationality simply refers to the success of goal attainment, whatever those goals may be. Sometimes, in this context, rationality is equated with behaviour that is self-interested to the point of being selfish. Sometimes rationality implies having complete knowledge about all the details of a given situation. Debates arise in these three fields about whether or not people or organizations are "really" rational, as well as whether it make sense to model them as such in formal models. Some have argued that a kind of bounded rationality makes more sense for such models.

Others think that any kind of rationality along the lines of rational choice theory is a useless concept for understanding human behaviour; the term *homo economicus* (economic man: the imaginary man being assumed in economic models who is logically consistent but amoral) was coined largely in honour of this view.

Rationality is a central principle in artificial intelligence, where a *rational agent* is specifically defined as an agent which always chooses the action which maximises its expected performance, given all of the knowledge it currently possesses.

Quality of Rationality

It is believed by some philosophers (notably A.C. Grayling) and experts, that a good rationale must be independent of emotions, personal feelings or any kind of instincts. Any process of evaluation or analysis, that may be called rational, is expected to be highly objective, logical and "mechanical". If these minimum requirements are not satisfied *i.e.*, if a person has been, even slightly, influenced by personal emotions, feelings, instincts or culturally specific, moral codes and norms, then the analysis may be termed irrational, due to the injection of subjective bias. It is quite evident from modern cognitive science and neuroscience, studying the role of emotion in mental function (including topics ranging from flashes of scientific insight to making future plans), that no human has ever satisfied this criterion, except perhaps a person with no affective feelings perhaps a with a massively damaged amygdala. Thus, such an idealized form of rationality is best exemplified by computers, and not people. However, scholars may productively appeal to the idealization as a point of reference.

Theories of Rationality

The German sociologist Max Weber proposed an interpretation of social action that distinguished between four different types of rationality. The first, which he called *Zweckrational* or purposive/instrumental rationality, is related to the expectations about the behaviour of other human beings or objects in the environment. These expectations serve as means for a particular actor to attain ends, ends which Weber noted were "rationally pursued and calculated."

The second type, Weber called *Wertrational* or value/belief-oriented. Here the action is undertaken for what one might call reasons intrinsic to the actor: some ethical, aesthetic, religious or other motive, independent of whether it will lead to success. The third type was affectual, determined by an actor's specific affect, feeling, or emotion-to which Weber himself said that this was a kind of rationality that was on the borderline of what he considered "meaningfully oriented."

The fourth was traditional, determined by ingrained habituation. Weber emphasized that it was very unusual to find only one of these orientations: combinations were the norm. His usage also makes clear that he considered the first two as more significant than the others, and it is arguable that the third and fourth are subtypes of the first two. These kinds of rationality were ideal types.

The advantage in this interpretation is that it avoids a value-laden assessment, say, that certain kinds of beliefs are irrational. Instead, Weber suggests that a ground or motive can be given – for religious or affect reasons, for example — that may meet the criterion of explanation or justification even if it is not an explanation that fits the *Zweckrational* orientation of means and ends. The opposite is therefore also true: some means-ends explanations will not satisfy those whose grounds for action are 'Wertrational'. Weber's constructions of rationality have been critiqued both from a Habermasian (1984) perspective (as devoid of social context and under-theorised in terms of social power) and also from a feminist perspective (Eagleton, 2003) whereby Weber's rationality constructs are viewed as imbued with masculine values and oriented towards the maintenance of male power·

An alternative position on rationality (which includes both bounded rationality, as well as the affective and value-based arguments of Weber) can be found in the critique of Etzioni (1988)· who reframes thought on decision-making to argue for a reversal of the position put forward by Weber. Etzioni illustrates how purposive/instrumental reasoning is subordinated by normative considerations (ideas on how people 'ought' to behave) and affective considerations (as a support system for the development of human relationships).

Theoretical and Practical Rationality

Kant had distinguished theoretical from practical reason. Rationality theorist Jesus Mosterin makes a parallel distinction between theoretical and practical rationality, although, according to him, reason and rationality are not the same: reason would be a psychological faculty, whereas rationality is an optimizing strategy. Humans are not rational by definition, but they can think and behave rationally or not, depending on whether they apply, explicitly or implicitly, the strategy of theoretical and practical rationality to the thoughts they accept and to the actions they perform.

Theoretical rationality has a formal component that reduces to logical consistency and a material component that reduces to empirical support, relying on our inborn mechanisms of signal detection and interpretation.

Mosterin distinguishes between involuntary and implicit belief, on the one hand, and voluntary and explicit acceptance, on the other. Theoretical rationality can more properly be said to regulate our acceptances than our beliefs.

Practical rationality is the strategy for living one's best possible life, achieving your most important goals and your own preferences in as far as possible. Practical rationality has also a formal component, that reduces to Bayesian decision theory, and a material component, rooted in human nature (lastly, in our genome).

EMOTIONAL CONTROL AND MATURITY

Controlling your emotion begins with a deliberate decision to keep composure and emotional retrain at all times. You should be very mindful of your feelings in every situation whether it is exuberant or sorrowful. You should avoid the belief that people are entitled to loose control in special occasions or that people have the right to let out their emotions in specific circumstances. You should stop adhering to the belief that "people are just humans" because what makes people humans is their ability to tame their desires and emotions.

You should understand that emotion is actually dependent on your behaviour and not the other way around. To put it simply, you feel sad because you realize that you are frowning, contrary to the popular notion that you frown because you realize that you are sad. In social psychology, it has been discovered behaviour influences emotions, and not the other way around. With this key information at hand, you are empowered with the ability to influence feelings by modifying behaviour.

A common cause of emotional outbursts is having problems. Oftentimes, when people are faced with difficulties, they react by panicking or by being angry. However, these are not solutions to the initial problem; they are sources of problems themselves. When you are in a state of panic or in a rage of anger, your mind is clouded with emotions that the situation is not seen clearly. This does make it harder to think of concrete solutions to the problem being dealt with. Whenever a problem comes, instead of immediately throwing tantrums or pacing restlessly, you should stop, breath, and evaluate the entirety of the circumstance. This way, the problem can be viewed in a clearer perspective, and solutions may be thought out more logically.

One very valuable tool in emotional control is the ability to pause. While there are no pause and play buttons in real life, people have the capability to stop themselves, and take a break in certain situations that usually cause emotional outbursts.

Before shouting and screaming out of a disappointing occurrence, you should first take a break to think and reflect. Pausing in itself is a form of emotional control. At this point, you can think if letting go of the emotions is indeed necessary. You should also think of the consequences that the emotional explosion would entail. If the emotions involved in the situation are too strong to withhold, you can think of reasonable emotional expressions such as crying instead of screaming when you are depressed, or smiling instead of jumping around when you are elated.

The problem with most people who are unable to control their emotions is that they dwell too much on the present situation. It is undeniable that the height of emotions experienced in certain circumstances could be overwhelming. However, these are also the times when you are vulnerable to do things that you might regret later on.

To avoid this, it is important to examine how things would go in the future. Emotional control is indeed difficult to master. But with will power and determination, it can be possibly achieved. You just have to be aware that emotions do not really have the power to overcome people. On the contrary, people have the ability to watch over their emotions and control them to what they think is necessary.

SIGNIFICANCE OF EMOTIONAL INTELLIGENCE

EMOTIONAL INTELLIGENCE AS AN APPROACH TO TEACHING AND LEARNING PROCESSES

Concerning emotional intelligence I believe that we need to reassess our thinking as we do in the world in general. We cannot put too much emphasis on one narrow or limited feature but should consider a person as a whole.

We need skilled people, professionals with scientific competence who can define what emotional intelligence is all about. In the past intelligence was defined in a limited way with emphasis on reading and arithmetic.

Many other features were left in the background until their importance was also acknowledged. What kind of educational system will be developed in Finland depends on what kind of all-round significance is attached to it.

I claim that all the big goals that Finland and Europe, for example, have concerning competitiveness, require a working educational system. It is no use talking about a competitive economy or society if it does not offer the kind of education that inspires and motivates people to work towards those goals. Education, on the other hand, is not only about economic competitiveness but also about contending with oneself and achieving one's moral goals.

In this realm we don't seem to have a vision. Everywhere people are waiting to see what others do. I don't mean that we would need another Leonardo da Vinci or Immanuel Kant or some accomplished pedagogue who would give us answers. It is not the world of one great man or woman that I'm after but some kind of school of thought, a cultural trend or, in formal language, ideology to define what we are as people.

If we get this kind of inspiration and realize that we exist and are inspired when our feelings are aroused, then we are talking about emotional intelligence. What is included is music and painting (not necessarily pompous music or imposing art), media and traditional theatre but also strong mathematical competence and capacity for abstract reasoning. All this is part of the Zeitgeist, the spirit of the age, which is our responsibility to create. If educational visionaries and school reformers cannot create this spirit, where else can it come from?

An individual teacher or professor may do their best in their own work but if their efforts are wrecked by bureaucratic machinery, nothing will be reformed. This is why it is good to point out and emphasize that emotions can be the basis of all this change. Then we should consider what that entails.

EDUCATIONAL PROGRAMMES AND CURRICULA FROM THE POINT OF VIEW OF EMOTIONAL INTELLIGENCE

They are of more importance than has been generally acknowledged so far. When it comes to education and educational programmes, the goal can be a good life or a good and just world, which is a world of knowledge and understanding. Finding a good life, on the other hand, requires the ability to perceive emotional life and its significance.

In fact I claim emotion can be an essential initiator in the birth of cognitive and traditional intellectual accomplishments. It is a source of motivation. The fact that a person becomes interested may be a sign of an ability to move in the realm of emotional life.

What media researchers have always known and what has been taken into account in the media education of some countries, are the dimensions of cognitive, emotional, moral, aesthetic education and so on.

It is a recognized fact that in the media, emotions are created. As a counter reaction young people and people in general should be able to perceive that emotions are being created *for* them, emotions whose origin is not in them. It is not bad in itself because when actors fall in love in a film, it can have a positive effect on the viewers. What is more important, however, is that people would fall in love in real life, too, and not only through films. In other words, I claim that to induce enthusiasm, motivation and persistent positive intellectual curiosity we need to be aware of emotional life and include it in the preparation of curricula.

EXAMPLES OF TAKING EMOTIONAL LIFE INTO CONSIDERATION

I don’t actually know of any existing programme which does this because I work in research which is not very close to the operational level. However, I can take an example of the international work that I am involved in, which perhaps shows how emotional life could be taken into account. I studied the life and work of Anton Chekhov in Taganrog in the southwest of Russia. His attitude to knowledge and technology was very positive and balanced because he also wanted to preserve nature and values instead of letting technology and knowledge change life too much. He was talking to a young lawyer about something when he suddenly, in the middle of the conversation, asked the young man if he liked the gramophone, which at that time was a new invention.

The young man answered that he did. Chekhov said that *he* didn’t. When the young man wondered why not, Chekhov replied that it did have moving sound and music but it was somehow dead, it was empty, it did not have emotions. Then they continued their conversation about photography and other things. This conversation crystallizes the concrete situation Chekhov lived in a hundred years ago and which is present in today's teaching situations. Young people have mechanical devices for reproducing art and music. Day in day out they wear their headphones. However, a machine is always a machine. No matter how interactive it is, it is devoid of all the sensitivity that is involved when you are making music in real life. A machine cannot identify the elements of the situation. If this is lacking, interaction between people is on a mechanical level. There is a difference between listening to a perfect CD and being at a live concert where one single cough is an expression of the tension in the air, let alone the actual singing and playing. The different elements of the situation regarding speed, tempo, emphasis or anything else that cannot be repeated are essential in the situation. They touch emotion. Emotion is involved. The same applies to speech. A person may speak well and learn mechanically to use good structures and express himself well, yet disregard the social situation. Without any other concrete example I claim that in order to be able to consider emotional intelligence sensibly, it has to be done in situational contexts. Of course the intellectual and background elements are also needed but it is only in the immediate social context that the emotions come into play.

EMOTIONAL INTELLIGENCE IN THE CO-OPERATION BETWEEN SCHOOLS AND HOMES

Co-operation between schools and homes must involve a sense of community but I don't think we can emphasize and use only this. I speak again from the ivory tower of university life because I am in no contact with the interaction between schools and homes in today's society other than through my grandchildren. But I am under the impression that many people are clueless about parenthood, at least when considered from the point of view of my generation.

Nowadays both parents pursue their own career and the home is different from what it was a few decades ago when schools were built and current educational thinking was developed. Nowadays responsibility is too easily shifted from homes on to institutions, such as schools, kindergartens, day-care centres and clubs. A prerequisite for a good life and well-being is that there are emotions and that they can be reciprocated. If there are no emotions in a home, it is impossible to develop them from the outside. Now I speak as an amateur with no factual knowledge of the matter but I would like to claim that *caring as a way of thinking* or a world where parents are made sensitive to what is happening in the world of children would be good. Whether something like that exists or not I have no way of knowing.

Chapter Six

Motivation and Learning

Basic principles of motivation exist that are applicable to learning in any situation.

1. The environment can be used to focus the student's attention on what needs to be learned.

   Teachers who create warm and accepting yet businesslike atmospheres will promote persistent effort and favourable attitudes towards learning. This strategy will be successful in children and in adults. Interesting visual aids, such as booklets, posters, or practice equipment, motivate learners by capturing their attention and curiosity.

2. Incentives motivate learning.

   Incentives include privileges and receiving praise from the instructor. The instructor determines an incentive that is likely to motivate an individual at a particular time. In a general learning situation, self-motivation without rewards will not succeed. Students must find satisfaction in learning based on the understanding that the goals are useful to them or, less commonly, based on the pure enjoyment of exploring new things.

3. Internal motivation is longer lasting and more self-directive than is external motivation, which must be repeatedly reinforced by praise or concrete rewards.

   Some individuals — particularly children of certain ages and some adults — have little capacity for internal motivation and must be guided and reinforced constantly. The use of incentives is based on the principle that learning occurs more effectively when the student experiences feelings of satisfaction. Caution should be exercised in using external rewards when they are not absolutely necessary. Their use may be followed by a decline in internal motivation.

4. Learning is most effective when an individual is ready to learn, that is, when one wants to know something.

   Sometimes the student's readiness to learn comes with time, and the instructor's role is to encourage its development. If a desired change in behaviour is urgent, the instructor may need to supervised directly to ensure that the desired behaviour occurs. If a student is not *ready to learn,* he or she may not be reliable in following instructions and therefore must be supervised and have the instructions repeated again and again.
5. Motivation is enhanced by the way in which the instructional material is organized.

   In general, the best organized material makes the information meaningful to the individual. One method of organization includes relating new tasks to those already known. Other ways to relay meaning are to determine whether the persons being taught understand the final outcome desired and instruct them to compare and contrast ideas.

None of the techniques will produce sustained motivation unless the goals are realistic for the learner. The basic learning principle involved is that *success is more predictably motivating than is failure*. Ordinarily, people will choose activities of intermediate uncertainty rather than those that are difficult (little likelihood of success) or easy (high probability of success). For goals of high value there is less tendency to choose more difficult conditions. Having learners assist in defining goals increases the probability that they will understand them and want to reach them. However, students sometimes have unrealistic notions about what they can accomplish. Possibly they do not understand the precision with which a skill must be carried out or have the depth of knowledge to master some material. To identify realistic goals, instructors must be skilled in assessing a student's readiness or a student's progress towards goals.

1. Because learning requires changed in beliefs and behaviour, it normally produces a mild level of anxiety.

   This is useful in motivating the individual. However, severe anxiety is incapacitating. A high degree of stress is inherent in some educational situations. If anxiety is severe, the individual's perception of what is going on around him or her is limited. Instructors must be able to identify anxiety and understand its effect on learning. They also have a responsibility to avoid causing severe anxiety in learners by setting ambiguous of unrealistically high goals for them.
2. It is important to help each student set goals and to provide informative feedback regarding progress towards the goals.

   Setting a goal demonstrates an intention to achieve and activates learning from one day to the next. It also directs the student's activities towards the goal and offers an opportunity to experience success.
3. Both affiliation and approval are strong motivators.

People seek others with whom to compare their abilities, opinions, and emotions. Affiliation can also result in direct anxiety reduction by the social acceptance and the mere presence of others. However, these motivators can also lead to conformity, competition, and other behaviours that may seem as negative.

4. Many behaviours result from a combination of motives.

   It is recognized that no grand theory of motivation exists. However, motivation is so necessary for learning that strategies should be planned to organize a continuous and interactive motivational dynamic for maximum effectiveness. The general principles of motivation are interrelated. A single teaching action can use many of them simultaneously.

Finally, it should be said that an enormous gap exists between knowing that learning must be motivated and identifying the specific motivational components of any particular act. Instructors must focus on learning patterns of motivation for an individual or group, with the realization that errors will be common.

KIND OF MOTIVES

A MOTIVE may be defined as meaning that which acts as an inducement to preference or choice. 'In other words it is a very strong influence towards some object to be attained.

Just what these influences are depends entirely upon the nature of the person and on the object to be attained. We often hear the expression, and doubtless use it as often ourselves, "What was your motive in doing this?" meaning what caused you or what influenced you to act in such a manner.

We may state with truth that every wilful act has its inducement or cause. Just as in mathematics there is a reason for every step, so also in every act there is a motive or cause. From the child on up the individual can give a reason for each and every one of his voluntary acts however small and mean the reason may be. Still it is a reason. It may only be a certain state of consciousness which happens to be uppermost in a man's mind as he acts.

Very often Intention and Motive are confused; in fact they are used interchangeably in society. Notwithstanding this fact there is a vast difference in their meaning. Intention is what a man means to do, while Motive is the personal frame of mind which indicates why he means to do it.

Therefore Intention has the stronger moral value, while Motive renders the consequences interesting and attractive. Motive is that which makes the difference between one act and another.

As we now know the difference between Intention and Motive, let us examine the different kinds of Motives. Motive may be divided into two classes, Egoistical and Altruistical. Both classes are absolutely necessary in our daily life. Let us first look at egoistic motives or the ones which concern the self only. Our original instincts are such that their objects are to look after the advantages of the self. However everything depends upon the sort of self maintained. What would become of a person if he would not struggle for food and strive against obstacles? What would become of society? Self-preserving instincts must be therefore socially conservative. No one has a right to neglect his own interests, hoping some one else will care for them. The man who takes exercise because he thinks of his health must be commanded, but the one who is thinking continually of his health and excludes other thoughts must be condemned.

THEORIES OF MOTIVATION

Ever wonder why some people seem to be very successful, highly motivated individuals? Where does the energy, the drive, or the direction come from? Motivation is an area of psychology that has gotten a great deal of attention, especially in the recent years. The reason is because we all want to be successful, we all want direction and drive, and we all want to be seen as motivated. There are several distinct theories of motivation we will discuss in this section. Some include basic biological forces, while others seem to transcend concrete explanation. Let's talk about the five major theories of motivation.

INSTINCT THEORY

Instinct theory is derived from our biological make-up. We've all seen spider's webs and perhaps even witnessed a spider in the tedious job of creating its home and trap. We've all seen birds in their nests, feeding their young or painstakingly placing the twigs in place to form their new home. How do spiders know how to spin webs? How do birds now how to build nests?

The answer is biology. All creatures are born with specific innate knowledge about how to survive. Animals are born with the capacity and often times knowledge of how to survive by spinning webs, building nests, avoiding danger, and reproducing. These innate tendencies are preprogrammed at birth, they are in our genes, and even if the spider never saw a web before, never witnessed its creation, it would still know how to create one. Humans have the same types of innate tendencies. Babies are born with a unique ability that allows them to survive; they are born with the ability to cry. Without this, how would others know when to feed the baby, know when he needed changing, or when she wanted attention and affection? Crying allows a human infant to survive. We are also born with particular reflexes which promote survival. The most important of these include sucking, swallowing, coughing, blinking. Newborns can perform physical movements to avoid pain; they will turn their head if touched on their cheek and search for a nipple (rooting reflex); and they will grasp an object that touches the palm of their hands.

DRIVE REDUCTION THEORY

According to Clark Hull (1943, 1952), humans have internal biological needs which motivate us to perform a certain way. These needs, or drives, are defined by Hull as internal states of arousal or tension which must be reduced. A prime example would be the internal feelings of hunger or thirst, which motivates us to eat. According to this theory, we are driven to reduce these drives so that we may maintain a sense of internal calmness.

AROUSAL THEORY

Similar to Hull's Drive Reduction Theory, Arousal theory states that we are driven to maintain a certain level of arousal in order to feel comfortable. Arousal refers to a state of emotional, intellectual, and physical activity. It is different from the above theory, however, because it doesn't rely on only a reduction of tension, but a balanced amount. It also does better to explain why people climb mountains, go to school, or watch sad movies.

PSYCHOANALYTIC THEORY

Remember Sigmund Freud and his five part theory of personality. As part of this theory, he believed that humans have only two basic drives: Eros and Thanatos, or the Life and Death drives. According to Psychoanalytic theory, everything we do, every thought we have, and every emotion we experience has one of two goals: to help us survive or to prevent our destruction. This is similar to instinct theory, however, Freud believed that the vast majority of our knowledge about these drives is buried in the unconscious part of the mind.

Psychoanalytic theory therefore argues that we go to school because it will help assure our survival in terms of improved finances, more money for health care, or even an improved ability to find a spouse. We move to better school districts to improve our children's ability to survive and continue our family tree. We demand safety in our cars, toys, and in our homes. We want criminal locked away, and we want to be protected against poisons, terrorists, and any thing else that could lead to our destruction. According to this theory, everything we do, everything we are can be traced back to the two basic drives

HUMANISTIC THEORY

Although discussed last, humanistic theory is perhaps the most well know theory of motivation. According to this theory, humans are driven to achieve their maximum potential and will always do so unless obstacles are placed in their way. These obstacles include hunger, thirst, financial problems, safety issues, or anything else that takes our focus away from maximum psychological growth.

The best way to describe this theory is to utilize the famous pyramid developed by Abraham Maslow (1970) called the Hierarchy of Needs. Maslow believed that humans have specific needs that must be met and that if lower level needs go unmet, we can not possible strive for higher level needs. The Hierarchy of Needs shows that at the lower level, we must focus on basic issues such as food, sleep, and safety. Without food, without sleep, how could we possible focus on the higher level needs such as respect, education, and recognition?

MASLOW'S HIERARCHY OF NEEDS

Maslow's hierarchy of needs is a theory in psychology, proposed by Abraham Maslow in his 1943 paper *A Theory of Human Motivation*, which he subsequently extended to include his observations of humans' innate curiosity.

Maslow studied what he called exemplary people such as Albert Einstein, Jane Addams, Eleanor Roosevelt, and Frederick Douglass rather than mentally ill or neurotic people, writing that "the study of crippled, stunted, immature, and unhealthy specimens can yield only a cripple psychology and a cripple philosophy." Maslow also studied the healthiest one percent of the college student population. In his book, *The Farther Reaches of Human Nature*, Maslow writes, "By ordinary standards of this kind of laboratory research... this simply was not research at all. My generalizations grew out of my selection of certain kinds of people. Obviously, other judges are needed."

REPRESENTATIONS

Maslow's hierarchy of needs is predetermined in order of importance. It is often depicted as a pyramid consisting of five levels: the first lower level is being associated with physiological needs, while the top levels are termed growth needs associated with psychological needs. Deficiency needs must be met first. Once these are met, seeking to satisfy growth needs drives personal growth.

The higher needs in this hierarchy only come into focus when the lower needs in the pyramid are met. Once an individual has moved upwards to the next level, needs in the lower level will no longer be prioritized. If a lower set of needs is no longer being met, the individual will temporarily re-prioritize those needs by focusing attention on the unfulfilled needs, but will not permanently regress to the lower level.

For instance, a businessman at the esteem level who is diagnosed with cancer will spend a great deal of time concentrating on his health (physiological needs), but will continue to value his work performance (esteem needs) and will likely return to work during periods of remission.

ROLE OF REWARDS AND PUNISHMENTS

REWARDS

Rewards can serve as effective incentives—if the person is interested in the reward. School grades are a case in point. The reward of a good grade is important to some students--but of no interest to others.

A reason why some students exert little effort for grades is that—to use Dr. William Glasser's vocabulary—grades are not in their "Quality World." If a good grade or any reward is not important to the person, it has little value as an incentive. Rewards can also serve as wonderful acknowledgements—ways of congratulating merit and demonstrating appreciation. Student of the week or employee of the month are examples of such acknowledgements. But notice that these are awarded after the behaviour—not as bribes beforehand.

As opposed to using rewards as incentives and acknowledgements, giving rewards for expected standards of behaviour is counterproductive. It is also based on the outmoded idea that all behaviour is modified thorough external approaches, similar to the techniques used to train animals. Internal approaches—such as self-talk—have no place in this mindset.

People who use this behavioural approach—often referred to as behaviour modification—have just one objective: changing behaviour. Practitioners of this approach do NOT promote ADULT values.

By rewarding kids with something youngsters value (candy, stickers, prizes, etc.), we simply reinforce their CHILDHOOD values. In the process, we lose opportunities to pass on OUR values—such as generosity, kindness, responsibility, perseverance, and integrity. What we really hope to do is to teach young people about values that will last a lifetime. So, while most kids will do what you want them to do to get the treat, and it might look as if they are becoming more mature, they have not moved one step further towards becoming more responsible.

PUNISHMENTS

No argument is presented here against such societal practices for adults who have been convicted of socially harmful behaviour. If you believe that an eight-year-old is an eighteen-yea-old, then it may seem natural to treat a young person with similar approaches. However, if you believe that young people are not yet adults, then the use of punishment to raise responsibility must be examined.

A common myth is that imposed punishments are necessary to change young people's behaviour. If this type of punishment worked, then once a youngster is punished the same behaviour would not be repeated. If you have used punishments to change behaviour and the same behaviour was repeated, read on.

Not too many of us remember that the horse pulling the buggy was urged on by a flick of the whip and that information was pounded into students by the cane, the strap, or fear of them. But since then, we have found better ways to teach; yet we are still using the horse and buggy approach to foster social responsibility in an era when both society and the nature of youth have dramatically changed.

Punishments operate on the theory young people must experience pain in order to grow into responsibility. We are expecting people whom we "intentionally hurt" to act constructively thereafter. But can you recall the last time you felt bad and did something good? People cannot think positively with negative feelings. People do "good" when they feel good. Imposed punishments can force compliance but never commitment. Have you ever seen anyone punished into commitment? Punishments kill the very thing we are attempting to do—change behaviour into something that is positive and socially appropriate.

One reason we keep thinking punishment works is that sometimes the behaviour stops. This may be the case with very young children if the behaviour is caught early before it becomes a habit and if the punishment itself is a novel experience. However, we must also consider whether or not the subject understands which action is being punished.

Punishments are ineffective with far too many young people. Escalation is a testament to its ineffectiveness. In schools when punishments fails to work, inevitably more punishments are prescribed, and the cycle perpetuates itself. By the time some students reach the secondary level, they have been talked to, lectured at, sent out of class, kept after school, referred to the office, referred to Saturday school, suspended in school, suspended from school—and they simply no longer care.

Punishments are temporary and transitory. Fear and force only produce changes in the short run. Once a punishment ends, the youngster has "served his time" and is "free and clear" from further responsibility. A coercive approach that works in the short run does not mean it is effective in the long run. Threatening a youngster with punishment may force compliance--but only so long as the threat is present. Needless to say, it does not change behaviour when the coercion is gone.

LEVEL OF ASPIRATION

BASUMALLIK AND BANERJEE, WEINSTEIN, AND RAY have reported findings from India, the United States, and Australia which call into question the once generally accepted relationship between high achievement motivation and intermediate levels of risk-preference (1). It now appears that the two variables are essentially unrelated. Wolk and Du Cette (2) also found no overall relationship but reported that when they studied only the “internals” (in Rotter’s (3) sense) in their sample, the predicted relationship could be found quite strongly. Their work in this respect has been praised by Lefcourt (4), but it is nonetheless hard to see how there could be a strong relationship in one half of the sample, no relationship in the other half, and also no relationship overall. One would surely have thought that the overall relationship would be some sort of weaker (in-between) relationship.

There is an opportunity for independent replication of their work with the use of data from Study II of the Ray paper (5) containing a "belief in luck" scale derived from and shown to measure the core concept of the Rotter Locus of Control (LOC) scale (6). The Ss in this study were therefore divided into those above and below the mean on belief in luck, and all relationships were reexamined. There were 55 "internals." They were categorized into those who expected to get 0, 1, or 2 rings on the peg in the Litwin (7) experiment; those who expected to get on 3 or 4 rings; and those who expected to get on 5 or 6 rings. This gave low-, intermediate-, and high-level aspiration groups (ns 26, 17, and 12). All Ss received personality inventories measuring achievement orientation, task orientation, success orientation, fear of failure, and fear of success. The means of the three groups on all inventories were compared, and the Fs were all found not to be significant at the <.05 level. The Wolk and Du Cette results in respect of "internals" were not confirmed.

Some reasons for the non-confirmation may be as follows: Rather than conducting an experiment in which preferred probability of success actually chosen could be ascertained, Wolk and Du Cette described an imaginary experiment and asked Ss what probability they would prefer. We may, in other words, be dealing with yet another instance of the attitude-behaviour discrepancy. Further, they did not directly compare motivation scores of people with different levels of aspiration but rather created for each S a "degree of intermediateness" score on level of aspiration. This artificial procedure equates very different levels of aspiration (*i.e.*, very low and very high ones). Their scale of achievement motivation contains at least two items that equate achievement motivation with preference for intermediate risk. The dependent and independent variables were thus artifactually related.

Also contrary to the considerations set out by Wolk and Du Cette, LOC itself in the present study showed no relationship with level of aspiration. They see "internality" and ambition as both leading to intermediate risk preference. In fact, neither did.

ACHIEVEMENT MOTIVATION—TECHNIQUES OF DEVELOPING ACHIEVEMENT MOTIVATION

Motivation can be defined as the driving force behind all the actions of an individual. The influence of an individual's needs and desires both have a strong impact on the direction of their behaviour. Motivation is based on your emotions and achievement-related goals. There are different forms of motivation including extrinsic, intrinsic, physiological, and achievement motivation. There are also more negative forms of motivation. Achievement motivation can be defined as the need for success or the attainment of excellence. Individuals will satisfy their needs through different means, and are driven to succeed for varying reasons both internal and external.

Motivation is the basic drive for all of our actions. Motivation refers to the dynamics of our behaviour, which involves our needs, desires, and ambitions in life. Achievement motivation is based on reaching success and achieving all of our aspirations in life. Achievement goals can affect the way a person performs a task and represent a desire to show competence. These basic physiological motivational drives affect our natural behaviour in different environments. Most of our goals are incentive-based and can vary from basic hunger to the need for love and the establishment of mature sexual relationships. Our motives for achievement can range from biological needs to satisfying creative desires or realizing success in competitive ventures. Motivation is important because it affects our lives everyday. All of our behaviours, actions, thoughts, and beliefs are influenced by our inner drive to succeed.

IMPLICIT AND SELF-ATTRIBUTED MOTIVES

Motivational researchers share the view that achievement behaviour is an interaction between situational variables and the individual subject's motivation to achieve. Two motives are directly involved in the prediction of behaviour, implicit and explicit. Implicit motives are spontaneous impulses to act, also known as task performances, and are aroused through incentives inherent to the task. Explicit motives are expressed through deliberate choices and more often stimulated for extrinsic reasons. Also, individuals with strong implicit needs to achieve goals set higher internal standards, whereas others tend to adhere to the societal norms. These two motives often work together to determine the behaviour of the individual in direction and passion (Brunstein & Maier, 2005).

Explicit and implicit motivations have a compelling impact on behaviour. Task behaviours are accelerated in the face of a challenge through implicit motivation, making performing a task in the most effective manner the primary goal. A person with a strong implicit drive will feel pleasure from achieving a goal in the most efficient way. The increase in effort and overcoming the challenge by mastering the task satisfies the individual. However, the explicit motives are built around a person's self-image. This type of motivation shapes a person's behaviour based on their own self-view and can influence their choices and responses from outside cues. The primary agent for this type of motivation is perception or perceived ability. Many theorists still can not agree whether achievement is based on mastering one's skills or striving to promote a better self-image (Brunstein & Maier, 2005). Most research is still unable to determine whether these different types of motivation would result in different behaviours in the same environment.

THE HIERARCHAL MODEL OF ACHIEVEMENT MOTIVATION

Achievement motivation has been conceptualized in many different ways. Our understanding of achievement-relevant effects, cognition, and behaviour has improved.

Despite being similar in nature, many achievement motivation approaches have been developed separately, suggesting that most achievement motivation theories are in concordance with one another instead of competing. Motivational researchers have sought to promote a hierarchal model of approach and avoidance achievement motivation by incorporating the two prominent theories: the achievement motive approach and the achievement goal approach. Achievement motives include the need for achievement and the fear of failure. These are the more predominant motives that direct our behaviour towards positive and negative outcomes. Achievement goals are viewed as more solid cognitive representations pointing individuals towards a specific end. There are three types of these achievement goals: a performance-approach goal, a performance-avoidance goal, and a mastery goal. A performance-approach goal is focused on attaining competence relative to others, a performance-avoidance goal is focused on avoiding incompetence relative to others, and a mastery goal is focused on the development of competence itself and of task mastery. Achievement motives can be seen as direct predictors of achievement-relevant circumstances. Thus, achievement motives are said to have an indirect or distal influence, and achievement goals are said to have a direct or proximal influence on achievement-relevant outcomes (Elliot & McGregor, 1999).

These motives and goals are viewed as working together to regulate achievement behaviour. The hierarchal model presents achievement goals as predictors for performance outcomes. The model is being further conceptualized to include more approaches to achievement motivation. One weakness of the model is that it does not provide an account of the processes responsible for the link between achievement goals and performance. As this model is enhanced, it becomes more useful in predicting the outcomes of achievement-based behaviours (Elliot & McGregor, 1999).

INTRINSIC MOTIVATION AND ACHIEVEMENT GOALS

Intrinsic motivation is defined as the enjoyment of and interest in an activity for its own sake. Fundamentally viewed as an approach form of motivation, intrinsic motivation is identified as an important component of achievement goal theory. Most achievement goal and intrinsic motivational theorists argue that mastery goals are facilitative of intrinsic motivation and related mental processes and performance goals create negative effects. Mastery goals are said to promote intrinsic motivation by fostering perceptions of challenge, encouraging task involvement, generating excitement, and supporting self-determination while performance goals are the opposite. Performance goals are portrayed as undermining intrinsic motivation by instilling perceptions of threat, disrupting task involvement, and creating anxiety and pressure (Elliot & Harackiewicz, 1996).

An alternative set of predictions may be derived from the approach-avoidance framework. Both performance-approach and mastery goals are focused on attaining competence and foster intrinsic motivation. More specifically, in performance-approach or mastery orientations, individuals perceive the achievement setting as a challenge, and this likely will create excitement, encourage cognitive functioning, increase concentration and task absorption, and direct the person towards success and mastery of information which facilitates intrinsic motivation.

The performance-avoidance goal is focused on avoiding incompetence, where individuals see the achievement setting as a threat and seek to escape it (Elliot & Harackiewicz, 1996). This orientation is likely to elicit anxiety and withdrawal of effort and cognitive resources while disrupting concentration and motivation.

A TRIPARTITE MODEL OF MOTIVATION FOR ACHIEVEMENT: ATTITUDE/DRIVE/STRATEGY

Motivating students to achieve in school is a topic of great practical concern to teachers and parents, and of great theoretical concern to researchers. New books on the topic appear with increasing frequency and relevant research is proliferating at a rapid rate. Higher education institutions are beginning to provide assistance to students, especially new ones, in developing so-called study skills and self-regulatory skills such as time management. One of the greatest *challenges and opportunities of the 21st century* will be for schools at all levels to focus more on assisting students to become motivated in order that they can succeed in school.

The purpose of this paper is to present a proposed model of motivation for achievement, as applied particularly to the educational setting. This model focuses on three generic variables: (1) *attitude* or beliefs that people hold about themselves, their capabilities, and the factors that account for their outcomes; (2) *drive* or the desire to attain an outcome based on the value people place on it; (3) *strategy* or the techniques that people employ to gain the outcomes they desire. Each of these variables will be described in more detail, and evidence will be provided to support the contention that each exerts an important influence on motivation to achieve in an academic environment.

Achievement outcomes have been regarded as a function of two characteristics, “skill” and “will” (McCombs and Marzano, 1990), and these must be considered separately because possessing the will alone may not insure success if the skill is lacking. The focus in this model is on will, or the motivation to achieve the outcome, and it will be considered separately from level of skill. Where achievement measures such as scores on course examinations or grades are used as criteria of motivation to achieve, measures of skill have to be separated out or controlled for. To measure motivation for achievement directly, measures of *engagement* must be examined.

Cognitive engagement represents the amount of effort spent in either studying or completing assignments. It is the result of motivation, not its source. Pintrich and Schrauben (1992) review a large body of research that suggests that (1) the value of an outcome to the student affects that student’s motivation, and (2) motivation leads to cognitive engagement, such engagement manifesting itself in the use or application of various learning strategies. Many of the studies Pintrich and Schrauben describe employed self-reports of learning strategy use as a measure of cognitive engagement. Such studies become dependent on what students claimed to be doing as a way of determining that they were indeed engaged in a task. To avoid this dependence on students' self-reports, the studies I carried out (either alone or in collaboration with Tom Sexton) and report on in this paper operationalized cognitive engagement as a manifestation of effort expenditure or actual performance on the homework task of writing test items on text chapters in which students had the option to perform assignments for extra credit. In studies I carried out and report on using achievement test scores as a measure of motivation to achieve, relevant data on students' skill level was used as a control.

MOTIVATION IN CLASSROOM CONTEXT

POWER IN THE CLASSROOM

Prior to presenting some of these motivational strategies, it would be of relevance to say a few things about the teacher/learner relationship. Whichever way we look at it, this relationship is riddled with power and status. For many, power plays a large part in the relationship. The rights and duties of teachers and learners are related to power.

For example, many teachers might assert that they have the right to punish those learners who misbehave. In any social encounter involving two or more people, there are certain power relationships 'which are almost always asymmetrical' (Wright, 1987: 17). Social psychologists distinguish between three different types of power—*coercive*, *reward-based*, and *referent* (ibid.). The basis of coercive power is punishment. Some individuals or institutions have the authority to punish others. The basis of the second type of power is reward. Some individuals or institutions have the power to reward what they deem appropriate behaviour. For example, business organisations reward employees with a salary, a bonus, etc. The basis of the third type of power is motivation. In this case, individuals or institutions appeal to the commitment and interest of others. In view of this three-fold paradigm, it is of importance to concern ourselves with the fostering of learner motivation, as it is considered to be the most effective and proactive, so to speak, power relationship.

GROUP PROCESSES AND MOTIVATION

A discussion of motivation and motivational strategies would not be complete without a consideration of group processes, inasmuch as there is usually a group of people that we as teachers are called on to motivate. Tuckman (1969, quoted in Argyle, 1969) established that a group went through four stages from its formation, which has important implications for the study of the classroom and the use of group activities during teaching.

- Stage 1 *Forming*: At first, there is some anxiety among the members of the group, as they are dependent on the leader (that is, the teacher) and they have to find out what behaviour is acceptable.
- Stage 2 *Storming*: There is conflict between sub-groups and rebellion against the leader. Members of the group resist their leader and the role relations attending the function of the group are questioned.
- Stage 3 *Norming*: The group begins to develop a sort of cohesion. Members of the group begin to support each other. At this stage, there is co-operation and open exchange of views and feelings about their roles and each other.
- Stage 4 *Performing*: Most problems are resolved and there is a great deal of interpersonal activity. Everyone is devoted to completing the tasks they have been assigned.

Experience shows that almost every group goes through these four (or even more) stages until it reaches equilibrium and, thus, taps into its potential. In reality, this process may go on forever, since student lethargy and underachievement norms in the classroom are considered to be basic hindrances to effective teaching and learning (Daniels, 1994). Against this background, we will try to design a framework for motivational strategies.

COMPETITION AND CO-OPERATION

When it comes to interaction and group dynamics, there are two particular factors that can hinder or facilitate group achievement: competition of Co-operation (Deutsch, 2003). Within a group or team setting, the two factors of Co-operation and competition can either propel the team to success or consign the group's effort to failure depending on which factor is dominant. Co-operation is "work[ing] or act[ing] together or jointly for a common purpose or benefit (cooperate). Competition, on the other hand, takes place when people "strive to gain or win something by defeating or establishing superiority over others who are trying to do the same thing" (Oxford American Dictionaries).

Deutsch suggests six "cooperative relations" in which he describes six positive characteristics that make Co-operation within a group more likely. These six factors include: 1. Effective communication. 2. Friendliness, helpfulness, and less obstructiveness. 3. Coordination of effort, divisions of labour, orientation of task achievements, orderliness in discussion, and high productivity. 4. Feelings of agreement with the ideas of others and a sense of basic similarity in beliefs in one's own ideas and in the value that other members attach to those ideas. 5. The willingness to enhance the other's power (as others capabilities are strengthened, you are strengthened). 6. Defining conflicting interests as a mutual problem to be solved by collaborative effort (Deustch).

In his six points, Deustch articulates the ideal features one would expect to find in an environment of a cooperative organization. And just as these factors contribute towards the cooperativeness of a group, one would find the opposite characteristics prevalent in a group where internal or in-group competition prevails. Poor communication, obstructiveness and lack of helpfulness (Deustch, 2003), and group disorder, disagreement and the critical rejection of ideas (Deustch), and striving to "enhance one's own power" (Deutsch, 2003) sum up the "conflicting relations."

An example of Co-operation versus competition can be found in a study called The Prisoner's Dilemma (Dixit and Nalebuff).

In this tradeoff, two criminals (we'll call them 1 and 2) are arrested on the "suspicion of having committed armed robbery and sure enough they are found to be carrying concealed weapons" (Dixit and Nalebuff). The police are able to talk to the suspects separately and both are given an opportunity to confess or "rat out" the other person. In this situation, there are four plausible and equally likely outcomes. In the first, both parties could cooperate with each other and refuse to confess or turn his or her partner in. The outcome of this condition is most beneficial for both individuals in that they are most like to receive a light sentence as a result of being armed, but not a long sentence for the actual crime. In the second and third situations, one of the individuals "rats out" the other and therefore he receives a light sentence while his partner receives a longer sentence for the crime. In this state of affairs, there is a sort of competition where there are conflicting interests amongst the two individuals. In the final circumstance both individuals rat each other out, are convicted and found guilty, and both serve equally long sentences in prison.

LEADERSHIP TRAITS

One of the most important contributions psychology has made to the field of business has been in determining the key traits of acknowledged leaders. Psychological tests have been used to determine what characteristics are most commonly noted among successful leaders. This list of characteristics can be used for developmental purposes to help managers gain insight and develop their leadership skills. The increasing rate of change in the business environment is a major factor in this emphasis on leadership. Whereas in the past, managers were expected to maintain the status quo in order to move ahead, new forces in the marketplace have made it necessary to expand this narrow focus. The new leaders of tomorrow are visionary. They are both learners and teachers. Not only do they foresee paradigm changes in society, but they also have a strong sense of ethics and work to build integrity in their organizations. Raymond Cattell, a pioneer in the field of personality assessment, developed the Leadership Potential equation in 1954. This equation, which was based on a study of military leaders, is used today to determine the traits which characterize an effective leader. The traits of an effective leader include the following:

- Emotional stability. Good leaders must be able to tolerate frustration and stress. Overall, they must be well-adjusted and have the psychological maturity to deal with anything they are required to face.

- Dominance. Leaders are often times competitive and decisive and usually enjoy overcoming obstacles. Overall, they are assertive in their thinking style as well as their attitude in dealing with others.
- Enthusiasm. Leaders are usually seen as active, expressive, and energetic. They are often very optimistic and open to change. Overall, they are generally quick and alert and tend to be uninhibited.
- Conscientiousness. Leaders are often dominated by a sense of duty and tend to be very exacting in character. They usually have a very high standard of excellence and an inward desire to do one's best. They also have a need for order and tend to be very self-disciplined.
- Social boldness. Leaders tend to be spontaneous risk-takers. They are usually socially aggressive and generally thick-skinned. Overall, they are responsive to others and tend to be high in emotional stamina.
- Tough-mindedness. Good leaders are practical, logical, and to-the-point. They tend to be low in sentimental attachments and comfortable with criticism. They are usually insensitive to hardship and overall, are very poised.
- Self-assurance. Self-confidence and resiliency are common traits among leaders. They tend to be free of guilt and have little or no need for approval. They are generally secure and free from guilt and are usually unaffected by prior mistakes or failures.
- Compulsiveness. Leaders were found to be controlled and very precise in their social interactions. Overall, they were very protective of their integrity and reputation and consequently tended to be socially aware and careful, abundant in foresight, and very careful when making decisions or determining specific actions. Beyond these basic traits, leaders of today must also possess traits which will help them motivate others and lead them in new directions. Leaders of the future must be able to envision the future and convince others that their vision is worth following. To do this, they must have the following personality traits:
- High energy. Long hours and some travel are usually a prerequisite for leadership positions, especially as your company grows. Remaining alert and staying focused are two of the greatest obstacles you will have to face as a leader.
- Intuitiveness. Rapid changes in the world today combined with information overload result in an inability to "know" everything. In other words, reasoning and logic will not get you through all situations. In fact, more and more leaders are learning to the value of using their intuition and trusting their own instincts when making decisions.
- Maturity. To be a good leader, personal power and recognition must be secondary to the development of your employees. In other words, maturity is based on recognizing that more can be accomplished by empowering others than can be by ruling others.

- Team orientation. Business leaders today put a strong emphasis on team work. Instead of promoting an adult/child relationship with their employees, leaders create an adult/adult relationship which fosters team cohesiveness.
- Empathy. Being able to put yourself in the other person's shoes is a key trait of leaders. Without empathy, you can't build trust. And without trust, you will never be able to get the best effort from your employees.
- Charisma. People usually perceive leaders as larger than life. Charisma plays a large part in this perception. Leaders who have charisma are able to arouse strong emotions in their employees by defining a vision which unites and captivates them. Using this vision, leaders motivate employees to reach towards a future goal by tying the goal to substantial personal rewards and values. Personal traits play a major role in determining who will and who will not be comfortable leading others.

However, it is important to remember that people are forever learning and changing. Leaders are rarely (if ever) born. Circumstances and persistence are major components in the developmental process of any leader. If your goal is to become a leader, work on developing those areas from the list above that you are weak in. For instance, if you have all of the basic traits, but do not consider yourself very much of a people person, take classes or read books on empathy. Get feedback from others on how they see you and what they think you can do to develop those traits. There are many leadership training programmes around. Contact your local business school to find one near you.

LEADERSHIP STYLES

The study of leadership styles take into consideration what a leader does, says, and how he acts. It has to do with the study of the leader's approach to the use of authority and the resultant participation of others in decision-making. A closer examination of some selected leadership styles reveal the magnitude of the leader's responsibility those with whom he works. Lall and Lall (1979) listed these five styles of leadership with their definitions as follows:

AUTOCRATIC LEADERSHIP

The autocratic leader gets vested authority through the office more than from personal attributes. He seeks little group participation in decision-making.

Advantage: The leader generally gets things done.

Disadvantage: The follower becomes dependent on the leader and his personal development is jeopardized.

BUREAUCRATIC LEADERSHIP

This style leadership is based on the utilization of a system of files to solve problems. It can be styled as leadership by centralization.

Advantage: There is a ready system on hand to embark upon the solution of problems.

Disadvantage: It is too well organized and tends to depersonalize the organization.

CHARISMATIC LEADERSHIP

Here the leader focuses attention on himself. He appears to possess a certain charisma and followers are converted to and are champions of the cause.

Advantage: The leader gets a lot of individuals to accept his views and defend his cause.

Disadvantage: These leaders tend to lean towards authoritarian or bureaucratic styles of leadership.

LAISSEZ-FAIRE LEADERSHIP

Complete permissiveness is allowed in this style of leadership. The group lacks direction because the leader does not help in making-decision.

Advantage: Every follower has the opportunity to make decisions.

Disadvantage: This style can easily lead to anarchy if allowed to function for a long period of time.

DEMOCRATIC LEADERSHIP

In this style of leadership most policies derive from group decision. The leader is involved in policy formation but does not dominate group action.

Advantage: Individual growth is enhancing through participation in the organization's operations.

Disadvantage: The possibility of the sidelining of leadership initiative as a result of majority group decision.

One of the first and most famous studies of leadership style was conducted by Lewin, Lippit, and White (1939) where they placed selected individuals into various groups with different leadership styles. The styles chosen were democratic, in which group decisions were made by majority vote, equal participation was encouraged and criticism and punishment were minimal; autocratic, in which all decisions were made by the leader and participants were required to follow prescribed procedures under strict discipline; and laissez-faire, in which the actual leadership activity of the group leader was kept at a minimum, allowing the participants to work and play essentially without supervision.

The groups with democratic style of leadership were the most satisfied and functional in the most orderly and positive manner. The number and degree of aggressive acts were greatest in the autocratically led groups. From the review of leadership theories, it is obvious that there is no one best leadership style. Hein and Nicholson (1968) argue that leaders are rarely totally people—or task-oriented; leader, followers, situation all influence leadership effectiveness and therefore an integrating of leadership theories seems appropriate. This type of approach reflects an escapist mentality that will settle for anything as long as he gets out of the present dilemma.

Is there a more excellent style of leadership? There must be another style of leadership that can provide for the upward mobility of followers void of the superordinate-subordinate syndrome; facilitatory leadership with vision. In my own search for a leadership style that makes a difference, I have discovered another model, which has broadened my vision of ministry: servanthood.

CLASSROOM CLIMATE

In all classroom settings with which the writer is familiar, the teacher is the designated leader and the students are the followers. A set of rules and regulations, as well as designated punishments for breaking those rules, is placed in a highly visible position in front of the students. It has generally been assumed that formal structures permit the leaders to function effectively, help members perform their tasks, and see that the purpose of the group has been achieved (Bany & Johnson, 1964). In classes where rule infractions are overly stressed, and the pupils criticized, the writer has observed a large amount of reporting and blaming others, on the part of the students. The children also seem unable to work together collaboratively.

In classes where the need for rules is discussed with the children, and the children are encouraged to understand why rules are needed and how they can help each other follow the rules, there is a minimum of scapegoating and tale-telling.

In order to be able to follow rules, children have to develop a sense of responsibility. This is a maturational task that includes emotional, social, and psychological growth.

Factors of Learning: Personal and Environmental

Learning, as we know, can be considered as the process by which skills, attitudes, knowledge and concepts are acquired, understood, applied and extended. All human beings, whether grown ups or children engage in the process of learning, either consciously, subconsciously or subliminally. It is through learning that their competence and ability to function in their environment get enhanced. It is important to understand that while we learn some ideas and concepts through instruction or teaching, we also learn through our feelings and experiences. Feelings and experiences are a tangible part of our lives and these greatly influence what we learn, how we learn and why we learn.

Learning has been considered partly a cognitive process and partly a social and affective one. It qualifies as a cognitive process because it involves the functions of attention, perception, reasoning, analysis, drawing of conclusions, making interpretations and giving meaning to the observed phenomena. All of these are mental processes which relate to the intellectual functions of the individual. Learning is a social and affective process, as the societal and cultural context in which we function and the feelings and experiences which we have, greatly influence our ideas, concepts, images and understanding of the world.

These constitute inner subjective interpretations and represent our own unique, personalized constructions of the specific universe of functioning. Our knowledge, ideas, concepts, attitudes, beliefs and the skills which we acquire are a consequence of these combined processes. You may be wondering why we are discussing the concept of learning, since our concern in this unit is with factors affecting learning.

We are doing this to enable you to appreciate the fact that the process of learning involves cognition, feeling, experience and a context. Individuals vary greatly with regard to their ability, capacity and interest in learning. You must have noticed such variations among your friends and students. In any family, children of the same parents differ with respect to what they can learn and how well they can learn. For example, a particular child may be very good at acquiring practical skills such as repairing electrical gadgets, shopping for the household, etc., while his brother or sister may in contrast be very poor on these, and good at academic tasks, instead.

Even for yourself, you may be perplexed why you can do some tasks well, but not others given the same competence level. For example, learning the tunes of songs and even their lyrics is often found to be easier than learning a formula or a poem. Do4 you ever wonder why this is so? You may have observed that for some people,

learning driving or swimming or cooking is achieved easily, while for some others it is 1 a nightmare. Why this happens, what could be the underlying reasons and why individuals differ with respect to how and what they learn, are the key questions addressed in the present unit. To find some answers to these questions, we will try to identify and understand the various factors affecting learning.

There are some personal factors affecting learning that deal with the innate aspects of an individual and are unique to hirn/her. These are extremely significant as they influence what the individual can learn, how much time, effort and energy helshe is required to put in and how well helshe is likely to learn. The environmental factors are other factors which mediate the learning process. Research has shown that the following are the key factors that affect learning:

- Intelligence
- Aptitude
- Goals
- Interests
- Readiness and Maturation
- Motivation
- Self Concept
- Attitudes and Values
- Level of Aspiration
- Learning style
- Socio cultural determinants.

Each of these factors play a significant role in learning. We will now examine them one by one a little later, in terms of what they mean, how they vary across individuals and how they influence the process of learning. CLASSIFICATION OF FACTORS To understand how we categories the factors affecting learning, let us begin by considering the following examples:

1. Ravi is sixteen years old and wants to please his mother by getting good results in his board examinations. He is so eager to please her, that he spends long hours of concentrated time and energy on his studies. He consciously tries to control other sources of distraction in his life and reduces the time spent on watching television, playing games and chatting with his friends.
2. Rita Williams wants to be a famous tennis player. To achieve her goal, she practices tennis whenever she can, even though she gets no encouragement from her family. She makes it a point to watch tennis matches and maintain a good rapport with her sports teacher.
3. Yuvraj is a good student, but lately he has been scoring very low marks at school. He is not able to concentrate or pay attention and his class work and home assignments reflect a very poor quality. Sources revealed that his parents fight a lot with each other and are about to get divorced.
4. Arti and Kavita are two sisters. Arti is very good at art and craft and can sketch just about anything she sees. Kavita has a ear for music. She knows most songs and can sing them even if she has heard them only once. Both of them spend hours together pursuing their respective interest areas.
5. Sayeeda is tall, attractive and has a very good figure. She wants to be a model or an air-hostess and nurtures this secretly as her dream. She is too scared to share her wishes with her family, since she belongs to an orthodox family, where girls at best can pursue teaching as a career. When she tries telling her mother what she wants, she is firmly told that she can only do her B.Ed and can go to the coaching classes for these.

The above cited examples illustrate that learning is a universal phenomenon mediated by a number of factors, both personal and environmental in nature. The dictum "everybody learns" is as true as its corollary, *i.e.*, everybody learns in accordance with his/her unique, individualized blend of personal and environmental factors. For example, in case of Ravi, the desire to please his mother, striving to do well in his board exams and managing his Life situations appropriately constitute the key factors which influence him. For Rita Williams, it is her intrinsic desire to be a good tennis player which is paramount. She is not deterred by the lack of family support and continues to make efforts to promote her love for tennis on her own and fulfil her desire to be successful.

In case of Yuvraj, in spite of his innate capacity to study and perform well, his lack of achievement can be attributed to the emotional insecurity stemming from his parents' divorce. As far as Arti and Kavita are concerned, their special interests and talent in art and music respectively, seem to guide their activities. For Sayeeda, the home environment and family culture and values determine her professional choice. Her own inner interests, desires and wishes are not to be taken into cognizance.

In all the examples cited, you can find evidence of both personal and environmental factors influencing the process of learning. Learning can thus be defined as a function of the interaction of personal and environmental factors. L = f (EF x PF) L = learning; f = function; EF = environmental factors; PI? = personal factors. Personal factors are the intra individual factors like motivation, interests, abilities, etc. which predispose an individual towards learning as in the case of Rita Williams, Arti and Kavita. Environmental factors on the other hand, are those contextual factors which highlight the role of the environment in learning, such as the socio-emotional, societal and cultural factors as seen in the case of Yuvraj and Sayeeda. Although the two factors represent different categories, they operate in a common system. The environmental factors provide the context within which the personal factors, operate. The learner and the learning process can only be completely understood with reference to the interaction of both environmental and personal factors.

Intelligence

One of the key factors which affects our ability to learn effectively is intelligence. There are wide variations across individuals and cultures as to what actually constitutes intelligence. Let us take a deeper look at what this statement means by engaging in the following analytical task.

Imagine a classroom in which there are ten students, each with a distinctive characteristic highlighting his/her ability. You have been asked to state which one among them you consider to be the most intelligent and why. How will you proceed in your judgement. The profiles of the ten students given to you are as follows:

- Manisha always comes first in class.
- Zubin is an ace chess player.
- Neha is the most popular girl in her class.
- Ashish can remember statistics about cricket from any time period. * Mumtaz never loses her cool and remains calm and even tempered even in difficult situations.
- Mark gets always called to arbitrate in peer disputes in his class.
- Zarina can organize any event, even at a short notice.
- Niloufer choreographs dances very well.
- Shrikant is a wizard in mental mathematics.
- Ayesha always gives novel and unique answers to routine questions.

In all probability, you will choose to identify some criteria which define intelligence and then, go case by case to consider who is the most intelligent. You will realize herein however lies your major difficulty of how to select the criteria? How to priorities their importance? How to assign ranking? In all likelihood, you will conclude that each student is intelligent in some way or the other, because some ability, competence or skill has been demonstrated by each one of them. Furthermore, this ability has optimised their performance or their existence in many ways. Thus two things may clearly strike you. First that there is no universal definition of the term intelligence. You will begin to realize that many possible definitions coexist. Secondly, it will become apparent to you, that irrespective of what intelligence is, it certainly augments learning, achievement and performance. You are thus likely to conclude that learning is greatly facilitated by intelligence.

To understand this you may like to take a brief look at some of the prominent views which attempt to conceptualize intelligence. The earliest among these was the psychometric tradition beginning with the work of Alfred Binet. This tradition is based on statistical analyses of conventional tests of intelligence, requiring students to show basic vocabulary, mathematical ability, reasoning and other skills. In a broad sense, intelligence is understood as a combination of several specific, component abilities. Binet for instance, suggested three main elements of intelligence: (1) direction, which involves knowing what has to be done and how to do it; *2)* adaptation or figuring out how to perform a task and (3) criticism, or the ability to critique one's own thoughts and actions.

British psychologist Charles Spearman suggested that intelligence can be understood in terms of two dimensions or factors. A single general factor (g factor), general mental ability, which would affect all tasks and activities, and specific factors (s factors) which affect performance in a single discrete area of mental ability, such as vocabulary, mathematics, reasoning, etc.

The American psychologist Thurstone, however, argued that a single factor was an inadequate index of intelligence and suggested seven inter-related factors which he called primary mental abilities. According to him learning and performance were the outcome of these seven primary mental abilities which, he labeled as: (i) Verbal comprehension, (ii) Verbal fluency, (iii) Inductive reasoning, (iv) Spatial visualization, (v) Number, (vi) Memory, and (vii) Perceptual speed. Other approaches to intelligence and learning view them as abilities to process information meaningfully (Sternberg) to adapt to the demands of the environment (Piaget), to construct knowledge based on one's experiences (constructivism), and to be emotionally well adjusted and secure (Goleman). The conceptualization of intelligence, thus in these different frameworks is much broader and recognizes that children as well as adults can be intelligent in many different ways. The most dominant contemporary view in intelligence which has a marked influence on educational practice is the one proposed by Howard Gardener. He has advanced a theory of multiple intelligences in which there are eight distinct and relatively independent intelligences. Each is a separate system of functioning, although *i.* the various systems can interact to produce overall intelligent performance. The eight intelligences are:

- Linguistic intelligence
- Logical-mathematical intelligence

- Spatial intelligence
- Musical intelligence
- Bodily-kinesthetic intelligence
- Interpersonal intelligence
- Intrapersonal intelligence
- Naturalist intelligence.

The intelligence on which one is high in terms of manifestation will thus determine what one learns, how well one learns, and also influence the various ways in which one seeks engagement. Thus, irrespective of how we define intelligence, in relating it to learning, there is no doubt that it affects what, when and how we learn. Further, in operational terns, it definitely is a capacity or an ability for problem solving, thinking, reasoning, relating to others, dealing with emotions, developing interests, sense of right and wrong and living in consonance with our environment.

Chapter Seven

Studying Human Behaviour

Some of the important methods of Studying Human Behaviour as Formulated by Psychologists are as follows:

Introspection Method

This method was introduced by EB Titchener. This is also known as self-observation method. Introspection means 'to look within'.

It is not possible to understand the inner feelings and experiences of other persons. But the individual himself can observe and report.

Example: A patient can report about his pains and other disturbances in a better way than by a nurse. He will look within himself and explain how he is feeling. This will help for a better treatment. Though Introspection is a useful method, it has some demerits also.

These are:

(a) We cannot verify the reports given by the observer hence we have to accept his report. At times even if he is reporting correctly there may be distortions

(b) This method cannot be used to study children, animals and persons suffering from mental disorders. But this is a cheap and easy method.

Observation Method

This method is very useful in the areas where experiments cannot be conducted. In this method the observer will observe and collect the data. Example: In the hospital the nurse will make observation of patient's temperature, pulse, BP, facial expressions, etc. This method is very useful to study the children, mentally ill, animals and unconscious patients. At times the observer will go to the natural settings, situations, etc. in order to get the objective data.

Because, in natural settings the person being observed will not be aware that he is being observed, his behaviour will be natural/ original. Hence, this method is also known as 'naturalistic observation' or 'objective observation' method.

This is a very good and useful method. But the disadvantages here are: (a) There are chances of subjective report, and prejudices of observer may creep in. Sometimes to observe the natural behaviour the observer may have to spend more time, energy and money.

Experimental Method

This is the most objective way of studying the behaviour. In this method, experiments are conducted in the laboratories under controlled conditions. In experiments, usually the effect of independent variable on the dependent variable is studied.

Hence, there will be two variables, *viz.*, Independent and dependent variable. There will be some other variables which are not wanted by the experimenter, and their interference may affect the results of the experiment.

Such variables are to be controlled. These unwanted variables are called 'extraneous' or 'intervening' variables. Experiments are conducted under controlled conditions in order to control the effect of these extraneous variables.

Examples: The effect of music on the level of blood pressure can be studied in laboratory settings. Here, the music is independent variable and the BP is dependent variable. All other sounds other than music are extraneous variables that are to be controlled, so that the effect of only music can be assessed. Similarly the effect of different drugs, food, etc. can also be studied.

Experiments may also be conducted by using two groups called experimental group and control group. In such experiments, independent variable is operated only on experimental group and the control group is kept constant.

Otherwise, the experiment may be conducted on the same group under two conditions, *viz.*, experimental and controlled conditions. Generally the following steps are followed in an experiment:

a. Identification of the problem
b. Formulation of hypothesis
c. Designing the experiment
d. Testing the hypothesis by experiment
e. Analysis of results
f. Interpretation of results.

The advantage of this method is that, the results of the experiment may be verified by repetition of the same experiment. But this method has some demerits also.

They are:

(a) Conducting experiment is very expensive and time consuming;

(b) another feature is that the experiments cannot be conducted outside the laboratory.

Clinical Method/Case History Method

This method is used very commonly in hospitals and also in educational settings. In hospitals, when a patient is admitted, the nurse can collect the detailed information pertaining to the disease of the patient. The information includes the past history of the disease, treatment taken already, changes if any like-improvement, present condition, probable causes, signs and symptoms, etc. This information may be obtained from the patient, his close relatives like parents, siblings or others who accompany him or from his friends, neighbours, etc.

Survey Method

This is used to gather the information from large number of people. Questionnaires, checklists, rating scales, inventories are used to collect the required information.

This method is usually used to gather information about political opinion, customers' preferences, etc. It may also be used to know the information pertaining to medical profession—like awareness about diseases and remedial programmes, malnutrition, opinions about health needs, health facilities available, etc.

Genetic Method

This method is also called as developmental method. Most of our behaviours are the result of earlier experiences. In some cases when we need to understand some behaviour we need to know their developmental aspects also. For example, in order to understand the behaviour of adults we need to know their childhood development.

This can be done by two ways:

(a) Cross-sectional study in which, the children of different age groups will be studied simultaneously,

(b) Longitudinal study in which, the same individual will be studied in different stages of life. This is more useful method to understand the behaviour from point of view of hereditary and environmental influences.

Testing Method

Different tests are developed by psychologists to study various aspects of behaviour. The attitudes, interests, abilities, intelligence, adjustments, personality and such other factors which influence behaviour, can be studied by administering the suitable tests.

HUMAN PERSONALITY IN PSYCHOLOGY

The answers which We have given up to the present still call for a survey of the basic principles from which they are derived and on the basis of which, in each specific case you will be able to form a fully justified personal judgement. We will only refer to the principles of a ethical order which concern both the personality of the person who practices psychology and that of the patient, to the extent that the latter intervenes through a free and responsible step. Certain actions are contrary to ethicals because they only violate the norms of a positive law; others are in themselves of an imethical character; among these the only ones which We will deal with-some will never be ethical: others will become imethical because of determined circumstances. Thus, for example, it is imethical to penetrate into the conscience of someone; but this act becomes ethical if the person involved gives his valid consent. It can also happen that certain actions lay a person open to the dangers of violating a ethical law: thus, for instance, the use of tests can in certain cases engender imethical impressions, but this action becomes ethical when proportionate motives justify the danger incurred.

One can therefore establish three kinds of imethical actions, which can be judged as such by referring to the three basic principles: whether they are imethical either in themselves, or because the person who enacts them lacks the right to do so, or because of the dangers they provoke without sufficient motive. Imethical actions in themselves are those where the constitutive elements are incompatible with ethical order, that is to say with healthy reasoning; where conscious and free action is contrary either to the essential principles of human nature or to the essential relations which it has with the Creator and with other men, or to the rules governing the use of material things, in the sense that man must never become their slave but must remain their master. It is therefore contrary to ethical order that man should freely and consciously submit his rational faculties to inferior instincts. When the application of the tests, or of psycho-analysis or of any other method reaches this extreme, it becomes imethical and must be refuted without discussion. It is naturally up to your conscience to determine in the individual cases, the lines of conduct to be rejected.

Actions which are imethical because the person who enacts them has no right to do so, do not in themselves contain any essential imethical element but, if they are to be licit, they must suppose the existence of an explicit or implicit right as will be the case in the majority of instances for the doctor and the psychologist. Since a right cannot be taken for granted, it must first of all be established through positive proof by the person who assumes it and based on a juridical reason. As long as the right has not been obtained, the action is imethical. But if, at a specific time, an action appears to be imethical, it does not still follow that it will always remain such, because it can happen that the right shown to be lacking is acquired later.

Nevertheless, the right in question can never be taken for granted. As We said previously, again in this instance, it is up to you to decide in concrete cases, many examples of which are quoted in the publications of your specialization, whether this principle is applicable to such or such an action. Thirdly, certain actions are imethical because of the danger incurred without a proportionate motive. We naturally refer to ethical danger for the individual or the community, either regarding the personal property of the body, of life, of reputation, of customs or with respect to material assets.

It is obviously impossible to avoid danger completely and such a demand would paralyze all enterprise and would seriously harm every one's interests; hence, ethical law permits this risk to be taken on the condition that it is justified by a motive proportionate to the importance of the assets at stake and to the proximity of the danger which threatens them. You refer several times in your works to the danger engendered by certain techniques, by certain procedures used in applied psychology. The principle which We have laid before you will help you solve in each case the difficulties that may arise.

The norms which We have formulated are above all of a ethical order. When psychology discusses a method or the effectiveness of a technique on the theoretical plane, it only considers their aptitude to achieve the specific aim psychology pursues and does not deal with the ethical aspect. In the practical application one must also take into account the spiritual values involved both in the psychologist and the patient and add to the scientific and medical point of view that of the human personality in general. These fundamental norms are obligatory because they are engendered by the nature of things and belong to the essential order of human action, the supreme and immediately evident principle of which is that one must do good and avoid evil.

At the beginning of this address, we described personality as the "psychosomatic unity of man insofar as determined and governed by the soul" and We have specified the meaning of this definition. Then, We endeavoured to answer your questions on the use of certain psychological methods and on the general principles which determine the ethical responsibility of the psychologist. One does not expect the psychologist to have only a theoretical knowledge of abstract norms, but also a deep ethical and pondered sense formed by constant loyalty to his conscience. The psychologist who really wishes to seek only the welfare of his patient will be all the more careful to respect the limitations placed upon his actions by ethicals, since one can say that he holds in his hands the psychic faculties of a man, his capacity of acting freely, of attaining the highest values of his personal destiny and of his social vocation.

It is Our wholehearted wish that your work may ever increasingly penetrate into the complexities of the human personality, that it may help it remedy its weaknesses and meet more faithfully the sublime designs which God, its Creator and Redeemer, formulates for it and proposes to it as its ideal.

As a token of this We call upon you, your collaborators and your families the most abundant heavenly favours, and heartily grant you Our apostolic benediction.

PREMATURE APPLIED PSYCHOLOGY

In recent years, knowledge about the historical developments of psychological ideas has substantially increased. This fact may partly be explained as an after-effect of the widespread Kuhnian model of science. The idea that scientists deal with phenomena through the handling of theoretical models of only temporally limited validity, brought to the fore the salience of historical views of science as an indispensable means for an in-depth understanding of theories and praxis in the scientific world. In addition, many centennial celebrations have enabled us to recover the memory of our past. Although a lot of work may be seen as ceremonial, large pieces of the past have been focused on, and all of us are benefiting from such recoveries.

When contemplating the past development of our science, theory and application have been kept apart from one another as if they were two unrelated disciplines. In our view applied psychology has not been-or has not always been-the mere application of previous theoretical knowledge to the realm of daily life. As Frisby put it in the closing lecture at the Rome congress in 1958, *'basic and applied research... may well go hand in hand, but they can quite easily be prosecuted separately. Nor is it necessarily the case that applied research must wait upon basic research, since it is sometimes the study of applied problems which permits the formulation of a fundamental problem for separate attack'* (in Gundlach, 1998b, XIII, 711).

In many cases, concrete demands have preceded the intervention plan, and its authors were forced to create new concepts and instruments in answer to the proposed questions. Although many scientists did not feel themselves completely secure as they left laboratories and entered the real world, they could not remain in the ivory tower, and stood up to the challenge. They felt the contradiction of building a science that seemed unable to solve real problems in its own field. And little by little applied knowledge was built, which in the long run has largely determined the present day situation in the field.

In what follows, we will mainly focus on the early developments of the IAAP, taking into thought its recently reissued proceedings (Gundlach, 1998b). This is part of a long-term project of studying its history, in the context of a broader view of applied psychology.

PREMATURE DEVELOPMENTS OF APPLIED PSYCHOLOGY

At the turn of the 20th century, a positivistic mentality dominated science and culture in western countries. Positivism had been conceived by the French philosopher Auguste Comte as a system of ideas (or philosophy), based on scientific knowledge and looking for general laws in the natural and social world. Its motto was 'see (or to know) in order to foresee, foresee in order to provide'. Knowledge without practical consequences was not worthy of being sought at all. As a result, 'scientism' overflowed into the rest of culture. Science, mainly conceived as quantitative knowledge of objective phenomena permitting both explanation and prediction, became the latest word to talk about everything. It was accompanied by technical and industrial expansion, which changed all aspects of human life, for both good and bad. Positivism (and its close companion, pragmatism) approached human life in search of facts and laws. Its ideals permeated all natural and social sciences: physiology, anthropology, criminology, were deeply influenced by the new outlook. It was pretty much the same in literature, where 'naturalism' invaded the field with its romances and fictions. The French novelist Emile Zola, a follower of the teachings of physiologist Claude Bernard, wrote: 'I tell what I see'; and a character from Dickens' Hard Times repeats in those pages: 'Facts, facts, give me facts'.

Psychology had been created as a new science for the study of the human mind. It had been conceived as a science of conscious experience, not centred on objective, empirical facts, as positivists were demanding, nor had it generated a body of technical procedures to solve practical questions, as sciences were supposed to bring forth. So its place as a science was frequently placed in doubt, and many people felt that a fundamental change was needed in the field. It is worth mentioning here Titchener's opposition to applied psychology as he considered theory not yet prepared for such a task. Nevertheless, the transition from science towards techniques soon became irresistible. It is interesting to point out Münsterberg's attitude, who in his psychology for industry, (Psychology and industrial efficiency) claimed imperiously the transition from theory to practice:

'The time has come for mental sciences, in which practice and theory must complement each other' (Münsterberg, 1914, 5).

On the other hand, social problems were there, calling at the door of the laboratories. Western societies were experiencing the natural troubles of an expanding epoch determined by the industrial revolution and tremendous social and political changes. Psychologists had to face a growing and demanding world. What follows here is some evidence related to it. In many fields new demands had appeared with force. Education, at the end of the 19th century, was in need of an in-depth reform. Democracy extended school education to all social classes. The 'New school', a wide movement spreading all over Europe, promoted active programmes in school, and stressed the relevance of motivation for engaging children in individual and collective work (Luzuriaga, 1923). Many voices were raised in society demanding school to be closer to life.

A French educator wrote: 'It is a common topic to repeat that our teaching, on the whole or very nearly, needs to be changed... Workers accuse it of being too much abstract, not taking the real life into thought... In 1866, the International association of workers gathering in Geneva, socialists asked for... a living knowledge that would substitute a dead one, and for a progressive transformation that would change the play instinct into an instinct of work...' (Fontègne, 1923,1)

Everywhere it was felt that populace had to have an opportunity to contribute something from their own experience, and to give their opinion about issues under debate. Individual differences were becoming increasingly important in all fields: in industries, courts of justice, car driving, workers' training and rehabilitation. Each time human behaviour of groups and individuals was examined, demand for a scientific understanding of it arose. First, teachers, physicians and physiologists were consulted; then little by little, psychologists became involved.

At the beginning, individual interventions were made, which did not follow a common model. For instance, Binet wanted to help teachers acquire real and predictive knowledge about their students' ability to learn; Decroly and Ferrari became interested in taking care of mentally retarded subjects; Lightner Witmer took care of a child suffering from language difficulties under the demand of one of his students. Such interventions were usually the fruit of some casual contacts with people in need of help. Creative scientists accepted human problems emerging from daily life, not from laboratories. The social value of their science was at stake, and conscientious people tried to do their best.

EARLY GROWTH OF PSYCHOTECHNOLOGY

Let us remember here some of the milestones of the early growth of psychotechnology in the first decades of the 20th century which began in Europe. At the end of the 19th century, interest in individual differences gave rise to the construction of instruments for testing people with various purposes. The pioneer work of Cattell in the US. was soon followed by further investigation in Europe by Ebbinghaus (completion test) and Kraepelin, in Germany, or Ferrari and Lombroso, in Italy; but the masterpiece came from the hands of Binet and Simon, in France, who were able to create an intelligence test that has generated endless research until now. Soon afterwards, some test collections were prepared by Whipple (1914), Decroly and Buyse (1928), and Burt (1923/1933).

It is interesting to note that perhaps during the time that elapsed between the two first editions of Whipple's book a change related to testing took place:

> *'One need not be a close observer to perceive how markedly the interest in mental tests has developed during the past few years. Not very long ago attention to tests was largely restricted to a few laboratory psychologists; now tests have become objects of attention for many workers whose primary interest is in education, social service, medicine, industrial management and many other fields in which applied psychology promises valuable returns' (Whipple, 1914, v).*

While some people worked on creating new testing instruments, others began to build specialized centres and institutions where due care would be provided for retarded children, injured workers, while others specialized in road traffic problems, vocational guidance, management or consumer and industrial behaviour. The need for a network of social facilities and institutions arose in countries under strongly progressive critical movements, which were demanding protective measures for workers and socially disadvantaged groups. World War I (1914-1918) 'put the applications of psychology on the map and on the front page', wrote James McKeen Cattell in 1937. It is true that the war brought many opportunities for psychologists to show the potential possibilities of their science, from personnel selection to clinical rehabilitation. The assessment that was carried out in the American army was a real success, with more than one and a half million people tested and placed in a more suitable place. In some countries-France, Germany, Belgium, the United States-new centres and institutions began to spread out at the very beginning of the 20th century. Above all, the War forced all the countries involved to face serious problems of selection and placement of military people, especially with regard to higher officers and specialized jobs.

Selection of pilots, drivers or officers became a matter demanding new psychological techniques, that were then put into question, and the utility of which was clearly demonstrated. Cattell wrote that the War put psychology on the map in the US., as well as in Europe. Laboratories such as Moede's in Charlottenburg, or Mira's Institute for Professional Guidance in Barcelona, (Spain), or the British National Institute of Industrial Psychology, headed by Myers, that had been preceded by the Health of the Munition Workers Committee during the war (Wilpert, 1990), began to spread the 'good news' of psychological intervention all over the continent. The need for joint action was soon felt. Claparède, Lahy, Myers, Moede, Christiaens, Mira and many others began to join together to discuss different topics in applied psychology. And we are still here, following that impulse, putting it into accordance with modern times.

THE MEANING OF EDOUARD CLAPARÈDE

It is possible to put back all these names with one, that of the Swiss psychologist Edouard Claparède, soul and inspiration of the first international society of applied psychology. He was to become the founder of the Institute Jean Jacques Rousseau, in Geneva, a well known centre for educational research that also turned into the early working home of Jean Piaget and other significant researchers. Trained as a physician, interested in biology, Claparède was a relative of Theodore Flournoy, the Swiss experimentalist, friend of William James and professor at Geneva University. He soon became interested in psychology as a means for a better education that would 'broaden' the lives of the students (Claparède, 1931/1946, 8). One of his most influential books, Psychologie de l'enfant et pédagogie expérimentale, clearly shows the two concepts he wanted to combine in a common project: better education for people, and better people for society to live in peace and progress.

If his very complex, very rich thought could be reduced to a single word, it would undoubtedly be 'functionalism'. He considered 'psychical phenomena primarily from the point of view of their function in life' (Claparède, 1930, 79). Mental phenomena always had to be considered in connection with human behaviour, and behaviour was understood under the law of need-reduction activity, each need putting the subject into activity to keep himself or herself alive.

Claparède promoted the 'new school' through his very influential centre, the J.J. Rousseau Institute, in Geneva. This was a school largely based on an up-to-date psychology. The task of education should be carried out keeping in mind that the child, like the adult, has his/her own personality and his/her own structure, and must be conceived as an autonomous being governed by the laws of need and interest. The child should not be seen as a mere project of an adult, but as a person, as a being complete in his/her own right, with his/her own world, needs and desires. The motto of the school was 'Discat a puero magister', that is, teachers must adapt themselves to children's nature.

Brought up in Geneva, he acquired an internationalist view of the world. He promoted congresses in search of creating a strong community of colleagues who would work together in peace. His ideal of such events was a 'Congress without reports', a meeting place to get acquainted with masters and colleagues who discussed problems and created personal bonds between them. As he confessed, 'are not bonds of friendship between men of all nations the indispensable affective substructure of the League of Nations work?'

Claparède was well aware of the negative consequences that an uncontrollable growth of tests and other instruments, and chaotic rules and recommendations made by psychologists could have on their science during its initial growth. By promoting the applied psychology conferences, he tried to reach some basic agreements on professional matters, and strengthen bonds between groups. In a broken post-war Europe, with German colleagues off the scene and widespread tensions, it is clear that a Swiss psychologist was in the best position to liaise between people of different psychological trends. In his undertaking he was supported by some French colleagues. Above all, Jules Fontègne should be mentioned, a technician specialized in educational guidance, who attended the Institute in Geneva and later founded with Piéron the National Office for Professional Guidance in Paris. The story begins with the First Geneva Conference, held in 1920.

THE FIRST CONFERENCE. GENEVA 1920

The first International Conference on Psychotechnology (French 'psychotechnique', as it was called) is a provisional starting point that may help to approach the field of applied psychology in its initial stages. This First Congress for Psychotechnology was held in Geneva, September 27-28th, 1920, sponsored by both the Institute for Educational sciences 'Jean Jacques Rousseau' and by the Laboratory for Experimental Psychology of Geneva University.

It was Edouard Claparède, then professor at the Geneva University, and Pierre Bovet, psychologist and educator, head of the Rousseau Institute in Geneva, who seem to have first had the idea of holding an international meeting for the applications of psychology, in 1920. The opportunity came as a newly created Swiss Association for Vocational Guidance and Apprentice Welfare decided to run a course in its speciality in Geneva, 24-25 September 1920 (Gundlach, 1998b, I, 18).

About 60 people attended the meeting. They came from Belgium, France, Italy, the Netherlands, Greece, Switzerland, Spain, Bulgaria; three countries were absent: the

United Kingdom, Germany and the United States. Among the participants, we find the names of Arthur G. Christiaens, Ovide Decroly and V. Brabant, from Belgium; Jean M. Lahy and Jules Fontègne (France); Gerard van Wayemburg (Netherlands), E. Claparède and Pierre Bovet (Switzerland); Emilio Mira and José Ruiz Castellá (Spain); and Giulio C. Ferrari (Italy).

This congress became a landmark in the history of psychotechnology. Here the creation of an International Association for Psychotechnology took place. This was strongly supported by the French 'Ligue d'Hygiène Mentale de Paris', an association founded by Dr. Toulouse, and of which Dr. J.M. Lahy was an important member. The President was Henri Piéron, and JM. Lahy the secretary general, and the society was established in Paris (from where it moved to Bern at the beginning of World War II). They also decided to meet the following year in Barcelona, where an exemplary centre for applied psychology was being run by Emilio Mira. The meeting served to discuss some aspects related to ability assessment and training, as well as to the usefulness of instruments for studying and recording individual performance.

Claparède was convinced that a very important point for vocational guidance was 'that of the possibility of modifying original aptitudes by practice'; he wondered: 'Can a certain individual, whose initial output is superior, be surpassed, after a time of practice, by others, whose initial output was inferior?' (Claparède, 1930, 96). Are innate abilities always superior to the learned ones, or the other way round under convenient training?

Some measures-such as the percentile ranks method-were proposed by Claparède. This appears as a symbol of all the effort of standardization and unification that seem to have been inspired by that meeting.

THE FIRST THIRTEEN CONGRESSES OF THE IAAP

The history of applied psychology has been scarcely studied for more than half a century, but in the last decade there has been a new-found interest in it. One of its results has been the reprinted edition of the proceedings of the first 13 Conference/Congresses of the International Association of Applied Psychology (Gundlach, 1998b). This fact has provided us with valuable data that gives us first hand information on the origins and growth of institutionalized applied psychology, as well as about the roots of the IAAP organization. Between 1920 and 1958 thirteen congresses or conferences were held, at very irregular intervals, mainly because of political obstacles, some overlapping dates between IAAP and IUPsyS programmes, and above all the long term effects of the Second World War that prevented any meeting between 1934 and 1949. All the meetings took place in Europe until the Kyoto (Japan) Conference in 1990. The first Conference held in America was in San Francisco (USA) in 1998.

This review will put aside the first Geneva conference (1920), as no proceedings remain from it and information about it has been presented above.

Communications and papers: Data related to the global volume of presentations made at the conferences. A total of 1134 papers (presentations, lectures, invited lectures, etc.) have been found (20 contributions, that had been counted twice in the Moscow Congress, have been eliminated). When split into two halves, before and after the Second World War, the growth of production for the second period can be clearly seen: it doubled after the war: 470 works in the first half, with an average of 67.1 papers per meeting, compared to 664 in the second, with an average of 132.8. *Collaboration among authors.* Collaborative work in a field, identified through the presence of joint authors working on the papers, appears everywhere as a sure indicator of an experimental and objective approach to the topics in a scientific area. People doing experimental work are usually organized in research teams and groups, while single authors are prone to offer more personal and, so to speak, philosophical reflections.

If we look at all the conferences papers together, the average index of authorship is very low: 1.1 authors/paper (1263 signatures, against 1134 papers). Most of the papers are single author works (1026 papers, 90.5 %). This is a rough measure of the low degree of institutionalization reached by early psychotechnics during the studied period.

Piéron, then Director of the Institute, and soon after member of the Collège de France, where a chair for the physiology of sensations was to be created, strongly collaborated with M. Piéron, and with other colleagues who all appear in the group. Among them, Alfred Fessard, deputy director for the Practical School of Higher Studies, worked on nervous physiology and helped Piéron as coeditor of L'Année Psychologique; the physiologist Henri Laugier, professor at the 'Conservatoire des arts et métiers', and head of the laboratory for applied physiology at the Hospital Henri Rousselle, and his assistant S. Nouel; and J. Monnin, a collaborator of Piéron and Fessard. It is clearly a group centred on the physiological approach to aptitudes and psychotechnics research.

Jean M. Lahy was the head of the other sub-group, based at the Ecole Pratique des Hautes Etudes. Whithin the same group, work on concrete professions was carried out. Representing the link between both sub-groups, we find Dagmar Weinberg, assistant to Henri Laugier at the Sorbonne. Other members of this subgroup are S. Korngold, assistant at the Ecole Practique d'Autes Etudes, G. Heuyer, a Parisian hospital doctor, and S. Serin, an assistant doctor at psychiatric hospitals.

Another group is worthy of mention. This brings 12 authors together, but seems to be an artificial grouping, mainly due to the creation of an *ad hoc* committee on the subject 'What is Psychotechnics?', presented in Gothenburg in 1951 by P. H. Cook from Australia, J. Elmgren from Sweden, C.B. Frisby from England, F.A. Geldard, S. Pacaud from France, M. Ponzo from Italy and A. Rey from Switzerland; two of its members (Pacaud and Ponzo) also presented papers with other colleagues, in an unrelated way with the committee.

Most productive authors: The most productive authors in a field are usually considered as visible and influential contributors, and studying them gives us accurate insights of a certain research area.

In this case, the most productive authors come from six different countries: Lahy and Weinberg from France, Baumgarten and Heinis from Switzerland, Frisby from

England, Ponzo from Italy, Mira from Spain/Brazil and Spielrein from Russia. The list contains some of the researchers who either played a representative role at the Directive Committee of the Association or contributed to the organization of any of the congresses, so that some of their papers are opening or closing speeches, and/or periodical reports of the activities of the Association. Besides, in these papers some psychotechnical information can be found about the growth in different fields, and they offer interesting material for the study of the history of applied psychology. A significant fact is that two out of the nine most productive authors are women, which is perhaps indicative of the substantial role they took in the growth and expansion of applied psychology.

We will present here some traits or notes related to each of them. At the head of the list is Jean M. Lahy. His collaboration with Pieron has already been mentioned; apart from that, he was the General Secretary of the International Association and one of its founders, attending the congresses until his death during World War II. He undoubtedly played a relevant role in applied psychology and was a well-known name from France. Franzisca Baumgarten (1886-1970), a Polish-born psychologist, was professor of psychology at Bern University, and was referred to in the proceedings as a member of the Director Committee of the Congresses of Psychotechnics since 1927, when she was also elected director of the commission for the unification of psychotechnical vocabulary and, as we have already noted, was General Secretary of the IAAP from 1949 to 1958 after Lahy. Henri Piéron (1881-1964), who led the largest group analysed above with Lahy, was one of the leading figures of French psychology, and a well-known researcher in psychobiology; he was President of the Association from the first conference until 1950. Emilio Mira (1896-1964) was the Director of the Institute of Vocational Guidance in Barcelona until his exile after the Spanish Civil War, when he became one of the main persons to promote the institutionalization and professional growth of Brazilian psychology. He was also the organizer of two of these conferences (Barcelona 1921 and 1930). He was one of the founders of the Association and also member of its executive committee before and after his exile. Mario Ponzo (1882-1960), professor of Psychology at Rome University, a psychiatrist and psychologist, was for a time the president of the Italian Society of Psychology; he was also an executive committee member and co-organizer of the Bern Congress in 1949. Clifford B. Frisby, organizer of the London Congress in 1955 and president of the IAAP in 1958, was the Director of the National Institute of Industrial Psychology in London. Dagmar Weinberg collaborated with J.M. Jahy, with whom she kept a close, active working relationship; her contacts both with Lahy and Piéron placed her in a central position in the invisible college previously analysed. She had worked as assistant with H. Laugier, at a Centre for Biometrics in Paris (both Laugier and Weinberg seem to have suffered, as Jews, an intense persecution during the war) [Piéron, 1992]. Hugo Heinis was a professor of psychology at Geneva University, Director of the Bureau of Vocational Guidance and an expert in military psychology. Isaak Spielrein, professor of Psychology and Head of the Laboratory of Psychotechnics at the Institute for Work Protection in Moscow, was also part of the

IAAP directive committee between 1927 and 1934; he had to face many difficulties when the Soviet government banned the tests and psychotechnics in 1936, and he died during the war.

In general these productive authors emerged as active members of the Association, and as such attended the congresses regularly, presenting many papers, most of them of an empirical nature. We offer some data of their presence at these meetings. In the table, it may be noted that 3 members did not attend any meetings held after the war, Lahy, Weinberg and Spielrein; on the contrary, Frisby joined the association and became an active member from the Bern meeting in 1949 onwards. The rest of them linked the pre-and post-war periods with their presence, giving continuity to the association.

Languages employed in the contributions: Papers included in these proceedings are presented in a variety of languages. Seven languages were officially accepted. Their differential weights in the distribution indicate some interesting changes in the sociology of the meetings. French is the most frequently used, accounting for half of the papers presented. This fact fits well with the French origins of the association. The rest is mainly divided between English (26%), and German (17%).

As is well-known, German, English and French were the international languages of the Association, and in some cases the language of the host country was accepted (Spanish, Catalan, Russian and Italian); this was not, however, a general rule without exceptions; Czech, Dutch or Swedish do not appear in the proceedings, while Italian is to be found in Milan (1922), Paris (1927), Barcelona (1927) and Bern (1949). Furthermore, French and English were the only accepted languages in the congress in Rome in 1958. Besides that, it appears that the use of English increased continually over time, at the expense of French. German, for its part, faded away in the post-war congresses. These changes are consistent, as we know, with the social changes that took place over that time, but it may also be related to the growing participation of US psychologists in the congresses during the post-war period (Trombetta, 1998, 196). One example of this is that M.S. Viteles, professor at the University of Pennsylvania and well known industrial psychologist in those days, was elected president of the IAAP in Rome (1958) (Canestrelli, 1958), therefore becoming its first non-European leader.

Evolution of topics dealt with: A very important question is that of the subjects covered during the congresses. Their analysis will give us a more accurate idea of the interests and problems that have generated research in the field.

Our analysis will be based on the terms appearing in the titles of the contributions, for didactic purposes, we will split our results into two halves, that is, before and after the Second World War.

The first point to be noted is the extraordinary change that the term ‘psychotechnics,-ical’ seems to have undergone. It was at the top of the list during the pre-war period. This may suggest the relevance of the new field as felt by all its cultivators. But, after the war, the term lost its attractiveness; instead of that, ‘psychology’ (at the bottom of the list in the first half) rose to the top, assuming, so to say, the role played before by the term it had replaced. We could say then that the species has been absorbed by its genus; 'psychology' dominates the whole field.

One should not forget that discussions about the word 'psychotechnics' had accompanied the discipline since its birth; the word was not well accepted in English when translated from German (Gundlach, 1998b, I, 3); Münsterberg and his group were inclined to equate Psychotechnics with Practical psychology (Saiz and Saiz, 1998); the growing weight of the American influence upon this European-born association after the war favoured the change, and in 1955 the name for the old association changed from International Association of Psychotechnics to International Association of Applied Psychology (Gundlach, ibid.).

It is clear that a core of questions remained practically unchanged. Some methodological or technical terms kept their values at the same level: 'Method' and 'Study/-ies', indicating a common way of describing the paper-contents. Guidance, school and work psychology, continued to attract research as shown by their topics, while 'Tests' maintained its relevance, as the unrivalled instrument employed by applied psychologists. Related to 'Vocational Guidance' it is to be noted that this term was included in the titles of the first congresses (Geneva, 1920; Barcelona, 1921 and Milan, 1922; the spectrum of interests broadened as new meetings were taking place and new people added to the pioneers.

Let us turn now to the contrasting aspects found. The pre-War psychotechnics, if we examine the terms which were then in favour, appear to have been centred upon the study of 'aptitudes', especially that of 'intelligence', that psychologists wanted to examine ('examination'), in a 'scientific' way, to carry out a 'selection' process, (or perhaps to make a 'scientific selection'). There is no doubt that this model dominated this early period.

Modern applied psychology, however, turned towards 'personality', the rising theme in psychology; it also stressed the learning approach ('training'), and put more attention on early phases of growth, focusing upon the 'child'. It also took into account another rising 'star', the 'group' variable, that had appeared as relevant to many researchers in social problems. Interest in social dimensions of psychotechnics had risen dramatically at the Moscow meeting in 1931, where Spielrein clearly focused on it and it was also stressed by Piéron in Bern, 1949.

The new weight of the 'group' concept is clearly documented; in Murchinson's A Handbook of Social Psychology [1935] the term 'group' appears in only 2 of the 1175 pages, while in Lindzey's Handbook of social psychology (1954), a whole section, 6 chapters, was organized a under the heading 'Group psychology and phenomena of interaction'; it is clear that the foundation of Lewin's Research Centre for Group Dynamics (USA), in 1945, had changed the scene from top to bottom.

Moreover, terms such as 'research' and 'problem' seem to have become more attractive in recent times for people writing academic papers, a fact that perhaps all of us would acknowledge as a mode still affecting our own style and rhetorics at the present. A complementary view may be gained when the term-distribution is carried out along the main applied specialities. We will not repeat here what has just been said, but it will be noted that, with regard to specialities, Educational and Work psychology are considered the leaders, while Clinical psychology is practically non-existent.

Maybe this fact is indicating that this speciality has to be seen as a completely different kind of application, far removed from the other two. Clinical psychology, although a branch of the applied body of knowledge, in no way may be included in the field of psychotechnics. In the table, the term 'personality' clearly indicates a tendency to assess individuals in a holistic way, that necessarily automatically includes a study of personality.

Another minor question deserves attention. Among the General psychology terms, the importance of the term 'Mental' should be acknowledged. As these proceedings cover four decades (from the 20's to the 50's), the period when behaviourism was at its height in the US psychology, the high frequency of this term in these papers may be taken as an indicator of their distance from the behaviourist paradigm and also of the peculiarity of applied psychology in those days, a fact that should not be forgotten by historians and professionals.

CLOSING COMMENTS

From our research it may be seen that 'applied psychology' was born to fulfil a necessity largely felt by society, but it cannot be seen as the mere application of a previously existing theory. Instead, it must be considered as an idiosyncratic way of posing problems and questions, around the mental dimensions operating in person-situation interactions. We would now merely like to point out the main traits that in my view may serve to characterize the Psychotechnological approach. When looking at applied psychology from the point of view of its intellectual meaning, some idiosyncratic traits are to be acknowledged.

1. Its main objective has always been the study of human subjects behaving in concrete situations (business, schools, hospitals, traffic...) as conscious and purposeful agents.
2. The entire life cycle has been taken into account, and age always appears as a central factor in these studies.
3. Everywhere social and historical factors have been acknowledged as playing a large role in explaining behaviour.
4. The need for practical results, and useful interventions have always been well above theoretical orthodoxies.

In a certain sense, applied psychology should be considered more as a theoretical position of its own, with its specific characteristics, and also with its potentiality, well rooted in a personalistic and humanistic views, and open to the real world.

On its 80th anniversary, the IAAP must become conscious of the human roots of its own enterprise, furthering that inspiration that has been kept alive under so many tragic circumstances thanks to the effort of so many devoted, courageous and valiant people, that have fought to put psychology at the service of human life.

Merits and Limitations of the Methods

The introspection method has advantages as well as disadvantages. First, let us consider the advantages.

Advantages of Introspection Method

It is an easy and simple method and provides direct observation of mental processes. By other scientific methods, the mental processes cannot be directly observed since those are purely private and personal experiences. Introspection is the only method by which the person can be directly aware of his own experiences.

While using experimental method to study the mental activities, a laboratory and scientific instruments are required. But for the use of introspection method, no laboratory or test materials are required. So introspection method can be used at any time and at any place.

The subjective observation method provides an opportunity to check the results obtained through other methods. For example, the general finding is that the pleasant materials are better remembered than the unpleasant materials. Suppose, in an experiment the results suggested that the unpleasant materials were better remembered than the pleasant materials.

The reason for this unexpected finding can be found from the introspective report given by the subject. The subject might have reported that he was inattentive or mentally disturbed or feeling unwell when the pleasant materials were presented to him. Here the introspective report would be helpful in explaining the results.

In spite of some advantages, the method of introspection or subjective observation has some disadvantages.

Disadvantages of Introspection Method

i. The observer is expected to perform two mental activities simultaneously. He experiences his mental processes, and at the same time analyzes what these experiences were like. At the same time, he acts as the ‘observed’, and the ‘observer’. Obviously, his mental experiences would be distorted, and not reflect the true nature of mental activities.

ii. The subjective observational report provided lacks objectivity, as the verbal report of the subject cannot be verified by other scientists. Furthermore, the experiences of one person cannot be generalized to understand the mental activity of another person. A universal principle concerning the mental processes cannot be stated, as mental processes would differ from person to person. Thus, the method lacks scientific validity.

iii. Most persons would not like to reveal their private experiences such as the feelings of guilt and shame. The report in such cases would be distorted

iv. The method of subjective observation cannot be applied to the study of the mental activities of the animals, children, insane, and persons having language disabilities. The behaviours of these subjects are of interest to the psychologists. Thus, introspection has only a limited applicability.

v. The unconscious experiences cannot be accessed through introspection. The psychoanalytic school founded by Freud argues that most of human behaviours are influenced by the unconscious motives and urges. A person cannot observe his unconscious mental processes, which means that the method of subjective observation leaves out a large chunk of relevant mental experiences.

Some psychologists have suggested that the inherent difficulties with the method of subjective observation can be overcome by observing the mental process after it ends. This is known as "retrospection", or backward introspection.

In retrospection, one is asked to give a report about his mental process after the mental activity ends. In case of anger or fear, he will give a report about his experiences after the anger or fear responses end.

He will have to recall the experiences immediately after the anger or fear responses and will give a report. But Titchener and others did not give importance to this type of post-mortem examination. It was viewed that retrospection is not actual observation of the mental process, but simply the recall of experiences and analysis of memory. Such reports cannot be taken for granted as accurate.

In spite of the above-mentioned limitations, Introspection method is still used as a method in psychology, because it is the only method that provides direct observation of mental processes.

Observation: Meaning and Definitions

Observation is the active acquisition of information from a primary source. In living beings, observation employs the senses. In science, observation can also involve the recording of data via the use of instruments. The term may also refer to any data collected during the scientific activity. Observations can be qualitative, that is, only the absence or presence of a property is noted, or quantitative if a numerical value is attached to the observed phenomenon by counting or measuring.

Observation in Science

The scientific method requires observations of nature to formulate and test hypotheses. It consists of these steps:

1. Asking a question about a natural phenomenon

2. Making observations of the phenomenon
3. Hypothesizing an explanation for the phenomenon
4. Predicting logical, observable consequences of the hypothesis that have not yet been investigated
5. Testing the hypothesis' predictions by an experiment, observational study, field study, or simulation
6. Forming a conclusion from data gathered in the experiment, or making a revised/new hypothesis and repeating the process
7. Writing out a description of the method of observation and the results or conclusions reached
8. Review of the results by peers with experience researching the same phenomenon

Observations play a role in the second and fifth steps of the scientific method. However the need for reproducibility requires that observations by different observers can be comparable.

Human sense impressions are subjective and qualitative, making them difficult to record or compare. The use of measurement developed to allow recording and comparison of observations made at different times and places, by different people.

Measurement consists of using observation to compare the phenomenon being observed to a standard unit. The standard unit can be an artifact, process, or definition which can be duplicated or shared by all observers. In measurement the number of standard units which is equal to the observation is counted. Measurement reduces an observation to a number which can be recorded, and two observations which result in the same number are equal within the resolution of the process.

Senses are limited, and are subject to errors in perception such as optical illusions. Scientific instruments were developed to magnify human powers of observation, such as weighing scales, clocks, telescopes, microscopes, thermometers, cameras, and tape recorders, and also translate into perceptible form events that are unobservable by human senses, such as indicator dyes, voltmeters, spectrometers, infrared cameras, oscilloscopes, interferometers, geiger counters, and radio receivers.

One problem encountered throughout scientific fields is that the observation may affect the process being observed, resulting in a different outcome than if the process was unobserved. This is called the *observer effect*. For example, it is not normally possible to check the air pressure in an automobile tire without letting out some of the air, thereby changing the pressure. However, in most fields of science it is possible to reduce the effects of observation to insignificance by using better instruments.

Considered as a physical process itself, all forms of observation (human or instrumental) involve amplification and are thus thermodynamically irreversible processes, increasing entropy.

Observational Paradoxes

In some specific fields of science the results of observation differ depending on factors which are not important in everyday observation. These are usually illustrated with "paradoxes" in which an event appears different when observed from two different points of view, seeming to violate "common sense".

- Relativity: In relativistic physics which deals with velocities close to the speed of light, it is found that different observers may observe different values for the length, time rates, mass, and many other properties of an object, depending on the observer's velocity relative to the object. For example, in the twin paradox one twin goes on a trip near the speed of light and comes home younger than the twin who stayed at home. This is not a paradox: time passes at a slower rate when measured from a frame moving with respect to the object. In relativistic physics, an observation must always be qualified by specifying the state of motion of the observer, its reference frame.
- Quantum mechanics: In quantum mechanics, which deals with the behaviour of very small objects, it is not possible to observe a system without changing the system, and the "observer" must be considered part of the system being observed. In isolation, quantum objects are represented by a wave function which often exists in a superposition or mixture of different states. However, when an observation is made to determine the actual location or state of the object, it always finds the object in a single state, not a "mixture". The interaction of the observation process appears to "collapse" the wave function into a single state. So any interaction between an isolated wave function and the external world that results in this wave function collapse is called an *observation* or *measurement*, whether or not it is part of a deliberate observation process.

Types-Controlled and Uncontrolled, Steps Involved During Observation

Observation (watching what people do) would seem to be an obvious method of carrying out research in psychology. However, there are different types of observational methods and distinctions need to be made between:

1. Controlled Observations
2. Natural Observations
3. Participant Observations.

In addition to the above categories observations can also be either overt/disclosed (the participants know they are being studied) or covert/undisclosed (the research keeps their real identity a secret from the research subjects, acting as a genuine member of the group). In general observations, are relatively cheap to carry out and few resources are needed by the researcher. However, they can often be very time consuming and longitudinal.

Controlled Observation

Controlled observations (usually a structured observation) are likely to be carried out in a psychology laboratory. The researcher decides where the observation will take place, at what time, with which participants, in what circumstances and uses a standardised procedure. Participants are randomly allocated to each independent variable group.

Rather than writing a detailed description of all behaviour observed, it is often easier to code behaviour according to a previously agreed scale using a behaviour schedule (*i.e.*, conducting a structured observation).

The researcher systematically classifies the behaviour they observe into distinct categories. Coding might involve numbers or letters to describe a characteristics, or use of a scale to measure behaviour intensity. The categories on the schedule are coded so that the data collected can be easily counted and turned into statistics.

For example, Mary Ainsworth used a behaviour schedule to study how infants responded to brief periods of separation from their mothers. During the Strange Situation procedure infant's interaction behaviours directed towards the mother were measured, *e.g.*

1. Proximity and contacting seeking
2. Contact maintaining
3. Avoidance of proximity and contact
4. Resistance to contact and comforting

The observer noted down the behaviour displayed during 15 second intervals and scored the behaviour for intensity on a scale of 1 to 7.

Observations were recorded every 15 seconds and placed into behavioural categories

Intensity	Proximity and contact seeking	Contact maintaining	Proximity and interaction avoiding	Proximity and interaction resisting	Searching
1					
2		✓			
3					✓
4	✓✓				
5					✓
6	✓				
7		✓			12

Sometimes the behaviour of participants is observed through a two-way mirror or they are secretly filmed. This method was used by Albert Bandura to study aggression in children (the Bobo doll studies).

A lot of research has been carried out in sleep laboratories as well. Here electrodes are attached to the scalp of participants and what is observed are the changes in electrical activity in the brain during sleep (the machine is called an electroencephalogram – an EEG).

Controlled observations are usually overt as the researcher explains the research aim to the group, so the participants know they are being observed. Controlled observations are also usually non-participant as the researcher avoids any direct contact with the group, keeping a distance (*e.g.*, observing behind a two-way mirror).

Strengths

1. Controlled observations can be easily replicated by other researchers by using the same observation schedule. This means it is easy to test for reliability.
2. The data obtained from structured observations is easier and quicker to analyze as it is quantitative (*i.e.*, numerical) - making this a less time consuming method compared to naturalistic observations.
3. Controlled observations are fairly quick to conduct which means that many observations can take place within a short amount of time. This means a large sample can be obtained resulting in the findings being representative and having the ability to be generalized to a large population..

Limitations

1. Controlled observations can lack validity due to the Hawthorne effect/demand characteristics. When participants know they are being watched they may act differently.

Naturalistic Observation

Naturalistic observation (*i.e.*, unstructured observation) involves studying the spontaneous behaviour of participants in natural surroundings. The researcher simply records what they see in whatever way they can.

Compared with controlled/structured methods it is like the difference between studying wild animals in a zoo and studying them in their natural habitat.

With regard to human subjects Margaret Mead used this method to research the way of life of different tribes living on islands in the South Pacific. Kathy Sylva used it to study children at play by observing their behaviour in a playgroup in Oxfordshire.

Strengths

1. By being able to observe the flow of behaviour in its own setting studies have greater ecological validity.
2. Like case studies naturalistic observation is often used to generate new ideas. Because it gives the researcher the opportunity to study the total situation it often suggests avenues of enquiry not thought of before.

Limitations

1. These observations are often conducted on a micro (small) scale and may lack a representative sample (biased in relation to age, gender, social class or ethnicity). This may result in the findings lacking the ability to be generalized to wider society.
2. Natural observations are less reliable as other variables cannot be controlled. This makes it difficult for another researcher to repeat the study in exactly the same way.
3. A further disadvantage is that the researcher needs to be trained to be able to recognise aspects of a situation that are psychologically significant and worth further attention.
4. With observations we do not have manipulations of variables (or control over extraneous variables) which means cause and effect relationships cannot be established.

Participant Observation

Participant observation is a variant of the above (natural observations) but here the researcher joins in and becomes part of the group they are studying to get a deeper insight into their lives. If it were research on animals we would now not only be studying them in their natural habitat but be living alongside them as well!

This approach was used by Leon Festinger in a famous study into a religious cult who believed that the end of the world was about to occur. He joined the cult and studied how they reacted when the prophecy did not come true.

Participant observations can be either cover or overt. Covert is where the study is carried out 'under cover'. The researcher's real identity and purpose are kept concealed from the group being studied.

The researcher takes a false identity and role, usually posing as a genuine member of the group. On the other hand, overt is where the researcher reveals his or her true identity and purpose to the group and asks permission to observe.

Limitations

1. It can be difficult to get time / privacy for recording. For example, with covert observations researchers can't take notes openly as this would blow their cover. This means they have to wait until they are alone and reply on their memory. This is a problem as they may forget details and are unlikely to remember direct quotations.
2. If the researcher becomes too involved they may lose objectivity and become bias. There is always the danger that we will "see" what we expect (or want) to see. This is a problem as they could selectively report information instead of noting everything they observe. Thus reducing the validity of their data.

Recording of Data

With all observation studies an important decision the researcher has to make is how to classify and record the data. Usually this will involve a method of sampling. The three main sampling methods are:

1. Event sampling. The observer decides in advance what types of behaviour (events) she is interested in and records all occurrences. All other types of behaviour are ignored.
2. Time sampling. The observer decides in advance that observation will take place only during specified time periods (*e.g.*, 10 minutes every hour, 1 hour per day) and records the occurrence of the specified behaviour during that period only.
3. Instantaneous (target time) sampling. The observer decides in advance the pre-selected moments when observation will take place and records what is happening at that instant. Everything happening before or after is ignored.

Types of Observation

In summary, your observation options are:

Participant - Where the researcher deliberately joins in with the activities of a group while observing them

OR

Non-Participant - Where you observe people's behaviour without joining in any way

Covert - Where participants do not know that they are being observed

OR

Overt - Where you inform participants that you will be observing them

Structured - Where you may have a list of behaviours that you have decided you will observe

OR

Unstructured - Which can provide a more in-depth narrative account

Observation may take place in the natural or real life setting or in a laboratory. Observational procedures tend to vary from complete flexibility to the use of pre-coded detailed formal instrument. The observer may himself participate actively in the group he is observing or he may be an observer from outside or his presence may be unknown to the people he is observing.

We may thus classify scientific observation broadly, on three bases, as follows:

(1) controlled/uncontrolled observation.

(2) Structured/unstructured/partially structured observation.

(3) Participant/non-participant/disguised observation.

The type of observational technique to be chosen in a particular study depends on the purpose of the study. In an exploratory study, the observational procedure is most likely to be relatively unstructured and the observer is also more likely to participate in group activity.

On the other hand, for studies of the descriptive or experimental type, the observational procedures are more likely to be relatively structured and involve a minimum of participation on the part of the observer.

It should be noted, however, that the degree of structuredness and the degree of participation need not be together. For example, the researcher in an exploratory study may be a participant observer or a non- participant or a disguised observer. A particular research situation may demand the coupling of participant observation with a highly structured observational instrument.

The investigator, whatever be the purpose of his study, should advisedly answer four broad questions before setting out to observe, *i.e.*, he must be sure about:

(i) What should be observed,
(ii) How the observation should be recorded,
(iii) How to ensure accuracy of observation, and
(iv) What relationship should obtain between the observer and observed and, how the desired relationship should be established.

The questions cannot be answered uniformly since the above decisions depend on the nature of the study and the extent to which observational procedures can be structured. Let us now discuss the major types of observational procedures. One of the most useful bases for classification of observational procedures is the degree of structuredness.

MERITS AND LIMITATIONS OF OBSERVATION

Merits of Observation

In participant observation, the investigator becomes a member of the community being observed by him. The investigator need not carry out exactly the same activities as the subjects, suffice it, if he finds a role in the group which does not disturb the usual patterns of behaviour. Thus, one of the advantages of participant observation is that since the members of the community are unaware of the researcher's purpose their behaviour is least likely to be affected. Thus, the researcher is enabled to record the "natural" behaviour of the group.

Secondly, since the researcher actually participates in the group under observation, he normally has an access to a body of information which could not easily be obtained by merely looking on in a disinterested fashion.

He thus obtains a great depth of experience, while he is able to record the actual behaviour of other participants. Since his period of participation may continue for months, the range of materials collected is likely to be much wider than those gained from a series of quite lengthy interview- schedules.

Thirdly, in participant observation, the researcher is able to record the context which gives meaning to expressions of opinion surpassing in richness and depth the usual questionnaire. He can also check the truth of statements made by members of the group.

Some occurrences are rarely, if ever, accessible to direct observation. Sexual bahaviour, family crisis and underworld activities, etc., are examples of events that are not amenable to direct observation by an outsider. It is here that the participant observation helps.

Limitations of Observation

Participant observation has certain disadvantages, one being that the investigator who actually becomes a participant happens to narrow his range of experience. He takes on a particular position within a group with a definite clique or friendship circle. He learns and follows a pattern of activity which is characteristic of its members.

Hence, many avenues become closed to him. Further, the role he comes to occupy in the group may be important so that he may be instrumental in effecting changes in the group- behaviour.

The position of participant-observer is especially precarious when it comes to maintaining objectivity. The involvement in the situation may lessen the sharpness of observation not only because the investigator identifies himself with his informants but also because he becomes so used to certain kinds of behaviour.

In certain situations, the physical and emotional endurance as well as the patience of the researcher may be put to an acid test. Even the observation of routine day-to-day occurrences may become difficult in view of the possibility that unforeseeable factors might interfere with the observational task.

To the extent that he participates emotionally, the observer comes to lose objectivity which in scientific parlance is his single greatest asset. He may react in anger when he ought to be recording. He may seek prestige or ego-satisfaction within the group rather than observing this behaviour in others.

His heart may be moved by tragedy but he may forget to record its impact upon his fellow-members. In consequence, he may fail to note these important details which may appear to him so commonplace as not to merit any attention.

It is clear that in both, the participant and non-participant types of observation, the problem of observation-control is not solved. To the degree that the investigator becomes a participant his experience becomes unique, peculiarly his own. Thus, any other researcher would not be able to record the same facts. There is thus less standardization of data. In short, his role of observer is handicapped somewhat by his being a participant.

Non-participant observation does answer some of these objections. But a purely non- participant observation is difficult. We have standard set of relationships or role patterns for the 'non-member' who should be ever present but never participating.

Both the subject-group and the outsider are likely to feel uncomfortable. And, naturally, for many research situations it is almost impossible for the outsider to be a genuine participant in all ways.

The sociologist cannot, for example, become a criminal in order to study a criminal gang. Sometimes, it is possible to take part in a great many activities of the group, just to avoid the awkwardness of complete non-participation while taking the role of an observer for other activities.

This strategy was employed by Leplay a century ago in his study of European working class families. In certain studies, the investigators have participated as members of the family taking part in games and dances. They nevertheless made clear that their purpose, above anything else, was to gather facts.

Introspection: Meaning and Definitions

Introspection is the examination of one's own conscious thoughts and feelings. In psychology, the process of introspection relies exclusively on observation of one's mental state, while in a spiritual context it may refer to the examination of one's soul. Introspection is closely related to human self-reflection and is contrasted with external observation.

Introspection generally provides a privileged access to our own mental states, not mediated by other sources of knowledge, so that individual experience of the mind is unique. Introspection can determine any number of mental states including: sensory, bodily, cognitive, emotional and so forth.

Introspection has been a subject of philosophical discussion for thousands of years. The philosopher Plato asked, "…why should we not calmly and patiently review our own thoughts, and thoroughly examine and see what these appearances in us really are?"

While introspection is applicable to many facets of philosophical thought it is perhaps best known for its role in epistemology, in this context introspection is often compared with perception, reason, memory, and testimony as a source of knowledge.

In Psychology

Wundt: It has often been claimed that Wilhelm Wundt, the father of modern psychology, was the first to adopt introspection to experimental psychology though the methodological idea had been presented long before, as by 18th century German philosopher-psychologists such as Alexander Gottlieb Baumgarten or Johann Nicolaus Tetens.

Also, Wundt's views on introspection must be approached with great care. Wundt was influenced by notable physiologists, such as Gustav Fechner, who used a kind of controlled introspection as a means to study human sensory organs.

Building upon the pre-existing use of introspection in physiology, Wundt believed the method of introspection was the ability to observe an experience, not just the logical reflection or speculations which some others interpreted his meaning to be.

Wundt imposed exacting control over the use of introspection in his experimental laboratory at the University of Leipzig making it possible for other scientists to replicate his experiments elsewhere, a development that proved essential to the development of psychology as a modern, peer-reviewed scientific discipline.

Such exact purism was typical of Wundt and he instructed all introspection observations be performed under these same instructions "1) the Observer must, if possible, be in a position to determine beforehand the entrance of the process to be observed. 2) the introspectionist must, as far as possible, grasp the phenomenon in a state of strained attention and follow its course. 3) Every observation must, in order to make certain, be capable of being repeated several times under the same conditions and 4) the conditions under which the phenomenon appears must be found out by the variation of the attendant circumstances and when this was done the various coherent experiments must be varied according to a plan partly by eliminating certain stimuli and partly by grading their strength and quality"

Titchener

Edward Titchener was an early pioneer in experimental psychology and student of Wilhelm Wundt. After earning his doctorate under the tutelage of Wundt at the University of Leipzig, he made his way to Cornell University where he established his own laboratory and research.

When Titchener arrived at Cornell, psychology was still a fledgling discipline, especially in the United States, and Titchener was a key figure in bringing Wundt's ideas to America. However, Titchener misrepresented some of Wundt's ideas to the American psychological establishment, especially in his account of introspection which, Titchener taught, only served a purpose in the qualitative analysis of consciousness into its various parts, while Wundt saw it as a means to quantitatively measure the whole of conscious experience.

Titchener was exclusively interested in the individual components that comprise conscious experience, while Wundt, seeing little purpose in the analysis of individual components, focused on synthesis of these components. Ultimately Titchener's ideas would form the basis of the short-lived psychological theory of structuralism.

Historical Misconceptions

American historiography of introspection, according to some authors, is dominated by three misconceptions. In particular, historians of psychology tend to argue 1) that introspection once was the dominant method of psychological enquiry, 2) that behaviourism, and in particular John B. Watson, is responsible for discrediting introspection as a valid method, and 3) that scientific psychology completely abandoned introspection as a result of those critiques. Yet, introspection has not been the dominant method. It is believed to be so because Edward Titchener's student Edwin G. Boring, in his influential historical accounts of experimental psychology privileged Titchener's views while giving little credit to original sources.

Introspection has been critiqued by many other psychologists, including Wilhelm Wundt, and Knight Dunlap who in his article "*The Case Against Introspection*", presents an argument against self-observation that is not primarily rooted in behaviourist epistemology. Introspection is still widely used in psychology, but under different names, such as self-report surveys, interviews and fMRIs. It is not the method but rather its name that has been dropped from the dominant psychological vocabulary.

Recent Developments

Partly as a result of Titchener's misrepresentation, the use of introspection diminished after his death and the subsequent decline of structuralism. Later psychological movements, such as functionalism and behaviourism, rejected introspection for its lack of scientific reliability among other factors. Functionalism originally arose in direct opposition to structuralism, opposing its narrow focus on the elements of consciousness and emphasising the purpose of consciousness and other psychological behaviour. Behaviourism's objection to introspection focused much more on its unreliability and subjectivity which conflicted with behaviourism's focus on measurable behaviour.

The more recently established cognitive psychology movement has to some extent accepted introspection's usefulness in the study of psychological phenomena, though generally only in experiments pertaining to internal thought conducted under experimental conditions. For example, in the "think aloud protocol", investigators cue participants to speak their thoughts aloud in order to study an active thought process without forcing an individual to comment on the process itself.

Already in the 18th century authors had criticized the use of introspection, both for knowing one's own mind and as a method for psychology. David Hume pointed out that introspecting a mental state tends to alter the very state itself; a German author, Christian Gottfried Schütz, noted that introspection is often described as mere "inner sensation", but actually requires also attention, that introspection does not get at unconscious mental states, and that it cannot be used naively - one needs to know what to look for.

Immanuel Kantadded that, if they are understood too narrowly, introspective experiments are impossible. Introspection delivers, at best, hints about what goes on in the mind; it does not suffice to justify knowledge claims about the mind. Similarly, the idea continued to be discussed between John Stuart Mill and Auguste Comte.

Recent psychological research on cognition and attribution has asked people to report on their mental processes, for instance to say why they made a particular choice or how they arrived at a judgement. In some situations, these reports are clearly confabulated. For example, people justify choices they have not in fact made.

Such results undermine the idea that those verbal reports are based on direct introspective access to mental content. Instead, judgements about one's own mind seem to be inferences from overt behaviour, similar to judgements made about another person. However, it is hard to assess whether these results only apply to unusual experimental situations, or if they reveal something about everyday introspection. The theory of the adaptive unconscious suggests that a very large proportion of mental processes, even "high-level" processes like goal-setting and decision-making, are inaccessible to introspection. Indeed, it is questionable how confident researchers can be in their own introspections.

One of the central implications of dissociations between consciousness and meta-consciousness is that individuals, presumably including researchers, can misrepresent their experiences to themselves. Jack and Roepstorff assert, '...there is also a sense in which subjects simply cannot be wrong about their own experiential states.'

Presumably they arrived at this conclusion by drawing on the seemingly self-evident quality of their own introspections, and assumed that it must equally apply to others. However, when we consider research on the topic, this conclusion seems less self-evident.

If, for example, extensive introspection can cause people to make decisions that they later regret, then one very reasonable possibility is that the introspection caused them to 'lose touch with their feelings'. In short, empirical studies suggest that people can fail to appraise adequately (*i.e.*, are wrong about) their own experiential states.

Another question in regards to the veracious accountability of introspection is if researchers lack the confidence in their own introspections and those of their participants, then how can it gain legitimacy? Three strategies are accountable: identifying behaviours that establish credibility, finding common ground that enables mutual understanding, and developing a trust that allows one to know when to give the benefit of the doubt. That is to say, that words are only meaningful if validated by one's actions; When people report strategies, feelings or beliefs, their behaviours must correspond with these statements if they are to be believed.

Even when their introspections are uninformative, people still give confident descriptions of their mental processes, being "unaware of their unawareness". This phenomenon has been termed the *introspection illusion* and has been used to explain some cognitive biases and belief in some paranormal phenomena.

When making judgements about themselves, subjects treat their own introspections as reliable, whereas they judge other people based on their behaviour. This can lead to illusions of superiority.

For example, people generally see themselves as less conformist than others, and this seems to be because they do not introspect any urge to conform. Another reliable finding is that people generally see themselves as less biased than everyone else, because they are not likely to introspect any biased thought processes. These introspections are misleading, however, because biases work sub-consciously.

One experiment tried to give their subjects access to others' introspections. They made audio recordings of subjects who had been told to say whatever came into their heads as they answered a question about their own bias. Although subjects persuaded themselves they were unlikely to be biased, their introspective reports did not sway the assessments of observers. When subjects were explicitly told to avoid relying on introspection, their assessments of their own bias became more realistic.

Process Involved in Introspection

Introspection is generally regarded as a process by means of which we learn about our own currently ongoing, or very recently past, mental states or processes. Not all such processes are introspective, however: Few would say that you have introspected if you learn that you're angry by seeing your facial expression in the mirror. However, it's unclear and contentious exactly what more is required for a process to qualify as introspective. A relatively restrictive account of introspection might require introspection to involve attention to and direct detection of one's ongoing mental states; but many philosophers think attention to or direct detection of mental states is impossible or at least not present in many paradigmatic instances of introspection.

For a process to qualify as "introspective" as the term is ordinarily used in contemporary philosophy of mind, it must minimally meet the following three conditions:

The mentality condition: Introspection is a process that generates, or is aimed at generating, knowledge, judgements, or beliefs about *mental* events, states, or processes, and not about affairs outside one's mind, at least not directly.

In this respect, it is different from sensory processes that normally deliver information about outward events or about the non-mental aspects of the individual's body. The border between introspective and non-introspective knowledge can begin to seem blurry with respect to bodily self-knowledge such as proprioceptive knowledge about the position of one's limbs or nociceptive knowledge about one's pains.

But it seems that in principle the introspective part of such processes, pertaining to judgements about one's mind—*e.g.*, that one has the feeling as though one's arms were crossed or of toe-ishly located pain—can be distinguished from the non-introspective judgement that one's arms are in fact crossed or one's toe is being pinched.

The first-person condition: Introspection is a process that generates, or is aimed at generating, knowledge, judgements, or beliefs about *one's own mind only* and no one else's, at least not directly. Any process that in a similar manner generates knowledge of one's own and others' minds is by that token not an introspective process.

The temporal proximity condition: Introspection is a process that generates knowledge, beliefs, or judgements about one's *currently ongoing* mental life only; or, alternatively (or perhaps in addition) *immediately past* (or even future) mental life, within a certain narrow temporal window (sometimes called the specious present; see the entry on the experience and perception of time). You may know that you were thinking about Montaigne yesterday during your morning walk, but you cannot know that fact by current introspection alone—though perhaps you can know introspectively that you currently have a vivid memory of having thought about Montaigne.

Likewise, you cannot know by introspection alone that you will feel depressed if your favoured candidate loses the election in November—though perhaps you can know introspectively what your current attitude is towards the election or what emotion starts to rise in you when you consider the possible outcomes.

Whether the target of introspection is best thought of as one's current mental life or one's immediately past mental life may depend on one's model of introspection: On self-detection models of introspection, according to which introspection is a causal process involving the detection of a mental state, it's natural to suppose that a brief lapse of time will transpire between the occurrence of the mental state that is the introspective target and the final introspective judgement about that state, which invites (but does not strictly imply) the idea that introspective judgements generally pertain to immediately past states.

On self-shaping and self-fulfilment models of introspection, according to which introspective judgements create or embed the very state introspected, it seems more natural to think that the target of introspection is one's current mental life or perhaps even the immediate future.

Few contemporary philosophers of mind would call a process "introspective" if it does not meet some version of the three conditions above, though in ordinary language the temporal proximity condition may sometimes be violated. However, many philosophers of mind will resist calling a process that meets these three conditions "introspective" unless it also meets some or all of the following three conditions:

The directness condition: Introspection yields judgements or knowledge about one's own current mental processes relatively *directly* or *immediately*. It's difficult to articulate exactly what directness or immediacy involves in the present context, but some examples should make the import of this condition relatively clear.

Gathering sensory information about the world and then drawing theoretical conclusions based on that information should not, according to this condition, count as introspective, even if the process meets the three conditions above.

Seeing that a car is twenty feet in front of you and then inferring from that fact about the external world that you are having a visual experience of a certain sort does not, by this condition, count as introspective. However, those who embrace transparency theories of introspection may reject at least strong formulations of this condition.

The detection condition: Introspection involves some sort of *attunement to* or *detection of* a *pre-existing* mental state or event, where the introspective judgement or knowledge is (when all goes well) *causally* but not *ontologically* dependent on the target mental state.

For example, a process that involved creating the state of mind that one attributes to oneself would not be introspective, according to this condition.

Suppose I say to myself in silent inner speech, "I am saying to myself in silent inner speech, 'haecceities of applesauce'", without any idea ahead of time how I plan to complete the embedded quotation.

Now, what I say may be true, and I may know it to be true, and I may know its truth (in some sense) directly, by a means by which I could not know the truth of anyone else's mind. That is, it may meet all the four conditions above and yet we may resist calling such a self-attribution introspective.

Self-shaping, expressivist, and transparency accounts of self-knowledge emphasize the extent to which our self-knowledge often does not involve the detection of pre-existing mental states; and because something like the detection condition is implicitly or explicitly accepted by many philosophers, some philosophers (including some but not all of those who endorse self-shaping, expressivist, and/or transparency views) would regard it as inappropriate to regard such accounts of self-knowledge as accounts of *introspection* proper.

The effort condition: Introspection is not *constant, effortless, and automatic*. We are not every minute of the day introspecting. Introspection involves some sort of special reflection on one's own mental life that differs from the ordinary un-self-reflective flow of thought and action.

The mind may monitor itself regularly and constantly without requiring any special act of reflection by the thinker—for example, at a non-conscious level certain parts of the brain or certain functional systems may monitor the goings-on of other parts of the brain and other functional systems, and this monitoring may meet all five conditions above—but this sort of thing is not what philosophers generally have in mind when they talk of introspection.

However, this condition, like the directness and detection conditions, is not universally accepted. For example, philosophers who think that conscious experience requires some sort of introspective monitoring of the mind and who think of conscious experience as a more or less constant feature of our lives may reject the effort condition.

Though not all philosophical accounts that are put forward by their authors as accounts of "introspection" meet all of conditions 4–6, most meet at least two of those. Because of differences in the importance accorded to conditions 4–6, it is not unusual for authors with otherwise similar accounts of self-knowledge to differ in their willingness to describe their accounts as accounts of "introspection".

Experimental : Meaning and Definitions

While psychologists use all of the research methods described so far, they often prefer the approach we will now consider: experimentation (or *the experimental method*). This involves efforts to determine if variables are related to one another by systematically changing one (or more) and observing the effects of such variations on the other (or others). There are several reasons why psychologists prefer this basic approach, but perhaps the most important is this: In contrast to the other methods, experimentation yields relatively clear-cut evidence on causality.

If systematic variations in one factor produce changes in another (and if additional conditions we'll soon consider are also met), we can conclude with reasonable certainty that there is a causal link between the factors: that changes in one caused changes in the other.

Establishing such causality is extremely valuable from the perspective of one major goal of science: *explanation*. Briefly, scientists do not wish merely to describe the world around them and the relation-ships between different variables or factors in it.

They also wish to be able to explain why such relationships exist, why, for example, people find some gauges easier to read than others, why individuals with certain personality traits are more likely than others to suffer heart attacks, why some people gain weight readily while others do not. Experimentation, because it often yields information useful in answering such questions, is frequently the method of choice in psychology.

But bear in mind that there are no hard and fast rules in this regard. Rather, most psychologists select the research technique that seems most suited to the topic they wish to study and the resources available to them.

Its Basic Nature

Reduced to its bare essentials, the experimental method involves two basic steps: (1) The presence or strength of some variable believed to affect behaviour is systematically altered, and (2) The effects of such alterations (if any) are measured. The logic behind these steps is as follows: If the factor varied does indeed influence behaviour or cognitive processes, then individuals exposed to different levels or amounts of that factor should differ in terms of their behaviour. Thus, exposure to a small amount of the variable should result in one level of behaviour; exposure to a larger amount should result in a different level, and so on.

The factor systematically varied by the researcher is termed the independent variable, while the aspect of behaviour or cognitive processes studied is termed the dependent variable. In a simple experiment, then, different groups of participants are exposed to contrasting levels of the independent variable (such as low, moderate, and high).

The participants' behaviour is then carefully measured to determine whether it does in fact vary with different levels or amount of the independent variable.

If it does, and if two other conditions described below are met the researcher can tentatively conclude that the independent variable does indeed cause changes in the aspect of behaviour being studied.

To illustrate the basic nature of experimentation in psychological research, let's return to the question we considered earlier: Are fast talkers more persuasive than slow ones? A researcher who decides to employ the experimental method to study, this topic might begin with the hypothesis that the faster people talk (at least up to a point), the more persuasive they are.

In such research, the independent variable would be the speed at which would be persuaders speak, and the dependent variable would be some measure of the persuaders' success in influencing their audiences.

There are many different ways of testing the hypothesis that fast talkers are more persuasive than slow ones, but for the sake of illustration, let's assume that the researcher arranges to have an assistant deliver a speech designed to alter listeners' views on a specific issue: legislation to limit the use of chemicals known to deplete the earth's ozone layer. The speaker would then present this speech to different groups of participants at contrasting speeds.

For example, for participants in one group, the speech would be presented at a slow pace (150 words per minute); for those in another group, it would be presented at a moderate pace (170 words per minute); and for those in the third group, it would be presented at a fast pace (190 words per minute). Then some measure of audience members' attitudes towards the legislation would be obtained.

We must assume by the way, that individuals in the three groups start out with similar attitudes towards the legislation; if they do not, serious complications can arise in the interpretation of any results that are obtained.

Two Requirements for Its Success

Earlier, we saw that before we can conclude that an independent variable has caused some change in behaviour, two conditions must be met. The first condition involves random assignment of participants to experimental conditions.

According to the principle of random assignment, all participants in an experiment must have an equal chance of being exposed to each level of the independent variable. The reason for this rule is simple: If participants are not randomly assigned to each group, it may later be impossible to determine whether differences in their behaviour stem from differences they brought with them, from the impact of the independent variable, or from both.

For instance, continuing with our study of speed of speech and persuasion, imagine that participants in the study are drawn from two different groups: first year law students and a troupe of high school dropouts enrolled in a special course designed to provide them with basic vocational skills. Now imagine that because of differences in the two groups' schedules, most of the participants exposed to the slow talker are law students, while most of the people exposed to the fast talker are high school dropouts.

Suppose that results indicate that participants exposed to the fast talker show much more agreement with the views expressed than participants exposed to the slow talker. What can we conclude? Not much, because it is entirely possible that the difference stems from the different mixes of participants in the two experimental conditions.

In the slow talker condition, 85 percent of the participants are law students and 15 percent are high school dropouts, while in the fast talker condition, the opposite is true. Since law students may be somewhat harder to persuade than high school dropouts, we can't really tell why these results occurred.

Did they derive from differences in the persuaders' rate of speech? From different pro-portions of the two groups of participants in each condition? Both factors? It's impossible to tell. To avoid such problems, it is crucial that all subjects have an equal chance of being assigned to each experimental group.

The second condition referred to above may be stated as follows: Insofar as possible, all other factors that might also affect participants' behaviour, aside from the independent variable, must be held constant.

To see why this is so, consider what will happen if, in the study on speed of speech and persuasion, different speakers are used in the two conditions. Further, imagine that one of these speakers the fast talker has a pleasant cultivated voice, while the slow talker has an irritating voice and a thick, unpleasant accent.

Now assume that participants express greater agreement with the fast talker, than the slow one. What is the cause of this result? The difference in the speakers' speed of speech? Differences in the pleasantness of their voices? Both factors? Obviously, it's impossible to tell. In this situation the independent variable of interest (speed of speech) is confounded with another variable pleasantness of the speaker's voice that is not a planned part of the research. That is, another factor changes as the independent variable changes, so we can't tell whether any effects observed are produced by the independent variable or this other factor. When such confounding occurs, the findings of an experiment are largely uninterpretable.

CHARACTERISTICS, STEPS INVOLVED IN EXPERIMENTAL METHOD

The Experimental Method

Throughout the laboratory portion of most Biology laboratories, you will be conducting experiments. Science proceeds by use of the experimental method. This handout provides a summary of the steps that are used in pursuing scientific research. This general method is used not only in biology but in chemistry, physics, geology and other hard sciences. To gather information about the biological world, we use two mechanisms: our sensory perception and our ability to reason. We can identify and count the types of trees in a forest with our eyes, we can identify birds in the rainforest canopy with our ears, and we can identify the presence of a skunk with our nose. Touch and taste help us experience the biological world as well. With the information we gather from our senses, we can make inferences using our reason and logic. For instance, you know that you see palm trees in tropical and subtropical regions and can infer that palm trees will not be found in central Maine because of the harshness of our winter.

Our reason allows us to make predictions about the natural world. Scientists attempt to predict and perhaps control future events based on present and past knowledge. The ability to make accurate predictions hinges on the seven steps of the Scientific Method.

Step 1. Make observations. These observations should be objective, not subjective. In other words, the observations should be capable of verification by other scientists. Subjective observations, which are based on personal opinions and beliefs, are not in the realm of science. Here's an objective statement: It is 58 °F in this room. Here's a subjective statement: It is cool in this room.

The first step in the Scientific Method is to make objective observations. These observations are based on specific events that have already happened and can be verified by others as true or false.

Step 2. Form a hypothesis. Our observations tell us about the past or the present. As scientists, we want to be able to predict future events. We must therefore use our ability to reason.

Scientists use their knowledge of past events to develop a general principle or explanation to help predict future events. The general principle is called a hypothesis. The type of reasoning involved is called inductive reasoning (deriving a generalization from specific details).

A hypothesis should have the following characteristics:

- It should be a general principle that holds across space and time
- It should be a tentative idea
- It should agree with available observations
- It should be kept as simple as possible.
- It should be testable and potentially falsifiable. In other words, there should be a way to show the hypothesis is false; a way to disprove the hypothesis.

Some mammals have two hindlimbs would be a useless hypothesis. There is no observation that would not fit this hypothesis!

All mammals have two hindlimbs is a good hypothesis. We would look throughout the world at mammals. When we find whales, which have no hindlimbs, we would have shown our hypothesis to be false; we have falsified the hypothesis.

When a hypothesis involves a cause-and-effect relationship, we state our hypothesis to indicate there is no effect. A hypothesis, which asserts no effect, is called a *null hypothesis*. For instance, the drug Celebra does not help relieve rheumatoid arthritis.

Step 3. Make a prediction. From step 2, we have made a hypothesis that is tentative and may or may not be true. How can we decide if our hypothesis is true?

Our hypothesis should be broad; it should apply uniformly through time and through space. Scientists cannot usually check every possible situation where a hypothesis might apply. Let's consider the hypothesis: All plant cells have a nucleus. We cannot examine every living plant and every plant that has ever lived to see if this hypothesis is false. Instead, we generate a prediction using deductive reasoning (generating a specific expectation from a generalization). From our hypothesis, we can make the following prediction: If I examine cells from a blade of grass, each one will have a nucleus.

Now, let's consider the drug hypothesis: The drug Celebra does not help relieve rheumatoid arthritis. To test this hypothesis, we would need to choose a specific set of conditions and then predict what would happen under those conditions if the hypothesis were true. Conditions you might wish to test are doses administered, length of time the medication is taken, the ages of the patients and the number of people to be tested.

All of these conditions that are subject to change are called variables. To gauge the effect of Celebra, we need to perform a controlled experiment. The experimental group is subjected to the variable we want to test and the control group is not exposed to that variable. In a controlled experiment, the only variable that should be different between the two groups is the variable we want to test.

Let's make a prediction based on observations of the effect of Celebra in the laboratory. The prediction is: Patients suffering from rheumatoid arthritis who take Celebra and patients who take a placebo (a starch tablet instead of the drug) do not differ in the severity of rheumatoid arthritis.

Step 4. Perform an experiment. We rely again on our sensory perception to collect information. We design an experiment based on our prediction.

Our experiment might be as follows: 1000 patients between the ages of 50 and 70 will be randomly assigned to one of two groups of 500. The experimental group will take Celebra four times a day and the control group will take a starch placebo four times a day. The patients will not know whether their tablets are Celebra or the placebo. Patients will take the drugs for two months. At the end of two months, medical exams will be administered to determine if flexibility of the arms and fingers has changed.

Step 5. Analyze the results of the experiment. Our experiment produced the following results: 350 of the 500 people who took Celebra reported diminished arthritis as the end of the period. 65 of the 500 people who took the placebo reported improvement.

The data appear to show that there was a significant effect of Celebra. We would need to do a statistical analysis to demonstrate the effect. Such an analysis reveals that there is a statistically significant effect of Celebra.

Step 6. Draw a conclusion. From our analysis of the experiment, we have two possible outcomes: the results agree with the prediction or they disagree with the prediction. In our case, we can reject our prediction of no effect of Celebra. Because the prediction is wrong, we must also reject the hypothesis it was based on.

Our task now is to reframe the hypothesis is a form that is consistent with the available information. Our hypothesis now could be: The administration of Celebra reduces rheumatoid arthritis compared to the administration of a placebo.

With present information, we accept our hypothesis as true. Have we proved it to be true? Absolutely not! There are always other explanations that can explain the results. It is possible that the more of the 500 patients who took Celebra were going to improve anyway. It's possible that more of the patients who took Celebra also ate bananas every day and that bananas improved the arthritis. You can suggest countless other explanations.

How can we prove that our new hypothesis is true? We never can. The scientific method does not allow any hypothesis to be proven. Hypotheses can be disproven in which case that hypothesis is rejected as false. All we can say about a hypothesis, which stands up to, a test to falsify it is that we failed to disprove it. There is a world of difference between failing to disprove and proving. Make sure you understand this distinction; it is the foundation of the scientific method.

So what would we do with our hypothesis above? We currently accept it as true. To be rigorous, we need to subject the hypothesis to more tests that could show it is wrong.

For instance, we could repeat the experiment but switch the control and experimental group. If the hypothesis keeps standing up to our efforts to knock it down, we can feel more confident about accepting it as true. However, we will never be able to state that the hypothesis is true. Rather, we accept it as true because the hypothesis stood up to several experiments to show it is false.

Step 7. Report your results. Scientists publish their findings in scientific journals and books, in talks at national and international meetings and in seminars at colleges and universities. Disseminating results is an essential part of the scientific method. It allows other people to verify your results, develop new tests of your hypothesis or apply the knowledge you have gained to solve other problems.

Merits and Limitations of Experimental

There are many benefits and limitations to experimental research and many of them have been alluded to in previous modules in this series. Following is more detailed discussion regarding both the advantages and the limitations or disadvantages.

Benefits and Advantages

- Experimental research is the most appropriate way for drawing causal conclusions, regarding interventions or treatments and establishing whether or not one or more factors causes a change in an outcome. This is largely due to the emphasis in controlling extraneous variables. If other variables are controlled, the researcher can say with confidence that manipulation independent variable caused a changed in the dependent variable.
- It is a basic, straightforward, efficient type of research that can be applied across a variety of disciplines.
- Experimental research designs are repeatable and therefore, results can be checked and verified.
- Due to the controlled environment of experimental research, better results are often achieved.
- In the case of laboratory research, conditions not found in a natural setting can be created in an experimental setting that allows for greater control of extraneous variables. Conditions that may take longer to occur in a natural environment may occur more quickly in an experimental setting.
- There are many variations of experimental research and the researcher can tailor the experiment while still maintaining the validity of the design.

Limitations and Disadvantages

- Experimental research can create artificial situations that do not always represent real-life situations. This is largely due to fact that all other variables are tightly controlled which may not create a fully realistic situation.

- Because the situations are very controlled and do not often represent real life, the reactions of the test subjects may not be true indicators of their behaviours in a non-experimental environment.
- Human error also plays a key role in the validity of the project as discussed in previous modules.
- It may not be really possible to control all extraneous variables. The health, mood, and life experiences of the test subjects may influence their reactions and those variables may not even be known to the researcher.
- The research must adhere to ethical standards in order to be valid. These will be discussed in the next module of this series.
- Experimental research designs help to ensure internal validity but sometimes at the expense of external validity. When this happens, the results may not be generalizable to the larger population.
- If an experimental study is conducted in its natural environment, such as a hospital or community, it may not be possible to control the extraneous variables.
- Experimental research is a powerful tool for determining or verifying causation, but it typically cannot specify "why" the outcome occurred.

Advantages and Limitations of Experimentation Method

Experimentation method is the highly systematic and scientific method. Researchers often use experimentation method in their research or in study, because it has several advantages which are as follows.

1. The most salient feature of experimental method is that it is very strong in determine the cause and effect relation between independent and dependent variable. We can accurately say through experiment that one (or more) variable(s) is causing another variable.
2. In experimental method the experimenter has control over his experiment. He can manipulate variable(s) the way he wants and can make some variable constant.
3. Replication of result (obtaining same result again and again) under similar condition is possible in experimental method.
4. This method is very useful in generalizing the outcomes. Because we can get same result again and again through experimental method we can accurately say that the results drawn through experiments can be drawn is other situations also.

Experimental method has also some limitations which are as follows.

1. Though some control is there, absolute control over variable is not possible. Because we are conducting experiments on human beings or animals, we can not control personal differences which may affect dependent variables.

2. Experimental situation is artificial and results may not be generalized well to the real world. The individual may show different behaviour in laboratory and real life.
3. Sometimes experimenters' biasness may affect the result. For example the experimenter is biased towards some people and if he is experimenting on those subjects chances are the result may be biased.
4. Though effects of extraneous variables are minimized, there may be still some effect of some of the extraneous variables in dependent variable.

Though there are some advantages and limitations, this method is very much used to find cause and effect and to generalize the findings.

Stages of Human Growth and Development

Have you ever brought home a new puppy and then watched it grow up? How did your dog change as it got older? You may have watched your dog grow and develop from a cute and cuddly puppy, to a bit of a troublemaker, to a confident companion, and finally to a lazy old dog who sleeps all day.

Each of these stages has different physical and emotional characteristics. Just like dogs, humans go through different developmental stages in their life, as well.

Typical human development is a pretty predictable process—most humans develop at similar rates. This pattern of development allows us to make generalizations about different stages, such as infancy, childhood, adolescence, and adulthood. Let's take a closer look at each stage.

Infancy, Childhood and Adolescence with Special Reference to Physical, Intellectual, Emotional and Social Change

Infancy: Infancy, typically the first year of life, is the first important stage of human development. Many physical milestones occur during this stage as an infant gains control over its body.

However, infants must rely on others to meet most of their needs. They learn to trust other people as needs are met. They need to feel this security in order to properly develop both physically and emotionally. Like your puppy, an infant needs to be loved and nurtured. As you met the needs of your puppy, he/she learned to trust and bond with you while he/she also developed physically.

The Social, Emotional and Physical Development of Infants

As you bring home your new infant, you may be surprised at how fast he seems to grow. Your baby will continue to develop socially, emotionally and physically and will develop a strong understanding of language. You can help your child grow by consistently interacting with him.

Social

The social development of your baby is what happens when she responds to human faces and voices. According to the American Pregnancy Association, she may learn to smile back at you when you hold her or she may even begin babbling as if trying to speak to you. Social development affects how your baby plays with other children and adults as she grows. If your baby is delayed in this particular developmental area, she may have a problem with eyesight or hearing, so she will be less able to learn from your cues.

Emotional

Your baby's emotions go hand in hand with his social development and they need to work together to mature. In the first six months of development, your baby will respond to your love and attention by developing a sense of trust, according to the National Network for Child Care. At the same time, your baby will express his emotions, anger, happiness, excitement or fright. Between six and 12 months, your child will get angry when his needs are not met but will also smile when content and relaxed.

Physical

At the time you give birth to your baby, she won't be able to sit up on her own nor will she have much control over her large muscles. Your baby will stay at this level of development until she reaches about four months of age, when she will begin to gain better control of her muscles and nerves. At this point, she will be able to sit up straight and hold her head up in addition to being able to slightly roll. As your infant continues to progress, her physical development will amaze you.

Language

According to Medline Plus, language development in your child is how he is starting to make sounds, learn words and understanding what people say. In the second month of life, your baby will begin to make sounds other than crying, a sign that his language is developing. By three months, your child will squeal, coo and babble. At the end of six months, your baby will be able to say one-syllable words such as "ma" and "pa." At the end of 12 months, your baby will say two or three words and imitate sounds. He will also understand many words.

Activities to Encourage Development

Begin the first six months by talking and singing to your baby as you change her diaper, bathe her or rock her. Put toys close enough to her face that she can see them. You can also try pointing to parts on your baby's body, such as her ears and nose, and say the name of each part. Put your baby in multiple positions to strengthen her muscles as she moves. The National Network for Child Care suggests that in months six to 12, start allowing your baby to crawl and move away from you, exploring her surroundings. Read books to her and give her toys she can chew. Begin teaching your child what is allowed or not allowed so that she will understand rules. As always, give your child love and attention, taking care of her needs as soon as they arise.

Childhood

The next stage of human development is childhood, during which children start to explore and develop a sense of independence. Eventually, children learn to make their own decisions and they discover that their actions have consequences. As they learn and grow, they develop a sense of self. Children need to be nurtured so that they develop self-confidence instead self-esteem issues. Achieving a healthy level of self-confidence helps children stay motivated to achieve. A child also needs guidance as they begin to test out new skills and gain confidence in their decision-making.

Do you remember when your puppy got big enough to start getting into things? You may have had to make sure to put your shoes up or your dog would chew them as he/she was learning what he/she should and shouldn't do.

During early childhood, children start to develop a "self-concept," the attributes, abilities, attitudes and values that they believe define them. By age 3, (between 18 and 30 months), children have developed their Categorical Self, which is concrete way of viewing themselves in "this or that" labels. For example, young children label themselves in terms of age "child or adult", gender "boy or girl", physical characteristics "short or tall", and value, "good or bad." The labels are used to explain children's self-concept in very concrete, observable terms. For example, Seth may describe himself this way: "I'm 4. I have blue eyes. I'm shorter than Mommy. I can help Grandma set the table!" When asked, young children can also describe their self-concept in simple emotional and attitude descriptions. Seth may go on to say, "Today, I'm happy. I like to play with Amy." However, preschoolers typically do not link their separate self-descriptions into an integrated self-portrait. In addition, many 3-5 year olds are not aware that a person can have opposing characteristics. For example, they don't yet recognize that a person can be both "good" and "bad".

As long-term memory develops, children also gain the Remembered Self. The Remembered Self incorporates memories (and information recounted by adults about personal events) that become part of an individual's life story (sometimes referred to as autobiographical memory). In addition, young children develop an Inner Self, private thoughts, feelings, and desires that nobody else knows about unless a child chooses to share this information.

Because early self-concepts are based on easily defined and observed variables, and because many young children are given lots of encouragement, Preoperational children often have relatively high self-esteem (a judgement about one's worth). Young children are also generally optimistic that they have the ability to learn a new skill, succeed, and finish a task if they keep trying, a belief called "Achievement-Related Attribution", or sometimes "self-efficacy". Self-esteem comes from several sources, such as school ability, athletic ability, friendships, relationships with caregivers, and other helping and playing tasks.

As with emotional development, both internal and external variables can affect young children's self-concept. For example, a child's temperament can affect how they view themselves and their ability to successfully complete tasks. Children with easy temperaments are typically willing to try things repeatedly and are better able to handle frustrations and challenges. In contrast, children with more difficult temperaments may become more easily frustrated and discouraged by challenges or changes in the situation.

Children who can better cope with frustrations and challenges are more likely to think of themselves as successful, valuable, and good, which will lead to a higher self-esteem. In contrast, children who become easily frustrated and discouraged, often quit or need extra assistance to complete a task. These children may have lower self-esteem if they start to believe that they can't be successful and aren't valuable.

External factors, such as messages from other people, also colour how children view themselves. Young children with parents, caregivers, and teachers providing them with positive feedback about their abilities and attempts to succeed (even if they aren't successful the first time) usually have higher self-esteem. On the contrary, when parents, caregivers, or teachers are regularly negative or punitive towards children's attempts to succeed, or regularly ignore or downplay those achievements, young children will have a poor self-image and a lower self-esteem.

Peers also have an impact on young children's self-concept. Young children who have playmates and classmates that are usually nice and apt to include the child in activities will develop a positive self-image. However, a young child who is regularly left out, teased, or bullied by same-age or older peers can develop low self-esteem.

As mentioned repeatedly throughout this document, each child is unique, and he or she may respond to different environments in different ways. Some young children are naturally emotionally "resilient" in certain situations. Resilient children experience or witness something seemingly negative or harmful, without experiencing damage to their self-esteem or emotional development. Resilience not only enables such individuals to withstand life stress, but quite often these children became high achievers. This ability also helps resilient children to maintain good health and to resist mental and physical illnesses. For example, many young children who are severely physically and/or emotionally bullied perform poorly in school, become aggressive or withdrawn, or depressed or anxious. Resilient children experience that same bullying and show no signs or symptoms that the experience has negatively impacted them.

Adolescence

During childhood, children begin to develop a sense of self and independence, and this process continues in the next stage of human development. During adolescence, young men and women are primarily concerned with finding their identity and expressing who they are in the world. Puberty causes many physical changes to take place, and adolescents must adapt to their changing bodies. All of these changes can make adolescence a confusing and stressful period. As adolescents try to find their place, they may experiment with different roles and make attempts to separate from authority figures. They are getting used to their bodies and trying to find out where they belong. They may try out different hairstyles and hobbies in an attempt to create an image of themselves they're comfortable with.

Social and Emotional Changes in Adolescence

Adolescence is a time of big social and emotional development for your child. It helps to know what to expect and how to support your child through the changes.

Social Changes and Emotional Changes: What to Expect in Adolescence

During adolescence, you'll notice changes in the way your child interacts with family, friends and peers. Every teen's social and emotional development is different. Your child's unique combination of genes, brain development, environment, experiences with family and friends, and community and culture shape development.

Social changes and emotional changes show that your child is forming an independent identity and learning to be an adult.

Social Changes

You might notice that your teen is:

- Searching for identity: young people are busy working out who they are and where they fit in the world. This search can be influenced by gender, peer group, cultural background, media, school and family expectations
- Seeking more independence: this is likely to influence the decisions your child makes and the relationships your child has with family and friends
- Seeking more responsibility, both at home and at school
- Looking for new experiences: the nature of teenage brain development means that teenagers are likely to seek out new experiences and engage in more risk-taking behaviour. But they're still developing control over their impulses
- Thinking more about "right" and "wrong": your child will start developing a stronger individual set of values and morals. Teenagers also learn that they're responsible for their own actions, decisions and consequences. They question more things. Your words and actions shape your child's sense of ''right'' and "wrong"
- Influenced more by friends, especially when it comes to behaviour, sense of self and self-esteem

- Starting to develop and explore a sexual identity: your child might start to have romantic relationships or go on “dates”. These are not necessarily intimate relationships. For some young people, intimate or sexual relationships don’t occur until later on in life
- Communicating in different ways: the internet, cell phones and social media can significantly influence how your child communicates with friends and learns about the world.

Emotional Changes

You might notice that your teen:

- Shows strong feelings and intense emotions at different times. Moods might seem unpredictable. These emotional ups and downs can lead to increased conflict. Your child’s brain is still learning how to control and express emotions in a grown-up way
- Is more sensitive to your emotions: young people get better at reading and processing other people’s emotions as they get older. While they’re developing these skills, they can sometimes misread facial expressions or body language
- Is more self-conscious, especially about physical appearance and changes. Teenage self-esteem is often affected by appearance - or by how teenagers think they look. As they develop, teens might compare their bodies with those of friends and peers
- Goes through a “invincible” stage of thinking and acting as if nothing bad could happen to him. Your child’s decision-making skills are still developing, and your child is still learning about the consequences of actions.

Changes in Relationships

You might notice that your teen:

- Wants to spend less time with family and more time with friends
- Has more arguments with you: some conflict between parents and children during the teenage years is normal as teens seek more independence. It actually shows that your child is maturing. Conflict tends to peak in early adolescence. If you feel like you’re arguing with your child all the time, it might help to know that this isn’t likely to affect your long term relationship with your child
- Sees things differently from you: this isn’t because your child wants to upset you. It’s because your child is beginning to think more abstractly and to question different points of view. At the same time, some teens find it hard to understand the effects of their behaviour and comments on other people. These skills will develop with time.

Supporting Social and Emotional Development

Here are some ideas to help you support your teen's social and emotional development.

- Be a role model for forming and maintaining positive relationships with your friends, children, partner and colleagues. Your child will learn from observing relationships where there is respect, empathy and positive ways of resolving conflict.
- Get to know your child's friends, and make them welcome in your home. This will help you keep in touch with your child's social relationships. It also shows that you recognize how important your child's friends are to your child's sense of self.
- Listen to your child's feelings. If your child wants to talk, stop and give your child your full attention. If you're in the middle of something, make a specific time when you can listen.
- Be explicit and open about your feelings. In particular, tell your child how you feel when your child behaves in different ways. Be a role model for positive ways of dealing with difficult emotions and moods.
- Talk with your child about relationships, sex and sexuality. Look for "teachable moments" - those everyday times when you can easily bring up these issues. Focus on the non-physical. Teenagers are often self-conscious and anxious about their bodies and appearance. So reinforce the positive aspects of your child's social and emotional development.

Staying connected with your teen can be an important part of supporting your child's social and emotional development.

Children with Special Needs

It's normal for parents to worry that their child with disability won't make friends easily or be accepted into a peer group. It helps to remember that the rate of social and emotional development varies widely for young people.

Teens who miss a lot of school because of a physical or mental illness, or who have a visible physical disability, might find it harder to make and keep friendships.

This doesn't mean that friendships won't happen. There might be other ways for your child to form friendships, such as joining community groups and online networks.

Give your child lots of love and support at home. Boost confidence and self-esteem by focusing on your child's strengths and interests.

Human Nature

Human nature is the concept that there is a set of inherent distinguishing characteristics, including ways of thinking, feeling and acting, that all humans tend to have.

The questions of what causes these distinguishing characteristics of humanity and in turn how fixed human nature is, are amongst the oldest and most important questions in western philosophy. They have particularly important implications in ethics, politics and theology because human nature is seen as providing standards or norms that humans can use when judging how best to live either as individuals, or members of a community. The complex implications of such discussion are also often themes which are dealt with in art.

For example the so-called "nature versus nurture" debate is a specific contemporary discussion topic within this theme. The branches of contemporary science associated with the study of human nature include sociology, sociobiology and psychology, particularly evolutionary psychology and developmental psychology.

In pre-modern scientific understandings of nature, human nature is understood with reference to final and formal causes. Such understandings imply the existence of an ideal, "idea," or "form" of a human which exists independently of individual humans. This in turn is sometimes understood to imply the existence of a divine interest in human nature, and indeed the concepts of formal and final cause are today often associated with attempts to reconcile religious and scientific understandings of nature.

The existence of an invariable human nature is a subject of much historical debate, continuing into modern times. Against this idea of a fixed human nature, the relative malleability of man has been argued especially strongly in recent centuries-firstly by early modernists such as Thomas Hobbes and Jean-Jacques Rousseau, and since the mid-19th century, by thinkers such as Hegel, Marx, Nietzsche, Sartre, structuralists and postmodernists. Still more recent scientific perspectives such as behaviourism, determinism, and the chemical model within modern psychiatry and psychology, claim to be neutral regarding human nature. (As in all modern science they seek to explain without recourse to metaphysical causation.) They can be offered to explain its origins and underlying mechanisms, or to demonstrate capacities for change and diversity which would arguably violate the concept of a fixed human nature.

Socratic Philosophy

Philosophy in classical Greece is the ultimate origin of the western conception of the nature of a thing. The study of human nature itself originated, according to Aristotle at least, with Socrates, who turned philosophy from study of the heavens to study of the human things. Socrates is said to have studied the question of how a person should best live, but he left no written works. It is clear from the works of his students Plato and Xenophon, and also what was said by Aristotle about him, that Socrates was a rationalist and believed that an the best life and the life most suited to human nature involved reasoning. The Socratic school was the dominant surviving influence in philosophical discussion in the Middle Ages, amongst Islamic, Christian and Jewish philosophers. Plato's main surviving writings are dialogues wherein Scorates, or some other wise person, appears as a character discussing questions such as the question of how humans should best live. The human soul was said by Plato to have a divided nature, divided in a specifically human way. For example, in the Timaeus it consisted of a rational part, resident in the human head, a spirited tendency, resident in the heart, and an appetitive beast, resident in the belly and genitals. The proper function of the 'rational' was to keep the other parts of the soul.

Aristotle, Plato's most famous student made some of the most famous and influential statements about human nature. His approach and understanding overlaps with Plato's but he wrote treatises, and not dialogues. In his works, some clear statements about human nature are made:-

- Man is a conjugal animal, meaning an animal which is born to couple when an adult, thus building a household (*oikos*) and in more successful cases, a clan or small village still run upon patriarchal lines.
- Man is a political animal, meaning an animal with an innate propensity to develop complex communities the size of a city or town. As a political animal, in contrast to his family and clan life, man thrives in his rationality-most fully in the making of laws and traditions.
- Man is a mimetic animal. Man loves to use his imagination, and not only to make laws and run town councils.

For Aristotle, reason is not only what is most special about humanity compared to other animals, but it is also what we were meant to achieve at his or her best. Much of Aristotle's position concerning human nature is still influential today, but the particular teleological idea that humans are "meant" or intended to be something, has become much less popular in modern times.

For the Socratics, human nature, and all natures, are metaphysical concepts. Aristotle developed the standard presentation of this approach with his theory of four causes. Human nature is an example of a formal cause according to Aristotle.

Modernism

One of the defining changes occurring at the end of the Middle Ages, is the end of the dominance of Aristotelian philosophy, and its replacement by a new approach to the study of nature, including human nature. In this approach, all attempts at conjecture about formal and final causes was rejected as useless. Under this approach a laws of nature is any regular and predictable pattern in nature, not literally a law, and in the same way "human nature" becomes not a special metaphysical cause, but simply whatever can be said to be typical tendencies of humans.

Although this new realism applied to the study of human life from the beginning, for example in Machiavelli's works, the definitive argument for the final rejection of Aristotle was associated especially with Francis Bacon, whose prime examples of the new approach returned philosophy or science to its pre-Socratic focus upon non-human things. David Hume, some centuries later, explicitly compared the time between Bacon and himself, with the time between the first Greek philosophers such as Thales and Socrates. In fact, Thomas Hobbes, the generation following Bacon, already claimed to be the first to use the approach of Bacon and Descartes in the study of human things. His controversial works, such as *Leviathan*, were accused of political and religious inappropriateness, but were very influential upon writers such as John Locke, Jean-Jacques Rousseau, and David Hume. In them, Hobbes famously described humanity as matter in motion, just like machines, and he described man's natural state (without science and artifice) as one where life would be nasty, short and brutish.

Jean Jacques Rousseau, was a contemporary and acquaintance of Hume, writing before the French Revolution, and long before Darwin. He shocked Western Civilization with his Second Discourse by proposing, building upon Hobbes, that humans had once been solitary animals, and had learned to be political. The important point about this was the idea that human nature was not fixed, or at least not anywhere near the extent previously suggested by philosophers. Humans are political, and rational, and have language now, but originally they had none of these things.

Rousseau still saw himself as a student of nature, and did not deny the existence of a human nature, but it was only now to be defined in terms of the instinctive passions of the original irrational and amoral human, such as those associated with self preservation. This has been seen as breaking ground for shocking political developments of the 19th and 20th centuries, such as totalitarianism and brain washing.

He was an important influence upon Kant, Hegel and Marx, but he himself made it clear that he was partly developing ideas based on the description of the "state of nature" in the *Leviathan* by Thomas Hobbes.

David Hume in Britain whose major works include *A Treatise of Human Nature* (1739-1740), and the two volumes developed to present the same material in a more popular way, *An Enquiry Concerning Human Understanding* (1748) and An Enquiry Concerning the Principles of Morals (1751). As a major figure in British empiricism and the development of modern economics, his influence on contemporary thinking is large. Despite treating this subject in a similar way to Rousseau, developing from the "state of nature" idea in Hobbes and Locke, Hume is often treated as a political conservative, and was in any case much less influential in inspiring violent political revolution.

Like Rousseau, he did not reject the idea that there were basic consistencies in humanity which tended to cause what we think of as human nature. He wrote of "human nature" itself, meaning the social aspects of it as it is often conceived, as something which humans did not even necessarily have, but rather something developed during the lifetime of individuals. It could be destroyed, for example if people did not associate in just societies. Nevertheless, like Rousseau he did not reject the concept of human nature entirely and rejected what he called the "paradox of the sceptics" saying that no politician could have invented words like "'honourable' and 'shameful', 'lovely' and 'odious' 'noble' and 'despicable'", unless there was not some natural "original constitution of the mind".

Hume was controversial in his own time for his modernist approach, following the example of Francis Bacon and Thomas Hobbes, of avoiding even considering metaphysical explanations for any type of cause and effect. He was accused of being an atheist. Concerning human nature also, he wrote for example:

We needn't push our researches so far as to ask 'Why do we have humanity, *i.e.*, a fellow-feeling with others?' It's enough that we experience this as a force in human nature. Our examination of causes must stop somewhere

After Rousseau and Hume, the nature of philosophy and science changes, branching into different disciplines and approaches, and the study of human nature changes accordingly.

Human Factors

Human factors science or human factors technologies is a multi-disciplinary field incorporating contributions from psychology, engineering, industrial design, statistics, operations research and anthropometry.

It is a term that covers:

- The science of understanding the properties of human capability (Human Factors Science).
- The application of this understanding to the design, development and deployment of systems and services (Human Factors Engineering).
- The art of ensuring successful application of Human Factors Engineering to a programme (sometimes referred to as Human Factors Integration). It can also be called ergonomics.

In general, a *human factor* is a physical or cognitive property of an individual or social behaviour which is specific to humans and influences functioning of technological systems as well as human-environment equilibriums. In social interactions, the use of the term *human factor* stresses the social properties unique to or characteristic of humans.

Human factors involves the study of all aspects of the way humans relate to the world around them, with the aim of improving operational performance, safety, through life costs and/or adoption through improvement in the experience of the end user.

The terms *human factors* and *ergonomics* have only been widely used in recent times; the field's origin is in the design and use of aircraft during World War II to improve aviation safety. It was in reference to the psychologists and physiologists working at that time and the work that they were doing that the terms "applied psychology" and "ergonomics" were first coined. Work by Elias Porter, Ph.D. and others within the RAND Corporation after WWII extended these concepts. "As the thinking progressed, a new concept developed-that it was possible to view an organization such as an air-defence, man-machine system as a single organism and that it was possible to study the behaviour of such an organism. It was the climate for a breakthrough."

Specialisation within this field include cognitive ergonomics, usability, human computer/human machine interaction, and user experience engineering. New terms are being generated all the time. For instance, "user trial engineer" may refer to a human factors professional who specialises in user trials. Although the names change, human factors professionals share an underlying vision that through application of an understanding of human factors the design of equipment, systems and working methods will be improved, directly affecting people's lives for the better.

Human factors practitioners come from a variety of backgrounds, though predominantly they are psychologists (engineering, cognitive, perceptual, and experimental) and physiologists. Designers (industrial, interaction, and graphic), anthropologists, technical communication scholars and computer scientists also contribute. Though some practitioners enter the field of human factors from other disciplines, both M.S. and Ph.D. degrees in Human Factors Engineering are available from several universities worldwide.

The Formal History of American Human Factors Engineering

The formal history describes activities in known chronological order.

This can be divided into 5 markers:

Developments prior to World War I: Prior to WWI the only test of human to machine compatibility was that of trial and error. If the human functioned with the machine, he was accepted, if not he was rejected. There was a significant change in the concern for humans during the American civil war. The US patent office was concerned whether the mass produced uniforms and new weapons could be used by the infantry men. The next development was when the American inventor Simon Lake tested submarine operators for psychological factors, follow by the scientific study of the worker. This was an effort dedicated to improve the efficiency of humans in the work place. These studies were designed by F W Taylor. The next step was the derivation of formal time and motion study from the studies of Frank Gilbreth, Sr. and Lillian Gilbreth.

Developments during World War I: With the onset of WWI, more sophisticated equipment was developed. The inability of the personnel to use such systems led to an increase in interest in human capability. Earlier the focus of aviation psychology was on the aviator himself. But as time progressed the focus shifted onto the aircraft, in particular, the design of controls and displays, the effects of altitude and environmental factors on the pilot. The war saw the emergence of aeromedical research and the need for testing and measurement methods. Still, the war did not create a Human Factors Engineering (HFE) discipline, as such. The reasons attributed to this are that technology was not very advanced at the time and America's involvement in the war only lasting for 18 months.

Developments between World War I and World War II: This period saw relatively slow development in HFE. Although, studies on driver behaviour started gaining momentum during this period, as Henry Ford started providing millions of Americans with automobiles. Another major development during this period was the performance of aeromedical research. By the end of WWI, two aeronautical labs were established, one at Brooks Airforce Base, Texas and the other at Wright field outside of Dayton, Ohio. Many tests were conducted to determine which characteristic differentiated the successful pilots from the unsuccessful ones. During the early 1930s, Edwin Link developed the first flight simulator. The trend continued and more sophisticated simulators and test equipment were developed. Another significant development was in the civilian sector, where the effects of illumination on worker productivity were examined. This led to the identification of the 'Hawthorne Effect', which suggested that motivational factors could significantly influence human performance.

Developments during World War II: With the onset of the WW II, it was no longer possible to adopt the Tayloristic principle of matching individuals to preexisting jobs. Now the design of equipment had to take into account human limitations and take advantage of human capabilities. This change took time to come into place. There was a lot of research conducted to determine the human capabilities and limitations that had to be accomplished. A lot of this research took off where the aeromedical research between the wars had left off. An example of this is the study done by Fitts and Jones (1947), who studied the most effective configuration of control knobs to be used in aircraft cockpits. A lot of this research transcended into other equipment with the aim of making the controls and displays easier for the operators to use. After the war, the Army Air Force published 19 volumes summarizing what had been established from research during the war.

Developments after World War II: In the initial 20 years after the WW II, most activities were done by the founding fathers: Alphonse Chapanis, Paul Fitts, and Small. The beginning of cold war led to a major expansion of Defence supported research laboratories. Also, a lot of labs established during the war started expanding. Most of the research following the war was military sponsored. Large sums of money were granted to universities to conduct research. The scope of the research also broadened from small equipments to entire workstations and systems. Concurrently, a lot of opportunities started opening up in the civilian industry. The focus shifted from research to participation through advice to engineers in the design of equipment. After 1965, the period saw a maturation of the discipline. The field has expanded with the development of the computer and computer applications.

The Cycle of Human Factors

Human Factors involves the study of factors and development of tools that facilitate the achievement of these goals. In the most general sense, the three goals of human factors are accomplished through several procedures in the human factors cycle, which depicts the human operator (brain and body) and the system with which he or she is interacting. First it is necessary to diagnose or identify the problems and deficiencies in the human-system interaction of an existing system. After defining the problems there are five different approaches that can be used in order to implement the solution. These are as follows:

- *Equipment Design:* changes the nature of the physical equipment with which humans work.
- *Task Design:* focuses more on changing what operators do than on changing the devices they use. This may involve assigning part or all of tasks to other workers or to automated components.
- *Environmental Design:* implements changes, such as improved lighting, temperature control and reduced noise in the physical environment where the task is carried out.
- *Training the individuals:* better preparing the worker for the conditions that he or she will encounter in the job environment by teaching and practicing the necessary physical or mental skills.
- *Selection of individuals:* is a technique that recognizes the individual differences across humans in every physical and mental dimension that is relevant for good system performance. Such a performance can be optimized by selecting operators who possess the best profile of characteristics for the job.

Human Factors Science

Human factors are sets of human-specific physical, cognitive, or social properties which either may interact in a critical or dangerous manner with technological systems, the human natural environment, or human organizations, or they can be taken under consideration in the design of ergonomic human-user oriented equipment. The choice or identification of human factors usually depends on their possible negative or positive impact on the functioning of human-organizations and human-machine systems.

The Human-Machine Model

The simple human-machine model is a person interacting with a machine in some kind of environment. The person and machine are both modeled as information-processing devices, each with inputs, central processing, and outputs. The inputs of a person are the senses (*e.g.*, eyes, ears) and the outputs are effectors (*e.g.*, hands, voice). The inputs of a machine are input control devices (*e.g.*, keyboard, mouse) and the outputs are output display devices (*e.g.*, screen, auditory alerts). The environment can be characterized physically (*e.g.*, vibration, noise, zero-gravity), cognitively (*e.g.*, time pressure, uncertainty, risk), and/or organizationally (*e.g.*, organizational structure, job design). This provides a convenient way for organizing some of the major concerns of human engineering: the selection and design of machine displays and controls; the layout and design of workplaces; design for maintainability; and the design of the work environment.

Example: Driving an automobile is a familiar example of a simple man-machine system. In driving, the operator receives inputs from outside the vehicle (sounds and visual cues from traffic, obstructions, and signals) and from displays inside the vehicle (such as the speedometer, fuel indicator, and temperature gauge). The driver continually evaluates this information, decides on courses of action, and translates those decisions into actions upon the vehicle's controls—principally the accelerator, steering wheel, and brake. Finally, the driver is influenced by such environmental factors as noise, fumes, and temperature.

No matter how important it may be to match an individual operator to a machine, some of the most challenging and complex human problems arise in the design of large man-machine systems and in the integration of human operators into these systems. Examples of such large systems are a modern jet airliner, an automated post office, an industrial plant, a nuclear submarine, and a space vehicle launch and recovery system. In the design of such systems, human-factors engineers study, in addition to all the considerations previously mentioned, three factors: personnel, training, and operating procedures.

- *Personnel* are trained; that is, they are given appropriate information and skills required to operate and maintain the system. System design includes the development of training techniques and programmes and often extends to the design of training devices and training aids.
- *Instructions, operating procedures, and rules* set forth the duties of each operator in a system and specify how the system is to function. Tailoring operating rules to the requirements of the system and the people in it contributes greatly to safe, orderly, and efficient operations.

Behaviour Therapy

Behaviour therapy is a type of psychotherapy that focuses on changing undesirable behaviours. Behaviour therapy involves identifying objectionable, maladaptive behaviours and replacing them with healthier types of behaviour. This type of therapy is also referred to a behaviour modification therapy. Cognitive therapy focuses primarily on the thoughts and emotions that lead to certain behaviours, while behavioural therapy deals with changing and eliminating those unwanted behaviours.

However, some therapists practice a type of psychotherapy that focuses on both thoughts and behaviour. This type of treatment is called cognitive-behavioural therapy.

Behaviour therapy can be used to treat a wide range of psychological conditions including, but not limited to depression, Attention Deficit Disorder, Attention Deficit Hyperactive Disorder, Obsessive-Compulsive Disorder, and certain addictions. Behaviour therapy may also be used to treat insomnia, chronic fatigue, and phobic behaviour.

This type of therapy may require fewer treatment sessions than cognitive therapy. However, the length of therapeutic treatment varies with each individual patient. In some cases, behaviour therapy is used as a treatment for obesity. When used for obesity, behaviour therapy starts with analysing eating and activity patterns, as well as dieting methods and other habits. The therapist then uses information gained through such analysis to identify positive strategies for promoting weight loss, healthier eating habits, and a more positive self-image. Behaviour therapy typically begins with the analysis of a trained therapist. The therapist analyses the behaviours of the patient that cause stress, reduce the patient's quality of life, or otherwise have a negative impact on the life of the patient. Once this analysis is complete, the therapist chooses appropriate treatment techniques. Treatments can include such techniques as assertiveness training, desensitization, environment modification, and relaxation training.

The therapist may also use exposure and response prevention to work towards controlling the patient's actions. Other commonly used techniques include positive reinforcement, modeling, and social skills training. In some cases, paradoxical intention techniques may be used in behaviour therapy. This type of technique involves encouraging the patient to continue adverse behaviours temporarily. Therapists who use this technique report that it is useful in identifying and removing a wide range of undesirable behaviours.

Another technique commonly used in behaviour therapy is called aversive therapy. Aversive therapy involves associating maladaptive behaviours with unpleasant stimuli. In the past, electric shock therapy was commonly used as a type of aversive therapy. Today, however, many psychotherapy experts consider this method unethical.

Operant Conditioning

Operant conditioning is the use of a behaviour's antecedent and/or its consequence to influence the occurrence and form of behaviour. Operant conditioning is distinguished from classical conditioning in that operant conditioning deals with the modification of "voluntary behaviour" or operant behaviour. Operant behaviour "operates" on the environment and is maintained by its consequences, while classical conditioning deals with the conditioning of reflexive behaviours which are elicited by antecedent conditions. Behaviours conditioned via a classical conditioning procedure are not maintained by consequences.

Four Contexts of Operant Conditioning

Here the terms positive and negative are not used in their popular sense, but rather: positive refers to addition, and negative refers to subtraction. What is added or subtracted may be either reinforcement or punishment. Hence positive punishment is sometimes a confusing term, as it denotes the "addition" of a stimulus or increase in the intensity of a stimulus that is aversive.

The four procedures are:

(1) *Positive reinforcement*: occurs when a behaviour is followed by a stimulus that is appetitive or rewarding, increasing the frequency of that behaviour. In the Skinner box experiment, a stimulus such as food or sugar solution can be delivered when the rat engages in a target behaviour, such as pressing a lever.

(2) *Negative reinforcement*: occurs when a behaviour is followed by the removal of an aversive stimulus, thereby increasing that behaviour's frequency. In the Skinner box experiment, negative reinforcement can be a loud noise continuously sounding inside the rat's cage until it engages in the target behaviour, such as pressing a lever, upon which the loud noise is removed.

(3) *Positive punishment*: occurs when a behaviour is followed by a stimulus, such as introducing a shock or loud noise, resulting in a decrease in that behaviour.

(4) *Negative punishment*: occurs when a behaviour is followed by the removal of a stimulus, such as taking away a child's toy following an undesired behaviour, resulting in a decrease in that behaviour.

Also:

- Avoidance learning is a type of learning in which a certain behaviour results in the cessation of an aversive stimulus. For example, performing the behaviour of shielding one's eyes when in the sunlight will help avoid the aversive stimulation of having light in one's eyes.
- Extinction occurs when a behaviour that had previously been reinforced is no longer effective. In the Skinner box experiment, this is the rat pushing the lever and being rewarded with a food pellet several times, and then pushing the lever again and never receiving a food pellet again. Eventually the rat would cease pushing the lever.
- Noncontingent reinforcement refers to delivery of reinforcing stimuli regardless of the organism's behaviour. The idea is that the target behaviour decreases because it is no longer necessary to receive the reinforcement. This typically entails time-based delivery of stimuli identified as maintaining aberrant behaviour, which serves to decrease the rate of the target behaviour. As no measured behaviour is identified as being strengthened, there is controversy surrounding the use of the term noncontingent "reinforcement".
- Shaping is a form of operant conditioning in which the increasingly accurate approximations of a desired response are reinforced.
- Chaining is an instructional procedure which involves reinforcing individual responses occurring in a sequence to form a complex behaviour.

Chapter Eight

Theory of Intelligences

SINGLE, TWO FACTOR AND MULTIFACTOR THEORIES

The theory of multiple intelligences was proposed by Howard Gardner in 1983, to more accurately define the concept of intelligence and address whether methods which claim to measure intelligence (or aspects thereof) are truly scientific.

Gardner's theory argues that intelligence, particularly as it is traditionally defined, does not sufficiently encompass the wide variety of abilities humans display. In his conception, a child who masters multiplication easily is not necessarily more intelligent *overall* than a child who struggles to do so.

The second child may be stronger in another *kind* of intelligence, and therefore may best learn the given material through a different approach, may excel in a field outside of mathematics, or may even be looking through the multiplication learning process at a fundamentally deeper level that hides a potentially higher mathematical intelligence than in the one who memorizes the concept easily.

DEVELOPMENT OF THE THEORY

"Howard Gardner, multiple intelligences and education" by given field (child prodigies, autistic savants);

- Neurological evidence for areas of the brain that are specialized for particular capacities (often including studies of people who have suffered brain damage affecting a specific capacity);
- The evolutionary relevance of the various capacities;
- Psychometric studies; and
- Susceptibility to encoding in a symbol system or notation (*e.g.*, written language, musical notation, choreography).
- An identifiable core operation or set of operations.
- A distinctive development history, along with a definable set of 'end-state' performances.

Gardner originally identified seven core intelligences: linguistic, logical-mathematical, spatial, bodily-kinesthetic, musical, interpersonal and intrapersonal. In 1997 at the symposium "MIND 97" (Multiple Intelligences New Directions) he added an eighth, the "Naturalist" Intelligence. Investigation continues on whether there are Existentialist (existential) and Spiritualist (spiritual) Intelligences.

The theory has been widely criticized in the psychology and educational theory communities. The most common criticisms argue that Gardner's theory is based on his own intuition rather than empirical data and that the intelligences are just other names for talents or personality types.

Despite these criticisms, the theory has enjoyed a great deal of popularity amongst educators over the past twenty years. There are several schools which espouse MI as a pedagogy, and many individual teachers who incorporate some or all of the theory into their methodology. Many books and educational materials exist which explain the theory and how it may be applied to the classroom.

CREATIVITY AND INTELLIGENCE

There has been debate in the psychological literature about whether intelligence and creativity are part of the same process (the conjoint hypothesis) or represent distinct mental processes (the disjoint hypothesis). Evidence from attempts to look at correlations between intelligence and creativity from the 1950s onwards, by authors such as Barron, Guilford or Wallach and Kogan, regularly suggested that correlations between these concepts were low enough to justify treating them as distinct concepts.

Some researchers believe that creativity is the outcome of the same cognitive processes as intelligence, and is only judged as creativity in terms of its consequences, *i.e.*, when the outcome of cognitive processes happens to produce something novel, a view which Perkins has termed the "nothing special" hypothesis.

A very popular model is what has come to be known as "the threshold hypothesis", proposed by Ellis Paul Torrance, which holds that a high degree of intelligence appears to be a necessary but not sufficient condition for high creativity. This means that, in a general sample, there will be a positive correlation between creativity and intelligence, but this correlation will not be found if only a sample of the most highly intelligent people are assessed. Research into the threshold hypothesis, however, has produced mixed results ranging from enthusiastic support to refutation and rejection.

An alternative perspective, Renzulli's three-rings hypothesis, sees giftedness as based on both intelligence and creativity. More on both the threshold hypothesis and Renzulli's work can be found in O'Hara and Sternberg. Kinds of intelligent performance that matter in everyday life, such as social intelligence. Most of theories of multiple intelligences are relatively recent in origin, though Louis Thurstone proposed a theory of multiple "primary abilities" in the early 20th Century.

TRIARCHIC THEORY OF INTELLIGENCE

Robert Sternberg's triarchic theory of intelligence proposes three fundamental aspects of intelligence — analytic, creative, and practical — of which only the first is measured to any significant extent by mainstream tests. His investigations suggest the need for a balance between analytic intelligence, on the one hand, and creative and especially practical intelligence on the other.

EMOTIONAL INTELLIGENCE

Daniel Goleman and several other researchers have developed the concept of emotional intelligence and claim it is at least as "important" as more traditional sorts of intelligence. These theories grew from observations of human development and of brain injury victims who demonstrate an acute loss of a particular cognitive function — *e.g.*, the ability to think numerically, or the ability to understand written language — without showing any loss in other cognitive areas.

EMPIRICAL EVIDENCE

IQ proponents have pointed out that IQ's predictive validity has been repeatedly demonstrated, for example in predicting important non-academic outcomes such as job performance (see IQ), whereas the various multiple intelligence theories have little or no such support. Meanwhile, the relevance and even the existence of multiple intelligences have not been borne out when actually tested. A set of ability tests that do not correlate together would support the claim that multiple intelligences are independent of each other.

FACTORS AFFECTING INTELLIGENCE

Intelligence is an ill-defined, difficult to quantify concept. Accordingly, the IQ tests used to measure intelligence provide only approximations of the posited 'real' intelligence. In addition, a number of theoretically unrelated properties are known to correlate with IQ such as race, gender and height but since correlation does not imply causation the true relationship between these factors is uncertain. Factors affecting IQ may be divided into biological and environmental.

BIOLOGICAL

Evidence suggests that genetic variation has a significant impact on IQ, accounting for three fourths in adults. Despite the high heritability of IQ, few genes have been found to have a substantial effect on IQ, suggesting that IQ is the product of interaction between multiple genes.

Other biological factors correlating with IQ include ratio of brain weight to body weight and the volume and location of gray matter tissue in the brain. Because intelligence appears to be at least partly dependent on brain structure and the genes shaping brain development, it has been proposed that genetic engineering could be used to enhance the intelligence of animals, a process sometimes called biological uplift in science fiction.

Experiments on mice have demonstrated superior ability in learning and memory in various behavioural tasks.

ENVIRONMENTAL

Evidence suggests that family environmental factors may have an effect upon childhood IQ, accounting for up to a quarter of the variance. On the other hand, by late adolescence this correlation disappears, such that adoptive siblings are no more similar in IQ than strangers. Moreover, adoption studies indicate that, by adulthood, adoptive siblings are no more similar in IQ than strangers, while twins and full siblings show an IQ correlation.

Consequently, in the context of the nature versus nurture debate, the "nature" component appears to be much more important than the "nurture" component in explaining IQ variance in the general population. There are indications that, in middle age, intelligence is influenced by life style choices (*e.g.*, long working hours).

Cultural factors also play a role in intelligence. For example, on a sorting task to measure intelligence, Westerners tend to take a taxonomic approach while the Kpelle people take a more functional approach. For example, instead of grouping food and tools into separate categories, a Kpelle participant stated "the knife goes with the orange because it cuts it"

ETHICAL ISSUES

Since intelligence is susceptible to modification through the manipulation of environment, the ability to influence intelligence raises ethical issues. Transhumanist theorists study the possibilities and consequences of developing and using techniques to enhance human abilities and aptitudes, and ameliorate what it regards as undesirable and unnecessary aspects of the human condition; eugenics is a social philosophy which advocates the improvement of human hereditary traits through various forms of intervention.

The perception of eugenics has varied throughout history, from a social responsibility required of society, to an immoral, racist stance. Neuroethics considers the ethical, legal and social implications of neuroscience, and deals with issues such as difference between treating a human neurological disease and enhancing the human brain, and how wealth impacts access to neurotechnology. Neuroethical issues interact with the ethics of human genetic engineering.

IDENTIFICATION AND PROMOTION OF CREATIVITY

Creativity is the condition for innovation. Creativity without realization, an idea without implemention has no effect. Creativity and innovation are essential factors for companies to succeed in business. But how do you get to a genious idea and how can you proceed from there to innovation and finally to realization?

This forum shall activate your creative potential and shall give you incentives, information about techniques, methodologies, etc., and finally also a view at the human factor in this area. Creativity, innovation, and initiative are psychological processes that facilitate the transformation of individual work roles, teams, and organisations into desired future states. Therefore, the present paper focuses on potential research trends in this increasingly important area.

Specifically, we identify three substantive gaps reflecting the needs for greater process differentiation, concept integration, and cross-cultural analysis. First, potential differential antecedents of specific creativity or innovation phases have received insufficient attention. Second, the creativity and innovation research domain may benefit from an integration of recently developed proactivity concepts such as personal initiative and voice behaviour. Third, cross-cultural differences in values, motivational orientations, and leadership preferences may determine how creativity and innovation are enacted and cultivated across the globe. With respect to each of these future challenges, we provide suggestions for theoretical and empirical advancements and discuss potential practical and methodological developments.

THINKING—CONVERGENT AND DIVERGENT THINKING

THINKING STYLES

Hudson (1967) studied English schoolboys, and found that conventional measures of intelligence did not always do justice to their abilities. The tests gave credit for problem-solving which produced the "right" answer, but under-estimated creativity and unconventional approaches to problems. He concluded that there were two different forms of thinking or ability in play here:

- One he called "convergent" thinking, in which the person is good at bringing material from a variety of sources to bear on a problem, in such a way as to produce the "correct" answer. This kind of thinking is particularly appropriate in science, maths and technology.
- Because of the need for consistency and reliability, this is really the only form of thinking which standardised intelligence tests, (and even national exams) can test
- The other he termed "divergent" thinking. Here the student's skill is in broadly creative elaboration of ideas prompted by a stimulus, and is more suited to artistic pursuits and study in the humanities.
- In order to get at this kind of thinking, he devised open-ended tests, such as the "Uses of Objects" test.

CONVERGENT AND DIVERGENT PRODUCTION

Convergent and **divergent production** are the two types of human response to a set problem that were identified by J. P. Guilford. Guilford observed that most individuals display a preference for either convergent or divergent thinking. (Others observe that most people prefer a convergent closure). As opposed to say TRIZ or Lateral Thinking divergent thinking is **not about tools** for creativity or thinking, but a way of categorizing what can be observed.

DIVERGENT THINKING

According to Guilford College, divergent or synthetic thinking is the ability to draw on ideas from across disciplines and fields of enquiry to reach a deeper understanding of the world and one's place in it. There is a movement in education that maintains divergent thinking might create more resourceful students. Rather than presenting a series of problems for rote memorization or resolution, divergent thinking presents open-ended problems and encourages students to develop their own solutions to problems.

Divergent production is the creative generation of multiple answers to a set problem. For example, *find uses for 1 metre lengths of black cotton*.

CONVERGENT THINKING

Convergent thinking is oriented towards deriving the single best (or correct) answer to a clearly defined question. It emphasizes speed, accuracy, logic, and the like, and focuses on accumulating information, recognizing the familiar, reapplying set techniques, and preserving the already known. It is based on familiarity with what is already known (*i.e.*, knowledge), and is most effective in situations where a ready-made answer exists and needs simply to be recalled from stored information, or worked out from what is already known by applying conventional and logical search, recognition and decision-making strategies. (OWAIS)

USE AND IMPLEMENTATIONS

The terms Divergent/Convergent thinking is widely used to describe a creative development process. In some "schools" the model is even used for facilitation of meetings, etc. many facilitators and leaders thus seek to end any facilitation with convergent thinking.

CRITIC OF THE ANALYTIC/DIALECTIC APPROACH

While the observations made in psychology can be used to analyse the thinking of humans, such categories may also lead to oversimplifications and dialectic thinking. The systematic use of convergent thinking may well lead to what is known as Group think-thus one should probably combine systematic use with critical thinking. Categorizing thinkers as "divergent" or "convergent" may seem appropriate for the purpose of general analyzes.

GUILFORD'S 'STRUCTURE OF INTELLECT'

Structure of Intellect's System (SOI) is the basis of our academic/educational assessment. Based on J. P. Guilford's theory of multiple intelligence and developed by Mary Meeker for use in schools, SOI tests a range of abilities (intelligences) which are needed for success with academic studies as well as in the work place. Mary Meeker found that intelligence is not "fixed", it can be developed. Upon diagnosing strengths and weaknesses, SOI's follow-up programme builds potential abilities to enable success in school and in life. This programme has wide-ranging applications in reading readiness, academic assessment and remediation, and career counselling.

Structure of Intellect (SOI) is a theory of human intelligence that was developed from Dr. J.P. Guilford's work at the University of Southern California. A psychologist in the U.S. Air Force in mid 1900s, Guilford created his assessment tool to help the Air Force find pilots who would succeed in the field. Despite other types of screening tests such as those for IQ and aptitude, 30% of the trainees did not make it through their vocational training. Upon putting his theories to practice, the attrition rate dropped dramatically. Guildford's assessment was adapted from his theory of multiple intelligence, which proposed 150 different abilities, extrapolated from his three dimensional model.

Dr. Mary Meeker, at the time a student of Guilford's and a psychologist, saw the potential of Guilford's work for use in the educational field, and modified his model to become an assessment and remediation tool for students in the educational system. Her adapted version (SOI) also showed potential in career counselling and the development of foundational cognitive skills needed in the workplace.

The success of this model has proved itself. SOI is currently being used in schools and learning clinics in North America to diagnose and remediate learning disabilities in students, as well as for enrichment of gifted students. The programme is also implemented in employment training programmes and in business and industry.

Structure of Intellect's philosophy is that intelligence is not "fixed", as has been generally supposed. Intellectual abilities can be taught. With the advent of studies in the plasticity of the brain, researchers have brought this new awareness into the fields of neuroscience and education. IQ tests have traditionally been slanted towards measuring a narrow range of abilities. Structure of Intellect's test measures a wide range of abilities needed for academic success. Although the ability to pinpoint problem areas is highly valuable, the results would not be as effective without a corresponding system in place to remediate these areas. Structure of Intellect's diagnostic test leads directly to remediation by developing potential learning abilities.

GARDNER'S MULTIPLE INTELLIGENCE THEORY

The categories of intelligence proposed by Gardner are:

BODILY-KINESTHETIC

This area has to do with bodily movement. In theory, people who have Bodily-kinesthetic intelligence should learn better by involving muscular movement, *i.e.*, getting up and moving around into the learning experience, and are generally good at physical activities such as sports or dance. They may enjoy acting or performing, and in general they are good at building and making things. They often learn best by doing something physically, rather than reading or hearing about it. Those with strong bodily-kinesthetic intelligence seem to use what might be termed muscle memory-they remember things through their body such as verbal memory or images.

Careers that suit those with this intelligence include athletes, dancers, actors, surgeons, doctors, builders, and soldiers. Although these careers can be duplicated through virtual simulation they will not produce the actual physical learning that is needed in this intelligence.

INTERPERSONAL

This area has to do with interaction with others. In theory, people who have a high interpersonal intelligence tend to be extroverts, characterized by their sensitivity to others' moods, feelings, temperaments and motivations, and their ability to cooperate in order to work as part of a group. They communicate effectively and empathize easily with others, and may be either leaders or followers. They typically learn best by working with others and often enjoy discussion and debate. Careers that suit those with this intelligence include sales, politicians, managers, teachers, and social workers.

VERBAL-LINGUISTIC

This area has to do with words, spoken or written. People with high verbal-linguistic intelligence display a facility with words and languages. They are typically good at reading, writing, telling stories and memorizing words along with dates. They tend to learn best by reading, taking notes, listening to lectures, and discussion and debate. They are also frequently skilled at explaining, teaching and oration or persuasive speaking. Those with verbal-linguistic intelligence learn foreign languages very easily as they have high verbal memory and recall, and an ability to understand and manipulate syntax and structure.

This intelligence is highest in writers, lawyers, philosophers, journalists, politicians, poets, and teachers.

LOGICAL-MATHEMATICAL

This area has to do with logic, abstractions, reasoning, and numbers. While it is often assumed that those with this intelligence naturally excel in mathematics, chess, computer programming and other logical or numerical activities, a more accurate definition places emphasis on traditional mathematical ability and more reasoning capabilities, abstract patterns of recognition, scientific thinking and investigation, and the ability to perform complex calculations. It correlates strongly with traditional concepts of "intelligence" or IQ.

Many scientists, mathematicians, engineers, doctors and economists function in this type of intelligence.

NATURALISTIC

This area has to do with nature, nurturing and relating information to one's natural surroundings. This type of intelligence was not part of Gardner's original theory of Multiple Intelligences, but was added to the theory in 1997. Those with it are said to have greater sensitivity to nature and their place within it, the ability to nurture and grow things, and greater ease in caring for, taming and interacting with animals. They may also be able to discern changes in weather or similar fluctuations in their natural surroundings. Recognizing and classifying things are at the core of a naturalist. They must connect a new experience with prior knowledge to truly learn something new. "Naturalists" learn best when the subject involves collecting and analysing, or is closely related to something prominent in nature; they also don't enjoy learning unfamiliar or seemingly useless subjects with little or no connections to nature. It is advised that naturalistic learners would learn more through being outside or in a kinesthetic way.

The theory behind this intelligence is often criticized, much like the spiritual or existential intelligence, as it is seen by many as not indicative of an intelligence but rather an interest. Careers that suit those with this intelligence include scientists, naturalists, conservationists, gardeners and farmers.

INTRAPERSONAL

This area has to do with introspective and self-reflective capacities. Those who are strongest in this intelligence are typically introverts and prefer to work alone. They are usually highly self-aware and capable of understanding their own emotions, goals and motivations. They often have an affinity for thought-based pursuits such as philosophy. They learn best when allowed to concentrate on the subject by themselves. There is often a high level of perfectionism associated with this intelligence.

Careers that suit those with this intelligence include philosophers, psychologists, theologians, writers and scientists.

VISUAL-SPATIAL

This area has to do with vision and spatial judgement. People with strong visual-spatial intelligence are typically very good at visualizing and mentally manipulating objects. Those with strong spatial intelligence are often proficient at solving puzzles. They have a strong visual memory and are often artistically inclined. Those with visual-spatial intelligence also generally have a very good sense of direction and may also have very good hand-eye coordination, although this is normally seen as a characteristic of the bodily-kinesthetic intelligence.

Some critics point out the high correlation between the spatial and mathematical abilities, which seems to disprove the clear separation of the intelligences as Gardner theorized. Since solving a mathematical problem involves reassuringly manipulating symbols and numbers, spatial intelligence is involved in visually changing the reality. A thorough understanding of the two intelligences precludes this criticism, however, as the two intelligences do not precisely conform to the definitions of visual and mathematical abilities. Although they may share certain characteristics, they are easily distinguished by several factors, and there are many with strong logical-mathematical intelligence.

Careers that suit those with this intelligence include artists, engineers, and architects.

MUSICAL

This area has to do with rhythm, music, and hearing. Those who have a high level of musical-rhythmic intelligence display greater sensitivity to sounds, rhythms, tones, and music. They normally have good pitch and may even have absolute pitch, and are able to sing, play musical instruments, and compose music. Since there is a strong auditory component to this intelligence, those who are strongest in it may learn best via lecture. In addition, they will often use songs or rhythms to learn and memorize information, and may work best with music playing in the background.

Careers that suit those with this intelligence include instrumentalists, singers, conductors, disc-jockeys, orators, writers (to a certain extent) and composers and sales reps.

CONSTANCY OF IQ

INTELLIGENCE QUOTIENT

An **intelligence quotient** or **IQ** is a score derived from one of several different standardized tests attempting to measure intelligence. The term "IQ," borrowed from the German *Intelligenz-Quotient*, was coined by the German psychologist William Stern in 1912 as a proposed method of scoring early modern children's intelligence tests such as those developed by Alfred Binet and Theodore Simon in the early 20th Century. Although the term "IQ" is still in common use, the scoring of modern IQ tests such as the Wechsler Adult Intelligence Scale is now based on a projection of the subject's measured rank on the Gaussian bell curve with a centre value (average IQ) of 100, and a standard deviation of 15, although different tests may have different standard deviations. IQ scores have been shown to be associated with such factors as morbidity and mortality, parental social status, and to a substantial degree, parental IQ. While its inheritance has been investigated for nearly a century, controversy remains as to how much is inheritable, and the mechanisms of inheritance are still a matter of some debate.

IQ scores are used in many contexts: as predictors of educational achievement or special needs, by social scientists who study the distribution of IQ scores in populations and the relationships between IQ score and other variables, and as predictors of job performance and income.

The average IQ scores for many populations have been rising at an average rate of three points per decade since the early 20th century with most of the increase in the lower half of the IQ range: a phenomenon called the Flynn effect. It is disputed whether these changes in scores reflect real changes in intellectual abilities, or merely methodological problems with past or present testing.

IQ AND THE BRAIN

In 2004, Richard Haier, professor of psychology in the Department of Pediatrics and colleagues at University of California, Irvine and the University of New Mexico used MRI to obtain structural images of the brain in 47 normal adults who also took standard IQ tests. The study demonstrated that general human intelligence appears to be based on the volume and location of gray matter tissue in the brain also demonstrated that, of the brain's gray matter, only about 6 percent appeared to be related to IQ.

Many different sources of information have converged on the view that the frontal lobes are critical for fluid intelligence. Patients with damage to the frontal lobe are impaired on fluid intelligence tests (Duncan *et al.* 1995). The volume of frontal grey (Thompson *et al.* 2001) and white matter (Schoenemann *et al.* 2005) have also been associated with general intelligence. In addition, recent neuroimaging studies have limited this association to the lateral prefrontal cortex. Duncan and colleagues (2000) showed using Positron Emission Tomography that problem-solving tasks that correlated more highly with IQ also activate the lateral prefrontal cortex. More recently, Gray and colleagues (2003) used functional magnetic resonance imaging (fMRI) to show that those individuals that were more adept at resisting distraction on a demanding working memory task had both a higher IQ and increased prefrontal activity.

A study involving 307 children (age between six to nineteen) measuring the size of brain structures using magnetic resonance imaging (MRI) and measuring verbal and non-verbal abilities has been conducted (Shaw *et al.* 2006). The study has indicated that there is a relationship between IQ and the structure of the cortex—the characteristic change being the group with the superior IQ scores starts with thinner cortex in the early age then becomes thicker than average by the late teens.

There is "a highly significant association" between the CHRM2 gene and intelligence according to a 2006 Dutch family study. The study concluded that there was an association between the CHRM2 gene on chromosome 7 and Performance IQ, as measured by the Wechsler Adult Intelligence Scale-Revised. The Dutch family study used a sample of 667 individuals from 304 families. A similar association was found independently in the *Minnesota Twin and Family Study* (Comings *et al.* 2003) and by the Department of Psychiatry at the Washington University. Significant injuries isolated to one side of the brain, especially those occurring at a young age, may not significantly affect IQ.

Studies reach conflicting conclusions regarding the controversial idea that brain size correlates positively with IQ. Jensen and Reed claim no direct correlation exists in nonpathological subjects. A more recent meta-analysis suggests otherwise. An alternative approach has sought to link differences in neural plasticity with intelligence, and this view has recently received some empirical support.

Measurement of Intelligence

Schools and placement agencies have been interested in the subject of intelligence testing because of its implications for placement. Intelligence tests attempt to predict the individual's capacity to function generally in tasks that require scholastic aptitude. They measure the individual's ability to cope with situations requiring the exercise of mental processes and try to measure the ability to comprehend the situation, apply past experiences, and solve the problem presented.

Tests are designed on the premise that intelligence can not be measured directly but is inferred by the observation or evaluation of behaviour. Behaviour has been grouped at various chronological age levels, and decisions have been made as to what is intelligent behaviour for a given age level. The measurements are converted into norms for representative groups of a population, so that eventually individuals can be classified as having intelligence like a typical five year old or eight year old. Some tests, like the Kuhlmann-Anderson, used a median score in an attempt to avoid the averaging of scores in determining the norms. Most tests of intelligence place a premium on the ability to do abstract thinking through the use of symbols. Vocabulary also is of great importance in measuring intelligence. Many psychologists believe that as a child matures his vocabulary is the best single indicator of intelligence as related to scholastic aptitude. This measurement of vocabulary can be made in an individual test, such as the Stanford-Binet or the Wechsler Intelligence Scale for Children, or in group testing. For a long time, test makers have been concerned with the problem of avoiding measuring only differences in cultural opportunities or education. An effective test item combines a criterion of novelty and universality. It is novel in the sense that it has not been encountered before by individuals taking the test; it is universal to the extent that all, at a certain level, may have had some experience related to it. The group intelligence test has frequently been shown to have a high relationship to the extent, level, and quality of the child's educational experience. When we speak of measuring intellectual potential or "native capacity," therefore, we need to be aware of the overlapping of hereditary and environmental factors.

The teacher and school administrator must be aware of some of the problems involved in the measurement of intelligence. We must not assume that tests with similar titles measure similar behaviours. We must always determine whether the group on which the norms for this test were based is comparable to the children we are testing. It is also pertinent to recognise that tests with dissimilar titles may actually measure the same thing. It is quite applicable in testing to ask, "What's in a name"" Intelligence testing with children must hold to a definite concept about what the test items are seeking to predict.

Psychologists engaged in test construction also need to recognise the cyclic nature of mental development. At certain times mental development proceeds more rapidly than at other times. Those who use intelligence tests in the schools should recognise that children with identical IQ's seldom function identically. One reason for this could very well be that they had acquired their mental-age points on different kinds of abilities and tasks.

Inhelder and Piaget also found differences in the modal age of children at a given stage on two different tasks. The age level for attainment of a given stage appears to be specific to a task rather than general, so that the child can operate at stage two on one task and at stage one on another.

The first individual intelligence test was the Stanford-Binet. Originally devised for use in dealing with a specific problem in France, it was later revised by Terman and then Terman and Merrill, and has been used extensively in the United States. The test items in this scale have been arranged in the order of empirically determined difficulty from the age of two upward. Items are always scored on a pass or fail basis. There are six test items at each half-year level from age two to five and six items at each yearly age level thereafter until the age of fourteen. Thereafter, the test provides for average and superior adult levels. The items have been arranged so that the normal child of a given age is expected to perform at his chronological-age level. Retarded children perform below the level, gifted children above.

Another frequently used individual intelligence test is the Wechsler Intelligence Scale for children (WISC), which the psychologist uses diagnostically to determine areas of strength and weakness. The Wechsler test, developed and originally copyrighted by David Wechsler in 1949, abandoned the concept of mental age originally introduced by Binet in 1908. The Wechsler test has instead worked on the basis of a deviation intelligence quotient. The IQ is obtained by comparing the child's test performance with the scores earned by individuals in a single age group, not with the composite age group.

This was done in an attempt to keep the standard deviation of IQ's identical from year to year, so that a child's obtained IQ does not vary unless his actual test performance as compared with his peers varies. The Wechsler test also permits development of a verbal, performance, and total or full-scale intelligence quotient. The verbal factors include information, comprehension, arithmetic, similarities, vocabulary, and digit span. The performance factors include picture completion, picture arrangement, block design, object assembly, coding, and mazes. Group intelligence tests are frequently used in the schools to make educational decisions. Close analysis reveals that many mental-ability tests being widely used are really strictly measures of classroom achievement, while others are quite independent of classroom instruction. The person who selects or uses tests needs to know whit the test actually measures. Individual deviations due to motivation, fatigue, distraction and the like may not be noted in a group-testing situation.

The following list includes some of the tests now being used in the public schools:

1. The California Test of Mental Maturity. Authors, E. T. Sullivan, W. W. Clark, end E. W. Tiegs. Publisher, California Test Bureau. Grades four to twelve.
2. Cattell Culture-fair Intelligence Test. Author, R. B. Cattell. Publisher, Bobbs-Merrill Co., Inc. School-learned skills not required. Grades three to twelve.
3. Differential Aptitude Tests. Authors, C. Bennett, H. Seashore, and A. Wesman. Publisher, Psychological Corporation. Items generally well-constructed and frequently used in grades eight to twelve to make educational and vocational decisions.
4. Lorge-Thorndike Intelligence Test. Authors, I. Lorge, R. L. Thorndike. Publisher, Houghton Mifflin Company. A promising instrument for predicting achievement in grade one. Series considered to be a well-constructed set of growth scales.
5. Science Research Associates Tests of General Ability (TOGA). Author, J. C. Flanagan. Publisher, Science Research Associates. Measures cultural understandings and reasoning with geometric forms.

6. Henmon-Nelson Tests of Mental Ability. Authors, T. A. Lanken, M. J. Nelson. Publisher, Houghton Mifflin Company. Places a. great emphasis on reading; scores tend to correlate well with teacher grades. Grades three to twelve.
7. Otis Quick Scoring Mental Ability Tests. Author, A. S. Otis. Publisher, Harcourt, Brace & World, Inc. Grades four to twelve. Places a great emphasis on school skills.
8. School and College Ability Tests (SCAT). Author, Educational Testing Service Staff. Publisher, Educational Testing Service. Grades four to twelve. A new test, which provides both a verbal and a quantitative score. Considered to be a good measure of school-learned abilities.
9. Kuhlman-Anderson Intelligence Tests. Authors, F. Kuhlman and R. G. Anderson. Publisher, Personnel Press. Grades three to twelve. Well-constructed and considered to be one of the better measures of growth available. Emphasises both reading and achievement skills.
10. S.R.A. Primary Ability Test. Grades four to twelve. Authors, L. L. Thurstone and T. C. Thurstone. Publisher, Science Research Associates. Grades four to twelve. Stresses reading, arithmetic arid reasoning abilities.

    Because the testing held is frequently under attack from critics of education, it is important that the teacher and the parent understand more adequately the school testing programme and some of the significant concepts related to it. Robert Bauernfeind's *Building a School Testing Programme* (Boston: Houghton Mifflin Company, 1963) is recommended to those interested in investigating testing problems further.

9. What are the crucial questions to ask in setting up a school testing programme to measure intelligence and achievement? Plan an elementary school testing programme to cover just these two areas. Defend your programme and the instruments selected in terms of your school's educational objectives.

Nature of Intelligence

Intelligence is the *manipulation of motor mechanisms* to derive happiness for the organism – inclusive of surviving – and entails the gathering of information external to the organism so by nonverbal communication and wise *motor manipulation*, it can maximize happiness. And, at root, all living organisms are nothing more than a physical organization of their genotype: the initial genetic blueprint of an organism.

Ultimately, at its root, the genotype is actually nothing more than immaterial energy configuring atoms and molecules into genetic code. As it is immaterial – as a thought or idea is immaterial – the genotype is more like a "thought" or "idea" interacting with the other "thoughts" and "ideas" and set to interact in harmony with its environment – also a "thought." Interaction is a *"thought being thought-out."* This idea is not new nor original. Aristotle theorized that God was pure Thought and prior to the Greeks, the Judeo-Christian idea of the Logos, and Egyptian idea of their God, Amun. The physical is not the blueprint – the immaterial electromagnetic is.

The immaterial is idea-like – a living *thought*. As thinking is mediated by plus and minus ions ($NaCl^-$, KCl^-, NA^+, K^+, Cl^-) moving in and out of the nerve fibres of our brain and nervous system, *thought* and *thinking* is the sophisticated summation of the flows and interplay of positive and negative energies. Both the medium for thinking (*i.e.*, the brain) and the ability of thinking itself exist in "seed-like" form in the genotype and is "species-specific."

The nature of the genotype is that it is a *hand-glove model* of its environment both in the sense that all features of a given animal's environment are modelled in the genotype's blueprint and in that the genotype creates the animal's body and behaviour to fit *hand-glove* into its unique environment. In essence, in tiny cells, genotypes taken as a whole – at some snapshot of geological time – serve as a perfect multimedia picture-record written, imprinted in genetic code, of the heavens, the skies, the waters, and the living earth.

Paul Davies, writing of the ability of birds to use stars in their astro-navigation, comments that, "The necessary astro-navigational information is built in DNA... given a sufficient understanding of the nature of DNA, one could 'decode' this information and reconstruct a map of the stars." DNA is built on an immaterial blueprint. In the case of you and I, we are, in essence, just the immaterial blueprint of the one-celled genotype that became us: eyes, brain, charming smile, wit, and great brilliance included.

Language ability is nothing more than the ability of the genotype to associate random sounds with contemporary objects of evolution (*i.e.*, today's world is different from the first human's world). Reading and writing is nothing less than the ability of the genome to devise and associate random signs (otherwise known as written alphabets, words, and languages) with contemporary objects of evolution such as the words "ice cream" and the "Internet." Rats evolve right along with us finding great comfort in living structures and types of food the prehistoric world had no imagination would ever exist. Insects have also kept pace with human evolution. Even viruses: computer viruses.

We are living witnesses of co-evolution. We can observe the ability of organisms worldwide to harmoniously co-evolve ever upwards: *e.g.*, in the microscopic-level arm's race, viral innovations against medicines force humans to develop better medicines.

Harmonious tit for tat co-evolution finds origins in the organizational marvel of the immaterial dimension. One unfortunate effect of imbalance is that it creates new "harmonious" balances and the imbalances the human has produced have eliminated many life forms from the environment that man is destroying in 200 years that took 2 million years to create. Nevertheless, for living creatures, harmony and balance with nature are written and achieved in genotypes and together, genotypes make the whole living picture of life on earth – each specie of genotype one piece of the puzzle fitting perfectly with the rest.

With the right equipment, the genotype could play an organism's life like a video in a VCR. Through life – via the interaction of living things – Nature manifests on yet another level, the balance and harmony found throughout the universe. Balance is found, for instance, in every animal having a strength of defence against others, but a weakness making it available as fodder and the strong have the greatest likelihood of surviving, but death takes an old life away to make way for new.

While from the perspective of the individual, intelligence serves the purpose of striving for, realizing, and maintaining individual happiness, taken as a whole intelligence serves the greater purpose of bringing the whole of the living world and nature into harmonious oneness. Let us remember several things: living things have not the foggiest idea of the technologies that make their organs and forms and make them interact harmoniously within the organism and with the whole environment from starlight to raindrops.

Differences in Intelligence

In this module, we will consider the extremes of intelligence— from the very bright to the very dull. We will also look at a few of the many studies that have been done with IQ scores, correlating many things from sense of humour (Brumbaugh, 1939) to racial background. A few of these correlation studies and their results will be described in this module. As you read the text, try to answer the following questions.

- When is a child considered gifted? Retarded?
- What are the degrees of mental retardation?
- What were Terman's findings in his study of gifted children?
- How does intelligence correlate with sex? Race? Socioeconomic level?

Fostering the Growth of Intelligence

Factors have been identified as beneficial to the growth and development of intelligence. The psychological climate that the child is raised in is one of the most significant. The child needs to feel accepted and valued as an individual. His ideas; thoughts, and even idiosyncrasies should be listened to, accepted, and discussed in a mature fashion. He should not be laughed at as immature because he has an idea that is different. Instead, a climate should be created in which having unique ideas and making contributions to family living are rewarded.

The implication is that the child must be encouraged to make choices, solve problems, and become independent in his thinking. This type of behaviour should always be reinforced and encouraged. It is also vital that the child be provided with materials and information which permit problem solving. He should be allowed to develop new uses for old tools, to see new relationships, to do creative writing, and to have access to a wide variety of experiences and hooks.

The family needs to provide a system for the regular recognition of achievement in the intellectual area. There should be an atmosphere which provides stimulation for all kinds of broad intellectual growth and which permits the child to work on his developmental problems and developmental tasks. He should be made aware of the alternatives and then encouraged to solve the problem. Adults need to have faith that children can learn from the consequences of their decisions. The growth of intelligence is dependent upon a stimulating atmosphere and a creative relationship with the significant adults both in the home and in the school.

Formulate some specific recommendations for the promotion of intellectual development during the preschool, primary, intermediate, and junior high years. What are the respective roles of the child, parent, and teacher in these plans?

Factors Influencing Intelligence

A natural endowment unfolds maturationally and sets limits on potential intellectual functions. This native intellectual endowment varies among individuals, and its maturation is hindered or assisted by the type of stimulation available in the environment during the early years. By comparing increments in mental and physical growth in curves of mental and physical development, Abernethy found that rates of physical and mental growth were essentially unrelated within the normal range of individual differences. He believed that the entire period of development is involved in determining the small positive correlations typically found between physical and mental growth during childhood.

Olson and Hughes found a general correlation of all aspects of child growth when plotted according to derived age values. Shuttleworth, who worked with data from the Harvard Growth Study, also found that both boys and girls with early maximum-growth ages in standing height proved to be more intelligent than children with late maximum-growth ages. His findings support the evidence of Abernethy, Olson, and Hughes that growth proceeds in patterned fashion.

DISTRIBUTION OF INTELLIGENCE

People who score above 130 on intelligence tests are usually considered gifted. An important study of gifted children was undertaken by who devised the Stanford-Binet test. In the course of developing that test, Terman and his associates tested many thousands of children, and then did further research on those who had lO's of 140 or more. This group of more than 1500 children was in the top 1% of the population. Terman followed most of these children from 1921 until his death in 1956. Some of his associates are still continuing this detailed longitudinal study.

The study revealed an important fact related to the home environments of the children. Approximately one-third were children of professional people and about half were children of persons in business occupations. Only seven percent came from "working classes." These percentages do not agree at ail with the distribution of social class membership in the general population. They indicate that relatively more gifted children come from the higher socioeconomic classes, although they do not tell us why.

Another discovery in Terman's study was that gifted children are more successful in later life. Twenty-five years after the study began, about 700 of the original group were still in contact with Terman. Of these, 150 were considered very successful. This distinction was based on such criteria as being listed in Who's Who or having received recognition for outstanding intellectual or professional achievement. Most of the others were much more successful than people of average intelligence. On the other hand, some persons in the gifted group had committed crimes, dropped out of school early, or been unsuccessful at a number of jobs.

Those least successful were more poorly adjusted emotionally and less motivated than the others. On the whole, though, the most striking fact revealed in the study was that gifted children made generally outstanding contributions to society in the form of intellectual achievements. In their early years, Terman's gifted children were above average in height, weight, and physical development, and they were also better adjusted. It had ben generally believed that very gifted children were likely to be maladjusted and socially backward. Terman's study definitely disproved this notion.

The gifted child is liable to encounter serious difficulties in school. He may be bored by children of his own age and his knowledge may even exceed that of his teacher. Teachers may consider him rude or a show-off. He may become a real problem simply because he is bored and loses interest in a class designed for "average" children. Some schools today are facing this problem by planning appropriate programmes for gifted children.

ASSESSMENT OF INTELLIGENCE

On the dawn of the second century of scientific-based mental testing, the intelligence is viewed as a diffuse notion with a great number of theories, attempting to explain it. There’s no universal definition of what intelligence is and that’s why each of them tend to explain it in its own way. A commonly accepted definition is by Boring (1923), stating that “intelligence is what is measured by intelligence tests”, or I would say, it is what they try to measure. What is important for this essay is that the subject measured by intelligence tests is a variety of cognitive abilities (semantic relations, general reasoning, sensorimotor functioning, spatial orientation, etc.), with the values behind the intelligence term tending to express the overall score.

THE USES OF INTELLIGENCE TESTS

LEWIS M. TERMAN (1916)

Intelligence tests of retarded school children. Numerous studies of the age-grade progress of school children have afforded convincing evidence of the magnitude and seriousness of the retardation problem. Statistics collected in hundreds of cities in the United States show that between a third and a half of the school children fail to progress through the grades at the expected rate; that from 10 to 15 per cent are retarded two years or more; and that from 5 to 8 per cent are retarded at least three years. More than 10 per cent of the $400,000,000 annually expended in the United States for school instruction is devoted to re-teaching children what they have already been taught but have failed to learn.

The first efforts at reform which resulted from these findings were based on the supposition that the evils which had been discovered could be remedied by the individualizing of instruction, by improved methods of promotion, by increased attention to children's health, and by other reforms in school administration. Although reforms along these lines have been productive of much good, they have nevertheless been in a measure disappointing. The trouble was, they were too often based upon the assumption that under the right conditions all children would be equally, or almost equally, capable of making satisfactory school progress.

Psychological studies of school children by means of standardized intelligence tests have shown that this supposition is not in accord with the facts. It has been found that children do not fall into two well-defined groups, the "feeble-minded" and the "normal."

Instead, there are many grades of intelligence, ranging from idiocy on the one hand to genius on the other. Among those classed as normal, vast individual differences have been found to exist in original mental endowment, differences which affect profoundly the capacity to profit from school instruction. We are beginning to realize that the school must take into account, more seriously than it has yet done, the existence and significance of these differences in endowment.

Instead of wasting energy in the vain attempt to hold mentally slow and defective children up to a level of progress which is normal to the average child, it will be wiser to take account of the inequalities of children in original endowment and to differentiate the course of study in such a way that each child will be allowed to progress at the rate which is normal to him, whether that rate be rapid or slow.

While we cannot hold all children to the same standard of school progress, we can at least prevent the kind of retardation which involves failure and the repetition of a school grade. It is well enough recognized that children do not enter with very much zest upon school work in which they have once failed. Failure crushes self-confidence and destroys the spirit of work. It is a sad fact that a large proportion of children in the schools are acquiring the habit of failure. The remedy, of course, is to measure out the work for each child in proportion to his mental ability.

Before an engineer constructs a railroad bridge or trestle, he studies the materials to be used, and learns by means of tests exactly the amount of strain per unit of size his materials will be able to withstand. He does not work empirically, and count upon patching up the mistakes which may later appear under the stress of actual use. The educational engineer should emulate this example. Tests and forethought must take the place of failure and patchwork. Our efforts have been too long directed by "trial and error." It is time to leave off guessing and to acquire a scientific knowledge of the material with which we have to deal. When instruction must be repeated, it means that the school, as well as the pupil, has failed.

Every child who fails in his school work or is in danger of failing should be given a mental examination. The examination takes less than one hour, and the result will contribute more to a real understanding of the case than anything else that could be done. It is necessary to determine whether a given child is unsuccessful in school because of poor native ability, or because of poor instruction, lack of interest, or some other removable cause.

It is not sufficient to establish any number of special classes, if they are to be made the dumping-ground for all kinds of troublesome cases — the feeble-minded, the physically defective, the merely backward, the truants, the incorrigibles, etc. Without scientific diagnosis and classification of these children the educational work of the special class must blunder along in the dark. In such diagnosis and classification our main reliance must always be in mental tests, properly used and properly interpreted.

Intelligence tests of the feeble-minded: Thus far intelligence tests have found their chief application in the identification and grading of the feeble-minded. Their value for this purpose is twofold. In the first place, it is necessary to ascertain the degree of defect before it is possible to decide intelligently upon either the content or the method of instruction suited to the training of the backward child.

In the second place, intelligence tests are rapidly extending our conception of "feeble-mindedness" to include milder degrees of defect than have generally been associated with this term. The earlier methods of diagnosis caused a majority of the higher grade defectives to be overlooked. Previous to the development of psychological methods the low-grade moron was about as high a type of defective as most physicians or even psychologists were able to identify as feeble-minded.

Wherever intelligence tests have been made in any considerable number in the schools, they have shown that not far from 2 per cent of the children enrolled have a grade of intelligence which, however long they live, will never develop beyond the level which is normal to the average child of 11 or 12 years. The large majority of these belong to the moron grade; that is, their mental development will stop somewhere between the 7-year and 12-year level of intelligence, more often between 9 and 12.

The more we learn about such children, the clearer it becomes that they must be looked upon as real defectives. They may be able to drag along to the fourth, fifth, or sixth grades, but even by the age of 16 or 18 years they are never able to cope successfully with the more abstract and difficult parts of the common-school course of study. They may master a certain amount of rote learning, such as that involved in reading and in the manipulation of number combinations, but they cannot be taught to meet new conditions effectively or to think, reason, and judge as normal persons do.

It is safe to predict that in the near future intelligence tests will bring tens of thousands of these high-grade defectives under the surveillance and protection of society. This will ultimately result in curtailing the reproduction of feeble-mindedness and in the elimination of an enormous amount of crime, pauperism, and industrial inefficiency. It is hardly necessary to emphasize that the high-grade cases, of the type now so frequently overlooked, are precisely the ones whose guardianship it is most important for the State to assume.

Intelligence tests of delinquents*:* One of the most important facts brought to light by the use of intelligence tests is the frequent association of delinquency and mental deficiency. Although it has long been recognized that the proportion of feeble-mindedness among offenders is rather large, the real amount has, until recently, been underestimated even by the most competent students of criminology.

UNDERSTANDING AND ENHANCING THE CREATIVE PROCESS WITH NEW TECHNOLOGIES

A simple but accurate review on this new Human-Computer Interactions (HCI) angle for promoting creativity has been written by Todd Lubart, an invitation full of creative ideas to develop further this new field.

Groupware and other Computer Supported Collaborative Work (CSCW) platforms are now the stage of Network Creativity on the web or on other private networks. These tools have made more obvious the existence of a more connective, cooperative and collective nature of creativity rather than the prevailing individual one.

Creativity Research on Global Virtual Teams is showing that the creative process is affected by the national identities, cognitive and conative profiles, anonymous interactions at times and many other factors affecting the teams members, depending on the early or later stages of the cooperative creative process. They are also showing how NGO's cross cultural virtual team's innovation in Africa would also benefit from the pooling of best global practices online. Such tools enhancing cooperative creativity may have a great impact on society and as such should be tested while they are built following the Motto : "Build the Camera while shooting the film". Some European FP7 scientific programmes like Paradiso are answering a need for advanced experimentally-driven research including large scale experimentation test-beds to discover the technical, societal, and economic implications of such groupware and collaborative tools to the Internet.

On the other hand Creativity research may one day be pooled with a computable Metalangguage like IEML from the University of Ottawa Collective Intelligence Chair, Pierre Levy. It might be a good tool to provide an interdisciplinary definition and a rather unified theory of creativity. The creative processes being highly fuzzy, the programming of cooperative tools for creativity and innovation should be adaptive and flexible. Empirical Modelling seems to be a good choice for Humanities Computing.

If all the activity of the universe could be traced with appropriate captors, it is likely that one could see the creative nature of the universe to which humans are active contributors. After the web of documents, the Web of Things might shed some light on such a universal creative phenomenon which should not be restricted to humans. In order to trace and enhance cooperative and collective creativity, Metis Reflexive Global Virtual Team has worked for the last few years on the development of a Trace Composer at the intersection of personal experience and social knowledge.

Metis Reflexive Team has also identified a paradigm for the study of creativity to bridge European theory of "useless" and non instrumentalized creativity, North American more pragmatic creativity and Chinese culture stressing more creativity as a holistic process of continuity rather than radical change and originality. This paradigm is mostly based on the work of the German philosopher Hans Joas, one that emphasizes the creative character of human action. This model allows also for a more comprehensive theory of action. Joas elaborates some implications of his model for theories of social movements and social change. The connexion between concepts like creation, innovation, production and expression is facilitated by the creativity of action as a metaphore but also as scientific concept.

The Creativity and Cognition conference series, sponsored by the ACM and running since 1993 has been an important venue for publishing research on the intersection between technology and creativity. The conference now runs biannually, next taking place in 2009.

The Process of Creativity-So what is creativity and what is the mechanism through which people actually create new ideas, solutions or concepts? According to many theorists, creativity is about chance or serendipity or making discoveries by 'accident'. So the creative process, according to this explanation is an 'accident'. This means that while you're trying out several methods, a best method or a solution to your problem arises out of nowhere and by chance you discover something totally unique. Some people would suggest that the creative process is more of trying to find out new relations between older known concepts so this is less about originality and more about 'experience'. The more experienced you are in a particular subject area, the more likely you are to consider creative solutions. Creativity has also been described as a moment of 'insight'. It is almost like enlightenment and divine intervention and a flash and the trick is to prolong this moment and creative individuals are people who can develop their sudden insights. So the creative process can be about a sudden chance, novel use of the knowledge/experience or a sudden insight. The creative process thus involves using several possibilities/methods and past experiences to arrive at sudden solutions through insights or accidents.

In 1926, Graham Wallas described stages of creativity in which a creative idea is first prepared, then internalized through incubation, after which the creative individual uses the illumination or insight to finally go through the verification process of applying the idea. Psychologist JP Guilford explained creativity with his concept of convergent and divergent thinking and convergent thinking is about trying to find the single correct solution to a problem and divergent thinking is the generation of multiple creative solutions to a problem. Creativity is thus characterized by divergent thinking and generation of multiple possibilities.

According to the Geneplore model developed by Finke, Ward and Smith (1992), creativity involves two phases-the generative phase in which the individual generates constructs from pre-inventive structures or known processes/ideas and the exploratory phase in which pre-inventive structures are interpreted to come up with new creative ideas. Most of these psychological theories seem to be emphasizing on preexisting mental structures through knowledge and experience and using these structures for novel or unique solutions.

The creative process is thus all about insight, 'a sudden flash', almost like a moment of realization and it has been described as serendipity or divine intervention by scientists and artists alike who have tried to described their moment of discovery, although the role of previous knowledge and experience is an equally important background factor. The scientists and artists are able to realize the potential of these 'flashes' and are able to recognize, capture and prolong their moments of insight for better realization of their creative goals.

WHY IS IT EASIER TO MEASURE INTELLIGENCE THAN CREATIVITY?

To be intelligent one has or shows understanding, is clever and quick of mind, whereas to be creative is to bring something into existence, to give rise to something or to originate. It seems easier to assess levels of intelligence, through the use of IQ tests than it is to assess levels of creativity as there is no such equivalent test.

IQ tests usually select an activity or skill and then design tests to measure individual performances in the specified activity. A high score will indicate a high level of performance and a low score will indicate a low level of performance. However it is difficult to measure creativity in such a way because the skills are harder to define as they are usually unique and diverse.

"Divergent thinking" tests of "creativity" were designed without any substantial validation. They differ from "convergent" tests of "intelligence" in that they are open ended and do not have right or wrong answers. There are also problems in validating creativity because it is very difficult to define what terms such as "giftedness" and "genius" mean.

There is an apparent contradiction between the predictability of objective tests and the unpredictability of creativity. Confusion arises with the measurement of general and specific intellectual skills as there is no clear idea about how they interact together in creativity.

The Life Cycle affects measurements too as IQ-related skills and knowledge improve with age until adulthood, whereas creativity seems to occur randomly at different ages. It seems impossible to measure creativity unless it depends on a non creative, intellectual skill. On the other hand intelligence is relatively easy to assess through IQ tests, which give clear indications of levels of performance.

MODERN THEORIES OF INTELLIGENCE

Early intelligence theory emerged from an emphasis on a unitary concept of general ability, as can be seen in the definitions of Binet and Spearman. Spearman created a statistical technique called factor analysis to explore his approach. From his studies with this technique, he was able to report that about half of the variance in tests of mental ability was due to the general *(g)* factor. The remainder was due to the special ability (*e.g.*, numerical reasoning, vocabulary, mechanical skill) that was required of a person to enable performance of the specific tasks on the test. Later approaches tended to emphasize expanded abilities. For example, Cattell divided g into fluid (*gf*) and crystallized (*gc*) intelligence. *Fluid intelligence* encompasses abilities involved in thinking, reasoning, and in learning, while *crystallized intelligence* represents the knowledge and broader understanding that has developed through learning in the environmental setting.

Other theories further recognized the diversity of intelligent performance. In his 1967 book Joy Paul Guil-ford (1897–1987), using factor analysis, devised a model of intelligence he termed the *Structure of Intellect* in which he proposed three aspects of intelligence: operations, products, and contents. Each of these is broken down further into specific kinds of intellectual activity which Guilford considered interrelated to produce intelligent functioning of specific tasks.

Gardner (1993) proposed a theory of multiple intelligences based on the differential cognitive processing required for demonstration of intelligent or creative performance in different areas. Gardner's theory references eight intelligences. Linguistic and logico-mathematical intelligences are most often associated with academic performance, although others could be relevant depending on the task. Other intelligences identified by Gardner were musical, kinesthetic, spatial, naturalistic, and personal (intrapersonal and interpersonal).

Robert Sternberg also proposed a theory informed by research from cognitive psychology. Sternberg's model is named the *triarchic theory* of intelligence because it is composed of three kinds of components: memory-analytic, creative-synthetic, and practical-contextual. The first component is related to the academic view of intelligence and is most similar to what most intelligence tests measure. The second component is necessary for creative endeavours, including traditionally academic areas such as science and mathematics. Finally, the practical-contextual is necessary for success in an everyday environment like school or business. Daniel Golman (1995) has proposed an emotional intelligence that bears some similarity to the personal intelligences of Gardner and the practical intelligence of Sternberg, but goes further in tying emotion and personality to the capacity for intelligent behaviour. For example, fear, excitement, or anger may contribute to how one behaves regardless of one's knowledge or capacity to reason.

In the first half of the 20th century, research in intelligence was heavily influenced by factor analysis (first used by Spearman). Factor analysis is a statistical procedure that enables the systematic study of the relationships within a set of variables in order to find the common aspects. Research and theory emerging in the second half of the 20th into the 21st century has begun to have an impact on the methods of assessment of intelligence. For example, research in cognitive psychology, neuropsychology, cognitive science, biopsychology and evolutionary processes, and cultural psychology has begun to affect theory and research in intelligence, its measurement and applications.

MEASUREMENT OF INTELLIGENCE IN EDUCATIONAL SETTINGS

Two kinds of intelligence tests will be presented here. The first, group tests, are used to identify the range of IQ scores in a group, usually for research or administrative purposes. Group tests tend to have more specific content and question formats, and are designed for more specific purposes than individually administered tests, which aim for a relatively more comprehensive clinical picture of the individual's cognitive functioning. The individual tests are designed to provide much more information about the individual. They are only used by professionals who are trained and licensed to administer and interpret these assessment instruments as part of a comprehensive clinical assessment that contributes information useful for planning educational or therapeutic interventions. These tests provide a variety of scores and clinical information that licensed professional psychologists may use to plan interventions in schools, family, or other settings. In other words, the usefulness of individual intelligence tests goes well beyond the scores on the test. For this reason, group tests are not suitable substitutes for individual assessments when planning clinical or educational interventions.

Group Tests of Intelligence. Two examples of group intelligence test. The Cognitive Abilities Test, and the Multidimensional Aptitude Battery, Second Edition (MAB-II) were chosen because they may be used in school or educational settings.

The Multidimensional Aptitude Battery-II (MAB-II) is a multiple-choice assessment of aptitude and intelligence that can be administered to groups or individuals above the age of 16. The instrument was designed to obtain scores similar to that of the individually administered Wechsler Adult Intelligence Scale-Revised (WAIS-R) in a group, as opposed to individual administration. The MAB-II produces composite scores: Verbal Scale (Information, Comprehension, Arithmetic, Similarities, Vocabulary), Performance Scale (Digit Symbol, Picture Completion, Spatial, Picture Arrangement, and Object Assembly) and Full Scale. The test can be pencil and paper or computer-administered.

The MAB-II is considered to be a useful tool for assessing cognitive abilities when large numbers of students must be screened. It is a well-developed and empirically sound instrument that provides correlations between subtest scores and occupational strengths. The test can be administered by a proctor, rather than by a post-master's-level professional. As a result, the MAB-II should not be used for making clinical diagnoses about intelligence. Additionally, MAB-II should not be administered to students with a learning disability related to reading comprehension or whose reading level is below ninth grade because the test relies on the test-takers' reading ability.

The Cognitive Abilities Test—Multilevel Edition, Form 6 (CogAT-6) is one of the more widely used group ability tests for students in kindergarten through 12th grades. The test is intended to guide instruction to match the cognitive abilities and needs of each student, to provide an "alternative" measure of cognitive development, and to identify achievement-ability discrepancies. The test has a multitheoretical foundation as it is based on Vernon's model of hierarchical abilities, Cattell's model of crystallized and fluid abilities, and Carroll's work specifically on general abstract reasoning. The CogAT-6 is composed of two editions: the Primary and Multilevel Editions, both of which are intended to assess reasoning and problem-solving abilities and can be broken down into Verbal, Non-verbal, and Quantitative Batteries. The Primary Edition is designed for students in K–2nd grade and consists of six subtests. The Multilevel Edition is a nine-subtest instrument that is based on the Lorge-Thorndike Intelligence Tests and is appropriate for students in 3rd to 12th grade. The CogAT-6 has a mean of 100 and standard deviation of 16.

DiPerna's 2005 review of the CogAT-6 suggested that the strengths of the test include a large, representative standardization sample, co-norming with the Iowa Tests, and a theoretical basis. In spite of these positive attributes, significant weaknesses were cited relative to the CogAT's purposes as described earlier. Criticisms included insufficient empirical evidence to support basing instructional recommendations on test results, a lack of reliability and predictive validity to measure cognitive ability and to predict cognitive ability-achievement discrepancies.

Individual Intelligence Tests. The following individual intelligence tests will be reviewed:

- Wechsler Intelligence Scale for Children-IV (WISC-IV)
- Woodcock-Johnson Tests of Cognitive Abilities-III (WJ COG III)
- Stanford-Binet Intelligence Scales, Fifth Edition (SB5)
- Das-Naglieri Cognitive Assessment System (CAS)
- Kaufman Assessment Battery for Children (K-ABC)

The Wechsler scales continue to be the most widely utilized individually administered tests of intelligence. The Wechsler tests are based on the *g* factor or the "overall capacity of the individual to act purposefully, to think rationally, and to deal effectively with the environment". The Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV) is the current revision of the Wechsler scales for children 6.0 to 16.11 years of age. The WISC-IV comprises 15 sub-tests (5 of which are supplemental).

Scores on these tasks contribute to the four composite indices (Verbal Comprehension, Perceptual Reasoning, Working Memory, and Processing Speed) in addition to a Full Scale IQ (FSIQ) score that can range from 40 (very low) to 160 (very high), with a mean of 100 representing the average score, and a standard deviation of 15. A profile of the test taker's learning strengths and weaknesses are derived from the test performance. The FSIQ is derived from a combination of all subtest scores and is considered the most representative estimate of global intellectual functioning. However, when large discrepancies between Index scores exist, the FSIQ can be invalid and misleading. In this case, it is most helpful to describe the strengths and weaknesses in the profile and de-emphasize the FSIQ.

The WISC-IV continues to be a reliable and valid instrument. WISC-IV scores can be interpreted in combination with Wechsler Individual Achievement Test, Second Edition (WIAT-II) scores for comparisons between ability and achievement. Flanagan and Kaufman (2004) provide an extensive description of the strengths and limitations of this assessment tool.

They cite the WISC-IV's most significant strengths as being "a robust four-factor structure across the age range of the test, increased developmental appropriateness, de-emphasis on time, improved psychometric properties, and an exemplary standardization sample." Flanagan and Kaufman (2004) indicate that none of the WISC-IV's limitations are very serious, although they suggest ways in which its validity could be improved. Other Weschler scales have been developed to assess IQ for young children and adults. The Wechsler Preschool and Primary Scale of Intelligence, Third Edition (WPPSI-III) (Harcourt Assessment, 2002) can be administered to children 2.6–7.3 years old and the Wechsler and the Wechsler Adult Intelligence Scale, Third Edition (WAIS-III) (Harcourt Assessment, 1997) is given to adults between 16 and 89 years of age.

The Woodcock-Johnson Tests of Cognitive Abilities-III (WJ COG III) is also widely used instrument in school settings. The WJ COG III is often administered as the examiner's second choice, if the age-appropriate Wechsler scale has been administered less than three years from the current assessment date. The WJ COG III is based on Catell, Horn, and Carroll's (CHC) concept of intelligence. The WJ COG III has been normed on individuals age 2 through 90+. The standard battery includes 10 tests, while the extended battery consists of 20. Based on these scores, the examinee earns three Cluster Scores (Verbal Ability, Thinking Ability, and Cognitive Efficiency) and a General Intellectual Ability (GIA) score. Test scores in the standard and extended batteries are not weighted equally when the GIA score is computed.

The WJ COG III has high technical quality and is based on a well-respected theory of cognitive abilities. It assesses cognitive abilities for a wide age range and uses sophisticated scoring procedures to calculate scores and discrepancies (Sattler, 2001). Additionally, ability and achievement discrepancy norms are provided as the WJ COG III was co-normed with the Woodcock Johnson Tests of Achievement, Third Edition (WJ ACH III). Greater evidence is necessary to understand the utility of the clinical clusters. The WJ COG III may also overestimate abilities if interpreted incorrectly. For example, the Written Language test score emphasizes one's ability to write brief sentences rather than to develop and organize paragraphs.

The Stanford-Binet Intelligence Scales, Fifth Edition (SB5), a direct descendent of Terman's adaptation of the Binet test developed more than 100 years ago, is used occasionally in the educational setting. The SB5 is based on the CHC theory of intelligence. It is designed for assessing intelligence and cognitive abilities among individuals between the ages of 2 and 85+. The SB5 consists of ten subtests and these scores are used to calculate four composite scores: factor, domain, abbreviated, and full scale (score range 40–160, mean = 100, SD = 15). The five factors measured include

Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Reasoning, and Working Memory, each with verbal and Non-verbal components. The SB5 contains two domain scales: Non-verbal IQ (NVIQ) and Verbal IQ (VIQ). An Abbreviated Battery IQ (ABIQ) can be determined with two routing subtest scores and the Full Scale IQ (FSIQ) is calculated using all 10 subtests.

The SB5 is advantageous as it is emphasizes both verbal and Non-verbal abilities. The instrument is technically sound, according to Johnson and D'Amato (2005), although Kush's 2005 study cites technical limitations (*e.g.*, lower stability for young children and individuals with low cognitive abilities, problematically high correlations with achievement, uncertain factor structure). In spite of these weaknesses, the SB5 is referred to as an outstanding measurement instrument for the assessment of cognitive abilities of children, adolescents, and adults.

The Das-Naglieri Cognitive Abilities System (CAS) is a tool based on Luria's cognitive processing model that is intermittently used to evaluate cognitive abilities in schools. The CAS is useful for assessing Planning, Attention, Simultaneous, and Successive (PASS) abilities among students between the ages of 5.0 and 17.11. The instrument's basic battery comprises 8 subtests and the standard battery consists of 12 subtests.

The CAS is an innovative instrument and its development meets high standards of technical adequacy. When compared to other individually administered general ability tests, it takes less time to administer. Additional empirical research must be completed to support the PASS construct. Factor analyses of subtests support both a 4-factor PASS model as well as a 3-factor model, which suggests that Planning and Attention may or may not be separate factors. The Kaufman Assessment Battery for Children, Second Edition (KABC-II) is occasionally utilized in schools as a culture-fair assessment of cognitive abilities for students between the ages of 3 and 18.

The KABC-II is based on a dual theoretical framework, Luria's neuropsychological model and the CHC approach. The authors suggest that its multitheoretical base allows the examiner to select which model is most appropriate for interpretation of results depending on the culture and/or verbal skills of the examinee. This assessment instrument contains 20 subtests that contribute to 4 scale scores (Sequential Processing, Simultaneous Processing, Planning, Learning, and Knowledge). The KABC-II has a mean of 100 and standard deviation of 15.

The KABC-II is an acceptable option for measuring cognitive abilities as it provides a reasonable, well-normed, clinically appealing, and technically sound approach to measuring cognitive abilities and generating diagnoses. Other strengths of the KABC-II include smaller score discrepancies between ethnic groups and the ability to compare ability and achievement differences with the Kaufman Test of Educational Achievement, Second Edition, Comprehensive Form. However, it has been criticized for the suggestion that examiners can select the model (Luria or CHC) by which to interpret results. Braden (2005) and Thorndike (2005) suggest that the interchangeability of theoretical models is inappropriate and illogical.

Intelligence testing should not be conducted in a vacuum. In addition to using previously mentioned norm-referenced tests, an assessment should include a variety of other data from a multitude of informants.

Sattler (2001) suggests that norm-referenced testing should be accompanied by interviews with a parent, teacher, and student; observations of the student during both the formal testing and natural environment (*e.g.*, classroom, lunchroom, playground); and informal assessment procedures (*e.g.*, district-wide criterion-referenced tests, school records). Such an assessment will provide the most accurate information by which educators can most effectively serve the student. Group versus Individual Intelligence Tests. Whereas group-administered IQ tests can be very well constructed (Aiken 2003) and can provide valid and reliable information suitable for certain non-clinical applications, they do not provide the same kind of information as individual tests and should not be used for the same purposes.

One reason is that group tests primarily emphasize multiple-choice question format, whereas individually administered tests provide a variety of response formats across the test.

This allows clinicians to gather considerable clinically useful information about the person being tested, such as the approaches used in problem solving, quality of verbal expression, and other observational data.

Group tests were not designed to be used for clinical purposes and therefore should not be utilized in clinical settings or to substitute for individually administered intelligence tests for designing clinical interventions or individual educational programmes (IEPs) in school settings.

Individual intelligence tests are designed to be administered by highly trained professionals who interpret data through the lens of learning, cognition, emotion, language, culture, health, and development. The one-on-one setting provides advantages to the test-taker and the examiner. First, it allows the examiner to develop rapport with the test-taker, which can benefit the shy or anxious test-taker. Second, the examiner has the opportunity to observe important test-taking behaviour such as impulsiveness, compulsiveness, confidence, anxiety, and wandering attention, which may vary according to the task and contribute to the interpretation of the performance on the test.

Third, the examiner can observe specific problem-solving approaches, which may also vary with the task. The integration of these observations with test scores allows the examiner to take a holistic approach in interpreting scores and developing interventions.

Chapter Nine

Psychology of Learning and Human Development

PSYCHOLOGICAL THEORY OF DEVELOPMENT

Many theoretical perspectives attempt to explain development, among the most prominent are: Jean Piaget's Stage Theory, Lev Vygotsky's Social Contextualism (and its heir, the Ecological Systems Theory of Urie Bronfenbrenner), and especially the information processing framework employed by cognitive psychology.

Historical theories continue to provide a basis for additional research, among them are Erik Erikson's eight stages of psychosocial development and John B. Watson's and B. F. Skinner's Behaviorism. Many other theories are prominent for their contributions to particular aspects of development. For example, Attachment theory describes kinds of interpersonal relationships and Lawrence Kohlberg describes stages in moral reasoning.

Ecological Systems Theory: Generally regarded as one of the world's leading scholars in the field of developmental psychology, Bronfenbrenner's primary contribution was his Ecological Systems Theory, in which he delineated four types of nested systems, with bi-directional influences within and between systems.

1. *Microsystem:* Immediate environments (family, school, peer group, neighbourhood, and childcare environments)

2. *Mesosystem:* A system comprised of connections between immediate environments (*i.e.*, a child's home and school)
3. *Exosystem:* External environmental settings which only indirectly affect development (such as parent's workplace)
4. *Macrosystem:* The larger cultural context (Eastern vs. Western culture, national economy, political culture, subculture)

Each system contains factors that can powerfully shape development, as can the interaction of factors across systems.

The major statement of this theory, The Ecology of Human Development (1979) had widespread influence on the way psychologists and others approached the study of human beings and their environments. It has been said that before Bronfenbrenner, child psychologists studied the child, sociologists examined the family, anthropologists the society, economists the economic framework of the times and political scientists the structure. As a result of Bronfenbrenner's groundbreaking work in "human ecology," a field that he created, these environments-from the family to economic and political structures-were viewed as part of the life course from childhood to adulthood.

Role of Experience: A significant question in developmental psychology is the relation between innateness and environmental influence in regard to any particular aspect of development. This is often referred to as "nature versus nurture" or nativism versus empiricism. A nativist account of development would argue that the processes in question are innate, that is, they are specified by the organism's genes.

An empiricist perspective would argue that those processes are acquired in interaction with the environment. Today developmental psychologists rarely take such extreme positions with regard to most aspects of development; rather they investigate, among many other things, the relationship between innate and environmental influences. One of the ways in which this relationship has been explored in recent years is through the emerging field of evolutionary developmental psychology. One area where this innateness debate has been prominently portrayed is in research on language acquisition. A major question in this area is whether or not certain properties of human language are specified genetically or can be acquired through learning. The nativist position argues that the input from language is too impoverished for infants and children to acquire the structure of language. Linguist Noam Chomsky asserts that, evidenced by the lack of sufficient information in the language input, there is a universal grammar that applies to all human languages and is pre-specified. This has led to the idea that there is a special cognitive module suited for learning language, often called the language acquisition device. The empiricist position on the issue of language acquisition suggests that the language input does provide the necessary information required for learning the structure of language and that infants acquire language through a process of statistical learning. From this perspective, language can be acquired via general learning methods that also apply to other aspects of development, such as perceptual learning. There is a great deal of evidence for components of both the nativist and empiricist position, and this is a hotly debated research topic in developmental psychology.

On the other hand, Chomsky's critique of a specific empiricist position on this issue, radical behaviourist Burrhus Frederic Skinner's Verbal Behaviour written in 1957, is widely considered among developmental psychologists to have sparked the decline in influence of behaviourism and signaled the beginning of the cognitive revolution in psychology.

Mechanisms of Development: Developmental psychology is concerned not only with describing the characteristics of psychological change over time, but also seeks to explain the principles and internal workings underlying these changes. Understanding these factors is aided by the use of models. Developmental models are often computational, but they do not necessarily need to be. A model must simply account for the means by which a process takes place. This is sometimes done in reference to changes in the brain that may correspond to changes in behaviour over the course of the development. Computational accounts of development often use either symbolic, connectionist (neural network), or dynamical systems models to explain the mechanisms of development.

LEARNING THEORY AS COGNITIVE DEVELOPMENT

Traditionally, learning theory has focused on forms of human change that were by definition somewhat peripheral to human development. Learning theory was about relatively rapid associations between stimuli or between responses and reinforcements. When stimuli, responses and consequences were all simple enough and when conditions were properly controlled, the fast-paced (by developmental standards) stimuli and responses displayed lawful properties and allowed for testable predictions about (short-term) behaviour change. The resulting changes were called "learning." Connections between stimuli were named associative learning or "classical" conditioning, and those between a response and its consequence were named operant conditioning. Those between a response and consequence that were merely observed by an individual were called Modelling, observational learning or "vicarious conditioning."

Assessed in terms of developmental theory and of SBTL, the problem with behaviourally oriented learning theory had three blind spots: it did not say enough about long-term change, about the experience of learning, or about how learning and behaviour became organized in the minds and lives of learners. Observational learning theory was an early effort to remedy the last two of the three problems (Bandura, 1977), and represented an initial turn towards what SBTL theorists eventually called meaning-making. In observational learning, a person somehow "sees" relationships between a relatively complex behaviour and its consequences. A student does not merely see a classmate elevate an arm and vocalize, for example, but sees the classmate "answer a teacher's question." If the teacher then tightens facial muscles, the student sees not merely muscular action, but sees the teacher "smile in response to the student." In observational learning theory, observed concrete behaviours become expressions of meanings perceived as happening in others (Bandura, 1995). This approach is still far from the thorny issues of interpretation discussed by social constructivists, but it is nonetheless a nod towards students' lived experience and personally constructed meanings.

Recent versions of learning theory have moved even more towards meaning-making than did Bandura and his colleagues. In one relatively radical formulation of learning theory, for example, conditioning has been interpreted as not about behaviour change per se, but about making meaning (DeGrandpre, 2000). In this view, linkages among overt stimulus cues, responses and consequences are merely cases of this broader process. Operant conditioning is not merely about reinforcing or strengthening specific responses as a result of specific consequences, but about creating perceived relationships among responses, the stimuli prior to responses, and the consequences that follow. "Reinforcement" of a behaviour is therefore not simply the consequences of a behaviour, as in earlier versions of learning theory, but the relationship among all elements of a learning trial-stimuli, responses, and consequences. If a teacher smiles pleasantly at a young child's interesting comment during a circle-time discussion, for example, what is reinforced is not just the child's tendency to make verbal comments in the future. What is reinforced is actually the child's tendency to associate or connect the teacher's smile, the child's own verbal contributions, and the setting or conditions when doing so is appropriate. To say that "the teacher reinforces the child with her smile," in fact is misleading because it implies more control on the part of the teacher than the teacher really has. She does contribute to the reinforcement-relationship by smiling, but so do the classroom circumstances and the child's own behaviour. All must happen together, and together they constitute reinforcement-and the creation of meaning.

Learning theory has also become more constructivist by developing more complex models of how children learn, and therefore more able to explain concepts and skills that truly matter in the lives of children and caregivers. An example is the "overlapping waves theory" proposed recently by Robert Siegler (1996). This theory makes three assumptions: 1) that children normally use a variety of strategies for solving problems, 2) that strategies can co-exist for long periods even when some are suboptimal or incompatible, and 3) that experience changes the relative salience of particular strategies, rather than eliminating any outright.

Changes in salience alter the likelihood of particular strategies actually appearing as behaviour. When graphed across time, relative salience rises and falls like a series of overlapping waves; hence the name of the model. Learning theory-or at least Siegler's overlapping waves version of learning theory-is about studying these changes in salience and their eventual expression as altered behaviour.

The result, compared to older models of learning, are predictions about learning that are more realistic. In actual classroom practice, a child is indeed likely to try a variety of strategies in ways that are often somewhat trial-and-error. A child may work two-digit subtraction problems, for example, by using "borrowing" algorithms officially recommended by the teacher, but also persist in counting on fingers some of the time, and at other times also try personalized (often bug-ridden) one-digit procedures that work only with particular subsets of problems.

Even when a child gains skill with a new strategy-in this case the “borrowing” algorithm-he or she still often regresses to earlier methods now and then, just as adults do. The child is sometimes also likely to spend time perfecting existing, suboptimal approaches rather than trying new ones-learning to count on fingers faster or more surreptitiously, for example-since doing so usually has short-term payoff. New, more optimal strategies appear both suddenly and intermittently when and if they finally do.

The relative importance of a child’s alternative subtraction strategies will gradually change with practice, but suboptimal ones will not totally disappear for quite a while, if ever. The results are patterns of behaviour that resemble Siegler’s “overlapping waves” model of learning (Siegler, 2000). The educational value of a more meaning-oriented version of learning theory can be seen in the recent professional writing of early educators. Discussions of behaviour problems in young children, for example, still draw heavily on concepts of applied behaviour analysis. Teachers are often still urged to frame behaviour problems in terms of operant conditioning (or as antecedents, behaviour, and consequences, which amount to the same idea), in order to influence a child’s unwanted behaviour.

But though this literature is quite behaviourist, it often is also concerned with the meaning of antecedents, behaviour, and consequences for the child: with the relationships she perceives among these elements, and with how a teacher might influence the perceptions in desirable directions. One result is that teachers are urged to alter not just consequences, but also the connections (or lack thereof) among antecedents, behaviour, and consequences.

A temper tantrum may be highly disruptive, for example, but the solution may not be simply to avoid paying attention to it (withholding reinforcement in the narrow sense). It may also be necessary to avoid connecting antecedents to the tantrum behaviours-avoid the triggers for tantrums in the first place. Not only is the broader strategy more likely to alter behaviour, it is also more likely to help a child “see” his tantrum in new, more acceptable ways-to alter, that is, its personal meaning. Much of the literature on behaviour management also implies a learning theory of “overlapping waves.” In discussing how to deal with a tantrum, for example, teachers are reminded that relapses are likely to occur, and that teachers should persevere with altering relationships among antecedents, responses, and consequences in spite of them. Expecting relapses implies that a child is guided by competing responses that are gradually shifting in relative importance or salience. Helping a child respond to frustration in consistently desirable ways is really a matter of developing a high ratio of desirable responses to undesirable ones, and thereby also making the desirable responses highly “meaningful” to the child in the sense of strengthening their connections to antecedents and consequences.

In addition to becoming more compatible with early childhood practice, such revisions to learning theory have had significant influences both on learning theory itself and on traditional meanings of human development, including cognitive development. The ideas about meaningful reinforcement described earlier bring learning theory closer to SBTLs in that it becomes more concerned than before about how learning is experienced by the child and how the child can take responsibility for "constructing" new, learned behaviours. The shifting, overlapping strategies described by Siegler, on the other hand, blur the conventional distinction between short-and long-term change which has been central to the idea of cognitive development. At the heart of the model is a process that is continuous, yet never complete, and in this sense it is neither short-nor long-term, but always both and always lifelong.

Mingling the short-and long-term is consistent with the professional needs of teachers and others with direct responsibilities for children. As professionals, we are fundamentally concerned with connecting the past and the future to a child's present: with helping children to relate previous experience and future intentions to current skills and plans.

Classroom practice can be interpreted, in fact, as an active effort to combine the short-term (the current activity) with the long-term (the child's prior skills and knowledge with the child's goals), even though doing so confuses the conventional distinction between immediate learning and long-term growth implied by the term cognitive development. In addition, of course, good teaching of the young integrates children's actions in other ways.

Teachers often attend to a child's unstated intuitions and fully conscious reasoning simultaneously; or they deal with a child's individual academic achievements and social relationships at the same time. In order to understand these concerns in parallel, integrated ways, even the revised versions of learning theory are not fully helpful, and early childhood educators must again turn elsewhere, such as to information processing theory or to SBTL itself. The first of these alternatives, information processing theory, and its relation to cognitive development and to early childhood education. Following this discussion, we can look directly at what SBTLs have to offer.

AVOIDANCE LEARNING

An individual's response to avoid an unpleasant or stressful situation; also known as escape learning. Avoidance learning is the process by which an individual learns a behaviour or response to avoid a stressful or unpleasant situation. The behaviour is to avoid, or to remove oneself from, the situation. Researchers have found avoidance behaviour challenging to explain, since the reinforcement for the behaviour is to not experience the negative reinforcer, or punishment. In other words, the reinforcement is the absence of punishment.

To explain this, psychologists have proposed two stages of learning: in stage one, the learner experiences classical conditioning; a warning, or stimulus, paired with a punishment. The learner develops a fear response when he experiences the stimulus. In stage two, the learner experiences operant conditioning; whereby he realizes that an action response to the stimulus eliminates the stressful outcome. In a common laboratory experiment conducted to demonstrate avoidance learning, a rat is placed in a confined space with an electrified floor. A warning signal is given, followed by an electric current passing through the floor. To avoid being shocked, the rat must find an escape, such as a pole to climb or a barrier to jump over onto a non-electric floor. At first, the rat responds only when the shock begins, but as the pattern is repeated, the rat learns to avoid the shock by responding to the warning signal. An example of avoidance learning in humans is the situation when a person avoids a yard where there is a barking dog. This learning is particularly strong in individuals who have been attacked by a dog.

Auditory Learning

Auditory learning is a learning style in which a person learns through listening. An auditory learner depends on hearing and speaking as a main way of learning. Auditory learners must be able to hear what is being said in order to understand and may have difficulty with instructions that are written. They also use their listening and repeating skills to sort through the information that is sent to them. The Fleming VAK/VARK model, one of the most common and widely used categorizations of the various types of learning styles, categorized the various types of learning styles as follows: visual learners, auditory learners, reading/writing-preference learners, and kinesthetic learners (also known as "tactile learners").

Characteristics

Auditory learners may have a knack for ascertaining the true meaning of someone's words by listening to audible signals like changes in tone. When memorizing a phone number, an auditory learner will say it out loud and then remember how it sounded to recall it. Auditory learners are good at writing responses to lectures they've heard. They're also good at oral exams, effectively by listening to information delivered orally, in lectures, speeches, and oral sessions.

Proponents claim that when an auditory/verbal learner reads, it is almost impossible for the learner to comprehend anything without sound in the background. In these situations, listening to music or having different sounds in the background (TV, people talking, etc.) will help learners work better. Auditory learners are good at storytelling. They solve problems by talking them through. Speech patterns include phrases "I hear you; That clicks; It's ringing a bell", and other sound or voice-oriented information. These learners will move their lips or talk to themselves to help accomplish tasks.

Recommended Techniques

Proponents say that teachers should use these techniques to instruct auditory learners: verbal direction, group discussions, verbal reinforcement, group activities, reading aloud, and putting information into a rhythmic pattern such as a rap, poem, or song.

Proponents recommend techniques like these to auditory learners:

- Record class notes and then listen to the recording, rather than reading notes.
- Remember details by trying to “hear” previous discussions.
- Participate in class discussions.
- Ask questions and volunteer in class.
- Read assignments out loud.
- Study by reading out your notes
- Whisper new information when alone.

An auditory learner may benefit by using the speech recognition tool available on many PCs.

LACK OF EVIDENCE

Although learning styles have “enormous popularity”, and both children and adults express personal preferences, there is no evidence that identifying a student’s learning style produces better outcomes, and there is significant evidence that the widely touted “meshing hypothesis” (that a student will learn best if taught in a method deemed appropriate for the student’s learning style) is invalid. Well-designed studies “flatly contradict the popular meshing hypothesis”. Rather than targeting instruction to the “right” learning style, students appear to benefit most from mixed modality presentations, for instance using both auditory and visual techniques for all students.

BEHAVIOURIST VIEWS OF LEARNING

Behaviourism is one of the most dominant among the modern theories of learning. The behaviourist school is very comprehensive and it includes a variety of thoughts. However, all these thoughts suggest a common approach to learning in terms of the development of connections in the organism between stimuli(S) and response(R). Based on laboratory experiments with animals, behaviourists concluded that learning is a process by which stimulus and response bonds are established when a successful response immediately and frequently follows a stimulus. They assumed that people are similar to machines, and considered any reference to the rule of mind irrelevant. Behaviourism holds that the subject matter of human psychology is the behaviour or activities of human beings. The behaviourists have put forward three main laws of learning: Law of Effect, Law of Readiness and Law of Exercise. The Law of Effect stresses the importance of the effect of a response. Satisfying results reinforce the response while annoying results weaken it. Reward and punishment are, therefore, important ingredients of learning. The law of readiness indicates the student’s willingness to make S-R connection while the law of exercise relates to strengthening the connection through practice.

Behaviourists consider learning a formation of habit through conditioning which links desired responses to stimuli. The prominent theorist among them is B.F. Skinner who propagated the idea of operant conditioning.

Educational Implications

The Behaviourist approach to learning has significantly influenced modern educational practices. Behaviourists have conceived teaching as manipulation of environment to produce desired behavioural changes in learners and thus make education more effective.

They suggest the adoption of the following three principles in the teaching-learning process:

- Knowledge of result and use of positive reinforcement,
- Minimum delay in reinforcement, and
- Elaboration of complex behaviour by dividing learning into a series of small steps.

One of the major contributions of behaviourists to education is their emphasis on defining *teaching objectives in behavioural terms*. They have stressed the need for stating objectives in the form of overt behaviour which can be observed and measured. The role of teachers becomes very crucial in deciding the changes of behaviour in their students when they learn and teaching in such a way that can students make attain those behavioural changes.

Behaviourist principles have influenced the contemporary approaches to evaluation also. For instance, based on the hierarchy of learning outcomes, Bloom has suggested a model of 'taxonomy of educational objectives'. Another educational implication of the behaviourist approach is individualising instruction such as 'Personalised System Instruction (PSI)' based on the reinforcement theory that has been widely used in education.

Skinner's Theory of Operant Conditioning

Skinner propagated the theory related to stimulus-response behaviour and reinforcement. In his view, learning is a change in behaviour. As the student learns, his responses in terms of changed behaviour increase. He therefore, formally defines learning as a change in the likelihood or probability of a response. The operant conditioning is a learning force which effects desired response more frequently by providing reinforcing stimulus immediately following the response. The most important principle of this type of learning is that behaviour changes according to its immediate consequences. Pleasurable consequences strengthen behaviour while unpleasant consequences weaken it.

In operant conditioning, learning objectives are divided into many small steps/ tasks and reinforced one by one for teaching purpose. The operant — the response behaviour of act — is strengthened so as to increase the probability of its reoccurrence in the future. Three external conditions — reinforcement, contiguity and practice — must be provided in operant conditioning.

Reinforcement

The most important aspect of Skinner's theory of learning relates to the role of reinforcement. An organism is presented with a particular stimulus — a reinforcer — after it makes a response. In given situation, the organism will tend to repeat responses for which it is reinforced. Skinner distinguished between positive and negative reinforcements. Positive reinforcement is a stimulus, which increases the probability of desired response.

The positive reinforcement is a positive reward. Praise, smiles, prize money, a funny television programme, etc., are the examples of positive reinforcement.

In negative reinforcement, the desired behaviour is more likely to occur if such stimulus reinforcement is removed. For example, we can close windows and doors to avoid hearing loud noise; we can avoid wrong answers by giving right answers. Here 'noise' and 'wrong answers' are negative reinforcers. Thus a negative reinforcer is a negative reward the avoidance of which gives us relief from unpleasant state of affairs. Skinner did not equate negative reinforcement with punishment.

Educational Implications

The basic implication of operant conditioning to teaching/instructional activities is dependency on observable behaviour. For Skinner, reinforcement facilitates learning. Further, he thinks that the most effective control on human learning requires instrumental aids/teaching aids.

Broadly, Skinner's theory has made the following contribution to the practice of education in teaching:

- *Teaching Machine*: Teaching machine, in the sense of a systematic approach to teaching with the help of machines, deserves attention as it has strongly influenced education both in theory and practice. In this approach, machines present the individual students with programmes containing a set of questions to be answered, problems to be solved, or exercises to be done. In addition, they provide automatic feedback to the students. Teaching through machines and electronic gadgets encourages students to take an active part in the instruction process. Use of mechanical teaching devices has the following advantages:
  - Right answer is immediately reinforced. Machines encourage and force the students to come up with right answers.
  - Mere manipulation of the machines probably, will reinforce sufficiently to keep an average student busy at a task for a prescribed period.
  - Any student who is forced to leave a learning activity for a period of time may return at any time and continue from where he left off.
  - Each student may proceed with his learning task on an individual basis at his own pace.
  - The teacher is forced to arrange and design the course content carefully in a hierarchical order.

- There is constant interaction between the teaching material and the student, thus sustaining activities.
- After knowing about the progress of the student, the teacher can supply necessary supplementary reinforcement. Thus, machines make it compulsory that a given material be thoroughly understood before the student moves on to the next set of material.

- *Programmed instruction*: Programmed instruction is a self-learning system in which the subject matter is broken into small bits of information and presented in a logical sequences. Each step builds deliberately upon the preceding one. A student progresses through the theme that is being taught through the programme. At the end of each step there is a question to be answered by the learner. After the question is answered, the learner is expected to check his/ her answer with the correct answer supplied in the programme. This is an inbuilt feature of programmed material.

COGNITIVE APPROACH

Cognitive approaches mainly deal with the psychological aspects of human behaviour. 'Cognitive psychology' has taken an important place in the psychology of learning over the last three decades. While conducting experimental investigations, cognitivism takes into consideration activities such as perception, concept formation, language use, thinking, understanding, problem solving, attention and memory. Thus, the cognitive approach is concerned with the individual's inner psychological functioning. Cognitive theorists have investigated and shown that people learn by perceiving, comprehending and conceptualising the problem. The comprehension of concepts and rules, etc., is transferable to the solution of new problems.

The cognitive theorists argue that people grasp things as a whole, and therefore, oppose the Behaviourist approach to teaching which employed drills to memorise the information. They believe, learning is both a question of 'insight' formation and successful problem solving, and not a mechanical sequence of stimuli and responses. Thus, teaching according to cognitivists, should encourage understanding based on 'problem solving' and 'insight formation'.

Information Processing

The contemporary cognitivists equate human mental activities with the process that goes on in a 'computer' in operation. They conceptualise human beings as information processing system. The information processing system describes a psychological activity in terms of information being received by the senses and then information items being selected and passed on to short-term memory where encoding processes transfer them to the long-term memory. Long term memory provides a store room where information can be retrieved in order to make a response.

There are a number of elements, which are central to the cognitive theory of learning. To begin with, the individual is seen as one having active relationship with the environment. He has intentions and goals, and thinks of alternative strategies to achieve these goals. Thinking is essentially a purposive activity. Learning is, therefore, an intelligent and active process. Within this process, issues of perception are very important because perceptual activity is the first relationship between a person and his environment or situation. The individual interacts with the situation and this interaction leads to relativity in perception as he organises the stimulus into meaningful patterns. Thus an individual acquires knowledge through his interaction with the environment and stores it for using this in new situations.

METHODS OF STUDYING PSYCHOLOGY OF LEARNING

Students at many times, when you have experienced an emotion like anger or fear you begin to think reasons for the state of yours. You say, "Why have I been annoyed over this or that? Why been afraid of such things" The analysis of your emotional state may take place simultaneously with the emotion or it may be done after the emotional state is over. In whatever manner it is done, it gives you an understanding, though rudimentary of your mind. This method of probing into your mental processes is a method of introduction utilized by psychologists in a much-refined manner. Let us see in detail what do we mean by Introduction and its merits and demerits. What we mean by introspection Introduction is a method of self-observation. The word 'Introspection is made up of two Latin words. "Intro" meaning within and "Aspection" meaning looking. Hence it is a method where an individual is looking within one self. Angel considered it as "looking inward". In Introduction the individual peeps into his own mental state and observes his own mental processes. Stout considers that 'to introspect is to attend to the working of one's own mind in a systematic way'.

Introspection method is one of the oldest methods to collect data about the conscious experiences of the subject. It is a process of self – examination where one perceives, analyses and reports one's own feelings. Let us learn this process with the help of an example, suppose you are happy and in the state of happiness you look within yourself. It is said you are introspecting your own mental feelings and examining what is going on in your mental process in the state of happiness. Similarly, you may introspect in state of anger or fear; etc. Introspection is also defined as the notice, which the mind takes of itself. Let us see the stages distinguished in introspection. Students there are three clear stages in introspection.

- During the observation of external object, the person beings to ponder over his own mental states. For example While listening to the music, which is to him pleasant or unpleasant he starts thinking about his own mental state.
- The person begins to question the working of the his own mind. He thinks and analyses: Why has he said such and such thing? Why as he talked in a particular manner? And so on.

- he tries to frame the laws and conditions of mental processes: He thinks in terms of improvement of his reasoning or the control of his emotional stages. This stage of that of the scientific methods for the advancement of our scientific knowledge.

Characteristics of Introspection

Introspection being self- observation has the following characteristics:

- The subject gets direct, immediate and intuitive knowledge about the mind.
- The subject has actually to observe his own mental processes. He cannot speculate about them.

Students, Introduction Method was widely used in the past. Its use in modern time is being questioned. It is considered unscientific and not in keeping with psychology which has recently emerged out as a positive science however we may say that it is still being used by psychologists and though its supremacy is undetermined, yet it is not totally discarded.

Merits of Introspection Method

- It is the cheapest and most economical method. We do not need any apparatus or laboratory for its use.
- This method can be used any time and anywhere you can introspect while walking, travelling, sitting on a bed and so on.
- It is the easiest method and is readily available to the individual.
- The introspection data are first hand as the person himself examines his own activities.
- Introspection has generated research which gradually led to the development of more objective methods.
- It is still used in all experimental investigation.
- It is the only method with the help of which and individual can know his emotions and feelings.

William James has pointed out the importance of this method in these words. "Introspective observation is what we have to rely on first and foremost and always. The word introspection can hardly be defined-it means, of course, looking into our own minds and reporting what we there discover. Everyone agrees that we there discover states of consciousness. So far as I know, the existence of such states has never been doubled by my critic, however skeptical in other respects we may have been."

Limitations of Introspection Methods

- In introspection, one needs to observe or examine one's mental processes carefully in the form of thoughts, feeling and sensation. The state of one's mental processes is continuously changing therefore when one concentrates on introspecting a particular phase of one's mental activity that phase passes off. For example when you get angry at something and afterwards sit down to introspect calmly the state of anger is sure to have passed off and so what you try to observe is not what is happening at that time with yourself but what had happened sometime before.

- The data collected by introspection cannot be verified. An individual may not pass through the same mental state again. There is no independent way of checking the data.
- The data collected by introspection lacks validity and reliability. It is impossible to acquire validity and exactness in selfobservation of one's own mental processes.
- The data collected by introspection in highly subjective. It has danger of being biased and influenced by preconceptions of the individual.
- The observer and the observed are the same. Hence there is ample scope fro the individual to lie deliberately and hide the facts to mislead.
- Introspection cannot be applied to children, animal and abnormal people. It requires highly trained and skilled workers to introspect.
- Introspection us logically defective because one and the same person is the experiencer and observer. It is not possible for the same individual to act as an experiencer as well as an observer. There introspection is logically defective.

OBSERVATION METHOD

Student we observe so many things in nature. We also observe the action and behaviour of others and form our own notions about these person. We look at other persons, listen to their talks and try to infer what they mean. We try to infer the characteristics, motivations, feelings and intentions of others on the basis of these observations.

So let us study about Observation method employed by psychologists in detail. With the development of psychology as an objective science of learning behaviour, the method of introspection was replaced by careful observation of human and animal behaviour to collect data by research workers. In introspection we can observe the mental process of ourselves only, but in observation, we observe the mental processes of others. Hence Observation is the most commonly used for the study of human behaviour.

Meaning of Observation

Observation literally means looking outside oneself. Facts are collected by observing overt behaviour of the individual in order to locate underlying problem and to study developmental trends of different types. The overt behaviour is the manifestation of court conditions within the individual. The study of overt behaviour gives indirectly the clue to the mental condition of the individual. Observation means 'perceiving the behaviour as it is' In the words of Goods, "Observation deals with the overt behaviour of persons in appropriate situations."

Observation has been defined as "Measurements without instruments." For example students in classroom have been labelled as good, fair or poor in achievement and lazy or diligent in study, etc. on the basis of observation, observation is indirect approach to study the mental processes of others through observing their external behaviour. For example if someone frowns, howls, grinds his teeth, closes his fists, you would say that the person is angry by only observing these external signs of his behaviour. Students in the process of observation, following four steps are generally required.

Observation of Behaviour

The first step involved in the method of observation is directly perceiving or observing the behaviour of individuals under study. For example, if we want to observe the social behaviour of children we can observe it when they assemble and play.

Recording the Behaviour Observed

The observation should be carefully and immediately noted and recorded. Minimum time should be allowed to pass between happening and recording. It will make the observation more objective.

Analysis and Interpretation of Behaviour

When the notes of behaviour observed are completed, they are analysed objectively and scientifically in order to interpret the behaviour patterns.

Generalisation

On the basis of analysis and interpretation of the data collected with the help of observation method, it is possible to make certain generalization. Social –development and behaviour of children have been described by child Psychologists on the basis of generalization based upon analysis and interpretation of the data gathered through the observation method.

Types of Observation

Students you have just seen what is observation and how it is conducted.

Do you know there are different ways in which observation can be done, so let us see the different types of observation?

- Natural Observation: In natural observation we observe the specific behavioural characteristics of children in natural setting. Subject do not become conscious of the fact that their behaviour is being observed by someone.
- Participant – Observation: Here the observer becomes the part of the group, which he wants to observe. It discloses the minute and hidden facts.

- Non-Participant Observation: Here the observer observes in such a position, which is least disturbing to the subject under study, the specific behaviour is observed in natural setting without subjects geting conscious that they are observed by some one. Non-participant observation permits the use of recording instruments.
- Structure Observation: Here the observer in relevance sets up a form and categories in terms of which he wishes to analyse the problem. The observer always keeps in view
  - A frame of reference
  - Time units.
  - Limits of an act
- Unstructured Observation: This is also called as uncontrolled or free observation. It is mainly associated with participant observation in which the observer assumes the role of a member of the group to be observed. Here the individual is observed when he is in his class, playground or when he is moving about with his friends and class follows without knowing that he is being observed. Observation is very useful method to study child and his behaviour. Student's observation method, being commonly used method psychology has following merits:

Merits of Observation Method

- Being a record of actual behaviour of the child, it is more reliable and objective.
- It is an excellent source of information about what actually happens in classroom.
- It is a study of an individual in a natural situation and is therefore more useful than the restricted study in a test situation.
- The method can be used with children of all ages. Younger the child, the easiest it is to observe him. This method has been found very useful with shy children.
- It can be used in every situation, physical- activities, workshop and classroom situations as well.
- It is adaptable both to the individuals and the groups.

Although observation is regarded as an efficient method for psychological studies, students yet it suffers from the following drawbacks limitations:

Limitations

- There is great scope for personal prejudices and bias of the observer. The observers interest, values can distort observation.
- Records may not be written with hundred percent accuracy as the observations are recorded after the actions are observed. There is some time lag.

- The observer may get only a small sample of study behaviour. It is very difficult to observe everything that the student does or says. As far as possible observation should be made from several events.
- It reveals the overt behaviour only- behaviour that is expressed and not that is within.
- It lacks replicability as each natural situation can occur only once.

Students looking at the drawbacks an observation method has psychologists have suggested various guidelines to be followed for making good observation. So let us find out which are these essential guidelines for making good observation.

Essential Guidelines for Making Good Observation

- Observe one individual at a time. It is desirable to focus attention on just one individual at a time in order to collect comprehensive data.
- Have a specific criteria for making observations. The purpose of making observation should be clear to the observe before he or she begins to observe so that the essential characteristics or the behaviour of the person fulfilling the purpose can be noted.
- Observations should be made over a period of time. To have a real estimate of the true behaviour of a person it should be observed as frequently as possible. A single observation will not be sufficient to tell us that this is the characteristic of the individual.
- The observations should be made in differing and natural situations in natural settings to increase its validity. For example, a pupil's behaviour in the classroom may not be typical of him; therefore he should be observed in variety of settings to know the behaviour most typical of the person.
- Observe the pupil in the context of the total situation.
- The observed facts must be recorded instantly, that is just at the time of their occurrence otherwise the observer may forget some of the facts and the recording may not be accurate.
- It is better to have two or more observers.
- Observations should be made under favourable onditions. The observer should be in position to clearly observe what he or she is observing. There should not be any undue distraction or disturbances. One should also have an attitude free from any biases or prejudices against the individual being observed.
- Data from observations should be integrated with other data. While arriving at the final conclusion about the individual, one should put together all that we know about the individual from the other sources then we can give an integrated and comprehensive picture of the individual. These precautions must be borne in mind in order to have reliable observations.

Natural Process of Learning

Learning is a natural process of growth or change in a person which is manifested as new modes or patterns of behaviour. This change exhibits itself as a skill, a habit, an attitude an understanding, or as knowledge or an application. Learning is a relatively permanent change in behaviour and is the result of reinforced practice through the process of "stimulus and response". This definition of learning assumes that certain conditions in the environment bring about fundamental changes in our behaviour that persist for a long time.

The changes which result from learning are positive and active, not negative and inert. Learning is not directly observable but inferred from one's performance. We can infer that a person has learnt something when he does something, which he could not do before. A person may know some thing and yet may not have learnt it. You may know how a computer works, but may not be able to operate it. Thus the distinction between learning or acquisition of knowledge and performance is an important one. We use the term 'behavioural tendency' to maintain the distinction between learning and performance. The relatively permanent change in behaviour refers to a change in performance.

We can also define learning in terms of cognitive development. Cognitivists say that learning is a change or reorganisation of cognitive structures, which involves acquisition and transformation of new knowledge. Thus we may conclude that learning is a change in knowledge, skills, attitudes and values brought about through experiences and this change in knowledge may or may not be expressed in overt behaviour. However, it may be pointed that all behaviours cannot be related to learning. Some behavioural changes are due to biological development or maturation. In maturation, growth developments are independent of specific learning conditions. A child starts walking once his/her legs are strong enough to support his/her weight.

The child is born with the potential to mature and at successive age levels, grasp and learn language, ways of behaving, attitudes, and values of his/her cultures he is born also with potential for reorganising and remoulding many aspects of his culture in harmony with changing conditions and needs.

Thus, the child is a product of culture as much as he is of biology. Through maturation and learning, the child acquires a culture. The process through which the child is taught the cultural ways that society accepts him to follow is termed as 'enculturation'. In this process, the child adjusts his innate biological characteristics to the prevailing cultural practices in society.

Principles of Learning

Over the years, educational psychologists have identified several principles which seem generally applicable to the learning process. They provide additional insight into what makes people learn most effectively.

Readiness

Individuals learn best when they are ready to learn, and they do not learn well if they see no reason for learning. Getting students ready to learn is usually the instructor's responsibility. If students have a strong purpose, a clear objective, and a definite reason for learning something, they make more progress than if they lack motivation. Readiness implies a degree of single-mindedness and eagerness. When students are ready to learn, they meet the instructor at least halfway, and this simplifies the instructor's job. Under certain circumstances, the instructor can do little, if anything, to inspire in students a readiness to learn. If outside responsibilities, interests, or worries weigh too heavily on their minds, if their schedules are overcrowded, or if their personal problems seem insoluble, students may have little interest in learning.

Exercise

The principle of exercise states that those things most often repeated are best remembered. It is the basis of drill and practice. The human memory is fallible. The mind can rarely retain, evaluate, and apply new concepts or practices after a single exposure. Students do not learn to weld during one shop period or to perform crosswise landings during one instructional flight. They learn by applying what they have been told and shown. Every time practice occurs, learning continues. The instructor must provide opportunities for students to practice and, at the same time, make sure that this process is directed towards a goal.

Effect

The principle of effect is based on the emotional reaction of the student. It states that learning is strengthened when accompanied by a pleasant or satisfying feeling, and that learning is weakened when associated with an unpleasant feeling. Experiences that produce feelings of defeat, frustration, anger, confusion, or futility are unpleasant for the student. If, for example, an instructor attempts to teach landings during the first flight, the student is likely to feel inferior and be frustrated.

Instructors should be cautious. Impressing students with the difficulty of an aircraft maintenance problem, flight manoeuvre or flight crew duty can make the teaching task difficult. Usually it is better to tell students that a problem or manoeuvre, although difficult, is within their capability to understand or perform. Whatever the learning situation, it should contain elements that affect the students positively and give them a feeling of satisfaction.

Primacy

Primacy, the state of being first, often creates a strong, almost unshakable, impression. For the instructor, this means that what is taught must be right the first time. For the student, it means that learning must be right. Unteaching is more difficult than teaching. If, for example, a maintenance student learns a faulty riveting technique, the instructor will have a difficult task correcting bad habits and reteaching correct ones. Every student should be started right. The first experience should be positive, functional, and lay the foundation for all that is to follow.

Intensity

A vivid, dramatic, or exciting learning experience teaches more than a routine or boring experience. A student is likely to gain greater understanding of slow flight and stalls by performing them rather than merely reading about them. The principle of intensity implies that a student will learn more from the real thing than from a substitute. In contrast to flight instruction and shop instruction, the classroom imposes limitations on the amount of realism that can be brought into teaching. The aviation instructor should use imagination in approaching reality as closely as possible. Today, classroom instruction can benefit from a wide variety of instructional aids to improve realism, motivate learning, and challenge students.

Recency

The principle of recency states that things most recently learned are best remembered. Conversely, the further a student is removed time-wise from a new fact or understanding, the more difficult it is to remember. It is easy, for example, for a student to recall a torque value used a few minutes earlier, but it is usually impossible to remember an unfamiliar one used a week earlier.

Instructors recognize the principle of recency when they carefully plan a summary for a ground school lesson, a shop period, or a postflight critique. The instructor repeats, restates, or reemphasizes important points at the end of a lesson to help the student remember them. The principle of recency often determines the sequence of lectures within a course of instruction.

How People Learn?

Perceptions: Perceiving involves more than the reception of stimuli from the five senses. Perceptions result when a person gives meaning to sensations. People base their actions on the way they believe things to be. The experienced aviation maintenance technician, for example, perceives an engine malfunction quite differently than does an inexperienced student. Real meaning comes only from within a person, even though the perceptions which evoke these meanings result from external stimuli. The meanings which are derived from perceptions are influenced not only by the individual's experience, but also by many other factors. Knowledge of the factors which affect the perceptual process is very important to the aviation instructor because perceptions are the basis of all learning.

Factors which Affect Perception

There are several factors that affect an individual's ability to perceive. Some are internal to each person and some are external.

- Physical organism
- Basic need
- Goals and values
- Self-concept
- Time and opportunity
- Element of threat.

Physical Organism

The physical organism provides individuals with the perceptual apparatus for sensing the world around them. Pilots, for example, must be able to see, hear, feel, and respond adequately while they are in the air. A person whose perceptual apparatus distorts reality is denied the right to fly at the time of the first medical examination.

Basic Need

A person's basic need is to maintain and enhance the organized self. The self is a person's past, present, and future combined; it is both physical and psychological. A person's most fundamental, pressing need is to preserve and perpetuate the self. All perceptions are affected by this need.

Just as the food one eats and the air one breathes become part of the physical self, so do the sights one sees and the sounds one hears become part of the psychological self. Psychologically, we are what we perceive. A person has physical barriers which keep out those things that would be damaging to the physical being, such as blinking at an arc weld or flinching from a hot iron. Likewise, a person has perceptual barriers that block those sights, sounds, and feelings which pose a psychological threat. Helping people learn requires finding ways to aid them in developing better perceptions in spite of their defence mechanisms. Since a person's basic need is to maintain and enhance the self, the instructor must recognize that anything that is asked of the student which may be interpreted by the student as imperilling the self will be resisted or denied. To teach effectively, it is necessary to work with this life force.

Goals and Values

Perceptions depend on one's goals and values. Every experience and sensation which is funnelled into one's central nervous system is coloured by the individual's own beliefs and value structures. Spectators at a ball game may see an infraction or foul differently depending on which team they support.

The precise kinds of commitments and philosophical outlooks which the student holds are important for the instructor to know, since this knowledge will assist in predicting how the student will interpret experiences and instructions. Goals are also a product of one's value structure. Those things which are more highly valued and cherished are pursued; those which are accorded less value and importance are not sought after.

Self-Concept

Self-concept is a powerful determinant in learning. A student's self-image, described in such terms as confident and insecure, has a great influence on the total perceptual process. If a student's experiences tend to support a favourable self-image, the student tends to remain receptive to subsequent experiences.

If a student has negative experiences which tend to contradict self-concept, there is a tendency to reject additional training. A negative self-concept inhibits the perceptual processes by introducing psychological barriers which tend to keep the student from perceiving. They may also inhibit the ability to properly implement that which is perceived. That is, self-concept affects the ability to actually perform or do things unfavourable. Students who view themselves positively, on the other hand, are less defensive and more receptive to new experiences, instructions, and demonstrations.

TIME AND OPPORTUNITY

It takes time and opportunity to perceive. Learning some things depends on other perceptions which have preceded these learnings, and on the availability of time to sense and relate these new things to the earlier perceptions. Thus, sequence and time are necessary. A student could probably stall an aeroplane on the first attempt, regardless of previous experience. Stalls cannot really be learned, however, unless some experience in normal flight has been acquired. Even with such experience, time and practice are needed to relate the new sensations and experiences associated with stalls in order to develop a perception of the stall. In general, lengthening an experience and increasing its frequency are the most obvious ways to speed up learning, although this is not always effective. Many factors, in addition to the length and frequency of training periods, affect the rate of learning. The effectiveness of the use of a properly planned training syllabus is proportional to the consideration it gives to the time and opportunity factor in perception.

Element of Threat

The element of threat does not promote effective learning. In fact, fear adversely affects perception by narrowing the perceptual field. Confronted with threat, students tend to limit their attention to the threatening object or condition. The field of vision is reduced, for example, when an individual is frightened and all the perceptual faculties are focused on the thing that has generated fear.

Flight instruction provides many clear examples of this. During the initial practice of steep turns, a student pilot may focus attention on the altimeter and completely disregard outside visual references. Anything an instructor does that is interpreted as threatening makes the student less able to accept the experience the instructor is trying to provide. It adversely affects all the student's physical, emotional, and mental faculties. Learning is a psychological process, not necessarily a logical one. Trying to frighten a student through threats of unsatisfactory reports or reprisals may seem logical, but is not effective psychologically. The effective instructor can organize teaching to fit the psychological needs of the student. If a situation seems overwhelming, the student feels unable to handle all of the factors involved, and a threat exists. So long as the student feels capable of coping with a situation, each new experience is viewed as a challenge.

A good instructor realizes that behaviour is directly influenced by the way a student perceives, and perception is affected by all of these factors. Therefore, it is important for the instructor to facilitate the learning process by avoiding any actions which may inhibit or prevent the attainment of teaching goals. Teaching is consistently effective only when those factors which influence perceptions are recognized and taken into account.

Insight

Insight involves the grouping of perceptions into meaningful whole. Creating insight is one of the instructor's major responsibilities. To ensure that this does occur, it is essential to keep each student constantly receptive to new experiences and to help the student realize the way each piece relates to all other pieces of the total pattern of the task to be learned.

As an example, during straight-and-level flight in an aeroplane with a fixed-pitch propeller, the RPM will increase when the throttle is opened and decrease when it is closed. On the other hand, RPM changes can also result from changes in airplane pitch attitude without changes in power setting. Obviously, engine speed, power setting, airspeed, and airplane attitude are all related. True learning requires an understanding of how each of these factors may affect all of the others and, at the same time, knowledge of how a change in any one of them may affect all of the others. This mental relating and grouping of associated perceptions is called insight.

Insight will almost always occur eventually, whether or not instruction is provided. For this reason, it is possible for a person to become an electrician by trial and error, just as one may become a lawyer by reading law. Instruction, however, speeds this learning process by teaching the relationship of perceptions as they occur, thus promoting the development of the student's insight.

As perceptions increase in number and are assembled by the student into larger blocks of learning, they develop insight. As a result, learning becomes more meaningful and more permanent. Forgetting is less of a problem when there are more anchor points for tying insights together. It is a major responsibility of the instructor to organize demonstrations and explanations, and to direct practice, so that the student has better opportunities to understand the interrelationship of the many kinds of experiences that have been perceived. Pointing out the relationships as they occur, providing a secure and nonthreatening environment in which to learn, and helping the student acquire and maintain a favourable self-concept are key steps in fostering the development of insight.

Motivation

Motivation is probably the dominant force which governs the student's progress and ability to learn. Motivation may be negative or positive, tangible or intangible, subtle and difficult to identify, or it may be obvious. Negative motivation may engender fear, and be perceived by the student as a threat. While negative motivation may be useful in certain situations, characteristically it is not as effective in promoting efficient learning as positive motivation.

Positive motivation is provided by the promise or achievement of rewards. These rewards may be personal or social; they may involve financial gain, satisfaction of the self-concept, or public recognition. Motivation which can be used to advantage by the instructor includes the desire for personal gain, the desire for personal comfort or security, the desire for group approval, and the achievement of a favourable self-image. The desire for personal gain, either the acquisition of possessions or status, is a basic motivational factor for all human endeavour. An individual may be motivated to dig a ditch or to design a supersonic airplane solely by the desire for financial gain.

Students are like typical employees in wanting a tangible return for their efforts. For motivation to be effective, students must believe that their efforts will be suitably rewarded. These rewards must be constantly apparent to the student during instruction, whether they are to be financial, self-esteem, or public recognition.

Lessons often have objectives which are not obvious at first. Although these lessons will pay dividends during later instruction, the student may not appreciate this fact. It is important for the instructor to make the student aware of those applications which are not immediately apparent. Likewise, the devotion of too much time and effort to drill and practice on operations which do not directly contribute to competent performance should be avoided. The desire for personal comfort and security is a form of motivation which instructors often forget. All students want secure, pleasant conditions and a safe environment. If they recognize that what they are learning may promote these objectives, their attention is easier to attract and hold. Insecure and unpleasant training situations inhibit learning.

Everyone wants to avoid pain and injury. Students normally are eager to learn operations or procedures which help prevent injury or loss of life. This is especially true when the student knows that the ability to make timely decisions, or to act correctly in an emergency, is based on sound principles. The attractive features of the activity to be learned also can be a strong motivational factor. Students are anxious to learn skills which may be used to their advantage. If they understand that each task will be useful in preparing for future activities, they will be more willing to pursue it. Another strong motivating force is group approval. Every person wants the approval of peers and superiors.

Interest can be stimulated and maintained by building on this natural desire. Most students enjoy the feeling of belonging to a group and are interested in accomplishment which will give them prestige among their fellow students. Every person seeks to establish a favourable self-image. In certain instances, this self-image may be submerged in feelings of insecurity or despondency. Fortunately, most people engaged in a task believe that success is possible under the right combination of circumstances and good fortune. This belief can be a powerful motivating force for students. An instructor can effectively foster this motivation by the introduction of perceptions which are solidly based on previously learned factual information that is easily recognized by the student. Each additional block of learning should help formulate insight which contributes to the ultimate training goals. This promotes student confidence in the overall training programme and, at the same time, helps the student develop a favourable self-image. As this confirmation progresses and confidence increases, advances will be more rapid and motivation will be strengthened.

Positive motivation is essential to true learning. Negative motivation in the form of reproofs or threats should be avoided with all but the most overconfident and impulsive students. Slumps in learning are often due to declining motivation. Motivation does not remain at a uniformly high level. It may be affected by outside influences, such as physical or mental disturbances or inadequate instruction. The instructor should strive to maintain motivation at the highest possible level. In addition, the instructor should be alert to detect and counter any lapses in motivation.

Levels of Learning

Levels of learning may be classified in any number of ways. Four basic levels have traditionally been included in aviation instructor training. The lowest level is the ability to repeat something which one has been taught, without understanding or being able to apply what has been learned. This is referred to as rote learning. Progressively higher levels of learning are understanding what has been taught, achieving the skill for application of what has been learned, and correlation of what has been learned with other things previously learned or subsequently encountered.

For example, a flight instructor may explain to a beginning student the procedure for entering a level, left turn. The procedure may include several steps such as: (1) visually clear the area, (2) add a slight amount of power to maintain airspeed, (3) apply aileron control pressure to the left, (4) add sufficient rudder pressure in the direction of the turn to avoid slipping and skidding, and (5) increase back pressure to maintain altitude. A student who can verbally repeat this instruction has learned the procedure by rote. This will not be very useful to the student if there is never an opportunity to make a turn in flight, or if the student has no knowledge of the function of airplane controls.

With proper instruction on the effect and use of the flight controls, and experience in controlling the airplane during straight-and-level flight, the student can consolidate these old and new perceptions into an insight on how to make a turn. At this point, the student has developed an understanding of the procedure for turning the airplane in flight. This understanding is basic to effective learning, but may not necessarily enable the student to make a correct turn on the first attempt.

When the student understands the procedure for entering a turn, has had turns demonstrated, and has practiced turn entries until consistency has been achieved, the student has developed the skill to apply what has been learned. This is a major level of learning, and one at which the instructor is too often willing to stop. Discontinuing instruction on turn entries at this point and directing subsequent instruction exclusively to other elements of piloting performance is characteristic of piecemeal instruction, which is usually inefficient. It violates the building block concept of instruction by failing to apply what has been learned to future learning tasks. The building block concept will be covered later in more detail.

The correlation level of learning, which should be the objective of aviation instruction, is that level at which the student becomes able to associate an element which has been learned with other segments or blocks of learning. The other segments may be items or skills previously learned, or new learning tasks to be undertaken in the future. The student who has achieved this level of learning in turn entries, for example, has developed the ability to correlate the elements of turn entries with the performance of chandelier and lazy eights.

Domains of Learning

Besides the four basic levels of learning, educational psychologists have developed several additional levels. These classifications consider what is to be learned. Is it knowledge only, a change in attitude, a physical skill, or a combination of knowledge and skill? One of the more useful categorizations of learning objectives includes three domains: cognitive domain (knowledge), affective domain (attitudes, beliefs, and values), and psychomotor domain (physical skills). Each of the domains has a hierarchy of educational objectives.

The listing of the hierarchy of objectives is often called a taxonomy. A taxonomy of educational objectives is a systematic classification scheme for sorting learning outcomes into the three broad categories (cognitive, affective, and psychomotor) and ranking the desired outcomes in a developmental hierarchy from least complex to most complex.

Theory of the Original Human Nature

The Theory of the Original Human Nature is a study concerning the image of what the original human being, but for the human fall, would have been like. Because of the human fall, human beings have lost their original condition. They have lost not only their original selves, but also their original world. As, a result, up to the present time they have been endeavouring, sometimes even unconsciously, to restore the original human self and the original world. This means that, throughout their entire lives, human beings harbour the idea of becoming better selves and the hope of living in a better world. Yet, human history has continued, even until today, without the ideal being realized. Fish swim freely in the water; birds fly in the sky as they please. But what would happen if they were taken out of their natural environment? If fish were taken out of the water and thrown on land, they would stiffer tremendous pain. They would desire to go back to the water, their natural habitat. Similarly, if a bird is caught and put in a cage, it feels restrained and longs to go back to the open sky.

In the same way, people ardently desire the realization of the ideal, and at the same time feel disappointed in the world as it actually is. This means that human beings have lost their original selves and the ideal world. Since that ideal, even until today, could not be realized many have lived in constant disappointment, under dire hardships and difficulties; but having no other choice, they could not but live in this world as it is. Some people, however, have never ceased to pursue the original human way of life, especially religious people and philosophers. They seriously grappled with the question, “What is the human being?” and looked for ways to recover the original way of life. Buddha, for example, spent six years of his life in strict monasticism and asceticism, engaging in deep meditation. As a result, he came to realize that human beings originally possessed Buddhahood, but through ignorance came to be bound by worldly desires and fell into suffering. Buddha taught that the way to recover one’s original nature is through a life of spiritual discipline. Jesus, likewise, enquired deeply into the problems of human life while travelling through many places, until he started his public ministry at the age of thirty. Consequently, he preached that human beings are sinners and have a satanic blood lineage resulting from the human fall, and that everyone must be born again by believing in the Son of God, that is, in Jesus himself

Socrates said, that the true way of human life is to love true knowledge. In Plato, the supreme ideal of human life is to recognize the idea of the Good. For Aristotle, reason is what makes a person human, and he said that virtue is best realized in the communal life of the polis (city-state) and that the human being is a social animal (or polis-animal). Greek philosophers, broadly speaking, held the view that reason is the essence of human nature, and that if a person’s reason is allowed to operate fully, that person will become an ideal being.

In the Middle Ages, Christian theology reigned over philosophy, and the Christian view of human nature was that human beings are sinful beings and can be saved only by God’s grace. In this view, reason was regarded as ineffective. In the modern period, however, currents of philosophy that believed in human reason again came to appear. Descartes considered human beings to be rational beings, but he believed that people incur error or become confused because they do not know how to make proper use of reason. Therefore, Descartes discussed the method of how to use reason properly. Kant claimed that human beings are personal beings that obey the voice of moral obligation ordered by practical reason, and that human beings are personal beings that obey the voice of moral obligation ordered by practical reason, and that human beings should live according to reason, without succumbing to temptations or desires. Hegel, too, regarded human beings as rational beings. Reason was something that would self-actualize in the world. Freedom, the essence of reason, was to be realized along with the development of history. According to Hegel's theory, human beings and the world should have become rational beings with the establishment of the modern state (*i.e.*, the national state). In reality, however, people have remained deprived of their human nature just as they always had, and the world has continued as irrational as it had been before.

Kierkegaard opposed extreme types of rationalism such as the one offered by Hegel. Kierkegaard did not agree that humankind would become increasingly rational as the world progresses, as Hegel had claimed. In actual society, lie said, human beings are nothing but average people, whose true nature has been lost. Accordingly, only when a person carves out life independently as an individual, apart from the general public, can that person's true human nature be regained. Thus, the conceptual framework for dealing with people in actual society, who have lost their original nature, and for seeking to restore human nature independently, was subsequently developed as the thought of Existentialism.

Feuerbach, in opposition to Hegel's rationalism, regarded the human being as a sensuous being. According to Feuerbach, human beings alienated from themselves their essence as a species, objectified it, and came to revere it as a god. Therein lay the loss of human nature, lie thought. Thus, Feuerbach asserted that human beings must recover their original human nature, and that this can only be done through denying religion. Departing from Hegel's idea of actualizing freedom, Karl Marx called for the true liberation of human beings. In the society of Marx's time, the lives of labourers were indeed miserable. They were forced to endure long hours of labour, and were given wages that barely could sustain their lives. Diseases and crime were rampant among labourers, who were deprived of their human nature. In contrast, the capitalists were living in great affluence gained from their merciless exploitation and oppression of labourers. But in Marx's view, the capitalist themselves were deprived of their own original human nature.

Determined to liberate humankind, Marx first started with Feuerbach's humanism as a way to restore human nature; later, however, he came to realize that human beings were not only species beings, but also beings engaged in productive activity—and this led him to the view that the essence of humankind is the freedom of labour in capitalist society, labourers were deprived of all the products of their labour, and they laboured not by their own will, but by the will of the capitalists. Therein, precisely, lay the labourers' loss of human nature-according to Marx.

From that, Marx concluded that in order to liberate the working class, what must be done is to overthrow capitalist society, where labourers are exploited. When such liberation occurred, the capitalists themselves would regain their own human nature, Marx thought. Furthermore, based on the materialist view, Marx concluded that human consciousness is determined by the relations of production, which are the basis of society, and that the economic system must be changed by force. Nevertheless, the Communist nations, in which revolutions took place in accordance with Marx's theory, have become dictatorial societies where freedom is suppressed and human nature is violated and neglected. Those are societies in which people have increasingly been losing their original nature. This implies that Marx was mistaken both in his grasp of the cause of human alienation and in his method for solving the problem of human alienation. Human alienation, however, is not a problem of Communist society alone. In capitalist society as well, individualism and materialism are rampant, and a self-centred way of thinking-whereby people think they are permitted to do anything they want-has become pervasive. As a result, in capitalist society, too, human nature is quickly being lost.

In this way, numerous religious people and philosophers have developed their own views of human nature, devoting great effort to the recovery of the original human nature; yet they have been unsuccessful in actually liberating humankind. The difficult problem for everyone has always been how to determine what the human being is. The Reverend Sun Myung Moon has trod his entire life course trying to provide fundamental solutions to such unresolved questions in human history. He has proclaimed that, originally, every human being is a child of God, even though, having lost their original nature, people have become miserable.

Human beings were created in the image of God, but due to the fall of the first human ancestors, they have become separated from God. They can restore their original nature, however, by living in accordance with God's word, thus coming to receive God's love. The problems of the human fall and the way to restore the original human nature will not be discussed (these topics are entrusted to Divine Principle); our focus will be on describing the original human nature itself.

INTRODUCTION TO LEARNING THEORY AND BEHAVIOURAL PSYCHOLOGY

Learning can be defined as the process leading to relatively permanent behavioural change or potential behavioural change. In other words, as we learn, we alter the way we perceive our environment, the way we interpret the incoming stimuli, and therefore the way we interact, or behave. John B. Watson (1878-1958) was the first to study how the process of learning affects our behaviour, and he formed the school of thought known as *Behaviourism*. The central idea behind behaviourism is that only observable behaviours are worthy of research since other abstraction such as a person's mood or thoughts are too subjective. This belief was dominant in psychological research in the United Stated for a good 50 years.

Perhaps the most well known Behaviourist is B. F. Skinner (1904-1990). Skinner followed much of Watson's research and findings, but believed that internal states could influence behaviour just as external stimuli. He is considered to be a *Radical Behaviourist* because of this belief, although nowadays it is believed that both internal and external stimuli influence our behaviour. Behavioural Psychology is basically interested in how our behaviour results from the stimuli both in the environment and within ourselves. They study, often in minute detail, the behaviours we exhibit while controlling for as many other variables as possible. Often a gruelling process, but results have helped us learn a great deal about our behaviours, the effect our environment has on us, how we learn new behaviours, and what motivates us to change or remain the same.

The 4 Factors That Form The Definition of Learning:

1. Learning is inferred from a change in behaviour/performance*
2. Learning results in an inferred change in memory
3. Learning is the result of experience
4. Learning is relatively permanent

This means that behaviour changes that are temporary or due to things like drugs, alcohol, etc., are not “learned”.

* Behaviour Potential - once something is learned, an organism can exhibit a behaviour that indicates learning as occurred. Thus, once a behaviour has been “learned”, it can be exhibited by “performance” of a corresponding behaviour.

It is the combination of these 4 factors that make our definition of learning. Or, you can go with a slightly less comprehensive definition that is offered in many text books: **Learning is a relatively durable change in behaviour or knowledge that is due to experience.**

We are going to discuss the two main types of learning examined by researchers, classical conditioning and operant conditioning.

Classical Conditioning

Classical Conditioning can be defined as a type of learning in which a stimulus acquires the capacity to evoke a **reflexive** response that was originally evoked by a different stimulus.

A. Ivan Pavlov - Russian physiologist interested in behaviour (digestion).

1. Pavlov was studying salivation in dogs - he was measuring the amount of salivation produced by the salivary glands of dogs by presenting them meat powder through a food dispenser.

   The dispenser would deliver the meat powder to which the animals salivated. However, what Pavlov noticed was that the food dispenser made a sound when delivering the powder, and that the dogs salivated before the powder was delivered. He realized that the dogs associated the sound (which occurred seconds before the powder actually arrived) with the delivery of the food. Thus, the dogs had “learned” that when the sound occurred, the meat powder was going to arrive.

   This is conditioning (Stimulus-Response; S-R Bonds). The stimulus (sound of food dispenser) produced a response (salivation). It is important to note that at this point, we are talking about reflexive responses (salivation is automatic).

2. Terminology (if you are still confused by these definitions, please look in the non-Psychology jargon glossary on the AlleyDog.com homepage):
    a) Unconditioned Stimulus (US) - a stimulus that evokes an unconditioned response without any prior conditioning (no learning needed for the response to occur).
    b) Unconditioned Response (UR) - an unlearned reaction/response to an unconditioned stimulus that occurs without prior conditioning.
    c) Conditioned Stimulus (CS) - a previously neutral stimulus that has, through conditioning, acquired the capacity to evoke a conditioned response.
    d) Conditioned Response (CR) - a learned reaction to a conditioned stimulus that occurs because of prior conditioning.
    *These are reflexive behaviours. Not a result from engaging in goal directed behaviour.
    e) Trial - presentation of a stimulus or pair of stimuli.
    Don't worry, we will get to some examples that make this all much more clear.
3. Basic Principles:
    a) Acquisition - formation of a new CR tendency. This means that when an organism learns something new, it has been "acquired".

Pavlov believed in **contiguity** - temporal association between two events that occur closely together in time. The more closely in time two events occurred, the more likely they were to become associated; time passes, association becomes less likely.

For example, when people are house training a dog — you notice that the dog went to the bathroom on the rug,. If the dog had the accident hours ago, it will not do any good to scold the dog because too much time has passed for the dog to associate your scolding with the accident. But, if you catch the dog right after the accident occurred, it is more likely to become associated with the accident.

There are several different ways conditioning can occur — order that the stimulus-response can occur:

1. **Delayed conditioning (forward) - the** CS is presented before the US and it (CS) stays on until the US is presented. This is generally the best, especially when the delay is short.
   example - a bell begins to ring and continues to ring until food is presented.
2. **Trace conditioning -** discrete event is presented, then the US occurs. Shorter the interval the better, but as you can tell, this approach is not very effective.
   example - a bell begins ringing and ends just before the food is presented.
3. **Simultaneous conditioning -** CS and US presented together. Not very good**.**
   example - the bell begins to ring at the same time the food is presented. Both begin, continue, and end at the same time.
4. **Backward conditioning -** US occurs before CS.
   example - the food is presented, then the bell rings. This is not really effective.

b) Extinction - this is a gradual weakening and eventual disappearance of the CR tendency. Extinction occurs from multiple presentations of CS without the US.

Essentially, the organism continues to be presented with the conditioned stimulus but without the unconditioned stimulus the CS loses its power to evoke the CR. For example, Pavlov's dogs stopped salivating when the dispenser sound kept occurring without the meat powder following.

c) Spontaneous Recovery - sometimes there will be a reappearance of a response that had been extinguished. The recovery can occur after a period of non-exposure to the CS. It is called spontaneous because the response seems to reappear out of nowhere.

d) Stimulus Generalization - a response to a specific stimulus becomes associated to other stimuli (similar stimuli) and now occurs to those other similar stimuli.

For Example - a child who gets bitten by black lab, later becomes afraid of all dogs. The original fear evoked by the Black Lab has now generalized to ALL dogs.

Another Example - little Albert (I am assuming you are familiar with Little Albert, so I will give a very general example).

John Watson conditioned a baby (Albert) to be afraid of a white rabbit by showing Albert the rabbit and then slamming two metal pipes together behind Albert's head (nice!). The pipes produced a very loud, sudden noise that frightened Albert and made him cry. Watson did this several times (multiple trials) until Albert was afraid of the rabbit. Previously he would pet the rabbit and play with it. After conditioning, the sight of the rabbit made Albert scream — then what Watson found was that Albert began to show similar terrified behaviours to Watson's face (just looking at Watson's face made Albert cry. What a shock!). What Watson realized was that Albert was responding to the white beard Watson had at the time. So, the fear evoked by the white, furry, rabbit, had generalized to other white, furry things, like Watson's beard.

f) Stimulus Discrimination - learning to respond to one stimulus and not another. Thus, an organisms becomes conditioned to respond to a specific stimulus and not to other stimuli.

For Example - a puppy may initially respond to lots of different people, but over time it learns to respond to only one or a few people's commands.

g) Higher Order Conditioning - a CS can be used to produce a response from another neutral stimulus (can evoke CS). There are a couple of different orders or levels. Let's take a "Pavlovian Dog-like" example to look at the different orders:

In this example, light is paired with food. The food is a US since it produces a response without any prior learning. Then, when food is paired with a neutral stimulus (light) it becomes a Conditioned Stimulus (CS) - the dog begins to respond (salivate) to the light without the presentation of the food.

First order:

1. Light — US (food) —> UR (salivation)
2. Light — US (food) —> CR (salivation)

Second order:

3. Tone — light—> CR (salivation)
4. Tone — light —> CR (salivation)

B. Classical Conditioning in Everyday Life

One of the great things about conditioning is that we can see it all around us. Here are some examples of classical conditioning that you may see:

Conditioned Fear and Anxiety: Many phobias that people experience are the results of conditioning.

For Example - "fear of bridges" - fear of bridges can develop from many different sources. For example, while a child rides in a car over a dilapidated bridge, his father makes jokes about the bridge collapsing and all of them falling into the river below. The father finds this funny and so decides to do it whenever they cross the bridge. Years later, the child has grown up and now is afraid to drive over any bridge. In this case, the fear of one bridge generalized to all bridges which now evoke fear.

Advertising: Modern advertising strategies evolved from John Watson's use of conditioning. The approach is to link an attractive US with a CS (the product being sold) so the consumer will feel positively towards the product just like they do with the US.

US —> CS —> CR/UR

attractive person —> car —> pleasant emotional response

A Clockwork Orange: No additional information necessary! If you haven't seen this movie or read the book, do it. You will find it very interesting, and a wonderful example of conditioning in action.

Operant Conditioning

Operant conditioning is the use of consequences to modify the occurrence and form of behaviour. Operant conditioning is distinguished from classical conditioning (also called respondent conditioning, or Pavlovian conditioning) in that operant conditioning deals with the modification of "voluntary behaviour" or operant behaviour. Operant behaviour "operates" on the environment and is maintained by its consequences, while classical conditioning deals with the conditioning of respondent behaviours which are elicited by antecedent conditions. Behaviours conditioned via a classical conditioning procedure are not maintained by consequences.

Reinforcement, Punishment, and Extinction

Reinforcement and punishment, the core tools of operant conditioning, are either positive (delivered following a response), or negative (withdrawn following a response).

This creates a total of four basic consequences, with the addition of a fifth procedure known as extinction (*i.e.*, no change in consequences following a response)

It's important to note that organisms are not spoken of as being reinforced, punished, or extinguished; it is the response that is reinforced, punished, or extinguished. Additionally, reinforcement, punishment, and extinction are not terms whose use is restricted to the laboratory. Naturally occurring consequences can also be said to reinforce, punish, or extinguish behaviour and are not always delivered by people.

- **Reinforcement** is a consequence that causes a behaviour to occur with greater frequency.
- **Punishment** is a consequence that causes a behaviour to occur with less frequency.
- **Extinction** is the lack of any consequence following a behaviour. When a behaviour is inconsequential, producing neither favourable nor unfavourable consequences, it will occur with less frequency. When a previously reinforced behaviour is no longer reinforced with either positive or negative reinforcement, it leads to a decline in the response.

Four contexts of operant conditioning: Here the terms *"positive"* and *"negative"* are not used in their popular sense, but rather: *"positive"* refers to addition, and *"negative"* refers to subtraction.

What is added or subtracted may be either reinforcement or punishment. Hence *positive punishment* is sometimes a confusing term, as it denotes the addition of punishment (such as spanking or an electric shock), a context that may seem very negative in the lay sense. The four procedures are:

1. **Positive reinforcement** occurs when a behaviour (response) is followed by a favourable stimulus (commonly seen as pleasant) that increases the frequency of that behaviour. In the Skinner box experiment, a stimulus such as food or sugar solution can be delivered when the rat engages in a target behaviour, such as pressing a lever.
2. **Negative reinforcement** occurs when a behaviour (response) is followed by the removal of an aversive stimulus (commonly seen as unpleasant) thereby increasing that behaviour's frequency. In the Skinner box experiment, negative reinforcement can be a loud noise continuously sounding inside the rat's cage until it engages in the target behaviour, such as pressing a lever, upon which the loud noise is removed.
3. **Positive punishment** (also called "Punishment by contingent stimulation") occurs when a behaviour (response) is followed by an aversive stimulus, such as introducing a shock or loud noise, resulting in a decrease in that behaviour.
4. **Negative punishment** (also called "Punishment by contingent withdrawal") occurs when a behaviour (response) is followed by the removal of a favourable stimulus, such as taking away a child's toy following an undesired behaviour, resulting in a decrease in that behaviour.

Also:

- **Avoidance learning** is a type of learning in which a certain behaviour results in the cessation of an aversive stimulus. For example, performing the behaviour of shielding one's eyes when in the sunlight (or going indoors) will help avoid the aversive stimulation of having light in one's eyes.
- **Extinction** occurs when a behaviour (response) that had previously been reinforced is no longer effective. In the Skinner box experiment, this is the rat pushing the lever and being rewarded with a food pellet several times, and then pushing the lever again and never receiving a food pellet again. Eventually the rat would cease pushing the lever.
- **Noncontingent reinforcement** refers to delivery of reinforcing stimuli regardless of the organism's (aberrant) behaviour. The idea is that the target behaviour decreases because it is no longer necessary to receive the reinforcement. This typically entails time-based delivery of stimuli identified as maintaining aberrant behaviour, which serves to decrease the rate of the target behaviour. As no measured behaviour is identified as being strengthened, there is controversy surrounding the use of the term noncontingent "reinforcement".

THORNDIKE'S LAW OF EFFECT

Operant conditioning, sometimes called *instrumental conditioning* or *instrumental learning*, was first extensively studied by Edward L. Thorndike (1874-1949), who observed the behaviour of cats trying to escape from home-made puzzle boxes. When first constrained in the boxes, the cats took a long time to escape. With experience, ineffective responses occurred less frequently and successful responses occurred more frequently, enabling the cats to escape in less time over successive trials.

In his Law of Effect, Thorndike theorized that successful responses, those producing *satisfying* consequences, were "stamped in" by the experience and thus occurred more frequently. Unsuccessful responses, those producing *annoying* consequences, were *stamped out* and subsequently occurred less frequently. In short, some consequences *strengthened* behaviour and some consequences *weakened* behaviour. Thorndike produced the first known learning curves through this procedure. B.F. Skinner (1904-1990) formulated a more detailed analysis of operant conditioning based on reinforcement, punishment, and extinction.

Following the ideas of Ernst Mach, Skinner rejected Thorndike's mediating structures required by "satisfaction" and constructed a new conceptualization of behaviour without any such references. So while experimenting with some homemade feeding mechanisms Skinner invented the operant conditioning chamber which allowed him to measure rate of response as a key dependent variable using a cumulative record of lever presses or key pecks.

OPERANT CONDITIONING VS FIXED ACTION PATTERNS

Skinner's construct of instrumental learning is contrasted with what Nobel Prize winning biologist Konrad Lorenz termed "fixed action patterns," or reflexive, impulsive, or instinctive behaviours. These behaviours were said by Skinner and others to exist outside the parameters of operant conditioning but were considered essential to a comprehensive analysis of behaviour.

Fixed Action Patterns have their origin in the genetic makeup of the animal in question. Examples of "fixed action patterns" include ducklings that will follow any moving object if they see that object within the period of time when the behaviour will be released, or the dance that a bee performs. Characteristics of "fixed action patterns" include not needing to be learned or acquired; these behaviours are performed correctly the first time that they are performed.

Within operant conditioning, Fixed Action Patterns can be used as reinforcers for learned behaviours. Often, fixed action patterns such as predatory grabbing in dogs can be used as a reinforcer. In police and military dog training, the desire to engage in the predatory bite is often used as a reinforcement for successful completion of a search or an obedience exercise. The amount of desire that a dog might have to engage in the fixed action pattern is also known as "prey drive" although this may well be a misnomer as there is no quantification for how much a dog wants to engage in the predatory sequence.

Fixed Action Patterns can also get in the way of successful learning. Bailey and Breland note in their paper "The Mis-Behaviour of Organisms" note that raccoons cannot be taught to place an item in a jar due to the fixed action pattern that is released when they begin to place the item in the jar.

When a component of a learned sequence triggers the beginning of a fixed action pattern, it is difficult and sometimes impossible to interrupt that sequence before it is completed. In this way, teaching raccoons to place items in jars, pigs to fetch (fetching triggers routing behaviours) or young ducklings to sit and stay.

Criticisms

Thorndike's law of effect specifically requires that a behaviour be followed by satisfying consequences for learning to occur.

There are, however, cases in which learning can be shown to occur without good or bad effects following the behaviour. For instance, a number of experiments examining the phenomenon of latent learning showed that a rat needn't receive a satisfying reward (food, if hungry; water, if thirsty) in order to learn a maze; learning that becomes apparent immediately after the desired reward is introduced.

However, views claiming such research invalidates theories of operant conditioning are molecular to a fault. If the rat has a history of "searching behaviour" being reinforced in novel environments, the behaviour will occur in new environments. This is especially plausible in a species which scavenges for food and has thus likely inherited a propensity for searching behaviour to be sensitive to reinforcement. Behaving during initial extinction trials as the organism had during reinforcement trials is not proof of latent learning, as behaviour is a function of the history of the individual organism and its genetic endowment and is never controlled by future consequences. That an organism continues to respond during unreinforced trials has been well-established when studying intermittent schedules of reinforcement.

A different experiment, in humans, showed that "punishing" the correct behaviour may actually cause it to be more frequently taken (*i.e.*, stamp it in). Subjects are given a number of pairs of holes on a large board and required to learn which hole to poke a stylus through for each pair. If the subjects receive an electric shock for punching the correct hole, they learn which hole is correct more quickly than subjects who receive an electric shock for punching the incorrect hole. This cannot, however, be accurately described as punishment if it is increasing the probability of the behaviour.

Chapter Ten

Human Development

Meaning of Human Development

The term 'human development' may be defined as an expansion of human capabilities, a widening of choices, 'an enhancement of freedom, and a fulfilment of human rights. At the beginning, the notion of human development incorporates the need for income expansion. However, income growth should consider expansion of human capabilities. Hence development cannot be equated solely to income expansion. Income is not the sum-total of human life. As income growth is essential, so are health, education, physical environment, and freedom. Human development should embrace human rights, socio-eco-politico freedoms. Based on the notion of human development. Human Development Index (HDI) is constructed. It serves as a more humane measure of development than a strictly income-based benchmark of per capita GNP.

The first UNDP Human Development Report published in 1990 stated that: "The basic objective of development is to create an enabling environment for people to enjoy long, healthy and creative lives." It also defined human development as "a process of enlarging people's choices", "and strengthen human capabilities" in a way which enables them to lead longer, healthier and fuller lives.

From this broad definition of human development, one gets an idea of three critical issues involved in human development interpretation. These are: to lead a long and healthy life, to be educated, and to enjoy a decent standard of living. Barring these three crucial parameters of human development as a process enlarging people's choices, there are additional choices that include political freedoms, other guaranteed human rights, and various ingredients of self-respect.

One may conclude unhesitantly that the absence of these essential choices debars or blocks many other opportunities that people should have in widening their choices. Human development is thus a process of widening people's choices as well as raising the level of well-being achieved.

What emerges from- the above discussion is that economic growth measured in terms of per capita GNP focuses only on one choice that is income. On the other hand, the notion of human development embraces the widening of all human choices—whether economic, social, cultural or political. One may, however, contest GDP/GNP as a useful measure of development since income growth enables persons in expanding their range of choices.

This argument is, however, faulty. Most importantly, human choices go far beyond income expansion. There are so many choices that are not dependent on income. Thus, human development covers all aspects of development. Hence it is a holistic concept. "Economic growth, as such becomes only a subset of human development paradigm."

DEFINITION AND OBJECTIVES OF HUMAN DEVELOPMENT

In the traditional development economics, development meant growth of per capita real income. Later on, a wider definition of development came to be assigned that focused on distributional objectives. Economic development, in other words, came to be redefined in terms of reduction or elimination of poverty and inequality.

These are, after all, 'a goods-oriented' view of development. True development has to be 'people- centred'. When development is defined in terms of human welfare it means that people are put first. This 'people-oriented' view of development is to be called human development.

It is thus clear that per capita income does not stand as a true index of development of any country. To overcome this problem and to understand the dynamics of development, the United Nations Development Programme (UNDP) developed the concept of Human Development Index (HDI) in the 1990s. This index brought in revolutionary changes not only in development, but also in the policy environment in which the government was assigned a major role instead of market forces.

Economic development now refers to expanding capabilities. According to Amartya Sen, the basic objective of development is 'the expansion of human capabilities'. The capability of a person reflects the various combinations of 'doings and beings' that one can achieve. It then reflects that the people are capable of doing or being. Capability thus describes a person's freedom to choose between different ways of living.

Domains of Human Development: Biological, Cognitive and Psychosocial

Human development is a lifelong process beginning before birth and extending to death. At each moment in life, every human being is in a state of personal evolution. Physical changes largely drive the process, as our cognitive abilities advance and decline in response to the brain's growth in childhood and reduced function in old age. Psychosocial development is also significantly influenced by physical growth, as our changing body and brain, together with our environment, shape our identity and our relationships with other people.

In relation to human development, the word "domain" refers to specific aspects of growth and change. Major domains of development include social-emotional, physical, language and cognitive.

Kids often experience a significant and obvious change in one domain at a time, so it may seem that a particular domain is the only one experiencing developmental change during a particular period of life. In fact, however, change typically is also occurring in the other domains but it's occurring gradually and less prominently.

Biological Basis of Behaviour and Development

Like every other aspect of life, behaviour is determined largely by an individual's DNA makeup. DNA can be considered as a "blueprint" for every aspect of life. Evolutionary psychologists study the role of natural selection in the passing on of behavioural and other psychological traits. Genetic makeup is generally believed to account for much of an individual's temperament.

Structure and Function of DNA

The nucleic acids are the largest of the organic molecules. DNA stands for deoxyribonucleic acid. It is made up of units called nucleotides, which themselves are composed of a deoxyribose 5-carbon sugar molecule, a nitrogen base, and a phosphate group.

These nucleotides are bonded in columns by phosphate covalent bonds (very strong), and are laterally bound to other nucleotides with weaker hydrogen bonds. Nucleotides are arranged such that they form genes, which are specific biochemical units that code for a characteristic. In DNA replication, DNA is split and transcribed to form proteins, which are the basic structural units of life. Individual-to-individual variance, introduced in reproduction and meiosis, is the basis for different personality traits and differences in temperament between individuals.

Natural Selection in Behavioural Traits

The theory of natural selection may be applied to psychological, as well as biological traits. This is known as evolutionary psychology or the evolutionary perspective.

The basic idea of evolutionary psychology is this; genetic mutations are capable of altering not only an organism's physical traits, but also its behavioural traits. Like physical traits, these mutations may help the organism reproduce and pass its mutation on to the next generation. This theory explains how behaviours like mating rituals and migration came to be multi-generation rituals in some species.

Controversy Surrounding

Presently, the evolutionary psychology perspective is extremely controversial as it is unable to be scientifically proven in a laboratory setting and is, by definition, very susceptible to hind-sight bias.

Cognitive Development

Cognitive development refers to the acquisition of the ability to reason and solve problems. The main theory of cognitive development was developed by Jean Piaget, a Swiss developmental psychologist. Piaget broke childhood cognitive development into four stages spanning from birth through adolescence. A child who successfully passes through the stages progresses from simple sensorimotor responses to the ability to classify and create series of objects and eventually to engage in hypothetical and deductive reasoning, according to "The New Dictionary of Scientific Biography."

Stages of Cognitive Development

The Sensorimotor Stage: A period of time between birth and age two during which an infant's knowledge of the world is limited to his or her sensory perceptions and motor activities. Behaviours are limited to simple motor responses caused by sensory stimuli.

The Preoperational Stage: A period between ages two and six during which a child learns to use language. During this stage, children do not yet understand concrete logic, cannot mentally manipulate information and are unable to take the point of view of other people.

The Concrete Operational Stage: A period between ages seven and eleven during which children gain a better understanding of mental operations. Children begin thinking logically about concrete events, but have difficulty understanding abstract or hypothetical concepts.

The Formal Operational Stage: A period between age twelve to adulthood when people develop the ability to think about abstract concepts. Skills such as logical thought, deductive reasoning and systematic planning also emerge during this stage.

Psychosocial Development

The primary theory of psychosocial development was created by Erik Erikson, a German developmental psychologist. Erikson divided the process of psychological and social development into eight stages that correspond to the stages of physical development. At each stage, according to Erikson, the individual faces a psychological conflict that must be resolved in order to progress developmentally. Moving from infancy to old age, these conflicts are trust versus mistrust, autonomy versus shame and doubt, initiative versus guilt, industry versus inferiority, identity versus role diffusion, intimacy versus isolation, generativity—that is, creativity and productivity—versus stagnation, and ego integrity versus despair.

The importance of physical, cognitive and psychosocial development becomes apparent when a person does not successfully master one or more of the developmental stages. For example, a child who fails to achieve basic milestones of physical development may be diagnosed with a developmental delay. Similarly, a child with a learning disability may fail to master the complex cognitive processes of a typical adolescent. A middle-aged adult who does not successfully resolve Erikson's stage of generativity versus stagnation may experience "profound personal stagnation, masked by a variety of escapisms, such as alcohol and drug abuse, and sexual and other infidelities," as stated by Nursing Theories. Thus, the stakes are high for all humans as they tackle the developmental tasks they confront at every age.

COMPONENTS OF HUMAN DEVELOPMENT

The noted Pakistani economist Mahbub ul Haq considered four essential pillars of human development.

Equality

If development is viewed in terms of enhancing people's basic capabilities, people must enjoy equitable access to opportunities. Such may be called equality-related capabilities. To ensure equality-related capabilities or access to opportunities what is essential is that the societal institutional structure needs to be more favourable or progressive.

In other words, the unfavourable initial asset distribution, like land, can be made more farmer-friendly through land reform and other redistributive measures. In addition, uneven income distribution may be addressed through various tax-expenditure policies. Economic or legislative- measures that interferes with market exchange may enable people to enlarge their capabilities and, hence, well-being. Further, to ensure basic equality, political opportunities need to be more equal. In the absence of effective political organisation, disadvantaged groups are exploited by the 'rich' to further their own interests rather than social goals. However, participatory politics gets a beating by the inequality in opportunities in having basic education.

It is to be added here that basic education serves as a catalyst of social change. Once the access to such opportunity is opened up in an equitable way, women or religious minorities or ethnic minorities would be able to remove socio-economic obstacles of development. This then surely brings about a change in power relations and makes society more equitable.

Sustainability

Another important facet of human development is that development should 'keep going', should 'last long'. The concept of sustainable development focuses on the need to maintain the long term protective capacity of the biosphere. This then suggests that growth cannot go on indefinitely; there are, of course, 'limits to growth.'

Here we assume that environment is an essential factor of production. In 1987, the Bruntland Commission Report (named after the then Prime Minister Go Harlem Bruntland of Norway) defined sustainable development as '… development that meets the needs of the present without compromising the ability of future generations to meet their basic needs.'

This means that the term sustainability focuses on the desired balance between future economic growth and environmental quality. To attain the goal of sustainable development, what is of great importance is the attainment of the goal of both intra-generation and inter-generation equality.

This kind of inequality includes the term 'social well-being' not only for the present generation but also for the people who will be on the earth in the future. Any kind of environmental decline is tantamount to violation of distributive justice of the disadvantaged peoples. Social well-being thus, then, depends on environmental equality.

Productivity

Another component of human development is productivity which requires investment in people. This is commonly called investment in human capital. Investment in human capital—in addition to physical capital—can add more productivity.

The improvement in the quality of human resources raises the productivity of existing resources. Theodore W. Schultz—the Nobel Prize-winning economist—articulated its importance: "The decisive factors of production in improving the welfare of poor people are not space, energy, and crop land; the decisive factor is the improvement in population quality." Empirical evidence from many East Asian countries corroborate this view.

The Importance of Human Development in Terms of Living Conditions in Different Countries

Human development is very important in terms of living conditions in different countries. The statement "any society committed to improving the lives of its people must also be committed to full and equal rights for all" is true. The UN considers three factors to calculate human development in a country. These factors cover many aspects of a country, including social development in a country. Income, education, and healthy living are considered to be the most important factors in human development, which help to rid populations of poverty, and support human rights. First, the improvement of lives is directly related to human.

People in poverty are usually denied rights, which include civil, political, economic, social, and cultural rights. Human rights are directly related to human development. Discrimination against the poor is generally ignoring or denying their freedom of expression. Labour or communication rights, for example, may not be agreeable for the poor, which could mean unacceptable terms. Thus, the poor will become poorer, while the rich can thrive off of their cheap labour. This creates a larger gap in income distribution in a country. The larger the gap, the more unequal a country is. Without basic labour rights or freedoms, they may not be able to access social or economic rights. In addition to rights, a higher income generally means higher standards of living.

Principles and Factors Affecting on Growth and Development

Principles of Human Growth and Development: The process of human growth and development is described by various set of principles. These principles explain typical development as a predictable and orderly process. Therefore we can easily foretell how most children will develop even though there are differences in children's qualities, behaviour, activity levels, and timing of developmental milestones. To understand human growth and development, we need to understand nature and nurture, and the relationship between the two.

Factors Which Influence Human Growth and Development

The following are a list of factors which influence human growth and development:

- *Heredity:* Heredity and genes certainly play an important role in the transmission of physical and social characteristics from parents to off-springs. Different characteristics of growth and development like intelligence, aptitudes, body structure, height, weight, colour of hair and eyes are highly influenced by heredity.
- *Sex:* Sex is a very important factor which influences human growth and development. There is lot of difference in growth and development between girls and boys. Physical growth of girls in teens is faster than boys. Overall the body structure and growth of girls are different from boys.
- *Socioeconomic:* Socioeconomic factors definitely have some affect. It has been seen that the children from different socioeconomic levels vary in average body size at all ages. The upper level families being always more advanced. The most important reasons behind this are better nutrition, better facilities, regular meals, sleep, and exercise. Family size also influences growth rate as in big families with limited income sometimes have children that do not get the proper nutrition and hence the growth is affected.

- *Nutritional:* Growth is directly related with nutrition. The human body requires an adequate supply of calories for its normal growth and this need of requirements vary with the phase of development. As per studies, malnutrition is referred as a large-scale problem in many developing countries. They are more likely to be underweight, much shorter than average, and of low height for age, known as stunting.
- If the children are malnourished, this slows their growth process. There are nine different amino acids which are necessary for growth and absence of any one will give rise to stunted growth. Other factors like zinc, Iodine, calcium, phosphorus and vitamins are also essential for proper growth and deficiency of anyone can affect the normal growth and development of the body.
- Hormones: There are a large number of endocrine glands present inside our body. These glands secrete one or more hormones directly into the bloodstream. These hormones are capable of raising or lowering the activity level of the body or some organs of the body. Hormones are considered to be a growth supporting substance. These hormones play an important role in regulating the process of growth and development.
- Pollution: According to studies, air pollution not only affects the respiratory organs but also have harmful effects on human growth.

HUMAN GROWTH AND DEVELOPMENT

Definition

In the context of the physical development of children, growth refers to the increase in the size of a child, and development refers to the process by which the child develops his or her psychomotor skills.

Growth

The period of human growth from birth to adolescence is commonly divided into the following stages:

Microscope Stages

Infancy: From birth to weaning.
Childhood: From weaning to the end of brain growth.
Juvenile: From the end of childhood to adolescence.
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Personal advisors stay with your from application to graduation. Ranked by U.S. News and World Report among the best universities offering online degrees.

Adolescence: From the start of growth spurt at puberty until sexual maturity.

Growth curves are used to measure growth. The distance curve is a measure of size over time; it records height as a function of age and gets higher with age. The velocity curve measures the rate of growth at a given time for a particular body feature (such as height or weight). The height velocity curve is highest in infancy, up to two years of age, with more consistent annual growth afterwards and increases again at puberty. The height of the average infant increases by 30% by the age of five months and by 50% by the age of one year. The height of a five-year-old usually doubles relative to that at birth. The limbs and arms grow faster than the trunk, so that body proportions undergo marked variation as an infant grows into an adolescent. Different body systems grow and develop at different rates. For example, if infants grew in height as quickly as they do in weight, the average one-year-old would be approximately 5 ft (1.5m) tall. Thus, weight increases faster than height—an average infant doubles his birth weight by the age of five months and triples it by the age of one year. At two years of age, the weight is usually four times the weight at birth.

Physical Development

During the growth period, all major body systems also mature. The major changes occur in the following systems:

Skeletal system. At birth, there is very little bone mass in the infant body, the bones are softer (cartilagenous) and much more flexible than in the adult. The adult skeleton consists of 206 bones joined to ligaments and tendons. It provides support for the attached muscles and the soft tissues of the body. Babies are born with 270 soft bones that eventually fuse together by the age of 20 into the 206 hard, adult bones.

Lymphatic system. The lymphatic system has several functions. It acts as the body's defence mechanism by producing white blood cells and specialized cells (antibodies) that destroy foreign organisms that cause disease. It grows at a constant and rapid rate throughout childhood, reaching maturity just before puberty. The amount of lymphatic tissue then decreases so that an adult has approximately 50% less than a child.

Central nervous system (CNS). The CNS consists of the brain, the cranial nerves, and the spinal cord. It develops mostly during the first years of life. Although brain cell formation is almost complete before birth, brain maturation continues after birth. The brain of the newborn is not yet fully developed. It contains about 100 billion brain cells that have yet to be connected into functioning networks. But brain development up to age one is more rapid and extensive than was previously realized. At birth, the brain of the infant is 25% of the adult size. At the age of one year, the brain has grown to 75% of its adult size and to 80% by age three, reaching 90% by age seven. The influence of the early environment on brain development is crucial. Infants exposed to good nutrition, toys, and playmates have better brain function at age 12 than those raised in a less stimulating environment.

Psychomotor Development

During the first year of life, a baby goes through a series of crucial stages to develop physical coordination. This development usually proceeds cephalocaudally, that is from head to toe. For example, the visual system reaches maturity earlier than do the legs. First, the infant develops control of the head, then of the trunk (sitting up), then of the body (standing), and, finally, of the legs (walking). Development also proceeds proximodistally, that is from the centre of the body outward. For example, the head and trunk of the body develop before the arms and legs, and infants learn to control their neck muscles before they learn to direct their limbs.

This development of physical coordination is also referred to as motor development and it occurs together with cognitive development, meaning the development of processes such as knowing, learning, thinking, and judging.

The stages of motor development in children are as follows:

First year. The baby develops good head balance and can see objects directly in his line of vision. He learns how to reach for objects and how to transfer them from one hand to the other. Sitting occurs at six months of age. Between nine and 10 months, the infant is able to pull himself to standing and takes his first steps. By the age of eight to 24 months, the baby can perform a variety of tasks such as opening a small box, making marks with a pencil, and correctly inserting squares and circles in a formboard. He is able to seat himself in small chair, he can point at objects of interest, and can feed himself with a spoon.

Second year. At 24-36 months, the child can turn the pages of a book, scribble with a pencil, build towers with blocks up to a height of about seven layers, and complete a formboard with pieces that are more complex than circles or squares. He can kick a ball, and walks and runs fairly well, with a good sense of balance. Toilet training can be started.

Third year. The child can now draw circles, squares, and crosses. He can build 10-block towers and imitate the building of trains and bridges. He is also achieving toilet independence. Hand movements are well coordinated and he can stand on one foot.

Four years. At that age, a child can stand heel to toe for a good 15 seconds with his eyes closed. He can perform the finger-to-nose test very well, also with eyes closed. He can jump in place on both feet.

Five years. The child can balances on tiptoe for a 10-second period, he can hops on one foot, and can part his lips and clench his teeth.

Six years. The child can balance on one foot for a 10-second period, he can hit a target with a ball from 5 ft (1.5 m), and jumps over a rope 8 in (20 cm) high.

Seven years. He can now balance on tiptoes for a 10-second period, bend at the hips sideways, and walk a straight line, heel-to-toe for a distance of 6 ft (1.8 m).

Eight years. The child can maintain a crouched position on tiptoes for a 10-second period, with arms extended and eyes closed. He is able to touch the fingertips of one hand with his thumb, starting with the little finger and repeating in reverse order.

Changes and Factors Affecting Physical Development

Physical development and growth are influenced by both genetic and environmental factors. For example, malnutrition can delay a child's physical development significantly. On the other hand, according to the University of Minnesota, the role of some environmental factors, such as the amount of exercise the child is getting, has a much smaller effect on physical development than was previously thought.

Genetics

When a child is born, he has a unique set of genetic instructions that influence his physical growth. According to the University of Minnesota, genetics have a strong effect on rate of growth, the size of body parts and the onset of growth events. In one study, Dr. Stefan A. Czerwinski and colleagues followed their subjects for thirty years. By using such parental measurements as height and weight, these scientists were able to predict quite accurately the approximate height and weight of their subjects at the age of thirty. Other factors found to be closely linked to their parental values were blood pressure and body fat percentage, as well as muscle and total body mass. The study was published in the "American Journal of Human Biology" in September 2007.

Environment

Genetics alone, however, cannot determine the physical development of the child. The Minnesota Twin Studies have shown, for example, even identical twins who share the same genes can grow up to be of different height if they are raised in different environments. Such environmental factors as nutrition can affect physical growth significantly. According to the United Nations University, malnutrition can delay physical growth and development. It can also affect the quality and texture of bones and teeth, the size of body parts and delay the adolescent growth spurt. If the child gets better nutrition later on, she may be able to catch up, depending on how severe the malnutrition was. Besides diet, other environmental factors such as climate and toxins can also affect physical development.

Chronic Illnesses

Serious, chronic illness and surgeries have been shown to have a negative effect on a child's physical development. Dr. M. L. Cepeda and colleagues, for example, studied 30 subjects with homozygous sickle cell disease from age eight through 19. In their study, published in the "Journal of the National Medical Association" in January 2000, the authors reported that their subjects were significantly shorter and of lower weight than their healthy controls. The sexual development was also delayed in adolescents with sickle cell disease.

Factors That Affect: A Child's Growth And Development

Your child's growth and development involve all the changes that his/her body and mind go through. As a parent, it is important that you know about the factors that determine these changes:

Physical Development

Your child's physical development consists of the height, weight, development of muscles and bones. The following factors affect a child's physical development:

Genetics

Your child inherits your physical characteristics through genes. If your child is carrying a dominant gene related to height, he/she might achieve the same height as you. However, there are some disorders and health issues that are passed on genetically such as heart diseases, diabetes, obesity, etc. These problems can affect your child's weight, muscle and bone development, etc.

Nutrition

Your child's body needs the right amount of Carbohydrates, Proteins, Vitamins, and Minerals for his/her physical development. For example, Calcium is important for the development of bones, Vitamin E for developing the immune system, iron is important for immunity, proteins build body strength and Vitamin A is essential for vision in poor lighting. If not provided proper nutrition, your child's growth might get stunted.

Environment

Long time exposure to pollutants like lead, manganese, mercury, arsenic and pesticides through either land, water or food has been known to deteriorate growth, cause physical abnormalities, weaken the immune system and affect motor skills in children.

Mental And Cognitive Development

The mental and cognitive development of your child consists of the development of the brain, neurological processes, intellectual abilities, learning abilities, etc. The following factors affect a child's mental and cognitive development:

Nutrition

If your child doesn't get the right nutrition, it will adversely affect his/her mental development. For example, deficiency of Iodine, Iron, Vitamin B12 and Folic acid can cause mental retardation and in extreme cases, even neurological damage. Nutritional drinks like Horlicks can help you bridge the gap between nutritional requirements and consumption by the child.

Environment

Lead, manganese, arsenic poisoning has been known to cause cognitive dysfunction in children especially at the age of 6 years to 12 years. Mercury poisoning can lead to impaired language skills, attention and memory.

Emotional and Social Development

A child's emotional development consists of understanding, expressing and managing his/her emotions, while social development includes his/her behaviour with other people. The following factors affect a child's emotional and social development:

Parenting and Culture

Studies show that harsh parenting practices - like shouting, rough punishing, shaming - can cause the child to either become unusually indifferent or hypersensitive. On the other hand, a loving, caring, trustful relationship with your child can help him/her become an emotionally mature individual.

Cultural norms determine a child's social interactions. For example, in many cultures aggression is more acceptable in men than in women so growing in a certain cultural setting, a boy might grow more aggressive. While the above-mentioned factors play a crucial role in different aspects of your child's growth and development, you play an active role in facilitating. Hence, it is imperative that you take the necessary steps for his/her all round development.

Motor Development: Meaning and Skills

Motor Development: When babies are born, they are not able to move much on their own. Over time, a baby learns to move many parts of its body and control its muscles so it can hold its head up, sit up by itself, stand up, or pick up a toy. The process of motor development, however, does not happen overnight. Like many things, learning about the body and making it move takes time. Motor development is the process of learning how to use muscles in the body to move. The progression of acquiring motor skills goes from simple to complex.

Motor development happens in a predictable sequence of events for most children, but each child varies in age when each skill is mastered. For example, although most children begin to walk independently around twelve to fourteen months, some children are walking as early as nine months.

Further, children differ in terms of the length of time it takes to develop certain motor skills, such as the baby who sits up, virtually skips crawling, and begins walking.

Transition from Reflex Movement to Voluntary Movement

At birth, babies have very little control over their bodies. They spend most of their time curled up in what is called a fetal position. This position is how the baby lay in the womb during the nine months of the mother's pregnancy. In addition to the fetal position, primitive reflexes dominate virtually all of a newborn baby's movements. Babies are born with these reflexive movements as a means for basic life preservation. The reflexes, which are controlled by lower levels of the brain, eventually give way to more sophisticated voluntary movements monitored by higher levels of the brain. The first voluntary task of a newborn is learning to bring the arms and legs out straight in order to lie flat. This manoeuvre takes a lot of muscular energy, so the baby begins moving arms and legs around during the waking hours to develop coordination and strength. As babies move their bodies more in the first months of life, motor pathways begin to form in the brain. These pathways allow a baby to eventually perform motor movements without conscious thought.

Principles of Development

The process of motor development depends heavily on the maturation of the central nervous system and the muscular system. As these systems develop, an infant's ability to move progresses. The sequence of motor development follows an apparently orderly pattern. Arnold Gesell, a noted researcher in the field of child development, indicated through his studies that development does not proceed in a straight line. Instead, it swings back and forth between periods of rapid and slower maturation. Gesell and his colleagues also discovered from their infant observations made in the 1930s and 1940s that infant growth does indeed follow distinct developmental directions: cephalocaudal, proximal-distal, and general to specific.

Cephalocaudal Principle

First, most children develop from head to toe, or cephalocaudal. Initially, the head is disproportionately larger than the other parts of the infant's body. The cephalocaudal theory states that muscular control develops from the head downward: first the neck, then the upper body and the arms, then the lower trunk and the legs. Motor development from birth to six months of age includes initial head and neck control, then hand movements and eye-hand coordination, followed by preliminary upper body control. The subsequent six months of life include important stages in learning to control the trunk, arms, and legs for skills such as sitting, crawling, standing, and walking.

Proximal-Distal Principle

Second, children develop their motor skills from the centre of their bodies outward, near to far or proximal-distal. This principle asserts that the head and trunk develop before the arms and legs, and the arms and legs before the fingers and toes. Babies learn to master control of upper arms and upper legs, then forearms and legs, then their hands and feet, and finally fingers and toes. An example of this is an infant's need to control the arm against gravity before being able to reach for a toy.

General to Specific Principle

Lastly, the general to specific development pattern is the progression from the entire use of the body to the use of specific body parts. This pattern can be best seen through the learned process of grasping. Initially, infants can grossly hold a bottle with both hands at about four months of age. After practice and time, twelve-month-old infants can hold smaller toys or food in each hand using a pincher grasp. This finger and thumb grasp is more precise than the grasping skill of an infant at four months. Just as the child develops a more precise grasp with time and experience, many other motor skills are achieved simultaneously throughout motor development. Each important skill mastered by an infant is considered a motor milestone.

Motor Milestones

Motor milestones are defined as the major developmental tasks of a period that depend on movement by the muscles. Examples of motor milestones include the first time a baby sits alone, takes a step, holds a toy, rolls, crawls, or walks. As discussed previously, the timing of the accomplishment of each motor milestone will vary with each child. "Motor milestones depend on genetic factors, how the mother and father progressed through their own development, maturation of the central nervous system, skeletal and bone growth, nutrition, environmental space, physical health, stimulation, freedom and mental health" (Freiberg 1987; Paplia and Wendkosolds 1987). Within the motor milestones exist two forms of motor development: gross motor development and fine motor development. These two areas of motor development allow an infant to progress from being helpless and completely dependent to being an independently mobile child.

Gross Motor Development

Gross motor development involves skills that require the coordination of the large muscle groups of the body, such as the arms, legs, and trunk. Examples of gross motor skills include sitting, walking, rolling, standing, and much more. The infant's gross motor activity is developed from movements that began while in the womb and from the maturation of reflex behaviour. With experience, the infant slowly learns head control, then torso or trunk control, and then is rolling, sitting, and eventually walking. The first year of a baby's life is filled with major motor milestones that are mastered quickly when compared to the motor milestone achievements of the rest of the baby's development. In addition to the development of gross motor skills, a baby is simultaneously learning fine motor skills.

Fine Motor Development

Fine motor development is concerned with the coordination of the smaller muscles of the body, including the hands and face. Examples of fine motor skills include holding a pencil to write, buttoning a shirt, and turning pages of a book. Fine motor skills use the small muscles of both the hands and the eyes for performance. For the first few months, babies spend a majority of time using their eyes rather than their hands to explore their environment. The grasping reflex, which is present at birth, is seen when a finger is pressed into the baby's palm; the baby's fingers will automatically curl around the person's finger. This grasping reflex slowly integrates and allows the development of more mature grasping patterns. At four months, babies will begin to more frequently reach out for toys with their arms and hands. The reach looks more like a swipe because the baby is learning how to control the arm and hand. Over time, babies learn how to make smoother and coordinated movements with their arms and hands.

Assessment of Gross Motor and Fine Motor Development

Many assessment tools exist to measure a child's performance in regard to gross and fine motor skills. Each assessment requires good observational skills from the evaluator, who is typically a developmental paediatrician, nurse, educator, occupational therapist, or physical therapist. Some assessments call for each item to be administered in a formal standardized manner, so that each child is tested the same way every time. These tests are also called normative-based because they compare individual performance to that of other children. Other measures encourage professionals to ask parents questions about their child and are based on informal observations of the child at play. These more informal tests are referred to as criterion-based assessments because they compare individual performance to a criterion or standard. Regardless of the type of assessment, each measure has the common purpose of evaluating the child's current ability to perform motor-related tasks. Professionals use the results of these assessments to decide whether intervention is needed and also to guide goal setting and outcome measurement.

Delays in Gross Motor and Fine Motor Development

When children are not able to perform the motor skills at the appropriate milestones, their motor development may need to be evaluated by a professional. When motor skills do not progress along a normal trend, a child may be at risk for missing out on potential learning and social experiences.

Children who demonstrate potential motor delays are at risk for continuing these delays throughout later development. For example, a child who demonstrates weak hand strength and has difficulty coordinating finger movement may have trouble with handwriting in school.

Early Intervention

After 1986, legislation was passed at the state level to set up services that assist families who suspect their child may have some developmental delays. These services are called early intervention systems. The main purpose of early intervention is to offer evaluation and treatment to children from birth to age three and to their families.

The professionals involved with early intervention are members of a team who test a child's skills to see where the child's current skills are in relation to the chronological age. Children who are not doing many motor activities typical of their age may be considered at risk or delayed. These children may not have the strength, coordination, or balance to do most things that others of their age can do.

The professionals involved in early intervention include occupational therapists, physical therapists, speech language pathologists, special education teachers, nurses, doctors, social workers, and service coordinators. Each of these professionals help the child and family learn about ways to improve motor coordination so the child can function more independently.

Motor Skill

A motor skill is a learned ability to cause a predetermined movement outcome with maximum certainty. Motor learning is the relatively permanent change in the ability to perform a skill as a result of practice or experience. Performance is an act of executing a motor skill. The goal of motor skills is to optimize the ability to perform the skill at the rate of success, precision, and to reduce the energy consumption required for performance. Continuous practice of a specific motor skill will result in a greatly improved performance, but not all movements are motor skills.

Types of Motor Skills

Motor skills are movements and actions of the muscles. Typically, they are categorized into two groups:

- Gross motor skills – require the use of large muscle groups to perform tasks like walking, balancing, and crawling. The skill required is not extensive and therefore are usually associated with continuous tasks. Much of the development of these skills occurs during early childhood. The performance level of gross motor skill remains unchanged after periods of non-use. Gross motor skills can be further divided into two subgroups: locomotor skills, such as running, jumping, sliding, and swimming; and object-control skills such as throwing, catching and kicking.
- Fine motor skills – requires the use of smaller muscle groups to perform smaller movements with the wrists, hands, fingers, and the feet and toes. These tasks that are precise in nature, like playing the piano, writing carefully, and blinking. Generally, there is a retention loss of fine motor skills over a

period of non-use. Discrete tasks usually require more fine motor skill than gross motor skills. Fine motor skills can become impaired. Some reasons for impairment could be injury, illness, stroke, congenital deformities, cerebral palsy, and developmental disabilities. Problems with the brain, spinal cord, peripheral nerves, muscles, or joints can also have an effect on fine motor skills, and decrease control.

Development

Motor skills develop in different parts of a body along three principles:

- Cephalocaudal – development from head to foot. The head develops earlier than the hand. Similarly, hand coordination develops before the coordination of the legs and feet. For example, an infant is able to follow something with their eyes before they can touch or grab it.
- Proximodistal – movement of limbs that are closer to the body develop before the parts that are further away, such as a baby learns to control the upper arm before the hands or fingers. Fine movements of the fingers are the last to develop in the body.
- Gross to specific – a pattern in which larger muscle movements develop before finer movements. For example, a child only being able to pick up large objects, to then picking up an object that is small between the thumb and fingers. The earlier movements involve larger groups of muscles, but as the child grows finer movements become possible and specific things can be achieved.

In children, a critical period for the acquisition of motor skills is preschool years (ages 3–5), as fundamental neuroanatomic structure shows significant development, elaboration, and myelination over the course of this period. Many factors contribute to the rate that children develop their motor skills. Unless afflicted with a severe disability, children are expected to develop a wide range of basic movement abilities and motor skills. Motor development progresses in seven stages throughout an individual's life: reflexive, rudimentary, fundamental, sports skill, growth and refinement, peak performance, and regression. Development is age-related but is not age dependent. In regard to age, it is seen that typical developments are expected to attain gross motor skills used for postural control and vertical mobility by 5 years of age.

There are six aspects of development:

- Qualitative – changes in movement-process results in changes in movement-outcome.
- Sequential – certain motor patterns precede others.
- Cumulative – current movements are built on previous ones.
- Directional – cephalocaudal or proximodistal
- Multifactorial – numerous-factors impact
- Individual – dependent on each person

In the childhood stages of development, gender differences can greatly influence motor skills. In the article "An Investigation of Age and Gender Differences in Preschool Children's Specific Motor Skills", girls scored significantly higher than boys on visual motor and graphomotor tasks. The results from this study suggest that girls attain manual dexterity earlier than boys. Variability of results in the tests can be attributed towards the multiplicity of different assessment tools used. Furthermore, gender differences in motor skills are seen to be affected by environmental factors. In essence, "parents and teachers often encourage girls to engage in [quiet] activities requiring fine motor skills, while they promote boys' participation in dynamic movement actions". In the journal article "Gender Differences in Motor Skill Proficiency From Childhood to Adolescence" by Lisa Barrett, the evidence for gender-based motor skills is apparent. In general, boys are more skillful in object control and object manipulation skills. These tasks include throwing, kicking, and catching skills. These skills were tested and concluded that boys perform better with these tasks. There was no evidence for the difference in locomotor skill between the genders, but both are improved in the intervention of physical activity. Overall, the predominance of development was on balance skills (gross motor) in boys and manual skills (fine motor) in girls.

Components of Development

- Growth – increase in the size of the body or its parts as the individual progresses towards maturity (quantitative structural changes)
- Maturation – refers to qualitative changes that enable one to progress to higher levels of functioning; it is primarily innate
- Experience or learning – refers to factors within the environment that may alter or modify the appearance of various developmental characteristics through the process of learning
- Adaptation – refers to the complex interplay or interaction between forces within the individual (nature) and the environment (nurture)

Influences on Development

- Stress and arousal – stress and anxiety is the result of an imbalance between demand and the capacity of the individual. In this context, arousal defines the amount of interest in the skill. The optimal performance level is moderate stress or arousal. An example of an insufficient arousal state is an overqualified worker performing repetitive jobs. An example of excessive stress level is an anxious pianist at a recital.
- Fatigue – the deterioration of performance when a stressful task is continued for a long time, similar to the muscular fatigue experienced when exercising rapidly or over a long period. Fatigue is caused by over-arousal. Fatigue impacts an individual in many ways: perceptual changes in which visual acuity

or awareness drops, slowing of performance (reaction times or movements speed), irregularity of timing, and disorganization of performance.

- Vigilance – the effect of the loss of vigilance is the same as fatigue, but is instead caused by a lack of arousal. Some tasks include actions that require little work and high attention.
- Gender – gender plays an important role in the development of the child. Girls are more likely to be seen performing fine stationary visual motor-skills, whereas boys predominantly exercise object-manipulation skills. While researching motor development in preschool-aged children, girls were more likely to be seen performing skills such as skipping, hopping, or skills with the use of hands only. Boys were seen to perform gross skills such as kicking or throwing a ball or swinging a bat. There are gender-specific differences in qualitative throwing performance, but not necessarily in quantitative throwing performance. Male and female athletes demonstrated similar movement patterns in humerus and forearm actions but differed in trunk, stepping, and backswing actions.

Stages of Motor Learning

Motor learning is a change, resulting from practice. It often involves improving the accuracy of movements both simple and complex as one's environment changes. Motor learning is a relatively permanent skill as the capability to respond appropriately is acquired and retained.

The stages of motor learning are the cognitive phase, the associative phase, and the autonomous phase.

- Cognitive phase – When a learner is new to a specific task, the primary thought process starts with, "What needs to be done?" Considerable cognitive activity is required so that the learner can determine appropriate strategies to adequately reflect the desired goal. Good strategies are retained and inefficient strategies are discarded. The performance is greatly improved in a short amount of time.
- Associative phase – The learner has determined the most-effective way to do the task and starts to make subtle adjustments in performance. Improvements are more gradual and movements become more consistent. This phase can last for a long time. The skills in this phase are fluent, efficient, and aesthetically pleasing.
- Autonomous phase – This phase may take several months to years to reach. The phase is dubbed "autonomous" because the performer can now "automatically" complete the task without having to pay any attention to performing it. Examples include walking and talking or sight reading while doing simple arithmetic.

Law of Effect

Motor-skill acquisition has long been defined in the scientific community as an energy-intensive form of stimulus-response (S-R) learning that results in robust neuronal modifications. In 1898, Thorndike proposed the law of effect, which states that the association between some action (R) and some environmental condition (S) is enhanced when the action is followed by a satisfying outcome (O). For instance, if an infant moves his right hand and left leg in just the right way, he can perform a crawling motion, thereby producing the satisfying outcome of increasing his mobility. Because of the satisfying outcome, the association between being on all fours and these particular arm and leg motions are enhanced. Further, a dissatisfying outcome weakens the S-R association. For instance, when a toddler contracts certain muscles, resulting in a painful fall, the child will decrease the association between these muscle contractions and the environmental condition of standing on two feet.

Feedback

During the learning process of a motor skill, feedback is the positive or negative response that tells the learner how well the task was completed. Inherent feedback: after completing the skill, inherent feedback is the sensory information that tells the learner how well the task was completed. A basketball player will note that he or she made a mistake when the ball misses the hoop. Another example is a diver knowing that a mistake was made when the entry into the water is painful and undesirable. Augmented feedback: in contrast to inherent feedback, augmented feedback is information that supplements or "augments" the inherent feedback. For example, when a person is driving over a speed limit and is pulled over by the police. Although the car did not do any harm, the policeman gives augmented feedback to the driver in order for him to drive more safely. Another example is a private tutor for a new student in a field of study. Augmented feedback decreases the amount of time to master the motor skill and increases the performance level of the prospect. Transfer of motor skills: the gain or loss in the capability for performance in one task as a result of practice and experience on some other task. An example would be the comparison of initial skill of a tennis player and non-tennis player when playing table tennis for the first time. An example of a negative transfer is if it takes longer for a typist to adjust to a randomly assigned letter of the keyboard compared to a new typist. Retention: the performance level of a particular skill after a period of no use.

The type of task can have an effect on how well the motor skill is retained after a period of non-use:

- Continuous tasks – activities like swimming, bicycling, or running; the performance level retains proficiency even after years of non-use.
- Discrete tasks – an instrument, video game, or a sport; the performance level drops significantly but will be better than a new learner. The relationship between the two tasks is that continuous tasks usually use gross motor skills and discrete tasks use fine motor skills.

Brain Structures

The regions of the frontal lobe responsible for motor skill include the primary motor cortex, the supplemental motor area, and the premotor cortex. The primary motor cortex is located in the precentral gyrus and is often visualized as the motor homunculus. By stimulating certain areas of the motor strip and observing where it had an effect, Penfield and Rassmussen were able to map out the motor homunculus. Areas on the body that have complex movements, such as the hands, have a bigger representation on the motor homunculus.

The supplemental motor area, which is just anterior to the primary motor cortex, is involved with postural stability and adjustment as well as coordinating sequences of movement. The premotor cortex, which is just below the supplemental motor area, integrates sensory information from the posterior parietal cortex and is involved with the sensory-guided planning of movement and begins the programming of movement.

The basal ganglia are an area of the brain where gender differences in brain physiology is evident. The basal ganglia are a group of nuclei in the brain that is responsible for a variety of functions, some of which include movement. The globus pallidus and putamen are two nuclei of the basal ganglia which are both involved in motor skills.

The globus pallidus is involved with the voluntary motor movement, while the putamen is involved with motor learning. Even after controlling for the naturally larger volume of the male brain, it was found that males have a larger volume of both the globus pallidus and putamen.

The cerebellum is an additional area of the brain important for motor skills. The cerebellum controls fine motor skills as well as balance and coordination. Although women tend to have better fine motor skills, the cerebellum has a larger volume in males than in females, even after correcting for the fact that males naturally have a larger brain volume.

Hormones are an additional factor that contributes to gender differences in motor skill. For instance, women perform better on manual dexterity tasks during times of high oestradiol and progesterone levels, as opposed to when these hormones are lo such as during menstruation

ıAn evolutionary perspective is sometimes drawn upon to explain how gende differences in motor skills may have developed, although this approach is controversial For instance, it has been suggested that men were the hunters and provided food for th family, while women stayed at home taking care of the children and doing domesti work. Some theories of human development suggest that men's tasks involved gros motor skill such as chasing after prey, throwing spears and fighting. Women, on th other hand, used their fine motor skills the most in order to handle domestic tools an accomplish other tasks that required fine motor-control.

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